

The Swords of Armageddon

U.S. Nuclear Weapons Development since 1945

VOLUME III

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THERMONUCLEAR WEAPONS DEVELOPMENT, 1942 - SPRING 1952

Overview

The origins of modern U.S. thermonuclear weapons can be traced to the earliest days of the Manhattan Project. However, the successful development of a working, practical, and deliverable fusion weapon was not nearly so certain as the likelihood of a practical fission weapon, even as unlikely as that seemed at times. Fission on a very small scale had been demonstrated in laboratories as early as 1939.

An atom of uranium-235 could be split at room temperature and at sea level pressure. Several techniques to assemble a large mass of fissionable material to initiate and sustain a chain reaction were reasonably obvious, based on known conventional ordnance principles. Large-scale fusion, on the other hand, was a completely theoretical concept, with no clear or foreseeable method to accomplish it. In addition, very little was known about the fuels required for it.

Work on the hydrogen bomb started in the U.S. in 1942. It continued at the Los Alamos Scientific Laboratory on a very limited basis, due to wartime pressure to build an atomic bomb, through the end of World War II. An important review meeting was held at Los Alamos in April 1946, summarizing what was known about fusion at that time, and predicting the probability for the successful design of a weapon.

The project remained at a relatively low level of activity until January 1950, following the public announcement of the first Soviet atomic explosion. A major design breakthrough was discovered early in 1951; soon afterwards, a related experiment tested fusion principles in a small-scale test explosion.

The first large-scale U.S. fusion explosion occurred in November 1952. It was produced by a heavy, multistage liquid-fueled test device which was not a deliverable weapon by any definition of the word and which depended upon fission for the majority of its power. Sixteen months later, in the spring of 1954, the first solid-fueled U.S. "emergency capability" weapons were tested. These shots included the highest-yield American thermonuclear explosion ever announced by the United States; this high yield was produced largely by accident.

Approximately two years later, the first airdropped stockpiled U.S. hydrogen bomb was detonated over Bikini atoll at the U.S. Atomic Energy Commission's Pacific Proving Grounds. Subsequent test series in 1958 and 1962 further advanced and refined the state-of-the art of hydrogen weaponry.

The following thermonuclear weapons history is sequenced chronologically and traces several aspects of the subject in parallel, including the development of hydrogen weapons in general; "clean" low-fission and "dirty" high-fission devices; high-yield thermonuclear weapons; fuel materials and the construction of special plants to produce those materials; "enhanced radiation" thermonuclears; the corresponding development of fission weapons; weapons testing; and coordination and cooperation between the U.S. and the United Kingdom regarding postwar weapons development.

1. THE SEARCH FOR FEASIBILITY, 1942 - 1949

Wartime Development

As early as 1922, the possibility of using some form of hydrogen to liberate vast amounts of energy was predicted. Even then, thermonuclear power was foreseen to be a risky proposition:

Should the research worker of the future discover some means of releasing...(the energy from hydrogen) in a form which could be employed, the human race would have at its command powers beyond the dreams of scientific fiction, but the remote possibility must always be considered that the energy, once liberated, will be completely uncontrollable and by its violence detonate a neighboring substance. If this happens, all of the hydrogen on earth might be transformed (into helium) at once, and this most successful experiment might be published to the rest of the universe in the form of a new star of extraordinary brilliance, as the earth blew up in one vast explosion.¹

An important milestone in the development of fusion weapons occurred ten years later when Dr. Harold C. Urey discovered "heavy water," formulated D₂O instead of H₂O to show the presence of deuterium, a heavy isotope of hydrogen with a neutron as well as a proton in its nucleus.

Deuterium constitutes about 0.02% of the world's seawater, a ratio of about one gram for every seven gallons of water, from which it can be separated by electrolysis at the relatively low cost of \$4,500 per kilogram. In 1952, deuterium cost about \$0.75 per gram.²

¹AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 175.

²September 12, 1952 Walker memorandum, p. 35.

In theory at least, nuclei of deuterium -- if raised to a high enough energy level (temperature) -- would fuse with one another a million times faster than the nuclei of ordinary hydrogen. This idea was intriguing, but of little value in the absence of a means to generate the requisite high ignition temperatures.

Another important discovery two years later, by British physicists Ernest Rutherford and Mark Oliphant, and an Austrian, P. Harteck, revealed a second thermonuclear fuel. This was tritium, an unstable radioactive isotope of hydrogen with a half life of 12.3 years. A tritium nucleus, or triton, was composed of a proton and two neutrons. This new isotope was found when deuterium was bombarded by high-energy deuterons (deuterium nuclei) forming tritium and protons, or helium nuclei (see Equation H under "Fusion Physics" in Volume I). That same year (1934), Urey received a Nobel Prize for his discovery of deuterium.

The first serious conjecture about the use of hydrogen fusion in a weapon dated from the late summer or early fall of 1941 when Edward Teller and Enrico Fermi were at a luncheon meeting at Columbia University in New York. Fermi rather casually raised the idea that an exploding fission bomb might be used to provide the energy necessary to initiate a fusion reaction in deuterium. This offhand comment became the basis of a ten-year-long messianic obsession by Teller with the concept of the "Super," as the nascent hydrogen bomb came to be known.³

Some preliminary discussion of the theoretical possibility of the "Super" (and the potentially cataclysmic consequences of the detonation of a thermonuclear device, as noted in 1922) occurred at a conference of nuclear physicists in Berkeley, California in the summer of 1942.⁴

One of the early steps away from the theoretical and towards the real development of a terrestrial thermonuclear reaction was taken in May 1942 when Arthur H. Compton requested J. Robert Oppenheimer to compile data covering the basic nuclear reactions produced by fast neutrons. Oppenheimer subsequently assembled a group of theoreticians at the University of California at Berkeley in July.

³Goodchild, p. 53; ENERGY & CONFLICT: THE LIFE AND TIMES OF EDWARD TELLER, Stanley A. Blumberg and Gwinn Owens, G. P. Putnam's Sons, New York, New York, 1976, pp. 109-110. Teller later claimed that this conversation with Fermi at Columbia University occurred in March 1942. (Letter dated January 20, 1953 to Dr. Donald Cooksey from Dr. Edwin M. McMillan.)

⁴Goodchild, p. 53; Blumberg and Owens, pp. 116-117.

This group, including Edward Teller, who would later play a decisive role in the development of the thermonuclear bomb, came to a somewhat pessimistic conclusion: it was believed that the amount of nuclear material needed for a weapon might be prohibitively large. However, this discouraging proposition was more than balanced by a startling possibility.

The discussions and calculations suggested that a reaction more powerful than nuclear fission might be generated by thermonuclear fusion of deuterium. This possibility was of such immediate concern that in July 1942 Oppenheimer visited Compton's summer retreat in Michigan to communicate the news, which soon spread among scientists at Berkeley and at the University of Chicago, despite conscientious efforts to suppress reference to a high-yield weapon that might use a more-easily acquired material than uranium-235 or plutonium-239.

Oppenheimer subsequently arranged for basic nuclear studies of the light elements, using cyclotrons at Harvard and Minnesota Universities, and interest was aroused in the construction of a heavy-water plant.⁵

Careful attention was given to the possibility first raised in 1922 that the thermonuclear bomb might ignite a fatally-cataclysmic reaction in the light elements of the earth's crust or atmosphere. The easiest reaction to initiate was between nitrogen atoms in the atmosphere. Calculations showed that no matter how high the temperature, energy loss would exceed production by a reasonable factor, and the reaction would not be self-propagating (this was paralleled by a problem that arose after the war: a self-propagating thermonuclear reaction in deuterium was prevented by energy loss due to radiation -- this would not be conclusively confirmed until 1950).⁶

One concept that was obvious from the moment the H-bomb was conceived was the need for the hydrogen fuel to be outside the atomic bomb. At the time, it was expected that the H-bomb would release at least 100 times as much energy as the atomic bomb. If the fusion fuel were inside the atomic bomb core, its potential power would be limited by the size of the core

⁵EARLY THERMONUCLEAR WEAPONS DEVELOPMENT: THE ORIGINS OF THE HYDROGEN BOMB, RS 3434/100, SC-WD-68334, Frederick C. Alexander, Jr., Information Research Division 3434, Sandia Laboratories, Albuquerque, New Mexico, May 1969, p. 7; Hewlett and Anderson, p. 104.

⁶THE HISTORY OF MODERN PHYSICS, Vol. II, p. 15.

(although small amounts of fusion fuels were later incorporated into the cores of "boosted" fission warheads).⁷

Another idea discussed at Berkeley was the use of solid and compact lithium deuteride as a fusion fuel, a much more convenient and practical reactant than cryogenic liquid deuterium (lithium deuteride was not used in an American thermonuclear device until early 1954).⁸

However, lithium has a distinct disadvantage. Lithium has a higher reaction rate at high temperatures than that of deuterium, even though its nuclear charge is three instead of one. However, atoms with higher nuclear charge radiate much more rapidly; in fact, this radiation rate proceeds at the square of the rate of increase in charge. Since tripling the charge increases the radiation rate by a factor of nine, lithium is far more difficult to ignite than deuterium.⁹

The Berkeley conference attendees also considered adding tritium to the deuterium to lower its ignition temperature, and the use of dense tampers to contain radiation entering and escaping the reacting deuterium. Two big unknown factors at this time were the rates of energy loss through the deuterium and the tampers.¹⁰

Work on the hydrogen bomb at Los Alamos during World War II proceeded along both theoretical and experimental lines. Theoretical development consisted of analyses of known thermonuclear principles and physics, and the determination of the feasibility and design of a possible experimental device. Small-scale laboratory experiments were conducted to determine the chemical and nuclear characteristics of fusion fuels. One of the first buildings erected at the new nuclear weapons laboratory was a hydrogen liquefaction facility to study the behavior of liquid deuterium.

Fundamental understanding of the fast thermonuclear reaction had been reached by the time the lab at Los Alamos was started in early 1943. In the first rough calculations, Teller had

⁷Letter dated 29 December 1980, from Dr. H. A. Bethe, Cornell University, to Representative Paul N. McCloskey, U.S. House of Representatives.

⁸"H-Bomb Dreams," Richard Rhodes, The New York Times Book Review, February 11, 1990, p. 3.

⁹"Excerpts from Manhattan District History, Book VIII - Los Alamos Project (Y) Volume 2 - Technical," para. 5.53.

¹⁰Hoddeson, et. al., pp. 45, 46.

ignored radiation, which drains off energy at a rate that increases rapidly with temperature. These preliminary calculations indicated nonetheless that a fusion reaction would occur if ignited by the explosion of a fission bomb as a "detonator."

One suggestion that arose even this early in the program was to lower the deuterium ignition temperature by admixture of artificially-produced tritium. A tritium-deuterium fusion reaction had approximately 150 times the cross section of deuterium-deuterium fusion and would occur at only 10% of the temperature (energy level) required for the deuterium-deuterium reaction.¹¹

The idea was not followed up immediately because of the obvious difficulty of manufacturing tritium and the mistaken hope of igniting pure deuterium directly from a fission explosion (later pursuit of some of Teller's ideas would make use of tritium mandatory).

By April 1943, the proposed hydrogen bomb had already been the subject of much discussion; some thought had even been given to proceeding directly to the thermonuclear bomb and perfecting the fission bomb only far enough to serve as a trigger for the device. This was a fantastically optimistic view: difficulties in developing a fission bomb would soon overshadow all thoughts of a fusion device before the end of the war (the hydrogen bomb itself would present nearly overwhelming technical problems even after the war).

Among available materials, deuterium had the lowest ignition temperature, estimated to be about 400 million degrees. Once ignited, deuterium was believed to be about five times as energy productive per unit mass as was U^{235} (it turned out to be closer to 8 1/2 times as effective).

Furthermore, one kilogram of deuterium is more cheaply produced in usable form than U^{235} or Pu^{239} . The proposed weapon, using a fission bomb as a detonator and deuterium as an explosive, could properly be called an atomic superbomb (hence the appellation of the "Super").

Development of this superbomb was necessarily secondary to that of the fission bomb, but its potential was so great that research on it could not be neglected completely. A planning committee report of May 10, 1943 recommended that investigation of the thermonuclear bomb be pursued, but mainly along theoretical lines and at a lower priority than the fission bomb. This confirmed established wartime Los

¹¹September 12, 1952 Walker memorandum, p. 34.

Alamos laboratory policy to build an atomic bomb as quickly as possible.¹²

There was no systematic theoretical work on the Super at Los Alamos until the fall of 1943. Liquid deuterium properties were studied at Ohio State University. A cryogenics laboratory was started at Los Alamos in the Chemistry and Metallurgy Division with the object of building a deuterium liquefaction plant. This laboratory had been included in plans for Los Alamos as early as the fall of 1942.¹³

In September 1943, Teller proposed more extensive investigation of the Super. Experimental cross sections had been revised upward, so that the Super would be feasible at lower temperatures. There was also some evidence that German interest in deuterium might be directed towards producing a similar bomb.¹⁴ (This may have been the case; however, the Germans were mainly trying to build a heavy-water nuclear reactor as a step toward developing their own fission weapon.) Work on the Super continued, but at a low intensity because Teller and his group were preoccupied with more urgent problems with the implosion fission bomb.

One potential obstacle foreseen at this time was the phenomenon known as "Compton cooling" or "inverse Compton scattering."¹⁵ This was a physical effect that caused photons, gamma rays, and x-rays to lose energy to longer-wavelength radiation. This radiation tended to dissipate energy that was intended to go into igniting the deuterium fuel.

Robert Serber later described this phenomenon:

¹²HISTORY OF MODERN PHYSICS, Vol. II, p. 24.

¹³Hoddeson, et. al., p. 58.

¹⁴Additional footnote 14a to "The H-Bomb: Who Really Gave Away the Secret," Daniel Hirsch and William G. Mathews, in Soviet Physics Usp. 34 (5) May 1991 (Uspekhi Fizicheskikh Nauk 161, 153-169), translated from the Russian (after originally appearing in English in the Bulletin of the Atomic Scientists, January/February 1990, pp. 22-30; "Excerpts from Manhattan District History, Book VIII - Los Alamos Project (Y) Volume 2 - Technical," para. 5.61; Hoddeson, et. al., p. 203.

¹⁵"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 271.

Edward first thought it (the classical Super) was a cinch. Bethe, playing his usual role, knocked it to pieces. Edward had figured the energy that would be released, how hot it would heat the (D-T plasma), and so forth. Everything looked fine until Bethe pointed out that you would get radiation; you had to be in equilibrium with black-body radiation, which goes up with the fourth power of temperature, drains the heat right off, and cools everything down.

You start feeding something and bingo, everything goes into electromagnetic radiation. Edward hadn't allowed for that. Bethe thought of a mechanism that really drained the energy off fast -- we called it the inverse Compton effect -- that knocked Edward's calculations into a cocked hat, and they never actually recovered.¹⁶

A later approximation of the operation of a Super took this energy loss into account, but with a model which assumed that the system of deuterons and electrons in hot deuterium would be in equilibrium with the radiation field. Because of the high heat capacity of the latter at high temperatures, the amount of energy that would have to be poured into the system to ignite it would be prohibitive.¹⁷

In the winter of 1943-44, LASL scientists discovered that although the temperature of a fission detonation would be enough to serve as a fusion trigger, the chain reaction would be complete in about a millionth of a second, and gaseous deuterium could not ignite within that brief moment, primarily because of cooling of the deuterium mass by loss of energy through radiation. One important addition to the Super project at this time was mathematician Stanislaw Ulam, who began calculating the effects of inverse Compton cooling.¹⁸

The problem thereafter was to find a means of igniting deuterium more rapidly.¹⁹ Because of theoretical problems and

¹⁶Serber, THE LOS ALAMOS PRIMER (1992 edition), p. xxxi.

¹⁷Excerpts from Manhattan District History, Book VIII - Los Alamos Project (Y), Volume 2 - Technical.

¹⁸Hoddeson, et. al., p. 204.

¹⁹AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 177; ATOMIC ENERGY COMMISSION THERMONUCLEAR WEAPONS PROGRAM CHRONOLOGY, U.S.

because the Super might have to be made with tritium, development time would be much longer than originally expected. However, it was decided that work on so portentous a weapon should be continued in every way possible without interfering with the main fission weapon program.²⁰

Cryogenic experiments at Los Alamos were limited locally to the design and construction of Building 29 (or Building Y), the Cryogenics Lab, at the northeast corner of the mesa. This lab contained a Joule-Thompson deuterium liquefier modeled after one at the University of California. It consisted of ethane, liquid air, and liquid hydrogen (or deuterium) cycles. The lab was completed by December 1943; at the time, it was one of the largest liquid hydrogen plants in the world.²¹

The first two cycles were completed by the beginning of 1944; the last in April 1944. Although designed for 35 liters/hour capacity, it produced only 25 liters/hour at Los Alamos' 7,300 foot elevation. This loss in capacity could be compensated for by additional compression, if necessary.²²

The problems of producing and storing liquid deuterium were investigated under a contract with Ohio State University in Columbus, Ohio (Ohio State continued research with liquid gases for the U.S. Atomic Energy Commission after the war ended). These experiments included hydrogen-deuterium exchange, the heat of vaporization of liquid deuterium, and long-term operation of hydrogen liquefaction equipment. The Ohio State contract began in May 1943 and continued through the end of the war.²³

Studies were conducted at Ohio State of technical problems associated with the production, storage, and transportation of large quantities of liquid hydrogen.²⁴

In May 1944, experimental quantities of tritium were produced in the reactors at Oak Ridge by the irradiation of lithium.²⁵

Atomic Energy Commission Secretariat, Washington, D.C., January 1955, p. 4.

²⁰HISTORY OF MODERN PHYSICS, Vol. II, pp. 86, 87.

²¹"Cryogenics and Nuclear Physics - Part 1," E. F. Hammel and J. D. Rogers, CRYOGENICS, February 1970, p. 6.

²²"Cryogenics and Nuclear Physics - Part 1," E. F. Hammel and J. D. Rogers, CRYOGENICS, February 1970, p. 6.

²³HISTORY OF MODERN PHYSICS, Vol. II, p. 154.

²⁴"Cryogenics and Nuclear Physics - Part 1," E. F. Hammel and J. D. Rogers, CRYOGENICS, February 1970, p. 6.

In June, Teller's small Super theory group was taken from the Theoretical Physics Division and placed in an independent position reporting directly to J. Robert Oppenheimer, director of the Los Alamos lab.

The Super program was re-evaluated in September 1944; theoretical difficulties indicated that ignition of deuterium might be prevented by energy dissipation. The idea of adding tritium to lower the ignition temperature was again considered. The practicability of using a tritium-deuterium mixture was limited because tritium was difficult to produce. It could be generated from the bombardment of lithium-6 by neutrons, yielding tritium and helium.

A tritium-deuterium fusion reaction would also release neutrons with a temperature-equivalent of a billion degrees, a distinct advantage when trying to ignite a surrounding mass of deuterium.²⁵

To produce tritium in any large quantity, reactors currently allocated to the manufacture of fissile plutonium from uranium would have to be taken out of service, re-engineered, and loaded with lithium-6. Li^6 itself would require further new facilities to separate it from the far-more abundant isotope Li^7 .

The small tritium samples used so far in cross section measurement experiments had been generated by cyclotrons. (By the end of the war, total tritium production amounted to only a few cubic centimeters in gaseous form, generated by the nuclear reactors at the Clinton Works at Oak Ridge.)

F Division was formed at Los Alamos in September 1944. As division leader, Enrico Fermi was directed to investigate potentially fruitful lines of development outside the laboratory's main program of immediate fission weapon development. This included the Super's theoretical and experimental aspects. Section F-1, The Super and General Theory, was created under the guidance of Edward Teller.

A second group, F-3 Super Experimentation, was created to continue laboratory experiments with potential fusion fuels. This group of 15 senior scientists and data processors

²⁵AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 178.

²⁶"From Reminiscences about Ya. B. Zel'dovich," S. S. Gershtein, Soviet Physics Usp. 34 (5) May 1991 (Uspekhi Fizicheskikh Nauk 161, 153-169), translated from Russian.

(tabletop calculator operators) concentrated on the design of the "runaway super."²⁷

Cryogenic work at Los Alamos was formally suspended in September when production capacity at Building 29 reached 50 liters of liquid deuterium per month, which could be boosted to 125 per liters per month with existing equipment. Building Y was converted to a warehouse at this time. (Cryogenic research at Ohio State was terminated a year later when the war ended.)²⁸

Also in September, J. Robert Oppenheimer suggested to Richard C. Tolman, scientific adviser to General Leslie Groves, that "the subject of initiating violent thermonuclear reactions be pursued with vigor and diligence, and promptly (after the war.)" One short-cut to such studies might be a "boosted" atomic bomb, using deuterium and tritium to improve its efficiency.²⁹

The first major technical-policy report involving the hydrogen bomb was written by Drs. Vannevar Bush and James B. Conant and was submitted to the Secretary of War on September 30. Drs. Bush and Conant and General Leslie Groves, commanding officer of the Manhattan Engineering District, also submitted this report to President Franklin Roosevelt.

The report stated, in part,

Some of our theoretical physicists believe that it is extremely probable that the energy generated by the fission (of an A-bomb) could under certain circumstances produce such a high temperature as to initiate a reaction which has never taken place on this earth, but is closely analogous to the source of energy of the sun. This reaction involves the transformation of heavy hydrogen into helium.

²⁷Letter dated November 30, 1953 to Honorable W. Sterling Cole, Chairman, Joint Committee on Atomic Energy, Congress of the United States, from Lewis L. Strauss, Chairman, U.S. Atomic Energy Commission.

²⁸LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1963, p. 121; "Cryogenics and Nuclear Physics - Part 1," E. F. Hammel and J. D. Rogers, CRYOGENICS, February 1970, p. 6.

²⁹"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 272.

Enormous amounts of energy are released in this reaction. A super bomb using heavy hydrogen (in the form of heavy water) and detonated by an atomic bomb ... would be of a (higher) order of magnitude in its destructive power from an atomic bomb itself. We may therefore designate it as a super-super bomb. While such a possibility lies in the future, it could even happen that a bomb of this type would prove feasible within six months or a year after the first atomic bomb is constructed.

....

... One cannot say with certainty that such a bomb can be constructed, but it seems as probable as was the atomic bomb development when this research was first undertaken by the government. But whether or not this particular line should prove profitable from a military point of view, one can be certain that there will be unexpected developments which would increase enormously the effectiveness of atomic energy for destructive purposes.³⁰

In a separate memorandum sent on the same day to the Secretary of War, Drs. Bush and Conant speculated on the future military potentialities of nuclear weapons; they struggled to define the possible power of the "Super" in terms of the capabilities of contemporary bombers and to warn that the effects on the U.S. by a "Super"-equipped enemy would be cataclysmic:

We are dealing with an expanding art and it is difficult to predict the future. At present, we are planning atomic bombs utilizing the energy involved in the fission of the uranium atom. It is believed that such energy can be used as a detonator for setting off the energy which would be involved in the transformation of heavy hydrogen atoms into helium.

If this can be done, a factor of a thousand or more would be introduced into the amount of energy released. This means that one such super-super bomb would be equivalent in blast damage to 1,000 raids of 1,000 B-

³⁰Borden-Walker H-bomb chronology, pp. 5, 6.

29 Superfortresses delivering their loads of high explosives on a target.

One must consider the possibility of delivering either the bombs at presently contemplated or the super-super bomb on an enemy target by means of a robot plane or guided missile. When one considers these possibilities, we see that very great devastation could be caused immediately after the outbreak of hostilities to civilian and industrial centers by an enemy (equipped) with a relatively few such bombs. That such a situation present a new challenge to the world is evident.

Conant and Bush also counseled against the dangers of secrecy and an international nuclear arms race:

It is our contention that it would be extremely dangerous for the United States and Great Britain to attempt to carry on in complete secrecy further developments of the military applications of this art. If this were done, Russia would undoubtedly proceed in secret along the same lines and so too might certain other countries, including our defeated enemies.

We do not believe that over a period of a decade the control of the supply (of fissionable material) could be counted on to prevent such secret developments in other countries. This is particularly true if the super-super bomb were developed, for the supply of heavy hydrogen is essentially unlimited and the rarer materials such as uranium and thorium would be used only as detonators.

If a country other than Great Britain and the United States developed the super-superbomb first, we should be in a terrifying situation if hostilities should occur. The effect on public reaction of the uncertainties in regard to an unknown threat of this new nature would be very great.³¹

³¹Memorandum dated September 30, 1944 to The Secretary of War from V. Bush and J. B. Conant, Subject: Salient Points

A week later, Dr. Edward Teller was very enthusiastic about the possibility of achieving a workable "Super" in the near future:

... the theoretical investigation of the super gadget has been carried out in considerable detail so that we can now predict with reasonable probability that such a gadget will become feasible a relatively short time after the first successful fission gadget is produced.

Teller estimated that only 100 scientists and technicians would be required to produce the "Super."³² With regard to both the number of persons and the time finally required, Teller was wildly over-optimistic.

Later in October, the status of the Super was summed up in a letter from Dr. Conant to Dr. Bush:

It seems that the possibility of inciting a thermonuclear reaction involving heavy hydrogen is somewhat less now than appeared at first sight two years ago. I had an hour's talk on this subject by the leading theoretical man at L. A. [Los Alamos] The most hopeful procedure is to use tritium (the radioactive isotope of hydrogen made in a pile) as a sort of booster in the reaction, the fission bomb being used as the detonator and the reaction involving the atoms of liquid deuterium being the prime explosive. Such a gadget should produce an explosion equivalent to 100,000,000 tons of TNT.³³

A second major technical-policy report was submitted to the War Department by a special committee of the Office of Scientific Research and Development on December 28. This

Concerning Future International Handling of Subject of Atomic Bombs.

³²"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), pp. 271, 272.

³³Letter dated October 20, 1944 from James B. Conant to Vannevar Bush, Subject: Possibilities of a Super Bomb. Vannevar Bush-James B. Conant Files, Records of the Office of Scientific Research & Development, S-1, Record Group 227, folder 3, National Archives of the United States.

paper recommended postwar development of both advanced fission and fusion weaponry and warned against the possibility of a future thermonuclear "Pearl Harbor" sneak attack:

The Government should continue the development of nuclear weapons as a project of overriding urgency so long as national security may be endangered. Nuclear fission bombs of greatly improved efficiency are certainly in sight, and thermo-nuclear (sic) bombs of ten-thousand-fold greater power may even be feasible.

... (such weapons) would permit an enemy in a single day preceding declarations of hostilities to carry out an action which might be decisive (to) the outcome of a war.³⁴

Theoretical work on the Super continued without surprises as analysis of the D-D thermonuclear reaction became more quantitative and concrete. Among other discoveries, the theoretical investigation showed that a steep rise in the rate of energy evolution could be expected with a rise in reaction temperature. At the critical temperatures for pure deuterium, the secondary tritium reactions provided about 1/3 of the energy developed. Given an initial admixture of only 0.3% tritium, the tritium reaction provided almost 3/4 of the energy.

A half-and-half mixture of tritium and deuterium provided 100 times the energy evolution of pure deuterium. The time scale of a reaction in this mixture was set by the fact that pure deuterium could reproduce the energy with which it was heated in a little less than 1/10th of a microsecond.

More attention was given to the theory of detonation mechanisms; work peaked in the spring of 1945 and continued for several months after the end of the war. Various physical models were investigated; the result sought was a bomb that burned about a cubic meter of deuterium, releasing energy equivalent to that of about 10 million tons of TNT (10 megatons).

In this proposed theoretical system, energy was transferred by high-velocity (high-energy) neutrons, spawned by triton-deuteron or deuteron-deuteron fusions, colliding with deuterium nuclei, and transferring some of their kinetic energy to the impacted deuterons.³⁵ In such a fashion, the D-D

³⁴Borden-Walker H-bomb chronology, p. 6.

reaction was expected to be self-propagating. At this time, no thought was given to using high densities of fusion fuels.

Early in 1945, a paper suggesting future study in nuclear physics directly applicable to the Super recommended the gathering of cross-section data on materials that might be used in the bomb. Specifically, the paper suggested study of cross-sections at various energy levels for collisions of deuterium and deuterium, tritium and tritium, deuterium and tritium, hydrogen and tritium, hydrogen and deuterium, neutrons and deuterium, helium-3 and deuterium, helium-3 and tritium, helium-3 and helium-3, neutrons and tritium, and helium-4 and deuterium, tritium, helium-4 and neutrons.

Good measurements had already been made of some of these reactions over rather limited energy regions. These tests needed to be duplicated and extended over all accessible energy levels that could be generated in a laboratory or in a particle accelerator.

The study also recommended that although ignition of the earth's atmosphere by thermonuclear reactions had been investigated in theory and was considered impossible, it should nonetheless be investigated further, preferably on an experimental basis to take it beyond the realm of theory.³⁶

In May 1945, Oppenheimer estimated that a thermonuclear bomb could be developed within just three years.³⁷ However, at the same time, he stated:

We are almost as far today as we were when we came to (Los Alamos) from knowing how to make a Super. The turn which the assembly techniques have taken, i.e., implosion, is one which does not lend itself readily to Super development.³⁸

Oppenheimer's doubt about the utility of implosion to thermonuclear weapons would turn out to be terribly mistaken, as developments six years later would prove: while the method

³⁵HISTORY OF MODERN PHYSICS, Vol. II, pp. 184, 185; "Excerpts from Manhattan District History, Book VIII - Los Alamos Project (Y) Volume 2 - Technical," paras. 5.55, 13.8.

³⁶"Recommendations for Research in Nuclear Physics," R. F. Bacher and R. R. Wilson, Los Alamos Scientific Laboratory, no date but ca. early 1945, pp. 7, 8.

³⁷Hewlett and Anderson, p. 356.

³⁸Letter dated May 7, 1945 to General Leslie R. Groves, USA, from J. Robert Oppenheimer, Director, LASL.

of implosion envisioned by Oppenheimer in 1945 was not the method finally applied, the basic principle was sound.

Other thermonuclear ignition methods were also considered, including the use of ordinary high explosives. By mid-June 1945, Manhattan District scientists believed that:

...significant thermonuclear reactions may be initiated without the use of fission bombs, by refinements in the application of ordinary high explosives and by other means not yet explored.³⁹

At this time, Oppenheimer continued to be an advocate of the Super:

We believe the subject of thermo-nuclear (sic) reactions among light nuclei is one of the most important that needs study. There is a reasonable presumption that with skillful research and development, fission bombs can be used to initiate the reactions of deuterium, tritium, and possibly other light nuclei. If this can be accomplished, the energy release of explosive units can be increased by a factor of a thousand or more over that of presently-contemplated fission bombs.⁴⁰

Experiments were also conducted in 1945 to determine the cross sections of D-D and D-T reactions. During these tests, water enriched to 25% to 50% deuterium was frozen onto metal plates, which were then bombarded by tritium atoms. Special neutron-detecting instruments measured the energy levels and quantities of neutrons and alpha particles resulting from these low energy-level fusion reactions.

The cross section for fusion of tritium and deuterium was found to be greater than that for fusion of deuterium and deuterium by as much as a factor of 100, which indicated a materially lower ignition temperature for tritium-deuterium mixtures.⁴¹

³⁹AEC THERMONUCLEAR WEAPONS CHRONOLOGY, U.S. Atomic Energy Commission Secretariat, Washington, D. C., 1955, p. 9A; Borden-Walker H-bomb chronology, p. 7.

⁴⁰"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 274.

These experiments showed that low-energy D-T reactions had a much-higher than expected reactivity; at relevant energy levels, this reactivity was about 30 times greater than the D-D reaction. This was in addition to the fourfold energy release in T-D reactions, as compared with the D-D reaction, i.e., 17.6 as opposed to 3.5 to 4 MeV.⁴²

Investigation of the Super continued at relatively low priority in F Division at Los Alamos until the end of the war. Technicians released from other work brought this research to partial completion. The results indicated, convincingly but not decisively, that such a weapon was indeed feasible.⁴³

One possible design pondered in July 1945 involved the fusion of about five tons of deuterium and several hundred grams of tritium, triggered by a gun-type atomic bomb. This configuration might yield as much as 100 megatons of explosive energy.⁴⁴

LASL scientists concluded that development of a suitable energy transfer method would require so much time and effort that the hydrogen weapon should not be considered for wartime use.

Priority was thus given to the fission weapon, as it represented the only real possibility for shortening the war. However, thermonuclear calculations were continued, and a fairly complete theory of Super operation, as then conceived, was prepared. Early computations were made on an ENIAC (Electronic Numerical Integrator and Calculator) computer at the University of Pennsylvania in Philadelphia, but assumptions made in the calculations were too limiting, and the results were inconclusive.⁴⁵

⁴¹Errata to "Nuclear Data," interviews with Ben C. Diven, John H. Manley, and Richard F. Taschek, in LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, p. 121; THE HISTORY OF MODERN PHYSICS, Vol. II, pp. 188, 189; AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 15; "Excerpts from Manhattan District History, Book VIII - Los Alamos Project (Y) Volume 2 - Technical," para. 13.6.

⁴²HISTORY OF MODERN PHYSICS, Vol. II, p. 185. One particular D-D reaction yields 23 MeV.

⁴³HISTORY OF MODERN PHYSICS, Vol. II, p. 169.

⁴⁴Notes on Meeting between Leslie R. Groves, Dr. Oppenheimer, and Dr. Tolman at Chicago, 24 July 1945.

These calculations were run during a six-week period in December 1945 and January 1946, using nearly a half-million punched cards; results seemed to establish the feasibility of the Super, and provided a strong psychological boost to the LASL weaponeers.⁴⁶

It was becoming apparent, even this early in the program, that more powerful computing capabilities were required to study the Super in any great detail. The problem of the propagation of the reaction involved the hydrodynamics of the expansion of hot material against cold material. Little headway could be made in a problem of such complexity without machine calculations, even after the problem was understood and stated clearly enough to be modeled mathematically.⁴⁷

One approach considered as early as the fall of 1945 was the ignition of a deuterium charge by ordinary high explosives, which would theoretically compress the fuel to densities sufficient to support fusion.⁴⁸

The possibility of a workable hydrogen bomb had waxed and waned throughout the war. In 1942 the outlook had been thought good. In 1943 the need for tritium and the impossibility of producing enough had killed hopes the Super might play a part in the war. In 1944 calculations suggested heat from fission could induce significant added heat from thermonuclear reactions in deuterium even where the

⁴⁵Alexander, p. 9. The ENIAC was constructed between 1942 and 1945 at the Moore School of Electrical Engineering at the University of Pennsylvania in Philadelphia. The goal of the project, supported by the U.S. Army Ballistics Research Laboratory at the Army's artillery proving ground in Aberdeen, Maryland, was to create a computer to solve ballistic trajectory problems. The result was the first large-scale electronic computer, using electronic vacuum tubes rather than relays or mechanical counters. The ENIAC weighed 30 tons, cost \$100,000, contained 18,000 vacuum tubes, and could perform about 5,000 additions per second, nearly 500 to 1,000 times as many per second as the Mark I, an electromechanical relay computer at Harvard University. The ENIAC was moved from Philadelphia to Aberdeen in 1947, where it continued to operate until 1955.

⁴⁶Rhodes, DARK SUN, p. 251.

⁴⁷"Excerpts from Manhattan District History, Book VIII - Los Alamos Project (Y) Volume 2 - Technical," para. 13.5.

⁴⁸Report of the Scientific Panel to the Interim Committee, September 28, 1945, as cited in THERMONUCLEAR WEAPONS CHRONOLOGY, p. 16.

thermonuclear reactions were unsustainable. In 1945 further calculations indicated this hope was premature.

By September 1945, the time estimated to construct an American thermonuclear bomb ranged from five to 15 years. The theoretical feasibility of the hydrogen weapon was not nearly so certain as the fission bomb was, on purely theoretical grounds, when LASL was established in 1942.⁴⁹ The postwar outlook for further development was grim, as the country demobilized and wartime Los Alamos employees returned to civilian life. (By the summer of 1946, the technical staff at LASL was down from a wartime peak of 3,500 to only 1,100.⁵⁰)

In October 1945, a report, with recommendations, was issued on the "super gadget." This document suggested future work in cryogeny, including large scale deuterium liquefaction, and more ordnance studies. Increased tritium production was proposed, by means of greater output from existing reactors at Oak Ridge and Hanford. A collection of technical data compiled during the Manhattan Project was issued as the "Super Handbook."⁵¹

Also in October, Norris Bradbury, soon to be LASL's second director, emphasized the importance of determining the feasibility of the Super (this search would occupy a significant portion of LASL's time and resources for the next

⁴⁹Borden-Walker H-bomb chronology, p. 8. Meeting in late September 1945, after the dropping of the two atomic bombs on Japan, Drs. Oppenheimer, Fermi, E. O. Lawrence, and A. H. Compton reversed their earlier endorsement of postwar development of the Super and instead advised Secretary of Commerce (and former Vice President) Henry Wallace that U.S. surrender in a future war would be preferable to the likely destruction caused by the employment of thermonuclear weapons. ("An Analysis of 'Two Cultures': Writing about the Making and the Using of the Atomic Bombs," review essay by Barton J. Bernstein in THE PUBLIC HISTORIAN, Vol. 12 No. 2 (Spring 1990), University of California Press, p. 104.; "In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 276.)

⁵⁰Address by Norris Bradbury to the Lawrence Livermore Laboratory on November 3, 1965. Lecture No. 3, "Los Alamos," LLNL Family Lecture Series.

⁵¹LAMS-290, October 8, 1945, cited in "Technical References Relating to Thermonuclear Program," Appendix E to September 12, 1952 Walker memorandum.

five years). Bradbury doubted that the hydrogen bomb was likely to be built anytime soon:

We will propose that the fundamental experiments leading to the answer to the question "Is or is not a Super feasible?" to be undertaken. These experiments are of interest in themselves in many cases; but even more, we cannot avoid the responsibility of knowing the facts, no matter how terrifying.

The word "feasible" is a weasel word -- it covers everything from laboratory experiments up to the possibility of actual building -- for only by building something do you finally determine feasibility. This does not mean we will build a Super. It couldn't happen in our time in any event. But someday, someone must know the answer: Is it feasible?⁵²

At this time, Dr. Edward Teller thought that the Super could be built and tested within five years.⁵³ In a letter to Enrico Fermi, Teller urged continuation and intensification of work on the explosive energy released by thermonuclear reactions from the "burning" of deuterium.⁵⁴ Teller's desires for the Super, and his desires for the continuation of fission weapon development, enjoyed little political support at this time, and virtually no economic support in the immediate postwar period.

Early Postwar Studies and Designs

The theoretical detonation of the "classical Super" (also known within the lab at this time as the "Runaway Super")⁵⁵

⁵²Notes on Talk Given by Comdr. N. E. Bradbury at Coordinating Council, October 1, 1945. Emphasis in original.

⁵³"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 277.

⁵⁴Letter dated October 31, 1945 to Dr. E. Fermi from Edward Teller, attachment to letter dated May 8, 1950 to Senator Brien McMahon from Edward Teller.

⁵⁵"Timetables of Nuclear Weapon Events," U.S. Department of Energy, Albuquerque, New Mexico, September 22, 1986, p. 26. The terminology "Runaway Super" does not appear in the

consisted of two parts: a tritium-deuterium initiating reaction, and a self-sustaining deuterium-deuterium "burn."⁵⁶ In this context, deuterium was treated as a super explosive that could be detonated in much the same way as a charge of conventional high explosives.

Ordinary explosives are usually set off by a detonator which heats up a small portion of the charge. The burning explosive at the point of ignition releases energy rapidly in the form of a shock wave. This energy heats up the layers of explosive surrounding the ignition point so that they release their energy and ignite more of the charge. This instantaneous burning continues, as an explosion, until all the fuel is expended, or until the container holding the explosive is disassembled.

Deuterium yields 100 million times as much energy per unit weight as does an ordinary explosive, but it also requires an ignition temperature on the order of several hundred million degrees, far beyond the capability of any detonator until fission bombs were developed.⁵⁷

This was the theory behind the Super. The major stumbling block was how to transmit the energy of the atomic bomb to the deuterium fuel in a form, such as a shock wave, that would ignite the deuterium.

In addition, other important questions needed to be explored and answered: Could an atomic bomb be made powerful enough to raise the temperature and density of the deuterium to levels sufficient to support fusion? Could a mass of reacting deuterium be physically contained long enough to produce a

censored version of AF ATOMIC ENERGY PROGRAM, but appears repeatedly in AEC internal documents cited in this book.

⁵⁶BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 40 No. 8, October 1984, p. 46. These are probably Parts 2 and 1, respectively, of "Method A" discussed by Dr. Hans Bethe in "Comments on the History of the Hydrogen Bomb," LOS ALAMOS SCIENCE, Vol. 3 No. 3, Fall 1982, pp. 42-53. Bethe states that the GREENHOUSE tests in 1951 included a test of Part 2 of Method A. In a public address at the Massachusetts Institute of Technology on November 14, 1990, Bethe identified "Method A" as the "Classical Super;" "Method B" as the "Alarm Clock;" "Method C" as the "Booster;" and "Method D" as the famous "three (Teller-Ulam) concepts." Bethe stated that the "Classical Super" was "a cylinder of deuterium, which you would heat at one end and thereby start a nuclear reaction." See also York, THE ADVISORS (1989), p. 163.

⁵⁷AEC THERMONUCLEAR WEAPONS PROGRAM CHRONOLOGY, p. 13; Ulam biography, 1991, p. x.

substantial yield? Would the fusion process spread (propagate) throughout the thermonuclear fuel mass or simply confine itself to a small part of it? Would it cool off, i.e., lose energy via radiation and physical expansion, after it was ignited, and die out? Since the reaction took place at a very high temperature, would the device radiate away that heat and cool off before it could explode?⁵⁸

The problems of igniting a mass of deuterium appeared to be very formidable. An understanding, in the greatest possible detail, of the mechanism of the preceding fission explosion was necessary to determine the initial conditions for the explosion of deuterium. The subsequent interaction of the effects and products of the fission explosion with the deuterium had to be predicted and calculated.

With respect to the process of the thermonuclear reaction itself, the behavior of the material as it heated and expanded -- the changing time rate of the reaction; the hydrodynamics of the motion of the material; and the interaction of the material with its own radiation, which would probably generate as much energy as the thermal content of the expanding material itself -- had to be formulated and calculated.

At any moment, calculations could be only as reliable as the understanding by the physicists of the processes involved. Such phenomena as the way energy flowed in dense hot material and the turbulent mixing of the burning fuel were critical to weapon design but were not well understood at the end of World War II.

To realize the magnitude of the problems involved, just the mathematical definition and computation of the ignition and explosion of a mass of deuterium presented a large number of separate problems. Each of these problems was quite difficult by itself, and all the problems were strongly interconnected.

The "chemistry" of the reaction, i.e., the production by fusion of new elements not originally present in the deuterium and the appearance of tritium, He^3 , He^4 , and other nuclei, together with the increasing density of neutrons of varied and variable energies in the deuterium and the plasma "cloud" surrounding it, directly influenced and altered the rate of the fusion reactions, by changing values of pressure, fuel density, and fuel temperature.

⁵⁸SOME SPECULATIONS ON THERMONUCLEAR WEAPONS, Aaron Meyers, June 23, 1979, p. 3.

Simultaneously, the radiation generated by the reaction was increasingly present and strongly influenced the motion of the mass.

By the end of 1945, preliminary calculations had been made on the temperatures that could be expected in a small deuterium mass located in the vicinity of an exploding fission bomb. Data had been gathered on the importance of a number of miscellaneous effects, including dispersion of heat (particle kinetic energy) by conduction by electrons, and on radiation cooling.⁵⁹

These were all very significant factors to define and calculate, especially in the absence of modern high-speed electronic computers.⁶⁰ LASL was in dire need of more information about the scattering of fission and fusion reaction products, properties of high-velocity neutrons, conduction in ionized gases, and other basic aspects of thermonuclear reactions.⁶¹

The most feasible design for the "classical Super" at this time was a very high-yield, advanced gun-assembly fission trigger located inside or at end of a cylinder containing a mixture of tritium and liquid deuterium.⁶² The gun-assembly trigger would yield several hundred kilotons and be enormously wasteful of scarce fissionable material (see comments on gun-type weapons under "Fission Explosion Principles" in Volume I).

This type of fission device was required because a more-efficient implosion-type trigger, with its partially outwardly-directed high explosives detonation, would only scatter the D-T fuel mixture before it could be ignited. In addition, the hydrogenous high explosives and the heavy tamper surrounding the fissile core of an implosion device would soak up neutrons that would be required to initiate fusion.⁶³

⁵⁹AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 15.

⁶⁰"Thermonuclear Devices," S. M. Ulam, Los Alamos Scientific Laboratory, in PERSPECTIVES IN MODERN PHYSICS, R. E. Marshak (ed.), Interscience, 1966, pp. 595, 596.

⁶¹"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 276.

⁶²Rhodes, DARK SUN, p. 252.

⁶³Rhodes, DARK SUN, p. 304.

The probable weight of a fission-fusion device of this design was estimated at around 10 tons; probable yield was between 10 and 40 megatons.⁶⁴

In theory, energy from the trigger would be transferred to the fusion fuel primarily by neutron-triton and neutron-deuteron collisions. The nuclear energy released in the reactions of deuterons or tritons with deuterons was transferred into the kinetic energy of the resultant particle products formed and generated by the reactions. This energy must, in turn, be used to heat the deuteron medium to produce more reactions.

The product particles would transfer their energy through collisions. In general, the energy was "deposited" throughout the reacting medium by means of a chain of collisions, during which the particle products dispersed themselves. Because of this dispersion, the volume to be initially heated had to be large enough to catch most of the energy deposits; otherwise the product particles would escape from the reacting system with too much kinetic energy, and the fusion reaction would not self-propagate.

Radiation loss at relatively low temperatures exceeded its production rate, but rose much less rapidly as the temperature of the reacting system rose. There was a minimum temperature, called the "ideal ignition temperature," at which the energy evolution reached equality with energy loss via radiation. The additional energy losses caused by escaping product particles was much more difficult to estimate and predict.

Another early idea proposed to limit energy loss via conduction to the walls surrounding the reacting mixture was to confine the plasma and fuel mass with a magnetic field (this notion later became the basis of an entire domain of controlled thermonuclear energy research). The drawbacks to this idea were that the magnetic field would be quickly overpowered by rising temperatures, and the bulky equipment required to maintain the field even briefly.

⁶⁴AEC THERMONUCLEAR WEAPONS CHRONOLOGY, pp. 6, 16, 17, 74. These pages discuss or cite "the possibility of a gun-type weapon employing approximately [deleted, but probably several tens of] kilograms of U-235 and giving an energy release of the order of that required to initiate a 'super'" (p. 6); the "unfavorable geometrical conditions" of "a fission bomb of the ordinary design" (p. 16); "uranium-235 for the primer" (p. 17); and "Implosion looked virtually impossible in such a geometry, and so a U-235 gun-type initiator looked most promising" (p. 74).

Since the problem of propagation of the reaction involved the hydrodynamics of the expansion of hot material against cold material, little progress could be made in this area without machine calculations. During the war, preliminary calculations had been undertaken at Columbia University in New York and at the Moore School of Engineering in Philadelphia, where the ENIAC was located.⁶⁵

The April 1946 Super Conference

An important early postwar meeting to review the status of the Super was held at Los Alamos between April 18 and April 20, 1946.⁶⁶ The work to date on thermonuclear processes was reviewed and a specific model of a thermonuclear bomb was presented. The conference centered on the feasibility of this model, which was chosen for its amenability to theoretical treatment rather than engineering practicability or efficient use of fissile material and tritium.

The purpose of the conference was to study the feasibility of thermonuclear bombs in principle, not to propose designs for actual weapons. An extensive series of computations on the ENIAC of one-dimensional burning of tritium and deuterium had been completed by this time. These calculations also excluded significant physical processes -- including the inverse Compton effect -- which would bleed off energy from the main fusion reaction.⁶⁷

An additional purpose of the meeting was to discuss the computational results and to assess the prospects for the physical realization of a thermonuclear device. Despite the simplified but relatively ambitious nature of the model, the general consensus was that the preliminary results were encouraging, given what was known at the time about the factors of radiation cooling and fuel dispersion.

The conclusion of the conference was:

It is likely that a super-bomb can be constructed and will work. Definite proof of this can hardly ever be expected and a final decision can be made only by a test of the completely assembled superbomb... There is

⁶⁵Excerpts from Manhattan District History, Book VIII - Los Alamos Project (Y), Volume 2 - Technical.

⁶⁶Rhodes, DARK SUN, p. 252. (These dates are from LA-575.) HISTORY OF MODERN PHYSICS, Vol. II, p. 269, claims that the meeting lasted from April 17 to April 23, 1946.

⁶⁷Rhodes, DARK SUN, p. 254.

at present no indication that any of the basic physical processes has been neglected, nor is it considered likely that any additional basic processes will need to be taken into consideration... The detailed design submitted to the conference was judged on the whole workable. In a few points doubts have arisen concerning certain components of this design... In each case, it was seen that should the doubts prove well-founded, simple modifications of the design will render the model feasible... It seems therefore very likely that a super can be constructed...⁶⁸

More specifically, the conference concluded:

- (a) In a sufficiently rich mixture of tritium and deuterium, a reaction can take place and propagate, given sufficient initial temperature.
- (b) A self-sustaining and propagating reaction in pure liquid deuterium seems a likely possibility.
- (c) If successful, such a reaction can deliver energies equivalent to 1,000 fission bombs and more, from a device weighing not much more than an ordinary fission bomb and containing mainly cheap materials. This device is now known as the [deleted].
- (d) To initiate a reaction in deuterium, mixtures of deuterium and tritium are useful, and it is believed that amounts of [deleted] might be sufficient for the purpose.
- (e) The initial heating of the T-D mixture to the required temperature appears perhaps as the most difficult task because (1) it is questionable whether fission bombs of

⁶⁸"Report of Conference on the Super," LA-575, Los Alamos Scientific Laboratory of the University of California, Los Alamos, New Mexico, February 16, 1950, pp. 44, 45. To a large extent, this very optimistic outlook was a result of the ignorance of the conferees in the area of thermonuclear reactions; it was also due in part to Edward Teller's boundless enthusiasm for the Super, which influenced the authors of the report.

sufficient yield can be constructed and (2)
[deleted]⁶⁹

Several comprehensive documents were drafted as the first phase of the thermonuclear development program drew to a close.⁷⁰ Although the purpose of the conference had not been to specify actual testable designs, two concepts were proposed. One was simple and relatively amenable to analysis; it did not immediately appear to be feasible. A second concept, of unknown feasibility, was considered too complex for reliable quantitative analysis.⁷¹

The studies presented at the conference showed that there was only a slender margin for error: the mechanisms by which energy was created in the system and uselessly lost from it were quantitatively comparable.⁷² One of the most significant causes of energy loss via radiation was the so-called "inverse Compton effect," a physical effect that causes photons, gamma and x-rays to lose energy to longer-wavelength radiation.⁷³

In the case of the Super, neutrons from the fission trigger were transferring most of their kinetic energy to electrons in the atoms of deuterium and tritium, and not enough energy to the nuclei of these hydrogen isotopes. This was an unforeseen complication that would finally doom the original Super design. In order to overcome these radiation losses, it

⁶⁹Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, p. 2. The amount of tritium deleted from item (d) in this list was probably on the order of several tens of grams.

⁷⁰LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, p. 135. These reports included "Prima Facie Proof of the Feasibility of the Super," LA-551, Los Alamos Scientific Laboratory of the University of California, Los Alamos, New Mexico, April 15, 1946; and "Report of Conference on the Super," LA-575, Los Alamos Scientific Laboratory of the University of California, Los Alamos, New Mexico, February 16, 1950. Other documents on the Super are LA-344(1) - (6), LA-341, and LA-401.

⁷¹"The Development of Nuclear Weapons," Lowell Wood and John Nuckolls, article in ENERGY IN PHYSICS, WAR AND PEACE, Hans Mark and Lowell Wood, eds., Kluwer Academic Publishers, Norwell, Massachusetts, 1988, p. 314.

⁷²Ulam biography, 1991, p. xi.

⁷³Report of Hugh H. Clegg and Robert J. Lamphere covering Interviews with Klaus Fuchs in London, England, between May 20 and June 2, 1950, pp. 27, 34.

would be necessary to have a nuclear reaction of very large cross-section at a low level of nuclear energy, at least as large as the D-D reaction.

In addition, in order to maintain a thermonuclear reaction, the energy released by the reaction had to be retained near the site of its production. This would be possible only under very high pressures, on the order of tons per square inch.⁷⁴

Because of the great complexity and variety of the most important nuclear reactions and physical processes, the successful design of the Super would require one of the most difficult and extensive mathematical analyses ever contemplated. Even then, there would be no certainty that such an attempt could succeed in being conclusive.⁷⁵ The requirements for materials, engineering developments, and more detailed understanding of basic physical processes were impressive, and would require a considerable fraction of the Los Alamos lab's meager postwar resources.

The development of a high-yield fission device was mandatory, as well as a much more detailed picture than had yet been developed of the phenomena occurring in the immediate region of an exploding fission weapon core.⁷⁶ The first step in any thermonuclear system is a fission explosion. Sometime in the middle of the explosion's life, it provides the energy required to induce thermonuclear burning, so the starting conditions for an estimate of thermonuclear behavior required a clear and comprehensive picture of the conditions in an advanced stage of a fission explosion.

This picture was not at all clear in early 1946.⁷⁷ The emphasis at Los Alamos during the war had been on the rapid design and construction of a practical fission weapon, and not on the use of the energies and products of a fission explosion to trigger a fusion explosion, or on the "tailoring" of the outputs of such an explosion to enhance the likelihood of fusion.

Design of the Super would require computer time and mathematical analysis in excess of anything that had been done during the war in connection with fission bomb design. The very proof of feasibility required a fully detailed

⁷⁴"Ultimate Catastrophe?", H. A. Bethe, BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 32 No. 6, June 1976, pp. 36, 37.

⁷⁵Mark, LA-5647-MS, p. 1.

⁷⁶Mark, LA-5647-MS, p. 2.

⁷⁷Mark, LA-5647-MS, p. 10.

calculation of its behavior during an explosion. Without this, no conclusive small-scale experiment was possible short of a successful stab in the dark, since a failure would not necessarily establish infeasibility, but possibly only that the system chosen was unsuitable, or that the required ignition conditions had not been met.⁷⁸

Wartime studies had also indicated that while a large amount of tritium was likely to be required for ignition, it was not prohibitively large, and modifications of the design might appreciably reduce the amount.⁷⁹ The April 1946 conference concluded that tritium should be produced at a rate starting at one gram per day, and increasing as design work progressed.⁸⁰

Although the Super was not at all similar to a conventional high-explosive bomb, the papers written after the conference nevertheless used the conventional explosive ordnance terms of "detonator," "primer," "booster," and "charge;" however, a disclaimer noted that these words were "not to be construed as having their customary meanings in connection with ordinary explosives, except when so specified."⁸¹

Preliminary designs were considered for a possible explosive fusion test device, one that would be configured primarily for ease of experimentation. It would not be a weapon by any stretch of the imagination. Among the factors considered were the range of temperatures required to initiate and sustain a thermonuclear "burn" in deuterium, the amounts of material to be heated, the optimum configuration, and the hydrodynamical behavior of such a system.⁸²

⁷⁸Mark, LA-5647-MS, p. 10. Not until the early 1970s, after more than a quarter-century of nuclear testing, and the concomitant "hands on" education of weapons designers, and the development of several generations of computers, was an exhaustive study of the burning processes in the Super completed. This task was clearly beyond the technological capabilities of the U.S. in 1946.

⁷⁹Mark, LA-5647-MS, p. 8.

⁸⁰Borden-Walker H-Bomb chronology, p. 11.

⁸¹LA-575, p. 2. A "booster," in the context of high explosive shells, is a small HE charge located inside the mass of the larger HE charge. The detonator ignites the primer, which in turn ignites the booster which ignites the main charge.

⁸²LA-575, pp. 19, 20.

These problems did not lend themselves to either analytical methods or similarity treatment; advanced numerical computational methods were needed. Extensive calculations were done on the ENIAC to determine processes by which energy could be shared and spread within the proposed model.

The proceedings of the conference noted that "it seems likely that no other naturally-occurring isotope (besides deuterium) is a suitable thermonuclear fuel, at least until the deuterium thermonuclear reaction is itself available as an initiating device" and that "the initial cost of a minimum deuterium explosion is fixed, however, by the cost of initiating it, which under present knowledge is at least that of several normal fission bombs."⁸³

As a practical matter, igniting fissionable material for the Super was limited to U-235 in a gun-type device. Although plutonium and U-233 were mentioned in the report as possible substitutes, neither of these was a practical alternative in 1946. At the time, and for quite a while later, U-233 was undeveloped as an explosive. Plutonium, if used in a gun, had to be free of the Pu-240 isotope which caused predetonation and no way was then known to remove this contaminant.⁸⁴

The smallest Super, which might be deliverable by drone aircraft or rocket, was expected to weigh several tons and yield as much as five megatons of TNT equivalent explosive force.⁸⁵

The Super could also be used as a formidable radiological weapon. Since there was an average of about 0.4 neutrons per deuteron consumed, if a 10-ton Super burned with a 30% efficiency, about 3×10^{29} fast neutrons would be produced. If a 50-ton wall of hydrogenous material containing a small fraction of some suitably chosen metal surrounded the bomb at a distance of a few feet, at least a third of these neutrons would be absorbed in the wall and each absorbed neutron would generate a radioactive atom with a specific half-life. This radiation would then be blown over hundreds of thousands of square miles.⁸⁶ (This idea was one of the earliest suggestions of the principle of a so-called "dirty" or "salted" hydrogen bomb.)

⁸³LA-575, pp. 3, 4.

⁸⁴Memorandum for the Files dated June 16, 1952 from John S. Walker, subject: Thermonuclear Program -- General Groves.

⁸⁵LA-575, p. 5.

⁸⁶LA-575, p. 6.

One conclusion of one conference report was that the general principle of the Super was ignition in stages, each larger than the preceding one:

The final object of the Super is the burning of a large mass of deuterium with corresponding release of energy. In order to achieve a sufficiently fast reaction, temperatures in the neighborhood of [deleted] are required.

In order to prevent excessive losses by heat conduction, the volume of material to be heated initially to this temperature has to comprise several liters. The achievement of such conditions is made difficult by the presence of low-energy radiation such as would be produced by a fission bomb as detonator. This radiation causes cooling by the inverse Compton effect.

In the presently discussed design, the difficulties of ignition are overcome by making use of the greatly superior reaction rates of mixtures of D and T.⁸⁷

An idea discussed very briefly during the April 1946 conference was the principle of radiation implosion, a theoretical technique of using gamma and x-radiation from an exploding fission bomb to compress fusion fuels. The principle was suggested in the context of a proposed low-yield thermonuclear experiment.⁸⁸ This concept was among many

⁸⁷LA-575, p. 9.

⁸⁸Letter dated August 7, 1990 from John H. Nuckolls, Director, Lawrence Livermore National Laboratory, to Priscilla Johnson McMillan. There is still some controversy over whether or not radiation implosion was discussed at Los Alamos in the spring of 1946. Drs. J. Carson Mark and Philip Morrison, who were both at the meeting, claim now that radiation implosion was not discussed. (Letter dated July 25, 1990 from Priscilla McMillan to Dr. Hugh DeWitt.) However, in John Nuckoll's letter to McMillan dated August 7, 1990, Nuckolls claims that radiation implosion was discussed in 1946. In addition, Dr. Edward Teller's August 1952 reply to Hans Bethe's June 1952 H-bomb chronology, Teller stated that the 1951 application of radiation implosion was merely an extension of the application discussed in 1946. In June 1952, Teller said that the "three concepts" ultimately employed in the design of the first successful American hydrogen bomb were an "amazingly simple

discussed in the conference, and was not to be applied conclusively or extensively for another five years.

Among other ideas discussed at the April meetings were the use of a beryllium oxide component of the proposed Super; the ignition of the Super by a cylindrical implosion process of fissionable material; and the compression properties of various materials, including deuterium and tritium. Numeric calculations run on the ENIAC specifically excluded the inverse Compton effect, and indicated that in the proposed Super design, the deuterium-deuterium reaction would build up quickly to an appreciable rate.⁸⁹

Various suggestions were made for means of improving the efficiency of the use of tritium and fissionable material and to simplify the engineering of the weapon. No objection on fundamental grounds to the feasibility of a thermonuclear bomb was raised during the meeting.

One conclusion of the conference was that the experimental program, development, and testing of a thermonuclear Super bomb would require work by a laboratory similar to LASL at its wartime peak for a period of one or two years.⁹⁰

The April meeting was attended by Dr. Klaus Fuchs, a German émigré scientist who had come to Los Alamos with a group of British scientists during World War II; in early 1950, Dr. Fuchs would confess to British counterintelligence agents that he had been a Soviet spy during his days at Los Alamos.⁹¹

At the end of May, Drs. Klaus Fuchs and John von Neumann filed for a patent on a method to initiate burning of a deuterium-tritium mixture.⁹² This method may have involved

and direct and logical step from the information which was present at the end of the war" and that "the information was derived from, and was inherent in, the 1946 report." (Memorandum for the File dated June 19, 1952 from Bill Borden and John Walker, subject: Second Laboratory -- Dr. Teller.)

⁸⁹Notes on Interrogation of Klaus Fuchs from FBI serial file 65-58805-1246, dated June 12, 1950 (provided by Richard Rhodes), pp. 6-8; Rhodes, DARK SUN, p. 253.

⁹⁰Excerpts from "Supplement to Manhattan District History, Book VIII - Los Alamos Project (Y) Volume 2 - Technical," 15 October 1947, para. 3.27.

⁹¹COMMENTS ON BETHE'S HISTORY OF THE THERMONUCLEAR PROGRAM, Dr. Edward Teller, August 14, 1952, p. 2.

⁹²Letter dated July 25, 1990 from Priscilla McMillan to Dr. Hugh DeWitt; Borden-Walker H-bomb chronology, p. 11;

implosion; by what means is not yet clear. An FBI summary of an interrogation of Fuchs in London after his arrest states:

Concerning the indicated suggestion of Dr. von Neumann, who suggested the ignition of the super bomb by the implosion process, (Fuchs) stated laughingly that this was his, Fuchs', suggestion and that he did not furnish information (to the Soviets) concerning the ignition of the super bomb by the implosion process.⁹³

The patent described a "combination with a device for initiating a thermonuclear reaction which employs a quantity of fissile material adaptable to sustain a neutron-induced divergent chain reaction, and a massive quantity of material in which a thermonuclear reaction can be maintained."

Intensive activity (such as it was) on the Super was ended in June 1946 by serious loss of personnel as wartime employees and consultants at Los Alamos returned to their civilian vocations.⁹⁴ Among the scientists who departed at this time was Dr. Fuchs, who returned to England.

Near the end of July, a technical report entitled "Possibility of Initiating a Thermonuclear Reaction by Non-Nuclear Methods," co-authored by James L. Tuck and Stanislaw Ulam, was published by LASL.⁹⁵ This report was one of the first LASL would generate as it examined alternate thermonuclear ignition techniques.

Existing knowledge on relevant thermonuclear processes was summed up in six lectures given at Los Alamos by Enrico Fermi between July and September 1946.

attachment to letter dated June 30, 1952 to William L. Borden, Executive Director, JCAE, from Roland A. Patterson, Chief, Patent Branch, USAEC. The patent, or "disclosure of invention," was numbered S-5292X and was cryptically entitled "Method and Apparatus for Releasing Nuclear Energy."

⁹³Report of Hugh H. Clegg and Robert J. Lamphere covering Interviews with Klaus Fuchs in London, England, between May 20 and June 2, 1950, pp. 28, 34.

⁹⁴HISTORY OF MODERN PHYSICS, Vol. II, pp. 307, 308.

⁹⁵LA-560 in "Publications of Stanislaw M. Ulam," compiled by Barbara Hendry, Los Alamos Scientific Laboratory, April 1968; Borden-Walker H-bomb chronology, p. 13; cited in "Technical References Relating to Thermonuclear Program," Appendix E to September 12, 1952 Walker memorandum.

September 1946: The Advent of the ALARM CLOCK

In early September, Edward Teller proposed a simplified alternative Super design called ALARM CLOCK, so-named because Teller hoped that it would awaken people to the prospects of thermonuclear weapons.⁹⁶ Although the ALARM CLOCK was a thermonuclear device, it derived only a small part of its explosive force from fusion; it was similar to a later BOOSTER design in that thermonuclear reactions in the bomb only enhanced fission.⁹⁷

The ALARM CLOCK used a tamper made of alternating layers of fissionable materials and fusion fuel. In a memorandum dated October 1, 1946, Dr. Teller had proposed an implosion bomb

⁹⁶Alexander, p. 10; Hewlett and Anderson, p. 632; Hewlett and Duncan, p. 59; Bethe, "Comments on the History of the H-Bomb," pp. 47, 48 (this is the so-called "Method B" in the article); statement by Norris E. Bradbury, Director of the Los Alamos Laboratory, at a press conference held in Los Alamos, New Mexico, on September 24, 1954, and reprinted in LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, pp. 27, 28; Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, p. 12; Borden-Walker H-bomb chronology, pp. 2, 13. Gregg Herken, in "Mad About the Bomb," (HARPER'S, December 1983, p. 50) quotes Teller as allegedly stating that he was "building an alarm clock -- one that would wake up the whole world." Bowen and Little, in AF ATOMIC ENERGY PROGRAM, Vol. II, pp. 497 and 498, and Vol. IV, pp. 186 and 201, claim that ALARM CLOCK was the name of a 1944 thermonuclear weapon design, that the name was later transferred to a 1949 "booster" design, and was transferred again in 1952 to a Los Alamos design for a solid-fueled thermonuclear weapon, possibly the TX-14: the ALARM CLOCK designation would eventually be assigned to the TX-14. J. Carson Mark, in LA-5647-MS, consistently uses the designation TX-14 in place of the original ALARM CLOCK, even though these were finally two completely different thermonuclear designs. The code name ALARM CLOCK was made public as early as 1964 in Volume II of the journals of David E. Lilienthal (pp. 106, 180), where it was described only as the code word for "a design for a super A-bomb." During a JCAE meeting in late September 1952, Dr. H. D. Smyth noted that "the whole 'Alarm Clock' design has fluctuated; there have been many different ideas." In January 1953, Teller recollected that the idea for the ALARM CLOCK was proposed on August 31, 1946, the same day that Teller's daughter Wendy was born. (Memorandum for the File dated January 13, 1953 from John Walker.)

⁹⁷OPERATION SANDSTONE, Nuclear Explosions - 1948, Scientific Director's Report - Volume I, General Report, Eniwetok Proving Ground, p. 57.

with a heterogeneous tamper of fusion materials that "might well be a steppingstone leading to a thermonuclear explosion." The ALARM CLOCK was later described as "a scheme to obtain large energy release with relatively cheap materials."⁹⁸

This was a novel approach that suggested that a thermonuclear bomb might be well within the capabilities of Los Alamos,⁹⁹ although there was no way of putting this idea into practice at the time.

This design would require an initiating explosion at least two or three times as powerful as the wartime FAT MAN. Theoretical work on the ALARM CLOCK continued from its inception to the end of 1947; its design changed a number of times during this period.

The first comprehensive report on the ALARM CLOCK was issued in November; it was written by Edward Teller and R. D. Richtmyer and contained arguments for the feasibility of the principle, and rough estimates of efficiency and behavior. The report concluded: "At the present time it seems the proposal is entirely feasible, provided (some possible adverse effects are) not too serious."¹⁰⁰

Fundamental study also continued on the processes that occurred during the detonation of a fission device.¹⁰¹ Before a successful fusion bomb could be built, much more advanced fission triggers had to be developed and the products and energies of nuclear explosions better understood. In December, an experiment was proposed to check predicted features of thermonuclear burning in conjunction with a fission bomb test of moderate yield.¹⁰²

⁹⁸Minutes of the Twenty-Third Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, October 30, 31, November 1, 1950, p. 62.

⁹⁹Borden-Walker H-bomb chronology, p. 14; letter dated November 14, 1946 to U.S. Atomic Energy Commission from N. E. Bradbury, Director, LASL.

¹⁰⁰Mark, LA-5647-MS, pp. 5, 8, 9; LA-610, "A New Thermonuclear System," cited in "Technical References Relating to Thermonuclear Program," Appendix E to September 12, 1952 Walker memorandum.

¹⁰¹Alexander, pp. 10, 11.

¹⁰²Mark, LA-5647-MS, p. 5; "Proposed Bomb Tests for Study of Thermonuclear Principles," LAMS-560, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, April 22, 1947.

Unknown to the U.S., the Soviets began speculating as early as 1946 on the possibility of building a hydrogen bomb. Some of their early work may have been based on espionage by Klaus Fuchs at LASL;¹⁰³ however, the physicists involved had been reading open literature on the subject since 1934 and may also have benefited from some early experimental work in Germany during World War II, data which may have come to their attention after the war ended and the Red Army occupied East Germany.¹⁰⁴

In any case, the Soviets were at least as far advanced theoretically as the U.S. by the end of 1946.¹⁰⁵ The Russian physicists realized intuitively the importance of compressing the fusion fuel to enhance the likelihood of reactions; the Soviets also proposed to increase confinement time with a thick bomb casing and to examine methods of minimizing loss of reaction energy to radiation and via transfer of energy to particles that did not play a direct role in the fusion process:

Introduction

It is very desirable to broaden the circle of elements the nuclear reaction of which can be utilized practically for power production or explosive purposes. Up until now the known processes of practical utilization of a nuclear reaction were based on the realization of the fission chain reaction: the reaction here takes place under the action of neutrons, additional neutrons being produced in the reaction.

....

... In all cases for the utilization of nuclear energy of the light elements, it is necessary to carry out a nuclear reaction

¹⁰³Sakharov, p. 94. Rhodes, in DARK SUN, p. 257, says that by the end of 1946, thanks to Fuchs, the Soviets knew everything the Americans knew by then about the Super.

¹⁰⁴Additional footnote 14a to "The H-Bomb: Who Really Gave Away the Secret," Daniel Hirsch and William G. Mathews, in Soviet Physics Usp. 34 (5) May 1991 (Uspekhi Fizicheskikh Nauk 161, 153-169), translated from the Russian (after originally appearing in English in the Bulletin of the Atomic Scientists, January/February 1990, pp. 22-30.

¹⁰⁵Rhodes, DARK SUN, p. 257.

of similarly charged nuclei. The reaction of similarly charged nuclei always requires a definite minimum energy of the colliding nuclei; at a lower energy of collision the probability of the (fusion) reaction falls sharply.

On the other hand, a charged particle much more frequently exchanges energy with electrons and nuclei rather than by entering into a nuclear reaction. Therefore, under ordinary conditions (at not too high temperatures), only a small fraction of the charged particles to which an initial energy was provided will enter into a nuclear reaction, so that the reaction will be a damped one.

The (thermo)nuclear reaction will proceed without being damped only at very high temperatures of the entire (fuel) mass, since only in that case the average loss of energy of a charged particle is compensated by the inverse process of the transfer of energy from highly heated electrons and nuclei to the particles under consideration. (Author's note: the opposite of the inverse Compton effect.)

....

Conditions for Carrying Out the Reaction

Based on the foregoing we propose the realization of a (thermonuclear) reaction under conditions that are distinguished by:

- a) a high reaction energy per particle;
- b) a low energy required to carry out the reaction with a large cross section for the interaction; for this is it desirable to use reactions of nuclei with a low charge;
- c) the reaction being carried out in a system with the lowest coefficients of absorption and emission of x-rays with the aim of obtaining a nonequilibrium distribution in which the entire energy would be confined as much as possible in the

kinetic energy of the charged particles, while the amount of energy transferred to radiation would be a minimum; for this purpose it is also required to use nuclei with a low charge;

- d) the realization of a reaction of the detonation type. The meaning of this condition consists of the fact that a shock wave is propagated throughout the mass of the reacting substance; the energy of heating in the shock wave is of the same order (of magnitude) as the energy of the reaction. Heating in the shock wave occurs during a very short period of time of the order of the flight time of the charged particles. Following this, the material heated in the shock wave reacts, liberating energy and expanding, pushing the shock wave in front of itself. The process gives in principle the possibility of an explosion of an unlimited amount of the light material suitable for the reaction from a prescribed sufficiently powerful initial pulse.

We emphasize that forming an opinion on the possibility of an explosive (thermonuclear) reaction is associated with the application of the modern detonation theory that has been developed in the Institute of Chemical Physics.

A Specific Proposal

As a system satisfying the conditions outlined above we propose (the detonation of) deuterium

The minimum diameter of the long charge of the deuterium, according to the available data on the dependence of the cross section of the reaction (with) the energy of the particles, cannot exceed 30 centimeters (Author's note: 11.8"). For the basis of these numbers, see the Appendix.

The greatest possible density of deuterium is desirable, and this should be realized

by using it under high pressure. (Author's note: emphasis added.)

In order to facilitate the initiation of a (thermo) nuclear detonation, it is useful to employ massive enveloping shells delaying the dispersal. (Author's note: a thick exterior casing or tamper to prolong confinement time.)

The most difficult problem is the problem of ignition, since in uranium or in plutonium the explosion develops relatively slowly, and, correspondingly, quite a significant portion of the energy of the explosion has time to be transformed into radiation, as a result of which the temperature and the pressure of the nuclei and the electrons turn out to be relatively not too high.

At the present time, the question remains unclear concerning the effect of radiation on the process of expansion of uranium which transmits the pressure to the deuterium. In order to improve the conditions of ignition, it appears possible to use uranium charges of increase sizes and of a special shape (cumulation) and to introduce into the deuterium heavy elements near the initiator which might be capable of receiving the radiation pulse. (Author's note: an early conceptual approach to the "sparkplug" in the fusion fuel.)

However, even with the remaining uncertainties in the question of ignition, it appears to us to be very essential to discover a system in which a single powerful pulse can initiate a (thermo)nuclear detonation of an unlimited amount of matter.

Appendix

The possibility of carrying out a (thermo)nuclear reaction is determined by the ratio between the energy emitted as radiation of electromagnetic waves and the energy of nuclear disintegration. If this ratio is less than unity, the (thermo)nuclear reaction can develop.

....

The detonation wave can be propagated over the deuterium "charge" only if its dimensions are sufficiently (large). This minimum size is equal in order of magnitude to the product of the velocity of sound in the fuel multiplied by the reaction time.

...
....

The critical diameter is ... approximately a half meter. (Author's note: approximately 19.7".)

This value may turn out to be (smaller), if the (fuel) "charge" (is) enclosed in a massive shell, and also due to the fact that an alternate reaction ($D+D \rightarrow n + He^3$) will be taking place which will increase the energy release and decrease the time of the reaction.¹⁰⁶

Work proceeded at LASL in 1947 on both the Super and the ALARM CLOCK designs. Loss of energy and a reduction in reactant temperature due to radiative cooling continued to be a problem.¹⁰⁷ Theoretical study and a few experimental physics measurements continued, based on wartime findings.

A substantial portion of the time of the physicists working on the Super was devoted to making improvements in existing and proposed fission weapons.¹⁰⁸ A major development in early 1947

¹⁰⁶"Utilization of the Nuclear Energy of the Light Elements," I. I. Gurevich, Ya. B. Zel'dovich, I. Ya. Pomeranchuk, and Yu. B. Khariton, 1946, originally in Soviet Physics Usp. 34 (5) May 1991 (Uspekhi Fizicheskikh Nauk 161, 153-169), translated from the Russian by G. M. Volkoff, and reprinted by the American Institute of Physics. Rhodes, in DARK SUN, p. 256, interprets this paper as being the genesis of the later Soviet "layer cake" or "Alarm Clock" type concentrically-spherical design. However, the reference to "the minimum diameter of the long charge of the deuterium" seems to imply a cylindrical arrangement and not a sphere.

¹⁰⁷LAMS-566, T-Division Progress Report, March 20, 1947 - April 20, 1947, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, May 6, 1947, pp. 1, 2, 7, 8. Rhodes, in DARK SUN, p. 279, quotes Hans Bethe as saying that calculations on the ALARM CLOCK were set aside at the end of 1946.

¹⁰⁸A letter dated January 13, 1947, from Norris E. Bradbury, Los Alamos lab director, to Carroll Wilson, General Manager,

was the invention and design of the BOOSTER (See "Advances in Boosted Fission Weapons" in Volume II).¹⁰⁹

An "improved theory of efficiency" of the ALARM CLOCK was developed in January and February; the study was released in May. In early February, the GAC noted "there is an optimistic report by Dr. Teller and some suggestions of a later date on a different thermonuclear arrangement."

The GAC also recommended that development of the thermonuclear program should not be undertaken in competition with, but as a helpful adjunct to, the improvement of fission weapons at LASL. If thermonuclear work were to be strengthened at the laboratory, as the GAC strongly recommended, then a small number of theoretical physicists would have to added to the lab staff.¹¹⁰

The GAC concluded:

The encouraging results of preliminary study in this field, the existence of suggestive new proposals for thermonuclear explosives, but above all, our conviction that, properly understood, the pursuit of thermonuclear problems would strengthen rather than weaken the quality of work on atomic weapons, would make us now wish to assign a higher urgency to this work.¹¹¹

Atomic Energy Commission, Washington, D. C., stated that at the start of 1947, Los Alamos efforts were directed along the lines of basic research applicable to weapons (15% of man-hours); production of weapon components (25%); near-term weapon development (35%); long-term weapon development (2%); ordnance engineering (16%); power or propulsion applications of nuclear energy (1%); and basic scientific research not obviously applicable to atomic weapons (6%). Production of weapon components and ordnance engineering would soon be taken over by contractors and the new Sandia Laboratories of Western Electric.

¹⁰⁹The "Booster" is "Method C" in Hans Bethe's "Comments on the History of the H-Bomb," LOS ALAMOS SCIENCE, Vol. 3 No. 3, Fall 1982, pp. 42-53.

¹¹⁰Draft Minutes of the Second Meeting of the General Advisory Committee, February 3-4, 1947, pp. 2, 8.

¹¹¹Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953.

The competition between the Super and fission weapon programs for scarce LASL resources would continue for the next several years. Behind the Iron Curtain, the Soviets grappled with a similar dilemma: how much should fission weapon improvement be delayed or lessened to provide materials, time, and manpower for a far-less certain but potentially far-more powerful weapon?¹¹²

The GAC also noted at this time "that should it appear, as the Committee believes unlikely, that a strong group of men interested and qualified in the development of atomic weapons will be more willing to undertake this development at a site other than Los Alamos, this course should not be discarded as too costly or too dangerous."¹¹³ Within a few years, when a second U.S. nuclear weapons development laboratory was advocated, both the GAC and the AEC would put up fierce opposition.

During the third meeting of the GAC late in March, Enrico Fermi noted that thermonuclear work required a theoretical physicist of Edward Teller's ability, and that work should be tempered with a high degree of pessimism concerning the possibilities of successful thermonuclear devices. Fermi hoped that the current doubts about feasibility could be resolved within a year; he also stated that although there might in the future be a requirement for increased tritium production, it should continue to be generated at the current rate.¹¹⁴

The GAC noted that although a thermonuclear weapon was not imminent, work on it should continue:

Rapid progress in the development of thermonuclear explosives is not to be anticipated; nevertheless, we are convinced that at least for the present, studies bearing on this problem are contributing to the health of the Los Alamos project, and laying the foundation for further progress in the future. For these reasons, we would like to see included in the Los Alamos directive a statement along the following lines:

¹¹²Zaloga, p. 95.

¹¹³Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953.

¹¹⁴Draft Minutes of the Third Meeting of the General Advisory Committee, March 28-30, 1947, p. 8.

"Research directed towards the utilization of thermonuclear reactions in atomic explosions. Research of this character, while necessarily of a long-range nature, is considered of very great interest by the Commission.

Periodic status reports at least every six months on the two developments now under consideration are requested, specifically the 'Super' and the 'Alarm Clock.' Each report should include not only a description of the present status of the problem but (also) an estimate of the necessary experimental program, both in time and character required to bring such developments to fruition. There should finally be included brief reports of any experimental programs undertaken whose primary objective concerns thermonuclear weapons."¹¹⁵

LASL also began planning for the possible application of advanced electronic computing equipment, then in early stages of design at Princeton University, to its thermonuclear design studies.¹¹⁶

April 1947: A Proposal for Thermonuclear Testing

LASL proposed in April a series of tests for study of thermonuclear principles:

The proposed bomb tests have been given much consideration lately in the Theoretical Division. It seems to us clear that any test sponsored or participated in by this laboratory should be of such character that it will truly extend our knowledge about the basic processes in a nuclear explosion. With this in mind, we wish to call attention to a possibility of testing certain principles that would be of importance in thermonuclear systems, such as the ALARM CLOCK. The possibilities for exploring these principles by theoretical methods alone are inadequate to give a

¹¹⁵Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p. 5.

¹¹⁶Mark, LA-5647-MS, p. 5.

completely reliable picture because of the great complexity of the phenomena, and consequently an actual test of the principles under conditions prevailing in a bomb explosion is highly desirable.

In the test a [deleted] The high temperature produced by the fission explosion would cause a thermonuclear reaction to occur. In such a system, the energy release by the thermonuclear reaction would be negligible but the 14-Mev neutrons produced by the DT reaction should be easily detectable and the tritium produced may also be detectable if the system contained only deuterium initially. The success of this experiment would depend firstly on the attainment in the deuterium of high temperatures as indicated by radiation flow studies and secondly on the [deleted]

Conversely, a suitably designed series of experiments would provide tests of these two important points. Consider three tests as follows: [deleted]

Test C is much less sensitive to [deleted] so that comparison of these two will give information [deleted] than a comparison of predicted and observed numbers of 14-Mev neutrons in either of tests B and C will give information about the temperatures reached. Test A is, of course, necessary as a control.

.....

The calculation was done for the solid delta-phase plutonium core, since this would not only be simpler than for a composite core but an observed value of the time constant (α) is available for this instance and this case also should be at least as sensitive to the presence of deuterium as the composite model.¹¹⁷

¹¹⁷"Proposed Bomb Tests for Study of Thermonuclear Principles," Rolf Landshoff, Carson Mark, and R. D. Richtmyer, LAMS-560, Los Alamos Scientific Laboratory, April 22, 1947, pp. 2, 6, 8. The device design hypothesized for at least part of this report featured a "deuterium-containing region" between the tamper and the fissile core, as in the so-called layered ALARM CLOCK design. Test B was to be fueled

This proposal became the basis of the George and Item shots of Operation GREENHOUSE four years later.

Also in April, AEC commissioner Lewis Strauss recommended that the AEC, CIA, and DOD begin an upper-atmosphere radiosampling program to determine whether a foreign nation - - most likely the Soviets -- might have detonated an atomic bomb:

It would be interesting to know whether the intelligence arrangements made any provision in the past for continuous monitoring of radioactivity in the upper atmosphere. This would be perhaps the only means we have for discovering that a test of an atomic weapon has been made by any other nation.

It is to be presumed that any other country going into a large-scale manufacture of atomic weapons would be under the necessity of conducting at least one test to "prove" the weapon. If the Combined Intelligence Group has no such monitoring system in effect, it is incumbent upon us to bring to their attention the desirability of such an immediate step and, in default of their action, to initiate it ourselves at once.¹¹⁸

Strauss was authorized by the AEC to proceed with an inquiry, and upon discovering that in fact no atmospheric monitoring was underway, he took steps to have it started. A series of conferences was held between the AEC and DOD; Defense Secretary James Forrestal authorized expenditure of funds and use of Air Force aircraft.

At its fourth meeting at the end of May, the GAC recommended that "the behavior of materials of atomic number 90 or above under neutron bombardment, the spontaneous fission of such materials, and the number and energy of neutrons emitted in fission by such materials, shall be maintained secret."¹¹⁹ In

only by deuterium; test C was to use both deuterium and tritium. In both tests B and C, the fusion fuel was to be highly compressed.

¹¹⁸Memorandum dated 11 April 1947 to Chairman and Commissioners of the AEC from Lewis Strauss, quoted in "A Chronology of the Thermonuclear Weapon Program to November 1952, " draft dated August 14, 1953. This document portrays Strauss as the savior of the American hydrogen bomb program.

this recommendation, the GAC apparently overlooked the fact that between 1938 and 1947, fission reactions and their products from transuranic elements had been well-researched, measured, and documented not only in the U.S., but also in France, Great Britain, the Soviet Union, Germany, Italy, and Japan.

In the widely-read and noted 1945 Smyth report on the Manhattan Engineer District, Smyth had published the fact that U-235 fission occurred and yielded one to three neutrons, and that plutonium could be produced artificially and could be fissioned by thermal neutrons. This 1947 GAC statement looked like an attempt to classify information that was already well-known.

In the summer of 1947, a device burning a cubic meter of liquid deuterium and yielding about 10 megatons was proposed; this design would require a fission trigger yielding several hundred kilotons and as much as 600 grams of tritium to ignite the deuterium. (This ratio of fuel to yield indicates a fusion efficiency of about 25%).

The continuing question of whether the reaction in deuterium would actually propagate or whether the cooling through radiation might stop the reaction was investigated in more detail than ever before. Although there were many doubts and unknowns, the Super was still considered feasible.¹²⁰

By September 1947, despite study of a number of possible configurations, the proposed ALARM CLOCK looked less and less promising.¹²¹ The Super calculations continued to be hindered by a lack of adequate computers; even the greatly simplified computations that were done were left out many important factors and were therefore quite unreliable (the computations were simplified to avoid exceeding the capabilities of the ENIAC).

Los Alamos desperately needed data to predict and describe the behavior of materials at incredibly high temperatures, the methods by which energy was transferred from particle to particle, and the means by which energy was dissipated and

¹¹⁹Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p. 7.

¹²⁰AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 6; "In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 281.

¹²¹Mark, LA-5647-MS, p. 5; Bethe, "Comments on the History of the H-Bomb," p. 47.

lost.¹²² Without this information, it would be nearly impossible to determine the feasibility of a thermonuclear weapon.

The greatest handicap facing the physicists in T (Theoretical) Division was the lack of suitable computers. (Computer calculations on the original ALARM CLOCK configuration were finally made in 1953-1954 and showed that the device would not have worked in its proposed form.)¹²³

The most favorable calculation at this time indicated the possibility of obtaining 10 megatons of TNT-equivalent explosive force from a device weighing between 40 and 100 tons. Although a megaton had been used in the calculation to provide the initiating explosion, there were arguments that a smaller explosion might suffice, possibly as small as 200 kilotons (the most powerful fission bomb detonated by this time was the 21 kiloton FAT MAN weapon used at TRINITY, Nagasaki, and CROSSROADS).¹²⁴

A high-yield trigger was required because the relatively-heavy neutrons lost much of their kinetic energy in collisions with atoms of materials in bomb components between the exploding fissile pit and the deuterium-tritium fuel mass. The higher the yield of the fission trigger, the greater the energy of the neutrons radiating from it. The number of collisions between neutrons and bomb component atoms could be significantly reduced by "pipelining" the neutrons directly from the fission trigger core into the D-T fuel mass.

Teller Suggests a Two-year Delay

On September 26, Dr. Teller wrote a technical memorandum entitled "On the Development of Thermonuclear Bombs," a major treatise on the thermonuclear program. The paper pulled together all aspects of the hydrogen bomb program as of that time. Dr. Teller again suggested the possibility of using Li^6D as a solid thermonuclear fuel, and suggested that quantities running into hundreds of kilograms might be required.¹²⁵ Teller noted that tritium production was a factor delaying early construction of a thermonuclear bomb.¹²⁶

¹²²Hewlett and Duncan, p. 439.

¹²³Alexander, p. 12.

¹²⁴Appendix II to Annex O of ComJointTaskFor One Operation Plan 1-46, General Guide on Public Information Policies, p. O-II-2.

The use of lithium would simplify some problems but also required the production of separated lithium-6 and did not solve the ignition problem. A program of continuing research and development was proposed, including studies of neutron and light element cross-sections and tests to check predictions of thermonuclear burning on a small scale.

In this report, Teller reviewed the "state of the art" of the American hydrogen bomb and the heavy dependence of its future success upon the development of more powerful electronic computers and a better understanding of the propagation of a shock wave in the thermonuclear fuel mass. Despite many unknown factors, he reiterated his belief in the feasibility of the 1946 Super design and hypothesized the effects of a 1,000 megaton explosion.

Teller's report also suggested that two thermonuclear tests be undertaken as soon as possible after the Pacific test operations then scheduled for the spring of 1948 (Operation SANDSTONE) so that data from those tests could be used to determine feasibility of the ALARM CLOCK. The paper emphasized that tritium production would probably be the determining factor in the early construction of any thermonuclear bomb.¹²⁷

The report concluded that one Super design, possibly the proposed "Runaway Super," was "probably feasible," but that "its complex construction gives little hope that it can actually be made to work in the next three or four years; it requires, furthermore, considerable amounts of tritium." As regarded the ALARM CLOCK, the report concluded that its

¹²⁵Borden-Walker H-bomb chronology, p. 16; September 12, 1952 Walker memorandum, p. 20. According to Richard G. Hewlett and Jack M. Holl, in A HISTORY OF THE U.S. ATOMIC ENERGY COMMISSION, Volume III: ATOMS FOR PEACE AND WAR, 1953-1961 (University of California Press, Berkeley, California, 1989, pp. 162, 163), the inclusion of lithium in a thermonuclear weapon to produce tritium in situ was not considered until the famous Princeton conference of June 1951; however, this seems rather late for this idea to have been suggested. According to the September 12, 1952 Walker memorandum, production quantities of lithium-6 on the order that Teller suggested were not considered by the AEC until 1951.

¹²⁶Alexander, p. 14; "On the Development of Thermonuclear Bombs," Edward Teller, LA-643, Los Alamos Scientific Laboratory, February 16, 1950 (the original report was written 1n September 1947).

¹²⁷Borden-Walker H-bomb chronology, p. 16.

feasibility ultimately hinged upon whether or not there would be mixing of materials, and if so, to what extent.¹²⁸

Teller also suggested that development of the ALARM CLOCK or the Super be delayed for two years until further calculations and experiments were completed:

In the summer of 1947 the possibilities for constructing thermonuclear bombs were reviewed. This was done with the purpose of arriving at a concrete program for the further investigation of thermonuclear explosions. Such further work is to lead to a more definite knowledge of the feasibility of the various kinds of thermonuclear bombs.

In outlining further research, I have attempted to concentrate on objectives which the Los Alamos Laboratory can accomplish in the foreseeable future. I am, therefore, restricting the discussion to designs which seem to require the smallest number of new technical developments. Even so, the amount of work required for any thermonuclear gadget will prove to be very considerable.

So far, three types of thermonuclear bombs have been proposed and theoretically explored: the Super, the Alarm Clock, and the Booster. ...

[deleted] This is accomplished by a fission bomb. The reaction is then expected to spread rapidly. The feasibility of such an explosion is based [deleted] Sizable amounts of energy are nevertheless released [deleted] detonated. Thus the energy of these projected bombs could exceed the energy of the conventional fission bombs by several orders of magnitude.

The third design, the Booster, is an old and frequently discussed scheme. [deleted]

In the first part of the following report a specific model of the Super will be described and critically reviewed.

¹²⁸Borden-Walker H-Bomb Chronology, pp. 15, 16.

In the second part the feasibility and minimum size of the Alarm Clock are discussed.

In the third part the effects of the Super or the Alarm Clock are described. These effects are compared with the effects of fission bombs.

In the fourth section a brief discussion of the Booster is given.

In the fifth section a program of research and development is outlined with the purpose to further clarify the feasibility of thermonuclear bombs and to facilitate their eventual construction.

PART I: A DESIGN OF THE SUPER BOMB

The principle of operation and the main design features of a Super will be discussed in connection with Figure I. This figure represents a model which was first presented in the Los Alamos conference of April 1946. The advantages of this model correspond to the aims outlined in the introduction: a minimum of new technical developments is required and preference is given to a construction whose functioning can be predicted with relative ease. It is probable that other designs of the Super will require less of the materials which are difficult to produce.

In Figure I a cross section of the initiating mechanism of the Super is given.
[deleted]

The processes going on in the sphere D are complex and are, so far, not checked by direct experiments. There is reason, however, to trust the calculations which predict the reactions described above.
[deleted] Such problems can be, and in this case have been, treated exhaustively.

I believe that the energy-transmission between A and D is very likely to proceed as described. Furthermore this part of the design is sufficiently (amenable to modification) if some unexpected difficulties in this energy transmission

should be found. The functioning of sphere D depends on the reliability of the measurements of deuterium-tritium (reaction) cross sections and on the correctness of calculations. Both have been carried out with care. Furthermore, conditions are not marginal and one may thus have considerable confidence that region D will function as described.

The 14 million volt neutrons generated in [deleted] radiation shield F.

The functions of regions G and H is very hard to calculate because in these extended regions, variations in time and space must both be taken into account. An attempt at such a calculation was carried out in the spring of 1946 on the electronic computer, the so-called ENIAC. [deleted]

In the past months a more detailed investigation of this question has been carried out. This investigation is actually the only relevant progress with regard to the Super since the beginning of 1946. These considerations are summarized in Appendix I.

[deleted] remain the most doubtful points in the feasibility of the Super. The calculations performed during this summer are expected to continue; checks on numerical results and on the consistency of the approximations used will have to be carried out. Further calculations will probably require high-speed electronic computing equipment. I believe, however, that at the present stage not very much doubt remains that the Super in principle is feasible. (Author's emphasis added.)

The residual doubt is of three kinds [deleted] This question has been considered now by a considerable number of men for several years and no flaw has been so far detected.

Second, the propagation of the reaction in the charge requires the discussion of a non-linear partial integral-differential equation. The mathematical handling of this question is incomplete and may have to

remain not quite complete up to an actual test. Work carried out during this summer and described in Appendix I diminishes, in my opinion, the reason to doubt that the thermonuclear reaction can be made to propagate.

Third, the actual design of the Super is complex. It is possible that during actual construction further difficulties will arise which are hard to foresee at the present stage of long-range planning.

In order to give an overall picture of the Super we include the schematic Figure II. [deleted]

In conclusion it may be stated that the Super is probably feasible. The complex construction gives us little hope that it can actually be made to work in the next three or four years. It requires, furthermore, considerable amounts of tritium. [deleted]

PART II: THE ALARM CLOCK

[deleted] last part we shall describe an experimental method of getting information on this point. In the following I shall outline the normal working of the Alarm Clock and I shall give some qualitative arguments concerning mixing.

[deleted]

On the basis of the best available information on cross sections we have estimated that the total number of fissions caused by 2.5 MeV neutrons is approximately 0.2, and the total number of fissions due to 14 MeV neutrons is 0.7. In these estimates, neutrons slowed down below the U-238 fission threshold are not included because [deleted] The estimates are consistent with integral experiments in which D-T-spawned neutrons are caught in a uranium block and with similar experiments now in progress under the direction of Dr. (Richard F.) Taschek in which D-T neutrons are used. The effect of the deuterium could so far only be crudely estimated.

[deleted]

Actually the calculations carried out in the Appendix are crude approximations. The two most important simplifications introduced are: It is assumed that at any one time [deleted]

A complete calculation taking into account these neglected effects is likely to become feasible only when high speed computing equipment becomes available. Such complete calculations are likely to show that the real ignition [deleted] It should be remarked, however, that our present knowledge of cross sections is still so uncertain that a further considerable change of the required ignition energy might result [deleted]

Part III: EFFECTS OF A SUPER

....

If a Super is used instead of the Alarm Clock and if no special arrangements are made to utilize the radioactivity of the fission products, the radioactive damage will be insignificant in comparison with other damage done (by fire and blast). If, however, special arrangements are made to utilize the neutrons (to make) fission products or other radioactive materials, one gets effects similar to those in the case of the Alarm Clock. In fact, by absorbing the neutrons (produced by fusion) in appropriate materials and generating (radio)activities of the right (half life), one might obtain from the Super many times the radioactive effect produced by an Alarm Clock.

It is in the nature of the Super and the Alarm Clock that the energy released by either of these devices may be increased with hardly any practical limit. ... [deleted]

There seems to be, however, no reason to believe that such a great engineering development cannot be accomplished within a few years. It is also clear that delivery of such an object by aircraft is likely to remain impossible for quite some time to

come. We shall see, however, that delivery by boat or submarine is capable of producing disastrous effects. [deleted] This again is an object that one hardly will be able to transport by air. We have seen perhaps the most hopeful design for an Alarm Clock [deleted]

This would decrease the engineering difficulties. It seems to me that if an Alarm Clock can be made at all, a big scale Alarm Clock could be more easily engineered than a big scale Super. Even so, the engineering and transportation of a one-billion-ton yield Alarm Clock will remain quite difficult.

....

PART V: PROGRAM OF RESEARCH AND DEVELOPMENT

Experiments which are now under way in Los Alamos will greatly help to clarify how difficult it is to initiate the Alarm Clock. The most important pieces of information which at present are quite incomplete and need further study are: the elastic, inelastic, and particularly the fission cross sections of high energy neutrons up to 14 MeV in ordinary uranium.

The same information for (U-235) and (Pu-239) and possibly (U-233) would also be of interest, though to a considerably lesser extent. It would be, however, of quite basic importance to study for high energy neutrons, the elastic cross section of deuterium and also the cross section for the disintegration of deuterium into a proton and a neutron.

There is some interest in studying the neutron cross sections in the high energy range for beryllium, oxygen and aluminum; [deleted]

Further research of D+D and D+T cross sections continues to be of interest. While these cross sections have been measured carefully, it is not yet quite clear that their values are more accurate than 20 or 30% even in the [deleted] In order to evaluate the cross sections for the lower energies, it would be of interest to study

the ranges of tritium and deuterium at low energies.

These ranges cannot be calculated with sufficient reliability ... theoretical range calculations become extremely tricky and ranges may even depend on chemical composition. ...

It is of some interest to explore some of the reactions which might become significant in advanced stages of the reactions of the Alarm Clock or the Super. I refer in particular to further research on the reaction of $\text{He}^3 + \text{D}$ and also to the $\text{T} + \text{T}$ reaction. These problems, however, are of considerably less importance than those mentioned above, since they will help to determine the exact energy output of the Super or the Alarm Clock rather than their feasibility.

Fully as important as the most vitally needed differential cross sections are integral experiments. Some of these have already been performed, such as the number of fissions produced in a big mass of uranium by $\text{D} + \text{D}$ and $\text{D} + \text{T}$ neutrons. ...

Further laboratory research should be performed in preparation of tests which in turn will be of importance in determining the feasibility of the Alarm Clock. I refer in particular to the exploration of the fission spectrum in the neighborhood of and above 14 MeV. As will be mentioned further below, tests on the feasibility of the Alarm Clock may be carried out in connection with Trinity-type explosions.

The success of these testing procedures with regard to the feasibility of the Alarm Clock, however, depends on producing enough 14 MeV neutrons from a $\text{D} + \text{T}$ reaction, so that these neutrons should be clearly distinguished above the background of the fission (generated) neutrons. ...

The experiments outlined above can be carried out without interfering with (important short-range work). In particular, one can hope that the most important pieces of valuable information --

namely, fission cross sections and n-D cross sections in the high-energy neutron range, the integral experiments, the threshold studies, and the analysis of the fission spectrum in the high energy range - - may be carried out within a year.

It would be of particular interest to make fast progress with the two topics last mentioned, since the forthcoming Pacific tests could give us by next Spring interesting pieces of information, if appropriate threshold detectors were available and if we should know enough about the shape of the fission spectrum in the high-energy range. If such measurements could be carried out during the coming Spring in the Pacific, the results could be used as valuable points of comparison (against) which results of later tests could be (interpreted). [deleted]

....

I should like to propose that the two tests described above should be carried out as soon as feasible after the Pacific tests next Spring. With the help of these tests, we could form a judgment in 18 or 24 months from now whether the Alarm Clock is feasible or not. In the meantime, the experimental program described in the first portion of Part 5 of this report should have furnished us with enough information to make a (closer) estimate of the energy needed to ignite the Alarm Clock.

At approximately the same time, high-speed computational facilities are likely to become available so that we shall be in a much better position to solve the integro-differential equations which occur in the theory of the Alarm Clock. Thus, we are likely to be able to form at that time a realistic estimate of the work involved in making an Alarm Clock. I think that the decision whether considerable effort is to be put on the development of the Alarm Clock or the Super should be postponed for approximately 2 years; namely, until such time as these experiments, tests, and calculations have been carried out. (Author's emphasis added)

[deleted]

....

Calculations on the Super, Alarm Clock, and Booster bomb may be continued with considerable profit. Eventual use of fast computing equipment may be speeded if the theory of these bombs is not neglected in the near future.

APPENDIX

[deleted]

We have made an attempt to find a solution for the (shock wave) propagation which is stationary and stable in the same sense as these words are used in the theory of detonations in high explosives. It is, of course, not clear in principle that such a solution exists at all nor is it clear that a solution will be stable against all perturbations and whether it will be rapidly approached whenever the ignition has been successful. These questions are among the most difficult ones of hydrodynamics and even for ordinary high explosives, they have not been completely solved.

We have calculated a number of cases. In each of these, several of the relevant effects have been neglected so as to obtain a problem sufficiently simple to be treated analytically or by simple numerical methods. These several calculations certainly do not solve the problem of the propagation of the explosion in deuterium but (with) their help, we can form a better judgment as to the actual existence of a stable and steady solution.

More refined calculations, taking into account effects now believed to be of secondary importance, will be undertaken in the immediate future by the use of ordinary computational methods. A still more exhaustive treatment will require the use of high-speed computing machines.¹²⁹

¹²⁹"On the Development of Thermonuclear Bombs," Edward Teller, LA-643, Los Alamos Scientific Laboratory, February 16, 1950.

Further intensive study of the ALARM CLOCK was reduced after September 1947 in light of nearly overwhelming ignition problems; by the end of 1947, calculations on the design were not very promising and were stopped.¹³⁰

The name ALARM CLOCK was ultimately assigned to another type of thermonuclear weapon, although the design was not completely abandoned. Bombs of the original ALARM CLOCK design were believed feasible in principle, and capable of providing arbitrarily large yields, but the mechanism required to amplify the initiating yield was so large as to be of little practical value.¹³¹

Planning for a fully-detailed machine computation of the course of a fission explosion was started in October (this turned out to be a two-year program, and the first example was calculated early in 1950).¹³² One of the more discouraging discoveries at this time was that roughly twice as much tritium would be required to ignite the Super as had been predicted in April 1946.¹³³

Also in October, Teller's September report was reviewed by the GAC. The GAC noted that Teller's study had been classified "Top Secret;" after some discussion, the committee resolved that in view of the long-range character of the

The "materials which are difficult to produce" include tritium. Not all of Teller's colleagues agreed that "at the present stage not very much doubt remains that the Super in principle is feasible." Contrary to Teller's expectations, calculations in the summer of 1950 showed that the present Super design was not feasible and that thermonuclear reactions would not propagate in an uncompressed fuel mass. "Mixing" is the intermingling of non-nuclear materials with nuclear fuels. In the original document, U-235 is identified as "25," Pu-239 as "49," and U-233 as "23." Teller was quite correct in noting that D+D and D+T cross sections had not yet been measured fully.

¹³⁰Borden-Walker H-bomb chronology, p. 15; Bethe, "Memorandum on the History of the Thermonuclear Program," p. 12.

¹³¹Mark, LA-5647-MS, p. 9.

¹³²Mark, LA-5647-MS, p. 5. In 1947, Stanislaw Ulam and Robert Richtmyer had published a three-page study (LA-559) entitled "On the Nuclear Composition of the Implosion-Type Bomb During the Last Stages of the Explosion," LA-559 in "Publications of Stanislaw M. Ulam," compiled by Barbara Hendry, Los Alamos Scientific Laboratory, April 1968.

¹³³Mark, LA-5647-MS, p. 8.

subject of thermonuclear explosives, nothing known at this time justified the "peculiarly high security classification of papers on this subject."

The GAC concluded that in fact, the very difficult nature of the thermonuclear problem suggested a need for more widespread circulation of papers on this subject, and that future reports in this field should not be classified higher than "Secret."¹³⁴ (A month later, Dr. Lee Dubridge emphasized this opinion when he noted that "it would be far better to have a secret weapon in a decade than a 'Top Secret' paper forever.")¹³⁵

The GAC also commented favorably on Teller's September report and concluded that "the program of basic studies recommended in (Teller's) report should be encouraged as a healthy and useful part of the work of the Los Alamos Laboratory."¹³⁶

On the subject of thermonuclear energy, the GAC concluded:

The committee wished to study the matter further before endorsing proposals for greatly increased tritium production on the one hand, or for a weapons test of thermonuclear reagents on the other. These questions were made an order of business for the forthcoming meeting.

The Committee did, however, note that even with the most optimistic interpretation of present knowledge, the realization of thermonuclear weapons is many years in the future and, and that in many cases our knowledge of the subject is fragmentary and inadequate.¹³⁷

Serious consideration of a possible delivery vehicle for the Super started in 1948. Development of a weapon carrier would proceed in parallel with warhead development.¹³⁸

¹³⁴Draft Minutes of the Sixth Meeting of the General Advisory Committee, October 3-5, 1947, p. 7.

¹³⁵Draft Minutes of the Seventh Meeting of the General Advisory Committee, November 21-23, 1947, p. 36.

¹³⁶Letter dated October 10, 1947 to David E. Lilienthal, USAEC, from J. R. Oppenheimer, Chairman, GAC.

¹³⁷Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p. 9.

¹³⁸AF ATOMIC ENERGY PROGRAM, Vol. IV, pp. 184, 185.

An initial program of analytical study and attempts at numerical solutions of the "Super Problem", using only cumbersome electromechanical desk-top calculators, was brought to a close at the beginning of 1948. Between January and April, a considerable amount of computational work was required in preparation for the SANDSTONE nuclear weapon proof tests and analyses of the results.¹³⁹ This project briefly diverted research effort from the Super program. Among other projects interrupted was the hand-calculation of the progress of a fission explosion, work that was instrumental to the Super.¹⁴⁰

In February, the GAC recommended that the AEC begin nation-wide sampling of krypton-85 gas emitted by U.S. reactors, initially the Hanford plutonium reactors. Since Kr-85 was a by-product of plutonium generation, a current baseline of the amount in the atmosphere above the U.S. would be helpful in determining foreign emissions of the same material. The GAC further recommended "that any results from this monitoring be correlated with the most accurate Intelligence (sic) information."¹⁴¹

In March, a new method of numerical analysis was discovered as a result of studies of fission explosions. This so-called "viscosity treatment" reduced the problem of calculating the progress of shock fronts in explosion and implosion calculations to manageable proportions on automatic computing machines and was of profound value in many subsequent computations.¹⁴²

Two months later, an analytic solution of the first approximation on the propagation of a thermonuclear reaction was completed at LASL.¹⁴³ The AEC noted at this time that it was expecting the armed forces to complete a study of military requirements for a thermonuclear weapon; in the meantime, LASL's efforts were best directed toward the "booster," with a view to the possible planning of a test to

¹³⁹Mark, LA-5647-MS, p. 6.

¹⁴⁰T-Division Progress Report, 20 May 1948 - 20 June 1948, LAMS-753, 13 July 1948, p. 2.

¹⁴¹Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p. 13.

¹⁴²Mark, LA-5647-MS, p. 6.

¹⁴³LA-665, May 11, 1948, cited in "Technical References Relating to Thermonuclear Program," Appendix E to September 12, 1952 Walker memorandum.

both prove out the device and obtain information which might chart the way toward later developments.¹⁴⁴

Despite Operation SANDSTONE's deleterious effects on work at LASL, it did provide more data on the early stages of a fission explosion:

It can be said, then, that the Laboratory's empirical work and theory on the initiation and early phases of the nuclear explosion are good and point to good weapon design. But the theory which carries the explosion on through to its total energy release, an important consideration in studying the use of the weapon, is not too well developed. Therefore, an important result of the Eniwetok tests was to provide considerable experimental data from which the Laboratory may advance the theoretical treatment and prediction of total energy release.¹⁴⁵

Another beneficial effect of SANDSTONE was an opportunity to test the Air Force's long-range test detection system and to train aircrews in upper-atmosphere sampling techniques.¹⁴⁶

On June 6, the General Advisory Committee reviewed the thermonuclear program, and recommended that LASL efforts be concentrated on the more technically feasible BOOSTER:

Thermonuclear Weapons. In accordance with our expectation, we considered a development of thermonuclear weapons and the related problems of tritium production. Of the various models of such weapons, for instance the Super and the Alarm Clock, one and only one appears to be fairly capable of fairly rapid development; this is the so-called Booster.

¹⁴⁴Draft Memorandum from the General Manager to the Manager, Santa Fe Directed Operations, p. 5, Appendix "A" to AEC 99, Atomic Energy Commission, Weapons Program of the Los Alamos Laboratory, May 14, 1948.

¹⁴⁵REPORT TO THE ATOMIC ENERGY COMMISSION ON OPERATION SANDSTONE ATOMIC WEAPON PROOF TESTS, 1948, Part I, Volume I, Test Director, Joint task Force Seven, p. 2.

¹⁴⁶"A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, p. 2.

With the scope of effort now available, the other weapons are at best in the remote future; and in the absence of clear, well-defined, and over-riding military requirements therefor, and of an increase in the personnel available for their development, scheduled and active work upon them hardly appears practicable. On the other hand, the development of the Booster appears feasible, leading to a test in two to five years from now.¹⁴⁷

With regard to the overall thermonuclear program, the GAC noted that "at the same time, such weapons, while not offering any immediate promise of radical improvement in cheapness or in effectiveness over those now available, may have certain economic or tactical advantages whose value can be determined only when design and performance are better known."¹⁴⁸

Detailed studies of the probable behavior of the BOOSTER started in July. Development of the BOOSTER marked the first time in the history of the Los Alamos thermonuclear research project that the design of a specific device had reached a point sufficient to require engineering studies or development of concrete processes or techniques.¹⁴⁹ In spite of these advances, in mid-1948 an American hydrogen bomb was estimated to be at least ten years in the future.¹⁵⁰

In mid-August, a special long-range nuclear weapons planning committee, appointed to look ten years into the future, submitted its report to the DOD. The Panel concluded that thermonuclear weapons were at best a long-range objective, partly because of their dependence upon the production of large quantities of tritium.

The Panel recommended against a Manhattan District-type scientific and industrial effort, and suggested instead that work be concentrated to more fully determine the nuclear and mechanical consequences of a chain reaction; the design of a gun-type primary; development of the BOOSTER as a short-term thermonuclear weapon; development of a suitable delivery vehicle; and generation of tritium as an essential nuclear material, in conjunction with production of plutonium,

¹⁴⁷Borden-Walker H-bomb chronology, p. 18.

¹⁴⁸Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p. 14.

¹⁴⁹Mark, LA-5647-MS, p. 12.

¹⁵⁰AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 56.

uranium-235, and polonium. In a cover letter to the report to DOD, Dr. J. Robert Oppenheimer stated: "In order to detonate a super, the fission 'primer' can almost certainly not be an implosion (device)."¹⁵¹

The Los Alamos Technical Program for 1949

Los Alamos' technical program for 1949, submitted to the AEC in September, included plans for: "Continued theoretical research on the problems of achieving a nuclear reaction involving light elements (the 'Super') and radiative and propagative properties of such reactions as well as the nature of processes involved in arrangements of the 'Alarm Clock' and 'Booster' types."¹⁵²

The Los Alamos 1949 research program also stressed the utility of a gun-type initiator for the Super, and the likelihood of a test of such a device in 1951:

It may be noted that such a (gun-type) weapon will doubtless have to be tested ultimately for the following reasons:

(A) A weapon of the order of 100 to 300 kilotons is required for the initiation of either an ALARM CLOCK or Super type weapon.

(B) While such tonnages might be attained by implosion techniques, the geometry of an implosion weapon seems to be applicable only with the greatest difficulty to a thermonuclear weapon.

....

(D) The emission of radiation from such a (gun-type) device, of importance to further thermonuclear work, could be experimentally investigated (but only at the price of much

¹⁵¹Draft of The Long Range Military Objectives in Atomic Energy, Committee on Atomic Energy, Panel on Long-Range Objectives, pp. 6, 7, 8; attachment to letter dated 18 August 1948 from J. Robert Oppenheimer, Chairman, Panel on Long Range Objectives, to Donald F. Carpenter, Chairman, Committee on Atomic Energy Committee, Research and Development Board, Department of Defense.

¹⁵²Letter dated 7 September 1948 from N. E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Office of Santa Fe Directed Operations, U. S. Atomic Energy Commission, p. 3.

prior research and development work at Los Alamos).

(E) The gun type assembly forms the purest of atomic explosions and is freest from doubts as to conditions immediately prevailing at the start of the reaction. Such a weapon has never been tested under controlled experimental conditions.¹⁵³

Plans for full-scale calculations on the Super were also made in September (mathematical planning was not completed until December 1949), but adequate computing machines did not become available until mid-1952 (and by that time, most of them were engaged full time in work on the new Teller-Ulam configuration discovered in early 1951).

A much-simplified calculation, basically a subset of the "classical Super" problem, was begun at the end of 1948 to help determine the likely amount of tritium required to provide possible thermonuclear ignition.¹⁵⁴

A revised program for thermonuclear research at Los Alamos in 1949 submitted to the AEC in October 1948 included the following aspects:

- (1) Continued theoretical research on the Super
- (2) Increased emphasis on the theoretical study of nuclear interactions between light elements

¹⁵³Letter dated 7 September 1948 from N. E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Office of Santa Fe Directed Operations, U. S. Atomic Energy Commission, pp. 6, 7; Ulam biography, 1991, p. xi. By October 1948, George Gamow, then a consultant to the Los Alamos laboratory's Theoretical Division, had prepared a highly preliminary report on a device he called the "New Mexican Jumping Bean." This was a nuclear-powered rocket which was to be detonated as a gun-type weapon at the end of its trajectory. Two numerical examples were considered in the report, one for a flight from Los Alamos to the moon and the other from Los Alamos to Moscow. (T-Division Progress Report, 20 September 1948 - 20 October 1948, LAMS-804, 9 November 1948, p. 2.) A gun-type weapon was finally tested under more-or-less controlled conditions during the Grable shot of Operation UPSHOT-KNOTHOLE in May 1953. (See "Artillery Shells" in Volume VII.)

¹⁵⁴Mark, LA-5647-MS, p. 6.

- (3) Development of a weapon called the BOOSTER utilizing a small quantity of deuterium and tritium at the center of an ordinary atomic weapon to increase its efficiency
- (4) Study of the possibility of a gun-type weapon employing approximately [deleted] kilograms of U-235 and giving an energy release of the order of that required to initiate a "Super."¹⁵⁵

The proposed "multi-crit" gun initiator for the Super received mixed blessings. The AEC was willing at this time to "support the fabrication and testing of a multi-crit gun at such a time it is clear that the testing of such a weapon is the bottleneck in the development of a 'Super.'"¹⁵⁶

However, the Military Liaison Committee recommended that the gun was not justifiable, and since a test of it did not appear possible by 1951, LASL should conduct only theoretical and experimental work on it in 1949.¹⁵⁷ The GAC recommended that the multi-crit gun not be slated for test in 1951.¹⁵⁸

By 1949, the design and understanding of fission bombs had proceeded far enough to permit much better study of their application to thermonuclear systems. The development of fusion weapons in the U.S. was simultaneously inextricably linked to, and limited by, the development of fission weapons, and to a great extent to the development of high-speed large-scale electronic computers and advanced methods of mathematical analysis.¹⁵⁹

¹⁵⁵AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 6; U.S. Atomic Energy Commission, Los Alamos Laboratory Program For Calendar Year 1949, October 11, 1948, pp. 2, 4, 5, 6.

¹⁵⁶Report by the Division of Military Application, USAEC, on the Los Alamos Laboratory Program for Calendar Year 1949, USAEC, December 10, 1948, p. 5.

¹⁵⁷Memorandum dated December 2, 1948 from William Webster, Chairman, Military Liaison Committee to the Atomic Energy Commission, Subject: Los Alamos Laboratory Program.

¹⁵⁸Minutes of the Eleventh Meeting of the General Advisory Committee to the AEC, October 21-23, 1948, pp. 3, 5; letter dated October 27, 1948 to David E. Lilienthal, Chairman, USAEC, from J. R. Oppenheimer, Chairman, GAC.

¹⁵⁹Bradbury statement at September 24, 1954 press conference in Los Alamos.

In January 1949, LASL director Norris Bradbury reviewed the status of the Super program:

Before leaving this discussion of future developments in atomic weapons themselves, I wish to call your attention to another completely different field of atomic energy.

The fission process produces energy because the parent uranium nucleus is heavier than the sum of the weights of the fission products. At the other extreme of the periodic table of elements are the very light nuclei of hydrogen, deuterium, tritium, helium, lithium, and so on. If two hydrogen nuclei and two neutrons could be persuaded to join together to make a nucleus of helium and this could be done for one gram of helium, almost 200,000 KWH (kilowatt-hours) of energy would be produced (equivalent to more than 100 tons of TNT).

Such processes can be accomplished to in the laboratory, more or less atom by atom, in high voltage accelerating machines, and this general type of process -- the so-called thermonuclear reaction -- is undoubtedly responsible for solar and stellar energies.

Can such enormous energies be used for military weapons? Here one must proceed with the greatest caution. No known law of nature forbids this, for the process can be shown to be thermodynamically possible. For this reason, one will always to continue to keep such a possibility in mind.

However, at this time, we cannot say we know how to do it. The basic problem arises from the fact that essentially stellar temperatures and pressures are required to establish such reactions and that, if established on a small scale, they tend to cool themselves off and stop.

In 1949, one can only say certain things about this possibility. As I have indicated, it is not thermodynamically impossible -- the energy is there. We do

not know if it can be done with techniques which are available at present; but we do not know that it cannot be done. Such a weapon, if it could be made, would have TNT equivalents of the order of millions of tons of TNT.

Since there is no critical mass problem, the weapon could, in principle, be made as large as one wished. It could probably not be made smaller than one to ten million tons TNT. No fission products would be formed, but enormous radiation would occur.

As a gadget, it would make the present weapon trivial in size, weight, and complexity. As a research product, it is estimated that an effort equivalent to that expended in wartime by the Manhattan District for the fission bomb would have to take place for a time probably twice as long just to demonstrate whether or not such a weapon could actually be made.

Such devices are certainly not in the near future and may possibly never exist except on paper. No country can neglect this possibility, however, and such a source of energy, even if possessing fantastic difficulties to tap, must not be forgotten. The actual amount of research and development effort to be expended must, however, be related to the foreseeable national need, if any, for a weapon of these characteristics.¹⁶⁰

Contrary to Bradbury's 1949 expectations, early U.S. thermonuclear weapons, due to their design, produced many tons of fission products.

Another step toward the Super was taken later in 1949 when the wartime liquid hydrogen plant at LASL was reactivated and put back into full production status.¹⁶¹

On July 27, AEC Chairman David Lilienthal, in response to queries from the MLC, stated:

¹⁶⁰"The Potentialities of the Atomic Bomb," N. E. Bradbury, 4 January 1949, pp. 13, 14.

¹⁶¹"Cryogenics and Nuclear Physics - Part 1," E. F. Hammel and J. D. Rogers, CRYOGENICS, February 1970, p. 6.

In regard to thermonuclear assemblies, theoretical studies are continuing at Los Alamos at a pace which does not interfere seriously with more urgent elements of the laboratory program. A substantial research program is in progress on the fundamental nuclear properties of the light elements, and this will have a bearing on thermonuclear reactions.

As you know, considerable effort is also being expended in developing, for test in 1951, an experimental booster system employing the light elements tritium and deuterium along with fissionable material, although it is not possible to say at this time whether this will lead to a satisfactory weapon model. It should in any case increase the basic knowledge needed for developments in this field.¹⁶²

Also during 1949, steps were taken to produce relatively large amounts of tritium required for a BOOSTER experiment scheduled for 1951. Tritium was produced by the reactors at Hanford, Washington, using "excess" neutrons not required for plutonium production.¹⁶³ Calculations on the details of behavior of thermonuclear fuels in the BOOSTER were started in August 1949.¹⁶⁴

At this time, the proposed Super was so huge and unwieldy that delivery by boat or railroad train were the only choices (any probable enemy was unlikely to permit either).¹⁶⁵ It seemed impossible to package the Super into a droppable bomb, and even if it were possible, its yield would be so great that the airplane carrying it would have to be sacrificed.¹⁶⁶

One Super design in early 1949 included a fission trigger that in itself weighed 30,000 lbs.; the overall length of the bomb was estimated at approximately 30 feet, with a diameter in excess of 162 feet. Under these circumstances, the

¹⁶²Borden-Walker H-bomb chronology, p. 20.

¹⁶³AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 7.

¹⁶⁴Mark, LA-5647-MS, p. 6.

¹⁶⁵AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 21.

¹⁶⁶AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 70; Borden-Walker H-bomb chronology, p. 25.

scientists at Los Alamos preferred not even to estimate the gross weight: this configuration represented essentially a huge fission trigger in the middle of a container of liquid deuterium the size of a large oil storage tank.¹⁶⁷

In August, the Military Liaison Committee (MLC) noted in its annual report to the Secretary of Defense that the Super was "a long range project" and that "not only would this weapon be massive and complex, but large (material production) plant investments would be required for its development" and that predominantly because of the latter reason, "this objective cannot and will not be attacked at this time with the extensive scientific and industrial effort which characterized the wartime (Manhattan Project) effort."¹⁶⁸

The First Soviet Nuclear Explosion, September 1949

Such was the status of the American hydrogen bomb program when President Harry S Truman was informed on September 21, 1949 that the Soviets had apparently successfully fired a plutonium implosion device in Siberia sometime between August 26 and August 29; yield was estimated at 10 to 20 kilotons and its nuclear efficiency was good.¹⁶⁹ An early estimate placed the detonation site somewhere in an area roughly centered on the northeastern shore of the Caspian Sea.

The reasons for the long time between the detection of the Soviet test and a public announcement on September by Truman included prolonged Administration indecision over whether to make a public announcement, and a conscious intent to confuse the Soviets as to the method of detection.¹⁷⁰

¹⁶⁷AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 187.

¹⁶⁸Annual Report of the Military Liaison Committee to the Secretary of Defense for the Fiscal Year 1949, 1 August 1949, p. 9.

¹⁶⁹NUCLEAR WEAPONS DATABOOK, Vol. IV, p. 336. The minutes of a JCAE meeting on August 16, 1951, p. 1291, note that "if the report on the effectiveness of the first (Soviet test) scientifically assayed is correct, (the Soviets) don't need a new model (atomic bomb.)" One source placed the first Soviet nuclear explosion within an area bounded by the east 35th and 170th meridians, an area embracing eastern Europe, Asia, and the western Pacific Ocean.

¹⁷⁰"A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, p. 2.

A commercial radiochemistry laboratory in the U.S., under contract to the Air Force, estimated the firing time of the Soviet test to within an hour of its actual moment of detonation, and also correctly deduced that the device used a plutonium core and unenriched uranium tamper.¹⁷¹

By November 1947, U.S. Air Force intelligence officers had circulated reports of possible Soviet atomic energy facilities near the Lake Baikal area in Siberia and the Uzbek-Khazak area of central Asia.¹⁷²

As recently as July 1948, one Central Intelligence Agency (CIA) estimate had stated that "the earliest date by which it is remotely possible that the USSR may have completed its first atomic bomb is mid-1950, but the most probable date is believed to be mid-1953." The CIA also assumed that if the Soviets completed their first atomic bomb in 1950, they might be able to stockpile as many as 50 by 1995, and only 20 by then if the first bomb were completed in 1953.¹⁷³ This estimate of Soviet nuclear capability remained virtually unchanged for the next year.

By the time of the first Soviet test, the Joint Nuclear Energy Intelligence Committee (JNEIC), comprising representatives of the Departments of State, Army, Air Force, and Navy, and the AEC and CIA, was meeting weekly to consider new information and to prepare semiannual reports of its estimates of the status of the Soviet atomic energy program based on the best information available, which was now derived mainly from technical and human sources.¹⁷⁴ The best CIA estimate at this time was that the Soviets would not test a fission bomb before mid-1951.¹⁷⁵

¹⁷¹Rhodes, DARK SUN, p. 372.

¹⁷²A NEED TO KNOW: THE ROLE OF AIR FORCE RECONNAISSANCE IN WAR PLANNING, 1945-1953, PhD thesis by John T. Farquhar, Ohio State University, Columbus, Ohio, 1991, p. 67.

¹⁷³Memorandum for the President dated 6 July 1948 from RAdm. R. H. Hillenkoetter, USN, Director of Central Intelligence; "Status of the U.S.S.R. Atomic Energy Project, 1 July 1949," OSI/SR-10/49, OSI Central Intelligence Agency, Nuclear Energy Branch; letter dated 13 December 1948 to Bourke B. Hickenlooper, Chairman, JCAE, from RAdm. R. H. Hillenkoetter, USN, Director of Central Intelligence.

¹⁷⁴Letter dated August 31, 1949 to Honorable Brien McMahon, Chairman, JCAE, from Sumner T. Pike, Acting Chairman, USAEC.

¹⁷⁵Rhodes, DARK SUN, p. 363.

About a week after the Soviet test, the JCAE heard testimony on the status of the U.S. Super program. The AEC noted that LASL was working toward a 1951 test of a deuterium-tritium boosted fission bomb, described as "a step toward a possible thermonuclear bomb" and which would occupy most of LASL's resources between now and the test date:

We reviewed about a year ago .. the program leading toward a .. thermonuclear weapon, and as best as anyone could see then, the first and necessary step would be .. the booster .. and that would take two or three years to do. The thermonuclear weapon itself, according to our best scientific advice, is a really major endeavor and can certainly (span) ... a number of years. ...

We have to achieve temperatures greater than we have achieved in any atomic explosion thus far even to trigger the thing, if it can be triggered. This is a fact that we are trying to determine in the next test (series) --- can it be triggered? ... If it can, there is a huge development program ahead of us.¹⁷⁶

Near the beginning of October, AEC Commissioner Lewis Strauss, an outspoken advocate of the American hydrogen bomb, called for a commitment of U.S. scientific and industrial resources to the Super comparable in scope to that dedicated to the wartime Manhattan Project.¹⁷⁷

On October 7, Drs. Stanislaw Ulam, George Gamow, John Manley, and Edward Teller, meeting at LASL with Drs. Ernest O. Lawrence and Luis Alvarez, gave the Super project a good chance of success "if there (were) plenty of tritium available."¹⁷⁸ At this time, the AEC believed that if current production rates were sustained, there would by 1951 be only

¹⁷⁶Rhodes, DARK SUN, p. 379.

¹⁷⁷"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 285; "A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953.

¹⁷⁸"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 286.

enough tritium available for the planned "Booster" shot of Operation GREENHOUSE.¹⁷⁹

The Joint Congressional Committee on Atomic Energy (JCAE) was notified of the Soviet detonation, and a meeting was held with the Joint Chiefs of Staff (JCS) on October 14 to re-evaluate existing U.S. nuclear projects in light of the Russian event. (The JCAE served as a legislative watchdog and tried to shape weapons-development policy by lobbying the AEC and the President.)¹⁸⁰

In this meeting, the Chief of Staff for Air made a strong plea for the development of thermonuclear weapons, but the other Chiefs of Staff were noncommittal. In contrast, the congressmen felt that thermonuclear weaponry should be advanced as rapidly as possible and urged the U.S. Atomic Energy Commission to authorize a hydrogen bomb program that would be as bold and urgent as the recent wartime nuclear weapons program.¹⁸¹

Numerical analysis started in mid-October on a device having a new and different thermonuclear burning pattern. (This work eventually led to the design of the GREENHOUSE George device tested in the spring of 1951.)¹⁸² At this time, it was not clear that measurements of cross-sections and other calculations would, in the absence of better experimental data, enable scientists to choose the best of several proposed methods of thermonuclear ignition.¹⁸³

One of the effects of the Soviet atomic bomb detonation, which occurred several years before many LASL scientists and government officials expected it, was to spur Edward Teller to lobby harder for an accelerated American Super program.

¹⁷⁹Rhodes, DARK SUN, p. 394.

¹⁸⁰"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 284.

¹⁸¹Alexander, p. 12.

¹⁸²Mark, LA-5647-MS, p. 7. One paper written during this period was LAMS-831, "Outline of a Method for the Calculation of the Progress of Thermonuclear Reactions," by G. Foster Evans, John von Neumann, and Stanislaw M. Ulam. (LAMS-831 in "Publications of Stanislaw M. Ulam," compiled by Barbara Hendry, Los Alamos Scientific Laboratory, April 1968.)

¹⁸³"Feasibility of Frequent Bomb Tests," ADWD-63, LASL paper by Darol Froman, October 10, 1949, p. 4.

Teller urged LASL and the AEC to determine as quickly as possible whether or not the Super was feasible, and if so, that it be developed as rapidly as possible. Teller also stated his beliefs and opinions about the political utility of a superbomb, and the likely state of current Soviet development of this weapon:

... it is essential for us to develop a Super bomb at the earliest possible time or else be able to say with reasonable confidence that the Super is not feasible.

...

It is my conviction that a peaceful settlement with Russia is possible only if we possess overwhelming superiority. We do not now possess such superiority. The most promising prospect to acquire a great lead is by an early development of a Super Bomb. (sic) I am sure that such an accomplishment will in itself not solve the problem ... but early possession of the Super Bomb will give us another chance in the political field. Without a Super Bomb, such (a) chance is not likely to rise (again).

...

It seems that the Russian rate of progress (on the Super) is at least comparable to, if it does not exceed, the rate of progress in this country ... If the Russians continue to make actual progress faster and if we lose the atomic armament race, it will make little difference whether the reason has been the particular brilliance of Russian scientists or the exaggerated caution and thoroughness of our own (scientists).

...

... It is quite possible that the Russians will possess a Super Bomb within a short time. The reasons for this conclusion can be summarized as follows:

1. The Russians have been evidently working with considerable speed and efficiency and their efforts are not likely to slacken now.
2. The Russians are likely to have produced their plutonium in heavy water piles. Thus, they may even at

present have considerably more excess neutrons available than we do. Their tritium production may thus surpass ours in the very near future.

3. The Russians had greater assurance than we did during the war concerning the feasibility of the fission bomb. It is therefore not impossible that during their work on the fission bomb, they have made thorough preparation for the time after the fission bomb has exploded. This preparation might have included work on the Super.
4. Our discussions have so far concentrated on designs of the Super whose feasibility can be proven theoretically ... Simpler ways of detonating a Super may exist and may be discovered by the Russians.

If the Russians demonstrate a Super before we possess one, our situation will be hopeless.

In this letter, Teller outlined in some detail an accelerated tritium production program, using existing piles in Hanford, Washington, and at a Canadian facility at Chalk River, Ontario province.

Teller also urged construction of new reactors and an "early speed-up and reorientation of (LASL) effort ... (and that) the Super program become the number one objective of the Laboratory." He urged "an all-out effort" to build an American hydrogen bomb.

In a statement accompanying Teller's, Dr. John Manley claimed that LASL had misjudged the probable date of a Soviet atomic detonation by three years, and had thus "lost" three years of its own expected progress. Manley also argued that the Soviets' rate of progress on their own version of the Super was at least as great as that of the U.S. and "quite likely appreciably greater." Manley called for accelerated production of U.S. fission weapons, and closed by noting:

The Russian (A-bomb detonation) should teach us one at least one thing: that our state of ignorance of their efforts is so nearly complete that we should no longer assume any time scale for their

developments but rather choose our action so as to strengthen our position as rapidly as possible and maintain a rate of progress limited only by our resources for a relatively long period of time.¹⁸⁴

Teller's and Manley's fears were vastly overblown. The Soviets had, by late 1949, made little progress toward a thermonuclear weapon, and their first test device, a device along the lines of Teller's 1946 ALARM CLOCK, would not be fired until August 1953. A Soviet multistage device would not be tested until late 1955, three years after a similar U.S. test.

As Manley noted, reliable U.S. intelligence on the status of the Soviet nuclear weapons program was virtually nonexistent at this time. This often led to hugely-mistaken U.S. estimates of Soviet nuclear capabilities.

Teller and Manley had many allies in their push for accelerated development of the Super. Among them was AEC Commissioner Lewis L. Strauss who, soon after the Soviet fission test, wrote to AEC Chairman David Lilienthal and his fellow commissioners and urged a "crash" program:

The purpose of this memorandum is to raise a question for immediate consideration in the light of the information as to progress which has been apparently made in Russia.

In the days since the President's announcement, I have found a tendency in my own thinking to resort to the prospect of increased production of fissionable material and weapons as the logical procedure. Although this is very important, I feel strongly that it is not enough. The frequently expressed thought that we must "maintain our lead" is generally taken to mean that we must have a larger stockpile of weapons than the Russians because we began sooner and can make them faster.

Of these latter considerations, however, only the fact that we began sooner can be relied upon absolutely. And, in any case, we can only maintain our lead in some

¹⁸⁴"Letters of Drs. Teller and Manley re Thermonuclear Controversy," LAMD-166, October 13, 1949, selectively excerpted and summarized in September 12, 1952 Walker memorandum, Appendix A.

arithmetical difference since our relative lead is most likely to decrease.

It seems to me now that the time has come for a quantum jump in our planning (to borrow a metaphor from our scientist friends) -- that is to say, we should now make an intensive effort to get ahead with the Super. By intensive effort, I am thinking of a commitment in talent and money comparable, if necessary, to that which produced the first atomic weapon. This is the way to stay ahead.

I recommend that we immediately consult the General Advisory Committee to ascertain their views as to how we can proceed with expedition.¹⁸⁵

Another powerful supporter of the Super was Senator Brien McMahon, chairman of the MLC, who, on October 19, stated his belief that the Soviets had assigned top priority to their own Super and that American efforts to achieve its hydrogen bomb should be "as bold and urgent as our original atomic enterprise."¹⁸⁶

A major technical meeting was held at LASL in mid-October to review the entire H-bomb program. By the time the meeting ended, LASL had crafted a recommendation favoring the Super, and this was forwarded to the GAC.¹⁸⁷

At a meeting of a subcommittee of the JCAE at LASL on October 27, LASL Director Norris Bradbury described one possible Super design with a weight of 20,000 lbs. and a minimum yield of one megaton. This device might be ready by 1958 or 1960, although it might be tested as early as mid-1952. Its great weight limited delivery to the B-36; however, its blast force was expected to eliminate the use of manned aircraft, which would not have sufficient speed to escape the blast.

¹⁸⁵Memorandum dated October 5, 1949 to D. E. Lilienthal, S. T. Pike, H. D. Smyth, and G. Dean, from Lewis L. Strauss.

¹⁸⁶"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 28.

¹⁸⁷Memorandum for the File dated January 13, 1953 from John Walker.

To meet the project time schedule, LASL would require 30 additional scientists and an expansion of several hundred persons over its present size.¹⁸⁸

The B-36 was the most obvious delivery vehicle for the H-bomb: On June 30, 1948, a B-36 had dropped 72,000 lbs. of conventional bombs during a test flight. The B-36B could carry 86,000 lbs., including conventional high explosive bombs weighing up to 43,000 lbs. and measuring 364 inches long and 54 inches in diameter, the size of the American variants of a World War II British "Grand Slam" bomb. On January 29, 1949, a B-36B had dropped two 43,000 lb. bombs on a practice target; the first bomb was released at an altitude of 35,000 feet and the second from 40,000 feet.¹⁸⁹

The General Advisory Committee Votes Against the H-Bomb

The General Advisory Committee to the AEC met between October 28 and 30; a primary subject of consideration was the Super. The GAC at this time included J. Robert Oppenheimer, James B. Conant, Hartley Rowe, Cyril Stanley Smith, Lee A. DuBridge, Oliver E. Buckley, Isidor I. Rabi, Enrico Fermi, and Glenn Seaborg.

Based in part on continuing unresolved technical difficulties in the design of the Super, the majority of the members voted against prosecution of the thermonuclear project and concluded that if the U.S. refrained from developing a hydrogen bomb, the Russians would not undertake a similar program of their own. A minority report offered to renounce the Super if the Soviets did likewise.

In retrospect, these were artless and naive assumptions by the GAC: lacking reliable intelligence about what was going on behind the Iron Curtain, the Committee did not know that Soviet Premier Joseph Stalin had ordered research started on a hydrogen weapon in the summer of 1948, a full year before the Russians detonated their first atomic bomb and nearly a

¹⁸⁸Memorandum for the Files dated November 8, 1949, from Walter A. Hamilton, subject: Inquiry into the Aspects of a Superweapon Program, pp. 7, 8, 9, 10; Memorandum for the Files dated November 8, 1949 from William L. Borden, re: Super Weapon Program, p. 6.

¹⁸⁹ENCYCLOPEDIA OF U.S. AIR FORCE AIRCRAFT AND MISSILE SYSTEMS, Volume II, POST-WORLD WAR II BOMBERS 1945-1973, Marcelle Size Knaack, Office of Air Force History, United States Air Force, Washington, D.C., 1988, pp. 23, 24.

year and a half before President Truman would issue an equivalent order in the United States.¹⁹⁰

At its 328th meeting on November 3, the Atomic Energy Commission reviewed the report and recommendations of the GAC. Each commissioner summarized his tentative conclusions concerning the recommendations of the GAC and the desirability of either proceeding or not proceeding with acceleration of the Super program.

In particular, AEC Commissioner Lewis Strauss suggested that President Truman appoint a joint AEC-DOD-State Department committee to review the GAC recommendations. Strauss pointed out that while the AEC was competent to advise the President on the feasibility, probable development time scale, fissile material economy, and likely physical characteristics of the Super, the AEC was not qualified to judge the military usefulness or diplomatic consequences of either renouncing or building the weapon.

Strauss felt that Defense and State, in conjunction with the AEC, could give the President a more balanced opinion:

(1) Are the chances in favor of the feasibility of making the super (sic) sufficient, assuming all other questions are answered, to warrant the diversion to it of talent, fissionable material, and funds?

(2) Assuming for the moment an affirmative answer to (1), are the "economics" of having the super justifiable by comparison with "ordinary" fission weapons? I understand that the answer to this depends on certain variables, to wit, the median amounts of tritium and U-235 required, and the median figure for estimated yield in KT of TNT (equivalent). The calculation ... is based on the use of 1,000 KT as the median figure, and if this actually on the side by a sizable factor, it negates the conclusion based on a comparison of damage areas.

¹⁹⁰"Fusion Devices, Explosive," article by Dr. Edward Teller in Volume 5 of ENCYCLOPEDIA OF PHYSICAL SCIENCE AND TECHNOLOGY, Academic Press, San Diego, 1987, pp. 723-726; Zaloga, p. 95. Dr. Teller cites this date in Andrei Sakharov's SAKHAROV SPEAKS, published in 1974 by Random House in New York. (Dr. Sakharov is commonly identified as the "father" of the Soviet H-bomb.)

It is also questionable whether a proper comparison can be made without deducting from stockpiles in the target year selected the ordinary weapons in the stockpile we would have in any event. The necessity of having more than a minimum number of supers (i.e., not as many as calculated) may be questioned. This is a military determination.

(3) Does the military want the super at all? It seems to me that we do not know the answer to this question and that an answer to it is certainly an important component of a Presidential decision. If they want it, how many? The answer to that bears on question (2) above.

(4) What will be the effect on the free countries of Europe of unilateral renunciation of the super by the U.S.? We do not know the answer to this question, and while in any case it is of course only a matter of opinion, yet the State Department should be the source of the best informed guess. For instance, it might very well be taken in friendly European quarters that this marked the beginning of a program of U.S. withdrawal from the role of a strong and ready protector.

(5) The argument that, as armament, the super is qualitatively rather than merely quantitatively different from the A-bomb is made more difficult to support in light of the unanimous GAC recommendation to us to push development of the booster. The booster is designed to "improve" the "ordinaries" by a factor of, say _____. Would the GAC have disapproved the booster if the factor had been [deleted]

In other words, the logic of renouncing the means of killing millions while stimulating a means of killing hundreds of thousands leaves something to be desired from the standpoint of consistency. The booster is certainly not a pinpoint weapon.

....

My present feeling is this: That any recommendation which (the AEC) makes to the President at this juncture can only be

concerned with our views as to the feasibility of making the super, the economics of its manufacture and use (assuming we can resolve the disparity in figures), and to forward the recommendations of the GAC. Obviously, we are not competent to advise on the military "usefulness" of the super or the diplomatic consequences of either making it or renouncing it.

In view of the fact that those latter considerations, while not necessarily controlling, are of the greatest importance, it may well be that we should obtain a definite point of view on them from the departments of Defense and State and submit to the President a complete package.¹⁹¹

With regard to military desires, there were still many questions to be answered before DOD could make a commitment to the project. Would the Super supplement or replace fission bombs? Which Soviet targets required destruction by a Super? Would the Super, as J. Robert Oppenheimer often claimed, be used only to offset bombing errors or could airborne bombing and navigation systems be improved sufficiently so that lower-yield fission bombs would be adequate for use against most targets? How many fission bombs, or kilotons of explosive power, would each Super "cost"? Could adequate delivery vehicles be built to carry the Super?¹⁹²

The GAC reports, both majority and minority, were forwarded to President Truman on November 9 by the AEC. Opinion within the AEC was also split: two commissioners favored an immediate thermonuclear program; the other three sided with the General Advisory Committee and recommended against developing a hydrogen bomb.¹⁹³

¹⁹¹Atomic Energy Commission, Meeting No. 328, Thursday, November 3, 1949, pp. 429, 430; "Questions" by Lewis Strauss dated 3 November 1949. Emphasis in original. The first "booster" improvement factor in the paper is an underlined space to be filled in, as shown.

¹⁹²"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), pp. 290, 291.

¹⁹³Alexander, pp. 12, 13; letter dated November 9, 1949 to The President from David E. Lilienthal, Chairman, USAEC, with attached Memorandum for the President from the Atomic Energy Commission, subject: Development of a "Super" Bomb.

In lieu of developing the hydrogen bomb, the GAC recommended acceleration of research and development work on low-yield tactical fission weapons; the production of tritium for "boosted" fission weapons; and the enlargement of special nuclear materials production facilities.¹⁹⁴ A target goal of a gram of neutrons per day was established to use in production of U-233, tritium, radiological warfare agents, plutonium, and polonium.¹⁹⁵

The GAC noted that a full-scale test would be necessary, instrumental and finally decisive:

It is notable that there appears to be no experimental approach short of actual test which should substantially add to our conviction that a given model will or will not work, and it is also notable that because of the unsymmetric and extremely unfamiliar conditions obtaining, some considerable doubt will surely remain as to the soundness of theoretical anticipation.

Thus we are faced with a development which cannot be carried to the point of conviction without the actual construction and demonstration of the essential elements of the weapon in question.

This does not mean that further theoretical studies would be without avail. It does mean that they could not be decisive.

A final point that needs to be stressed is that many tests may be required before a workable model has been evolved or before it has been established beyond a reasonable doubt that no such model can be evolved.¹⁹⁶

¹⁹⁴York, THE ADVISORS, pp. 48-50, 108, 109, 152, 153; AEC THERMONUCLEAR WEAPONS CHRONOLOGY, pp. 26, 27; Minutes of the General Advisory Committee to the U.S. Atomic Energy Commission, in letter dated October 30, 1949 from J. Robert Oppenheimer to David E. Lilienthal, Chairman, USAEC, pp. 1, 2.

¹⁹⁵Borden-Walker H-Bomb Chronology, p. 30; Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p. 19.

A gram of neutrons is approximately 6×10^{23} neutrons.

¹⁹⁶"A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, pp. 4, 5.

The design of the Super and the requirements for its successful realization were summed up in part II of the General Advisory Committee report to the AEC dated 30 October 1949:

The basic principle of design of the super bomb is the ignition of the thermonuclear DD reaction by the use of a fission bomb, and of high temperatures, pressure, and neutron densities which accompany it. In overwhelming probability, tritium is required as an intermediary, more easily ignited than the deuterium itself and in turn, capable of igniting the deuterium.

The steps which need to be taken if the super bomb is to become a reality include:

- (1) The provision of tritium in amounts perhaps of several hundred grams per unit.
- (2) Further theoretical studies and criticisms aimed at reducing the very great uncertainties still inherent in the behavior of this weapon under extreme conditions of temperature, pressure and flow.
- (3) The engineering of designs which may on theoretical grounds appear hopeful, particularly with regard to the cryogenic problems presented.
- (4) Carefully instrumented test programs to determine whether the deuterium-tritium mixture will be ignited by the fission bomb, whether this ignition in turn ignites the deuterium, and, if possible, to determine what, in the event that such ignition did not take place, went wrong with the calculations.¹⁹⁷

¹⁹⁷York, THE ADVISERS, p. 154; AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 28; Minutes of the General Advisory Committee to the U.S. Atomic Energy Commission, in letter dated October 30, 1949 from J. Robert Oppenheimer, to David E. Lilienthal, Chairman, USAEC, p. 3.

At this time, the Super was believed to offer unlimited power: its explosive yield could be increased merely by adding more deuterium fuel, assuming a self-propagating explosive burn in the fuel mass. (As described in Volume I, the concept of staging offers the same option of theoretically-unlimited thermonuclear explosive power.) As in earlier designs, neutrons were still the principal means of energy transfer between the fission trigger and the deuterium.

The weapons scientists believed that if the temperature of part of a mass of deuterium could be raised to about 240 million degrees Centigrade by D-T fusion, a thermonuclear reaction would start and propagate through all the deuterium present (D-D fusion required a temperature of about 400 million degrees to propagate).

They also thought there was a better than even chance it could be made to work, although there was no certainty of success. They stressed that development could be completed no sooner than 1952 (as events developed, this was a remarkably prophetic statement, although for all the wrong reasons).

The requirement for tritium was crucial and ultimately decisive. The major thermonuclear explosive, deuterium, was plentiful and relatively inexpensive; tritium, which would be required in unknown but probably large quantities, was scarce and extremely expensive in terms of both money and lost plutonium production.¹⁹⁸

The neutrons and energy used to produce one kilogram (2.2 lbs.) of tritium in a reactor could alternatively be used to produce 70 kilograms of plutonium, an amount sufficient for up to 30 fission bombs.¹⁹⁹ If each Super required several

¹⁹⁸AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 40.

¹⁹⁹Hirsch and Mathews, p. 10. Test devices fired during Operation SANDSTONE in 1948 probably contained about two to two-and-a-half kilograms of plutonium; in addition to this amount of plutonium, each composite bomb core also contained a comparable amount of uranium. According to Dr. J. Carson Mark (in a conversation with the author in Los Alamos, New Mexico on October 2, 1991), the neutrons and energy used to produce one kg of ${}_1\text{H}^3$ could theoretically alternately generate as much as 80 kg of ${}_{94}\text{Pu}^{239}$, the ratio of the atomic weights of tritium and plutonium. Senator Brien McMahon, then Chairman of the Joint Committee on Atomic Energy, wrote a letter dated November 21, 1949 to President Truman; in that letter, McMahon stated:

kilograms of tritium, then it would cost many fission warheads and result in a significantly smaller national nuclear weapons stockpile.

The GAC also noted at this time that

it is of course true that super (sic) bombs which are not as big as those here contemplated could be made, provided the initiating mechanism works. In this case however, there appears to be no chance of their being an economical alternative to the fission weapons themselves. It is clearly impossible with the vagueness of design and the uncertainty as to performance as we have at present to give anything like a cost estimate of the super.

If one uses the strict criteria for damage area per dollar and if one accepts the limitations on (delivery aircraft) capacity likely to obtain in the years immediately ahead, it appears uncertain to us whether the super will be cheaper or more expensive than the fission bomb.²⁰⁰

Carson Mark later commented on Edward Teller's estimates of tritium for each Super at this time:

Edward promised people in Washington that (LASL) would get by with a certain amount. He had no particular basis for the amount

It is argued that if we build a half-dozen or more new reactors and use them to manufacture the tritium necessary for Supers, we would forego the plutonium otherwise obtainable from those same reactors -- thereby losing about as much as we gained. Yet the best figures I can assemble show that an annual production of one kilogram of tritium would be accomplished in lieu of about 80 kilograms of plutonium. (AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 56.)

This figure is repeated in "Inside Story of the Hydrogen Bomb: How the U.S. Almost Lost It," James R. Shepley and Clay Blair, Jr., U.S. NEWS AND WORLD REPORT, September 25, 1954, p. 95. Cochran, et. al., in NUCLEAR WEAPONS DATABASE, Vol. II, p. 63, cite congressional testimony equating one gram of tritium output to 72 grams of weapons-grade plutonium.

²⁰⁰Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p. 20.

he mentioned except that it didn't appall them. He chose an amount that had that property. It didn't necessarily have the property of starting the (fusion) reaction.²⁰¹

By early November, despite hopes for an early test of some type of thermonuclear system, LASL did not anticipate that fusion weapons would be available in quantity until 1959, ten years hence.²⁰² A major conference on thermonuclear weapons which had been scheduled to meet in Los Alamos on November 7 was canceled.²⁰³

On November 15, in a conversation with Senator Brien McMahon and Robert LeBaron of the Joint Committee on Atomic Energy, Dr. Edward Teller reportedly stated that "chances for developing the Super were good and that test of an appropriate Booster with supplemental mathematical work would provide an almost complete proof of the Super feasibility, and at the same time would accomplish the major engineering development of the Super."²⁰⁴

If Teller actually said this, it was a gross oversimplification of the present state of affairs: the test of the Booster would in no way demonstrate feasibility of the Super, nor would it solve any of the Super's many formidable engineering problems.

Truman Appoints a Special NSC Advisory Subcommittee

On November 18, partly as a result of Strauss' recommendation, President Truman appointed a special subcommittee of the National Security Council to assist him in reaching a decision on the future of the Super. This committee included the Secretary of State, the Secretary of Defense, and the Chairman of the U.S. Atomic Energy Commission.²⁰⁵ AEC Chairman David Lilienthal was opposed to

²⁰¹Rhodes, DARK SUN, p. 400.

²⁰²Borden-Walker H-bomb chronology, p. 32.

²⁰³"A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, p. 5.

²⁰⁴"Comments Concerning the H-Bomb Development," Department of Defense, Military Liaison Committee to the Atomic Energy Commission, 21 April 1954.

²⁰⁵Alexander, p. 13; AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 117. In the summer of 1949, Truman had assembled an identical group to advise him on other nuclear weapons-related issues. In the spring of 1979, the Secretaries of State, Defense, and

the Super; Secretary of Defense Louis Johnson favored it; and Secretary of State Dean Acheson was undecided.²⁰⁶

The special committee was to weigh all technical, military, and diplomatic factors and consequences surrounding the Super, and advise the President as soon as possible as to whether or not to proceed with the project.

To meet the estimated needs of the current Super design, minimum tritium facilities were estimated to cost between \$100 million and \$200 million. They would consume hundreds of kilograms of U-235 each year in the production of tritium. At this time, the annual production of one kilogram of tritium was expected to consume neutrons that could instead be used to manufacture 80 kilograms of plutonium.²⁰⁷

On November 23, the Atomic Energy Commission initiated further studies of reactor designs specifically tailored to large-scale tritium production. The Argonne National Laboratory, North American Aviation, the General Electric Company, and the DuPont Corporation were issued contracts to study different types of reactors, modifications to existing reactors, and other phases of the long- and short-range tritium production problem.

Argonne was to submit a proposal for designing a modified core for the Materials Testing Reactor (a high neutron flux reactor which was built in Idaho and went into use in March 1952) in case it had to be converted to neutron production. North American was to develop an engineering design of a heavy-water moderated, light-water cooled natural uranium reactor for neutron production.

GE was to study ways to modify existing Hanford reactors for maximum neutron production; and DuPont was requested to prepare a comprehensive study of possible reactor designs and associated chemical facilities to determine the most economical approach to neutron production.²⁰⁸

Energy (an AEC successor) led the charge against the Progressive magazine during the U.S. government's attempt to prevent publication of an article about thermonuclear weapons operating principles.

²⁰⁶"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 302.

²⁰⁷AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 56.

²⁰⁸Atomic Energy Commission, The Super, Staff Report attached to Note by the Acting Secretary, December 8, 1949, p. 9.

At the same time, the Commission urged further research and development in the field of low-temperature engineering, a vital requirement for the cryogenic Super.²⁰⁹

Also on the 23rd, the Joint Chiefs of Staff weighed in with their own recommendation: the U.S. should proceed with the Super program. It would be intolerable for the Soviets to have such a powerful weapon in the absence of a similar U.S. counterbalance. The JCS believed that the Super would not be prohibitively expensive, in terms of either dollars or fissionable materials, and that it would be more effective than fission bombs in terms of the use of fissionable uranium and plutonium, and in damage area per bomb.

The Chiefs also argued for the "highest classification" of the Super, and blasted the GAC's wishful thinking that if the U.S. did not develop thermonuclear weapons, then other countries would follow suit:

The Joint Chiefs of Staff have studied the subject matter of this memorandum and have reached the following conclusions:

That the United States military position with respect to development of the thermonuclear weapon should be:

a. Possession of a thermonuclear weapon by the USSR without such possession by the United States would be intolerable.

b. There is an imperative necessity of determining the feasibility of a thermonuclear explosion and its characteristics. Such determination is essential for U.S. defense planning, preparations for retaliation, and direction of research. It will have a profound effect on policy in the field of international affairs.

c. If a thermonuclear weapon is determined to be feasible, the following additional considerations pertaining to military requirements are currently evident:

(1) Possession of such weapons by the United States may act as a deterrent to war.

²⁰⁹AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 75.

(2) Possession of such a weapon by the United States will provide an offensive weapon of the greatest known power possibilities, thereby adding flexibility to our planning and to our operations in the event of hostilities.

d. The cost in money, materials, and industrial effort of developing a thermonuclear weapon appears to be within the capabilities of the United States. Available information indicates that such a weapon may likewise be within the capability of the USSR.

e. It is reasonable to anticipate, and in some cases it is known, that a number of thermonuclear weapons can substitute for a greater number of fission bombs. Further, the thermonuclear weapon promises in the high range of energy release to be more efficient in utilization of available ore and production capacity per unit area of damage.

f. The foregoing considerations decisively outweigh the possible social, psychological, and moral objections which may be considered to argue against research and development leading to a thermonuclear weapon by the United States.

g. Any decisions or actions pertaining to the United States' effort to develop a thermonuclear weapon or the determinations of its feasibility constitute a military secret of the highest classification. It should be possible to maintain secrecy on a subject of such importance to the security of the United States.

h. A unilateral decision on the part of the United States not to develop a thermonuclear weapon will not prevent the development of such a weapon elsewhere.

For the Joint Chiefs of Staff:
Omar N. Bradley, Chairman²¹⁰

²¹⁰"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein,

In the meantime, more technical problems arose: full calculations of the proposed BOOSTER model gave disappointing results, leading to much engineering redesign.²¹¹ More promising were the results of analyses started on the nascent GREENHOUSE George device design in October: successful operation of the proposed configuration looked quite likely.

One of the documents reviewed by the newly-appointed Special Committee of the National Security Council was a staff report prepared in early December by the AEC.

This report summarized the current status of the Super program and reviewed many aspects and ramifications of the design, testing, production, and use of thermonuclear weapons. The report was an important factor in the deliberations of the Committee, and it also contained further speculation on a "dirty" or "salted" hydrogen bomb deliberately loaded or "doped" with contaminating materials, an idea that had been raised at Los Alamos in April 1946.

The report also stated that more than a kilogram of tritium would be required just to test the Super, and that production of such a quantity of tritium was beyond the current capability of the U.S.:

In discussing the characteristics, effects, and technical problems of the Super, it must continually be kept in mind that no engineering design of the weapon exists or is likely to exist until after one or more tests of feasibility. A year or two of study and investigation would probably be necessary before the nature of even a test design would be clear. This situation is different from that applying to fission weapons even as early as 1942.

The work to date does suggest what the necessary elements of a Super are, at least within a rather broad range, and it is possible to discuss certain theoretical aspects on this basis. For example, the

HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 298; Memorandum by the Joint Chiefs of Staff to Secretary of Defense Louis Johnson dated 23 November 1949, reprinted in FOREIGN RELATIONS OF THE UNITED STATES 1949, Volume I, National Security Affairs, Foreign Economic Policy, U.S. Government Printing Office, Washington, D.C., 1976, pp. 595, 596.

²¹¹Mark, LA-5647-MS, p. 7.

energy release if ignition of the main charge takes place can be predicted with reasonable accuracy; the requirements for such ignition only within a range; and the appearance, size, and weight only most crudely. Such matters as ordnance engineering, production and stockpile problems can be no more than indicated.

....

... A fission-weapon initiator would be one component of the Super and would itself have to give an energy release of the order of 100,000 to 300,000 tons of TNT. Such a weapon does not exist at present and would have to be developed.

.....

... It is considered necessary to have a mixture of deuterium and tritium as a component of the Super in order to reach the temperature of [deleted] required for the ignition of a mass of deuterium. At the present time, sufficient information does not exist to eliminate entirely the possibility that deuterium might be ignited directly by some new type of fission weapon without the use of tritium, but this is believed to be very improbable.

The reaction once initiated would have to propagate itself through a mass of deuterium. The energy release of a Super would depend on the amount of deuterium contained and the efficiency of its utilization. For each cubic meter (380 pounds) of deuterium completely detonated, there would be an energy release of ten millions tons of TNT.

Deuterium does not have a critical mass like fissionable materials; if detonated at all, it can probably be detonated in any amount like regular explosives. Consequently there is no theoretical limit to the energy release that might be attained.

The deuterium would have to be in the liquid form, which can exist only below minus 400 degrees Fahrenheit. Provisions might have to be made for refrigerating equipment to keep the deuterium and tritium in a Super below this temperature.

The total weight of a Super containing one cubic meter of deuterium has been very tentatively estimated at ten tons, with guesses varying widely from this figure. The total weight would increase with the amount of deuterium but not in direct proportion.

A study is being made on the effects of an air-burst Super with an assumed energy release of forty million tons of TNT. This figure, which is about one thousand times the highest energy release of the fission bombs now in stockpile, has been chosen as a very tentative upper limit for a Super to delivered by aircraft. If the energy release were assumed to be much higher, the possibility would arise that the blast damage might be limited by part of the atmosphere being blown out.

....

... by absorbing the neutrons from a Super in appropriate materials placed around it for that purpose, a tremendous quantity of radioactive products could be generated and might have significant effects on personnel down-wind from the explosion.

....

There are numerous technical problems that would have to be investigated in determining the feasibility of the Super. The required energy release from the fission-type initiator may be a number of times greater than that from any present weapon and may be difficult to attain. A gun-type weapon, which utilizes U-235, would probably be suitable.

The use of plutonium would require an implosion weapon, which in its usual form would be applicable only with the greatest difficulty to a Super. New kinds of fission weapons may have to be devised.

It would be necessary to ascertain whether the deuterium-tritium reaction can ignite the deuterium mass. A vital question is whether the reaction in deuterium once initiated would actually propagate itself. These problems can be investigated to some extent by theoretical analysis and

laboratory experimentation, but field tests involving explosions equivalent to perhaps several hundred thousand tons of TNT would be necessary to determine the feasibility of various designs.

The production of tritium is a crucial problem. Of the order of a kilogram would be needed to determine the technical feasibility of the Super, since more than one large-scale field test would be needed. The provision of such an amount in a few years is far beyond the capacity of present production facilities and would call for modification of the Hanford reactors or design and construction of a number of new reactors.

Extensive ordnance engineering would be required in order to develop a usable weapon. For example, it would be necessary to find a method of maintaining the deuterium and tritium in the Super at extremely low temperatures without making aircraft delivery impractical.

The tritium would present a special problem because of the heat generation due to its radioactivity. Low-temperature research, development of refrigerating equipment, and engineering design of equipment for practical use in aircraft would be required.

At the present time, it is estimated that the Super has better than an even chance of being technically feasible. ...

...

... The test of the Booster scheduled for 1951 will show whether the deuterium-tritium reaction will take place under particular conditions, but will not determine whether the Super is technically feasible.

If work on the Super were given high priority at Los Alamos, members of the staff would be shifted from work on other projects when, if, and as they could be used to advantage on the Super. This would mean that work on a number of other projects would gradually be reduced and in some cases terminated.

Other weapon developments now underway, principally the lighter and smaller weapon [deleted] (aimed at improved "deliverability" by aircraft and guided missiles), could probably still be carried out, but not with the care and refinement originally planned. It would be difficult to undertake any new developments aimed at increasing the usability or efficiency of the stockpile of fission weapons or any developments at Los Alamos in the field of countermeasures.

It is believed that the minimum time for the determination of the technical feasibility of the Super is three years. On such a time schedule, the tritium requirements are roughly estimated to be 100 grams by the Spring of 1951 and 1,000 grams by the Fall of 1952.

The return to the Los Alamos staff of some of the people who were there during the war or the addition of men of equivalent abilities would be essential, especially in the theoretical field. The return of a British group would be very helpful.

Relaxation of certain administrative restrictions and use of a proving ground in the United States would contribute to the speed of the work. (Field tests involving explosions equivalent to perhaps several thousand tons of TNT would need to be made at the proving ground.)

A longer schedule than three years for the determination of the technical feasibility of the Super would require less rapid production of tritium and would put less strain on the Los Alamos Laboratory.²¹²

Also in early December, AEC Commissioner Lee Dubridge stated his belief that the Super would cost only the resources required for eight fission bombs: the fissionable materials for the equivalent of four atomic bombs, and tritium that would consume reactor time and uranium that otherwise would

²¹²Atomic Energy Commission, The Super (Unclassified Title: "Weapons Program"), Staff Report attached to Note by Acting Secretary, December 8, 1949, pp. 1-7.

be required for four more bombs.²¹³ This estimate was far lower than others being presented at this time.

In the middle of December, a memorandum circulated by a subcommittee of the Special Committee of the NSC appointed by Truman noted that the delivery vehicle problem for the Super, while formidable, was still much less of a problem than the design of the weapon itself:

The development of a vehicle to carry the thermonuclear weapon to its target is a problem which remains to be solved in conjunction with the development of the weapon itself. The trend of this vehicular development will depend largely upon the anticipated characteristics and physical dimensions and weight of the weapon.

It is impracticable at this time to anticipate the exact nature of the eventual carrier. It is apparent that, for the US, a large airplane will be a basic requirement initially. The following possibilities suggest themselves: a drone aircraft; an aircraft with a detachable parasite escape ship, the bomb to be exploded by a timer mechanism; an improved B-36 type with an underbelly weapon attachment; etc.

In consideration of the technical problems to be solved in the development of the weapon as compared to the carrier, it is believed that the carrier problem, although difficult, is the lesser of the two.²¹⁴

The Los Alamos Technical Program for 1950

The Los Alamos Laboratory Program for Calendar Year 1950 was submitted by lab director Norris Bradbury to the Atomic

²¹³"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 300.

²¹⁴Memorandum circulated by the Defense Members of the Working Group of a Special Committee of the National Security Council, reprinted in FOREIGN RELATIONS OF THE UNITED STATES 1949, Volume I, National Security Affairs, Foreign Economic Policy, U.S. Government Printing Office, Washington, D.C., 1976, pp. 607, 608.

Energy Commission for approval on December 20. The program for the Super included an experimental test of a deuterium-tritium mixture (this test eventually became the GREENHOUSE George shot in May 1951) and further work on the multi-crit gun-type trigger:

We propose to augment to the greatest extent possible the effort devoted to research on the problem of attaining a nuclear reaction involving the light elements. The goal of this effort (which determines the minimum level to which the research effort in this field must be raised) will be an experimental test in 1952 of the practicability of ignition of a deuterium-tritium mixture at least part of which is external to the region of a fission reaction.

....

We propose to augment the small current effort on the development of a gun-type weapon employing approximately [deleted] of U-235. This development will follow along the lines described in the 1949 program and will have as its goal the completion of such a device, together with auxiliary equipment, by 1952 and in time to be completely coordinated with the thermonuclear test proposed above.²¹⁵

Bradbury closed the letter with a few comments about the current status of the Super program, noting that tritium requirements for a proposed 1952 test would be on the order of hundreds of grams:

We are aware that the utilization of thermonuclear reactions for military purposes is subject to debate at this time, and we agree that there are many unsettled questions of practical utilization, economics, grand military strategy, production, engineering, stockpiling, and even morality.

²¹⁵AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 76; letter dated December 9, 1949, from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1950, pp. 1, 3.

It is the strong feeling of this Laboratory that such questions are not yet answerable until there are more experimental facts upon which to base such answers. The importance of these questions makes, in our opinion, an understanding and test of the basic phenomena at the fastest practicable rate imperative. Then, and only then, can the many issues be resolved without recourse to hypothesis or wishful thinking.

We wish to bring one further point to the specific attention of the Atomic Energy Commission: If our proposal to direct our efforts toward a goal of a thermonuclear test in 1952 are concurred in by the Commission, it will be necessary to have available at that time an amount of tritium on the order of [deleted] grams. While the precise amount cannot be specified at this time, and the actual amounts required for the ignition of a thermonuclear reaction present one of the greatest uncertainties in this field, nevertheless this amount is probably not in error by a factor of two and would reasonably permit at least one experiment to be conducted at that time.²¹⁶

By early January 1950, the CIA had received evidence that the Soviets were preparing to build and operate a large-scale gaseous diffusion plant for uranium enrichment. There was no evidence available showing that the Soviets could produce any significant quantities of U-235 by isotopic separation.

The CIA was watching for evidence of Soviet research into the Super in three main areas, including cryogenic (low-temperature) physics and construction of facilities for deuterium production.²¹⁷

On January 13, General Omar Bradley, Chairman of the Joint Chiefs of Staff, sent a memorandum to Secretary of Defense

²¹⁶Letter dated December 9, 1949, from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1950, pp. 4, 5.

²¹⁷Memorandum for Bill Borden dated January 4, 1950 from W. P. Brobeck, subject: Conference with Dr. Walter F. Colby and Dr. Henderson, AEC Intelligence Division, and Mr. W. P. Brobeck on January 3, 1950.

Louis Johnson laying out the views of the JCS on the hydrogen bomb. This memo expanded on the Chiefs' earlier views of November 23, 1949 and it included comments about the desirability of a "crash" program on the Super:

QUESTION: Is it necessary now to launch a 'crash' program for development of a super bomb?

COMMENT: The Joint Chiefs of Staff at this time believe that it is not necessary to launch a 'crash' program. However, they consider the following program to be the minimum effort which should be undertaken at this time:

- (1) The determination of the technical feasibility of a thermonuclear explosion as a matter of top priority.
- (2) Studies of the necessary delivery vehicle and ordnance problems should proceed concurrently with (1) above and should not necessarily await trial of a thermonuclear assembly.
- (3) Decisions pertaining to the production of thermonuclear weapons in any quantity should be deferred pending further determination of the ultimate feasibility of a thermonuclear explosion and the feasibility of an appropriate weapon carrier.²¹⁸

The JCS estimated that it might cost as much as \$100 to \$200 million to build and test a device that might prove or disprove feasibility of the Super.²¹⁹

²¹⁸AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 79.

²¹⁹"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 304.

Soon afterwards, Bradley supplied to the JCAE his views on probable Super targets, and on the effectiveness of the Super versus a number of fission bombs:

In studying the target system which we would think would hurt Russia more, there are many targets where a hydrogen bomb could be used instead of several fission bombs. In other words, there are several target areas where we figure now it will take -- I think the largest one is about seven or eight fission bombs to really hit that particular one, where possibly one of these (Supers) would do the same work, and possibly do it better.²²⁰

Work on the study of the progress of fission explosions had advanced significantly by January 1950: one calculation had given a picture of the events in the core and tamper of the 1945 TRINITY test weapon to a point about halfway through the explosion.²²¹

On January 21, the Weapons Development Committee met at LASL to continue its many discussions of the Super. At this meeting, Dr. Stanislaw Ulam outlined the initial conditions for his calculations on the device; he reiterated earlier conclusions that tritium requirements would probably be large, and that prospects for successful detonation looked "miserable".

The question of how to reach a temperature sufficient to maintain the reaction had, by this time, been investigated only by 1946 ENIAC calculations of S. P. Frankel and N. Metropolis; these calculations assumed one-dimensional infinite plane geometry and had not been confirmed by later work in 1950 by Ulam and C. J. Everett which showed widely different conclusions.²²²

Six days later, the congressional Joint Committee on Atomic Energy met in a closed hearing with five AEC commissioners and various other AEC staff members to review the status of

²²⁰Memorandum for Mr. Cole dated March 7, 1951 from Bill Borden, subject: Your inquiries relating to the hydrogen program.

²²¹Mark, LA-5647-MS, p. 8.

²²²LAMD-411, Minutes of January 21, 1950 meeting of the Weapon Development Group, cited in "Technical References Relating to Thermonuclear Program," Appendix E to September 12, 1952 Walker memorandum.

the Super. A topic of discussion at this meeting was the design and power of the current model:

Dr. Smyth then reviewed the characteristics and technical problems of thermonuclear weapons. He pointed out the assumption (Author's emphasis) that once the reaction could be started, the energy liberation in that part would raise the temperature of the rest of the mixture so that the reaction would go on. The extent of the explosion would be limited only by the amount of deuterium added to the weapon. It might be possible to set off a deuterium explosion without the use of tritium, but this was doubtful. It was uncertain whether tritium would be either necessary, or sufficient.

Los Alamos was using 40,000 kilotons TNT equivalent as a working figure for the yield of a Super. This figure was based on the maximum yield that anyone could conceive of as being useful, since it was assumed that a bomb of any size could be made by simply adding deuterium. Dr. Smyth pointed out the difficulty, however, of keeping deuterium in the necessary liquid form, at 400 degrees below zero Fahrenheit. He then made a "wild guess" at 10 tons as the possible weight of the weapon.

Dr. Smyth estimated that, taking all things into account, one super (sic) would cost on the order of ten ordinary fission weapons in terms of productive effort and capacity.²²³

The importance of the inexorable linkage of fission weapon development to the development of the Super was stressed at this meeting:

Senator McMahon expressed the fear that, if it should be possible to produce a thermonuclear explosion without the use of tritium, the Russians might soon succeed in this and might seriously endanger the United States within a matter of months. Dr. Fine indicated that it would be necessary for the Soviets to produce a

²²³AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 92. "Dr. Smyth" was Dr. Henry DeWolf Smyth, an AEC commissioner and author of the famous 1945 "Smyth Report" on the wartime Manhattan Project.

wholly different kind of fission bomb from any that had heretofore evolved [deleted]. Even should such a fission bomb be developed, it would probably not be possible to get by without the use of some tritium.²²⁴

The JCAE was also informed that even success of the BOOSTER was no assurance of thermonuclear plenty:

Dr. Fine, in response to a question, noted that there were dissimilarities in the Booster and Super reactions such that success or failure on one would not guarantee the same in the other.²²⁵

The AEC still thought that a 1952 test was probable:

Mr. Strauss was then asked how soon he thought a Super could be developed, and replied that, with proper emphasis on the program, his guess was the lower end of the two to five year scale predicted by the experts. Mr. Dean pointed out the Los Alamos estimate that a test might be held in early 1952, with production two to three years thereafter.²²⁶

The AEC believed that the U-235 that would be required for the "initiators" for 10 Supers would require a large fraction of a year's production of enriched uranium.²²⁷

²²⁴AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 92. "Senator McMahon" was Senator Brien McMahon, Chairman of the JCAE; "Dr. Fine" was Dr. Paul C. Fine, a member of the AEC's Division of Military Applications (DMA) and a physicist who specialized in weapon and production problems.

²²⁵AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 93.

²²⁶AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 98; "A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, p. 7. "Mr. Strauss" was Admiral Lewis L. Strauss; "Mr. Dean" was Gordon L. Dean. Both men were AEC commissioners at this time, and both went on to become AEC chairmen. The prediction of a 1952 test date and weapon production two to three years later were correct; however, the devices tested and put into production by 1955 bore little resemblance to the Super design contemplated in early 1950.

The NSC Subcommittee Reports to Truman

The group of three NSC members appointed by President Truman in November 1949 reported to him on January 31, 1950. Dean Acheson, the Secretary of State, and Louis Johnson, the Secretary of Defense, recommended that steps be taken to determine the feasibility of a hydrogen bomb. David Lilienthal, Chairman of the AEC, disapproved of this action. They stated their position regarding a "crash" development program:

By letter ... dated November 19, 1949, the President designated the Secretary of State, the Secretary of Defense, and the Chairman of the Atomic Energy Commission as a Special Committee of the National Security Council to make recommendations (a) "as to whether and in what manner the United States should undertake the development and possible production of 'super' atomic weapons," and (b) "as to whether and when any publicity should be given to this matter."

The nature of the decision on which the Committee has been called upon to make recommendation needs to be defined with some precision. Systematic theoretical investigations on the possibilities of a thermonuclear weapon were undertaken at Los Alamos in the fall of 1943, and work on this problem has been going on since that time. Assuming a continuation of the present program at the present rate, however, it would be many years before a test of a thermonuclear weapon would be possible.

The question presented is whether the United States should undertake at this time an accelerated program to determine the feasibility of a thermonuclear weapon, should continue its research at the present rate, or should place a moratorium on further work in this field.

An all-out effort leading to both a feasibility test and quantity production of "supers" would seriously impair the

²²⁷Letter dated January 25, 1950 to Honorable Brien McMahon, Chairman, JCAE, from H. D. Smyth, USAEC.

efficiency and output of the fission bomb program, but there appear to be no advocates for this type of effort. Technical studies of the Atomic Energy Commission indicate that an accelerated research and development program to test the feasibility of such a weapon (as distinguished from a quantity production program) would require a minimum time of three years; that with such a target date other weapon developments now under way, principally lighter and smaller weapons aimed at improved deliverability by aircraft and guided missiles and a penetrating weapon, could probably still be carried out, but not with the care and refinement originally planned; and that this probable decrease in refinement would not be sufficiently important to serve as a deterrent to an accelerated effort on thermonuclear research and development.

The important consideration from a military point of view appears to be that the most advantageous rate and scale of effort would be such as to produce a weapon for testing as soon as possible without significant impairment to the quantity output of fission weapons.

In the present state of knowledge, it appears that there is at least a 50-50 chance that a thermonuclear weapon will be feasible, but this cannot be determined except by actual test.

It is estimated on the basis of technical studies made by the Atomic Energy Commission and the Department of Defense that an accelerated program, including ordnance and carrier development, is within the capabilities of the United States from the point of view of money, materials, and industrial effort.

Knowledge as to whether the thermonuclear bomb is or is not feasible and knowledge as to its potentialities and limitations, if feasible, are of importance to military and foreign policy planning. It should be recognized, however, that the failure of any given test may not conclusively demonstrate that other methods might not be feasible.

It must be considered whether a decision to proceed with a program directed toward determining feasibility prejudices the more fundamental decisions (a) as to whether, in the event that a test of a thermonuclear weapon proves successful, such weapons should be stockpiled, or (b) if stockpiled, the conditions under which they might be used in war. If a test of a thermonuclear weapon proves successful, the pressures to produce and stockpile such weapons to be held for the same purposes for which fission bombs are then being held will be virtually irresistible.

The special committee also considered the likelihood of a parallel Soviet effort on the superbomb:

There is also the question of possible effect on Soviet attitudes and actions of a decision to proceed with a program to test the feasibility of thermonuclear weapons.

a. Would a decision on the part of the United States to go ahead with an accelerated program cause the Soviet Union to press ahead in this field more vigorously? The theoretical possibilities of a thermonuclear reaction have long been known; as early as 1932 there were suggestions by Russian scientists and others that thermonuclear reactions might release enormous amounts of energy.

The Soviet Union probably has felt it could not make any other assumption than that the United States is working on such a weapon, especially in view of the public discussion that has already taken place. It is difficult to escape the conclusion that the Soviet Union will make an intensive effort to produce thermonuclear weapons.

A decision to accelerate our program may cause the Soviet Union to increase the priority of (its own) efforts. Knowledge by the USSR that we had

successfully completed development of a thermonuclear weapon might have the effect of increasing the probability that the USSR would successfully develop a similar weapon. These are risks which are difficult to measure, but which we must frankly face up to if a decision is made to accelerate our development program.

b. It does not appear likely that the character of United States military developments will have a decisive effect on Soviet military developments or be the cause of an arms race. The Soviet decision to re-equip its armies and devote major energies to developing war potential, after the end of (World War II) and at a time when (the U.S. was) disbanding its armies, was based on considerations more profound than our possession of the atomic weapon.²²⁸

The effects of a 40 MT blast from the Super on its delivery aircraft would be cataclysmic, sufficient to require an unmanned vehicle:

The violence of the blast and thermal effects of the Super require any manned aircraft to be a minimum of 30 miles from the detonation point. This requirement will dictate an unmanned vehicle to traverse the final 50 to 100 miles to the target. The development of such a vehicle is a problem which remains to be solved in conjunction with the development of the weapon itself, being dependent largely upon the characteristics, physical dimensions and weight of the weapon.

²²⁸AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 118; NSC working paper dated January 24, 1950 attached to letter dated September 29, 1966 to John T. Conway, Executive Director, Joint Committee on Atomic Energy, from Hayward C. Brown, Jr., Assistant General Manager, USAEC. The "penetrating weapon" alluded to became the MK 8 bomb. See Volume VI.

It is impracticable at this time to anticipate the exact nature of the eventual carrier. It is apparent that eventually a supersonic unmanned vehicle will be necessary, depending upon the scientific advancement in the field of guided missile ground-to-air weapons. It is also apparent that under such conditions of opposition, a supersonic delivery vehicle is also indicated for fission bombs.

Thus a seemingly paradoxical condition may eventually develop wherein the larger, more cumbersome Super may eventually be easier to deliver by virtue of the fact that it may be less demanding for refinements in the guidance system of the final delivery missile. In any event, in consideration of the technical problems in the development of the weapon as compared to the carrier, it is believed that the carrier problem, although difficult, is the lesser of the two.²²⁹

The possible requirement for a supersonic delivery vehicle was significant. Assuming an unretarded bomb drop from an altitude of 40,000 feet, the delivery aircraft would have to be flying 2,160 MPH, or about Mach 3, to be 30 miles away at the time of detonation. If the time-of-fall could be prolonged, then delivery vehicle escape speed could be lowered. The most likely thermonuclear bomb carriers at this time, the B-36 and B-47, had top or "dash" speeds of 400 and 600 MPH, respectively. If these aircraft were to be used as manned delivery vehicles, then parachute retardation was mandatory for the bomb.

The NSC report recommended that public debate over the development of the hydrogen bomb be kept to a minimum for security reasons:

There are many facets to the question of whether it should be a matter of public knowledge that the United States is engaged in an active effort to develop a thermonuclear weapon. It is considered that public discussion once initiated is extremely difficult to control and inevitably leads to a greater disclosure than originally intended. An additional factor of military significance is the

²²⁹AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 124.

divergence of opinion among scientific circles in this country relative to the feasibility of a thermonuclear explosion.

It can be expected that such divergence of opinion exists in the USSR on at least an equivalent parity. If the United States announces that we are engaged in an active effort to develop a thermonuclear weapon, such positive knowledge would give added credence and ascendancy to the Soviet group sponsoring development in this field and may result in an earlier start or greater impetus to the Soviet program. For the above two reasons, it is considered that any decision or actions pertaining to a United States effort to develop a thermonuclear weapon or any determination of its feasibility is military information of the highest security classification.²³⁰

The three-man committee concluded:

(The Special Committee) recommends that the President:

a. Direct the Atomic Energy Commission to proceed to determine the technical feasibility of a thermonuclear weapon, the scale and rate of effort to be determined jointly by the Atomic Energy Commission and the Department of Defense; and that the necessary ordnance developments and carrier program be undertaken concurrently.

b. Direct the Secretary of State and the Secretary of Defense to undertake a re-examination of objectives in peace and war and of the effect of these objectives on our strategic plans, in the light of the probable fission bomb capability and

²³⁰AEC THERMONUCLEAR WEAPONS CHRONOLOGY, p. 127. The NSC did not know that work on a Soviet H-bomb had begun in the summer of 1948, and that Soviet Premier Joseph Stalin was not likely to permit a "divergence of opinion" in the U.S.S.R. on this matter. See also Andrei Sakharov, MEMOIRS, Alfred A. Knopf, New York, 1990, pp. 94, 96, 99 for Sakharov's comments on Soviet determination to build the hydrogen bomb, and DANGER AND SURVIVAL, McGeorge Bundy, Vintage Books, New York, 1988, p. 197.

possible thermonuclear bomb capability of the Soviet Union.

c. Indicate publicly the intention of this Government (sic) to continue work to determine the feasibility of a thermonuclear weapon, and that no further information on it be made public without the approval of the President.²³¹

After considering the NSC report, President Truman met with Secretaries Lilienthal, Johnson, and Acheson:

Afterward, Acheson told me what had happened. At his urging, Lilienthal had led off with a short recapitulation of arguments against the H-bomb, including his own moral qualms. But it quickly became apparent that Truman had already made up his mind. "Can the Russians do it?" he had asked. Assured by the committee that (the Soviets) did indeed possess the necessary capabilities, he had cut short further discussion and signed off on the committee's recommendations and a press release announcing his decision. The meeting had lasted less than ten minutes.²³²

After the meeting, the White House issued a public statement on the afternoon of January 31; it included the following passage:

²³¹Recommendations of the Special Committee in "Development of Thermonuclear Weapons" in "List of Policies Approved by the President in the Atomic Energy Field on the Recommendation of the National Security Council or the Special Committee of the National Security Council of Atomic Energy," attachment to memorandum dated December 1, 1952, to the Secretary of State, the Secretary of Defense, and the Chairman, Atomic Energy Commission, from James S. Lay, Executive Secretary, National Security Council, Subject: Atomic Energy Policies Approved by the President on Recommendation of the National Security Council or its Special Committee, p. 10.

²³²FROM HIROSHIMA TO GLASNOST, Paul H. Nitze, Grove Wiedenfeld, New York, 1989, p. 91. Another account puts the length of the meeting in the White House at only seven minutes. ("In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 305.)

It is part of my responsibility as Commander in Chief of the Armed Forces to see to it that our country is able to defend itself against any possible aggressor. Accordingly, I have directed the AEC to continue its work on all forms of atomic weapons, including the so-called hydrogen or Super bomb.²³³

Privately, Truman sent a letter to AEC Chairman David E. Lilienthal that was somewhat more specific and less ambiguous about the status of the Super; the letter also forbade any further unauthorized public disclosures:

My dear Mr. Lilienthal:

After consideration of the report of the Special Committee of the National Security Council ... I hereby direct the Atomic Energy Commission to proceed to determine the technical feasibility of a thermonuclear weapon, the scale and rate of effort to be determined jointly by the Atomic Energy Commission and the Department of Defense; and that the necessary ordnance developments and carrier program be undertaken concurrently.

I have also decided to indicate publicly the intention of this Government to continue work to determine the feasibility of a thermonuclear weapon, and I hereby direct that no further official information be made public on it without my approval.²³⁴

Lilienthal took Truman's directive on secrecy about the Super to heart: the AEC soon issued a directive to its employees and contractors that barred discussion or comment on even unclassified aspects of the hydrogen bomb, its technology and science, and on the decision-making and debate that led to Truman's statement.

On March 14 and 20, 1950, the AEC sent telegrams to its field office managers which described broad new restraints to be applied to written or spoken communication of atomic

²³³Alexander, p. 13.

²³⁴Letter dated January 31, 1950, from Harry Truman to Honorable David E. Lilienthal, Chairman, Atomic Energy Commission. Truman was as good as his word: he refused, in public, to go even one word beyond his January 31 statement until the end of his presidency. (Bundy, p. 197.)

information, especially concerning the Super. One of the telegrams demanded that AEC staff "avoid release of technical information which, even though itself unclassified, may be interpreted by virtue of the project connection of the speaker as reflecting the (AEC's) program with respect to thermonuclear weapons."

This policy was scrupulously enforced: when Dr. Hans Bethe prepared an H-bomb article for publication in spring 1950 issues of the BULLETIN OF THE ATOMIC SCIENTISTS and SCIENTIFIC AMERICAN magazine, he did not submit the document to the AEC for security review, since there was no technical data in the article that had not been previously published widely.

When the AEC got wind of this violation of Lilienthal's directive, it ordered SCIENTIFIC AMERICAN to stop its printing presses, which were then in the process of preparing the April issue containing the Bethe article, and to destroy type, printing plates, and about 3,000 copies of the magazine containing the offending article. After just four statements were deleted, including one that was deliberately false and three others that had appeared earlier in SCIENTIFIC AMERICAN, the AEC approved the distribution of the magazine and the censored article.²³⁵

Adding political impetus to Truman's decision and public announcement to proceed with the H-bomb was the fact that on February 2, German physicist Klaus Fuchs, then the director of the Theoretical Physics Division and Deputy Chief Scientific Officer of the British Atomic Weapons Research Establishment (AWRE) at Harwell, England, and formerly a member of the wartime "British Mission" to LASL, was arrested in London and charged with espionage.

Fuchs had worked at LASL on the wartime Super design, and information he gathered had been passed on to the Soviets.²³⁶

²³⁵"Concerning H-Bomb Reactions," Science and Citizen, BULLETIN OF THE ATOMIC SCIENTISTS, May 1950, reprinted in ARMS CONTROL, Readings from SCIENTIFIC AMERICAN, W. H. Freeman and Company, San Francisco, California, 1973, p. 62; SECRET SCIENCE: FEDERAL CONTROL OF AMERICAN SCIENCE AND TECHNOLOGY, Herbert N. Foerstel, Prager Books, Westport, Connecticut, 1993, pp. 79, 80. Among deletions ordered by the AEC was Bethe's statement that "(the H-bomb) is based on the nuclear reaction between heavy isotopes of hydrogen, namely deuterium of mass 2 and possibly tritium of mass 3." (Draft of Bethe article "The Hydrogen Bomb" sent to the JCAE on March 1, 1950.)

²³⁶Williams, pp. 155, 192; Sakharov, p. 94.

Fuchs' treachery added weight to the arguments of Edward Teller and others who were urging immediate development of an American hydrogen bomb.

2. RACING IN THE DARK, 1950 - 1954

The Status of the H-Bomb Program, January 1950

Prospects for achieving a hydrogen bomb actually deteriorated in the months between the GAC report in October 1949 and President Truman's announcement to continue development of the hydrogen bomb at the end of January 1950. Lack of a suitable computer to perform the required calculations, questions concerning a concrete technique to initiate and complete a fusion reaction, and problems of providing a deliverable weapon all presented serious difficulties.

Los Alamos scientists were probably never less sure of thermonuclear feasibility than in early 1950: even achieving ignition of pure deuterium did not mean that the reaction would necessarily propagate, and analyzing the flow of energy in the ignition and propagation of a thermonuclear reaction was an extremely difficult mathematical process.²³⁷ However, work continued on the experiments planned for Operation GREENHOUSE to be held in the spring of 1951.²³⁸

The H-bomb's physical size and its blast effects were two stubborn enigmas that doomed to failure any early effort to nail down firmly a set of fully valid delivery aircraft requirements. Consequently, there was always a great deal of uncertainty in the minds of Air Force planners. Could manned aircraft do the job? Was any aircraft currently in the Air Force inventory capable of lifting such a massive weapon, let alone successfully delivering it? Could the weapon fit into any bomber's bomb bay? Could the Air Force produce a suitable bomb carrier concurrently with the first Super?

The Air Force's Air Materiel Command (AMC) at Wright-Patterson Air Force Base in Dayton, Ohio, constantly questioned the Sandia Corporation for pertinent and exact information, trying to get some leads, some ideas, some clues as to the bomb's characteristics. But the atomic scientists themselves were far from certain. They could only offer guesses, or minimum and maximum dimensions, which only deepened the Air Force's quandary.²³⁹

²³⁷"Thermonuclear Weapon Development, January 1950-1951," F. Duncan, USAEC (no date), pp. 1, 2, 3.

²³⁸Alexander, pp. 13, 14; AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 195.

²³⁹HISTORY OF PROJECT BRASS RING, Volume I -- Text, Delmer J. Trester, Historical Division, Chief of Staff, Wright Air Development Center, Wright-Patterson Air Force Base, Ohio, November 1953, p. 58.

While little concrete information was yet available on the physical configuration of the bomb, its expected high yield - - from 10 to 40 megatons, compared to the kiloton-range yields of contemporary atomic bombs -- meant that its blast and thermal effects were of such magnitude that escape of the delivery aircraft was impossible with known methods.

The problem might be solved with unmanned aircraft, but since the AEC expected to produce its first H-bomb as early as 1953, the use of a guided missile was eliminated because contemporary USAF missile programs were not far enough advanced to produce a workable missile in time.

The Air Force's Air Materiel Command originated Project EAGLE in January 1950 to modify B-47 aircraft as directed drones to carry the bomb. Two B-47Bs were allocated as bomb carriers and one B-47A as a director aircraft for the test program. The Boeing Airplane Company in Seattle, Washington, was contractor for development. In its later stages, this program became known as Project BRASS RING.²⁴⁰

The use of unmanned aircraft was always considered as a stop-gap until methods of using manned aircraft were developed. One technique studied later in 1950 was the employment of a glide bomb principle similar to that used in the British "Blue Boar" bomb; the Glenn L. Martin Aircraft Company of Baltimore, Maryland was assigned to this development.²⁴¹

Following a meeting of the General Advisory Committee between January 31 and February 1, GAC chairman J. Robert Oppenheimer stated that with "Super" development approved by the President, the Los Alamos technical program for 1950 should be amended so that when the currently-proposed test of thermonuclear initiation was carried out, it should include, unless overwhelming technical arguments against it appeared, a study of the propagation of a detonation wave into pure

²⁴⁰CAUCASIAN and BRASS RING were USAF study projects to determine the feasibility of using drone or manned B-36 and B-47 bombers as hydrogen bomb carriers for weapons up to 40 megatons in yield. CAUCASIAN was a project to adapt a parachute-retarded TX-14 to manned B-36 and B-47 aircraft, and BRASS RING was a study of the feasibility of drone B-47s as high-yield H-bomb carriers; the latter project was canceled in April 1953. For a good summary of BRASS RING, see Knaack, POST-WORLD WAR II BOMBERS, 1945-1973, pp. 125-128.

²⁴¹HISTORY OF MODIFICATION OF USAF AIRCRAFT FOR ATOMIC WEAPON DELIVERY, 1948 -- 1954, Frederick A. Alling, Historian, Historical Division, Office of Information Services, Air Materiel Command, Wright-Patterson Air Force Base, February 1955, p. 112.

deuterium to such an extent as to provide a valid test of feasibility of the weapon.

Oppenheimer also noted that LASL's current requirements for a quantity of tritium thought necessary for thermonuclear testing within two years could not be met with the AEC's current tritium production reactors. Only about half of LASL's needs could be fulfilled by the Hanford reactors.²⁴² The most promising method of providing as much tritium as was thought necessary would be through diverting some of the AEC's Hanford reactor piles plutonium production capacity over to the production of tritium.²⁴³

On February 2, the Committee for Weapon Development met at LASL to discuss calculations then being conducted by Dr. Stanislaw Ulam and Dr. Cornelius J. Everett. The purpose of the work was to determine if the D+D reaction was self-sustaining after ignition with a reasonably small amount of tritium in good fuel geometry.²⁴⁴ Ulam, a LASL mathematician supporting Super calculations, believed that chances for a successful Super were still no better than even.²⁴⁵

By the middle of the month, the AEC had issued orders to load the plutonium-producing reactors at Hanford with 12,000 to 15,000 U-235 and aluminum alloy canned slugs by September 1950. From 6,000 to 8,000 lithium-alloy slugs were also to be loaded to make tritium.²⁴⁶ These reactor loadings would seriously reduce plutonium production.

²⁴²"Comments Concerning the H-Bomb Development," Department of Defense, Military Liaison Committee to the Atomic Energy Commission, 21 April 1954; letter dated February 1, 1950 to David E. Lilienthal, Chairman, USAEC, from J. R. Oppenheimer, Chairman, GAC; letter dated February 5, 1950 to David E. Lilienthal, Chairman, USAEC, from J. R. Oppenheimer.

²⁴³Letter dated February 13, 1950 to Robert LeBaron, Chairman, MLC, from Carroll L. Wilson, General Manager, USAEC.

²⁴⁴The Committee for Weapon Development, Minutes of Meeting dated February 2, 1950, transcribed by Darol Froman.

²⁴⁵Hewlett and Duncan, p. 439.

²⁴⁶Letter to Robert LeBaron, Chairman, MLC, dated February 13, 1950 from Carroll L. Wilson, General Manager, USAEC; letter to Dr. J. Robert Oppenheimer, Director, Institute for Advanced Study (sic) dated February 14, 1950 from David E. Lilienthal, Chairman, USAEC; letter to Senator Brien McMahon, Chairman, JCAE, dated February 14, 1950 from David E. Lilienthal, Chairman, USAEC.

The Establishment of the Family Committee

Also in February, the "Family Committee" was established at LASL to coordinate thermonuclear weapon development, including the Super, the ALARM CLOCK, the Booster, and others. This committee, also known as the Supervisory Thermonuclear Weapon Development Group, included Drs. Edward Teller (as chairman), J. Carson Mark (chief of the Theoretical Division), Eric R. Jette (chief of the CMR Division), Duncan P. MacDougall (chief of GMX Division), and Marshall G. Holloway (chief of the W Division). The Family Committee reported to Darol Froman, Technical Associate Director at LASL, who was lab director Norris Bradbury's second-in-command.²⁴⁷

Norris Bradbury wrote to Edward Teller and stated that the Family Committee was to have "direct responsibility for the thermonuclear programs listed in the Los Alamos Laboratory program and their accomplishment."²⁴⁸

The committee met with representatives of the MLC and the AEC and discussed the weight and scale of effort to be employed in the thermonuclear program with respect to research, production, ordnance engineering, and military plans and requirements.²⁴⁹ The Family Committee met for the first time

²⁴⁷Letter dated March 3, 1950 to Edward Teller from N. E. Bradbury, Subject: Laboratory Matters; letter dated March 15, 1950 to Edward Teller from N. E. Bradbury, Subject: Assignment of Responsibility; BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 9, 1954, p. 290; Report of the Manager, Santa Fe Operations, U. S. Atomic Energy Commission, July 1947 to July 1950, Book Two, p. 13. The CMR Division was the Chemical and Metallurgical Research division at Los Alamos which conducted chemical and metallurgical research for the purification, production, and fabrication of fissionable material and produced nuclear components for weapons (Hewlett and Duncan, p. 543; Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, p. 68). GMX Division was the "Gadget, Mechanics, and Explosives" division, a weapons physics group at Los Alamos that worked with CMR Division on liquid hydrogen isotopes and fission and thermonuclear device design and engineering (GMX was the successor to the wartime "X Division" at LASL). W Division was the weapons research division at LASL that was responsible for advanced weapons design. (Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, p. 68.)

²⁴⁸"Notes on Minutes of Meetings of the 'Family Group'," Appendix B to September 12, 1952 Walker memorandum.

early in the following month. Weapons postulated by the Family Committee included the "Daddy" (the classical Super) and its several offspring.²⁵⁰

Among items examined by the Family Committee at this time were two immediate ways of proceeding to a full-scale test to carry the 1946 Super model to the point where deuterium might be ignited.²⁵¹

During February, testing of a full-scale mockup of a core for the Booster continued; preliminary testing of a multi-crit gun weapon began; and calculations were begun on new HE systems in connection with the thermonuclear program.²⁵²

The H-bomb development effort was divisible into six phases by now; emphasizing its importance to the program, half of these phases were concerned with tritium production: (1) research and development on the theory; (2) research and development on ordnance problems; (3) research and development on delivery methods; (4) design, construction and production of reactors to generate test quantities of tritium; (5) design and engineering for large-scale production of tritium; and (6) construction of reactors for large-scale tritium production.

Phase 1, theoretical research, was to remain largely at LASL; since development of the hydrogen weapon was part of the

²⁴⁹Monthly Status and Progress Reports for February 1950, U.S. Atomic Energy Commission, March 23, 1950, p. 24.

²⁵⁰"Notes on Minutes of Meetings of the 'Family Group'," Appendix B to September 12, 1952 Walker memorandum; Borden-Walker H-Bomb Chronology, pp. 43, 44. According to the latter source, when the Family Committee met for the first time on March 10, 1950, the use of radiation was the first method discussed for solving an unspecified "first problem." Robert Serber, in the 1992 edition of his LOS ALAMOS PRIMER, notes (p. 4) that he once saw a blackboard in Edward Teller's LASL office upon which were listed ideas for weapons, including their capabilities and properties, and their potential delivery systems. Serber states that the delivery system for the last item on the list, the largest, the method of delivery was "Backyard:" since the weapon was expected to extinguish all life on earth, there was no point in moving it anywhere. Rhodes, in DARK SUN, p. 253, claims that Serber saw this inscription at LASL during World War II.

²⁵¹Memorandum for the File dated January 13, 1953 from John Walker.

²⁵²Monthly Status and Progress Reports for February 1950, U.S. Atomic Energy Commission, March 23, 1950, p. 25.

lab's 1950 research program, no major changes were anticipated (this situation would soon change). However, representatives of the MLC were to meet at LASL during the week of February 20 to re-evaluate the lab's entire fission and fusion weapons research efforts to determine which parts of the program, if any, might be dropped and which other parts might be accelerated and terminated within the current year so that persons assigned to them might be reassigned to the Super program.

The Sandia lab would handle ordnance research and development, while little could be done on delivery methods until the hardware was better defined.

As far as small-scale tritium production was concerned, present plans called for test quantities to be available to LASL by the beginning of 1952 for an H-bomb test at Eniwetok in the spring of that year. To provide this material, a number of holes in the Hanford "H" reactor were to be loaded with lithium slugs and enriched uranium (U-235).

Some slugs were already being irradiated, and were to be extracted in October 1950. About 30 to 40 kilograms of U-235 would be required to increase activity in the "H" pile sufficiently to provide for radiation of lithium. This tritium generation would come only at the cost of lost plutonium production.

Four different methods were being considered for the design and engineering of large-scale reactors to produce large amounts of tritium. Under this program, about one kilogram of tritium per year could be produced (corresponding to about one gram of neutrons per day); even this program would still not supply enough tritium for more than a few H-bombs each year, assuming that one bomb would require several hundred grams of tritium to explode efficiently.

These four techniques included a heavy water-moderated, uranium-fueled reactor under study by North American Aviation; an enriched-uranium fueled reactor designed by the Argonne National Laboratory; existing Hanford reactors loaded with enriched fuel; and a high-current linear particle accelerator under design by the University of California Radiation Laboratory in Berkeley, California.²⁵³

The production capabilities of these piles varied considerably: a medium power (50 megawatt) natural uranium-fueled, heavy water-moderated pile could produce about 0.15 grams of tritium per day, while a 250 megawatt graphite-

²⁵³Letter dated March 1, 1950 to Senator Brien McMahon, Chairman, JCAE, from Sumner T. Pike, Acting Chairman, USAEC.

moderated pile could produce only a third of that per day, or 0.05 grams of tritium.²⁵⁴ At these rates, a single reactor running 24 hours per day could produce in a year only between 18 and 55 grams of tritium. To make a kilogram of tritium, between 18 and 55 50 megawatt to 250 megawatt reactors would be required, running around the clock. (The three Hanford wartime reactors were rated at 250 megawatts.)

The AEC was expected to decide by the summer of 1950 which method to use. Unfortunately, how much tritium might be required per weapon would probably not be known until one or more successful test firings of actual devices to determine the minimum quantity for a given design. The AEC hoped to provide sources of a gram of neutrons per day to make tritium in large quantities.

The last phase, construction of large-scale tritium generators, would not be possible until better estimates of required quantities were available.²⁵⁵ These requirements were vital to the entire H-bomb program.

In mid-February, the MLC compiled its own intelligence estimate of contemporary Soviet thermonuclear weapon capabilities, based in part on its estimations of disclosures by Fuchs. This report was intended to be an independent assessment from current CIA estimates of Soviet weapons capabilities.

The report concluded that the Soviet weapons capabilities were much higher than the CIA estimated. The MLC report stated that by 1945 the Soviets had probably begun the construction of uranium isotope separation and plutonium-production reactors; that the Soviets were now no more than two years behind the U.S. in manufacturing and stockpiling atomic bombs and at worst might actually be ahead of the U.S.; that the Soviets had started by 1945 theoretical

²⁵⁴Memorandum dated 10 July 1950 from RAdm. R. H. Hillenkoetter, USN, Director of Central Intelligence, subject: Semi-Annual Estimate of the Soviet Atomic Energy Program (CIA/SCI-2/50), with attachment, Central Intelligence Agency Joint Atomic Energy Intelligence Committee, Status of the Soviet Atomic Energy Program, 4 July 1950, "Facts and Discussion," p. 4. The figures quoted in this attachment are for Soviet production, but they applied to American reactors as well.

²⁵⁵Memorandum for the Record dated February 13, 1950 from E. L. Heller, subject: Weekly Conference at the AEC with Mr. Shugg and Mr. Hollis and Mr. Borden and Mr. Heller of the Joint Committee staff, at 4 PM, February 10, 1950.

calculations on their own Super and might have already conducted a clandestine and undetected test of their Super and have put the weapon into production.²⁵⁶

This was a very alarmist view that then was not widely shared by most Truman administration officials or by most LASL scientists. While the MLC was correct in concluding that the CIA had badly underestimated Soviet capabilities before September 1949, the MLC had no better intelligence sources than those available to the CIA upon which to base its conclusions.

In early 1950, there were no U-2 or satellite reconnaissance overflights of Soviet territory. The only information available to the U.S. included long-range detection of krypton-85 emissions; various nuclear explosion detection methods; and human intelligence, primarily from defectors from Soviet satellite states.

Late in February 1950, the AEC summarized for the MLC the current status of AEC activities relating to the hydrogen bomb. The summary noted that the tritium production program at Hanford would require 40 to 50 persons, including several physicists, to be transferred from other AEC sites to Hanford. Loss of the physicists would delay AEC research programs on reactor shielding and exponential piles for a year. Diversion of engineers might also delay design of a new Hanford plutonium production reactor.

The report also noted that the thermonuclear weapons program would require the investment of substantial quantities of fissionable material both for the production of tritium and for the weapons themselves.²⁵⁷

The Executive Secretary of the NSC on March 1 requested the Special Subcommittee of the NSC to prepare recommendations to President Truman in light of a letter dated February 24 from Truman to the Secretary of Defense that recommended an all-out "crash" program to develop the hydrogen bomb.

Drs. Ulam and Everett published at LASL on March 7 a 50-page report entitled "Ignition of a Large Mass of Deuterium I." This study was based upon their earlier calculations and

²⁵⁶Memorandum dated 16 February 1950 for Robert LeBaron, Chairman, MLC, from Brig. Gen. H. B. Loper and Maj. Gen. K. D. Nichols, subject: A Basis for Estimating Maximum Soviet Capabilities for Atomic Warfare.

²⁵⁷Letter to Robert LeBaron, Chairman, MLC, dated February 20, 1950 from Carroll L. Wilson, General Manager, USAEC.

concluded that one of the proposed hydrogen bomb configurations was not practical.²⁵⁸

On March 10, the Atomic Energy Subcommittee (the "Special Subcommittee") of the NSC recommended to President Truman that the AEC be prepared to start quantity production of materials, especially tritium, needed for thermonuclear weapons as soon as a feasible device design became apparent:²⁵⁹

The Special Committee recommended that the President:

a. Note that the thermonuclear weapons program is regarded as a matter of the highest urgency. There is no need for additional funds beyond those currently estimated for the feasibility test of the thermonuclear weapon.

b. Note with approval the program for the test of the feasibility of a thermonuclear weapon and the necessary ordnance and carrier developments, as now envisaged by the Atomic Energy Commission and the Department of Defense.

c. Instruct the Atomic Energy Commission to continue making preparations looking toward quantity production of materials needed for thermonuclear weapons, especially tritium, to the extent necessary to avoid delay between the (de)termination of feasibility and the start of possible weapon production.²⁶⁰

²⁵⁸Letter from unknown correspondent dated 8 August 1961 to Senator Clinton P. Anderson, Chairman, Joint Committee on Atomic Energy; LA-1076 by Cornelius J. Everett and Stanislaw M. Ulam.

²⁵⁹Borden-Walker H-bomb chronology, p. 42.

²⁶⁰Recommendations of the Special Committee in "Development of Thermonuclear Weapons" in "List of Policies Approved by the President in the Atomic Energy Field on the Recommendation of the National Security Council or the Special Committee of the National Security Council of Atomic Energy," attachment to memorandum dated December 1, 1952, to the Secretary of State, the Secretary of Defense, and the Chairman, Atomic Energy Commission, from James S. Lay, Executive Secretary, National Security Council, Subject: Atomic Energy Policies Approved by

This report also recommended that a production goal of ten weapons per year be established. To achieve this, the AEC set a goal of a gram of neutrons per day, to make a kilogram of tritium per year.²⁶¹

Other parts of the NSC subcommittee report stated that:

... the current LASL research and development program contemplates maximum effort leading to the test of a thermonuclear weapon at the earliest possible date. It aims at tests of the first steps of the process in 1951, to be followed, if successful, by a test of the process as a whole in 1952. ... There are no known additional steps which might be taken to further accelerate the test program. ...

Bomb-quality U-235 in sizable quantities will be needed for the research and test program, by Los Alamos as well as for tritium production. In addition, there will be a loss to the fission bomb program (caused) by the diversion of Hanford production facilities from the production of plutonium to (the production of) tritium. The total estimated cost in weapons, spread over a period of two to three years, will be on the order of 30 or 40 fission bombs.

Proceeding on the premise that there should not be any serious delay between the completion of the feasibility test and the establishment of a weapon production capability, it will be necessary to reach a prompt decision on the production of tritium. ... The amount of money now envisioned for a construction program (to provide reactors to generate enough tritium) for 10 weapons a year is estimated at 200 million (dollars). Fissionable material will also be required in considerable quantity.

the President on Recommendation of the National Security Council or its Special Committee, pp. 10, 11.

²⁶¹Letter dated April 1, 1950 to Sumner T. Pike, Acting Chairman, USAEC, from J. R. Oppenheimer, Chairman, GAC.

It is pointed out that if the hydrogen ignition should prove infeasible, most of the expenditure toward (tritium) production preparations would not be lost since the production capacity could be employed for fission weapons, radiological warfare elements, or other purposes.

...

The time scale for a prototype weapon is on the order of three years, but it should be emphasized that this is an extremely uncertain figure. Unlike the situation in 1942, when it was fairly clear that the (fission) process would work, it is simply not known whether the thermonuclear process will work at all or under what conditions. Time estimates are therefore nearly futile.²⁶²

Truman endorsed the NSC recommendations, and secretly ordered essentially mass production of the Super, if it could be built.²⁶³

March 1950: Los Alamos Revises Its Technical Program

On the same day that these recommendations were made to President Truman, Los Alamos submitted a "New Technical Program for the Calendar Year 1950" to update its earlier plans in light of Truman's January decision to proceed with development of the hydrogen bomb.

The principal objective of the new plan was research and development of the hydrogen bomb, even at the expense of more efficient, more flexible, more deliverable and more useful fission weapons, although fission weapon development was not to be abandoned altogether. The revised program included the following recommendations for the thermonuclear weapons effort:

We propose to augment to the greatest extent practicable the theoretical effort

²⁶²Undated draft memorandum to the President from the Special Subcommittee of the NSC regarding the rate and scale of investment of money and fissionable materials production facilities into the thermonuclear weapons program.

²⁶³"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 312.

on the study of the circumstances under which a reaction can be initiated and propagated in a deuterium mass. It is clear that such studies are fundamental to the entire question of basic feasibility of ultimate thermonuclear weapons and their results might greatly alter the entire emphasis, nature, and direction of research and development in this field.

...

We propose to continue the development along the lines indicated in the 1949 program of an implosion device of the booster type in order to determine whether a test of such a weapon in 1951 would contribute enough fundamental information to make such a test appropriate.

....

We propose to augment the small current effort on the development of a gun type weapon employing approximately [deleted] of U-235. This development will have as its goal the preparation of a model suitable for test [deleted]

....

We propose to consider and possibly even to start preparation of one or more models containing a T-D mixture and a nominal amount of D₂ for consideration for actual ultimate detonation. The nature of such models cannot now be clearly foreseen and both their nature and time of test of any one will depend upon the results of any 1951 test program and upon the status of the accompanying theoretical program.

We propose to develop appropriate procedures and equipment for the liquefaction of the necessary deuterium and tritium for the maximum 1951 test program which might involve these materials. We suggest that the matter of liquefaction of considerable amounts of deuterium is a matter which must, in the near future, concern the Atomic Energy Commission, and it is not proposed that such work be actually conducted at Los Alamos.²⁶⁴

Among other things, LASL proposed to determine whether or not deuterium, if ignited, would burn. (This was in accordance with Oppenheimer's suggestion at the beginning of February.) Theoretical calculations to date had been only rough approximations and results did not clearly indicate whether or not a thermonuclear reaction would propagate in a mass of fuel.

LASL also proposed a tritium-deuterium gun-type device to test thermonuclear burning, although such a gun design might not be ready for test by 1951.²⁶⁵ Teller's ALARM CLOCK would also be given further consideration, but a test of the device in its present form did not appear to be desirable.

Even now, after several years of study, the feasibility of a thermonuclear weapon was still not as certain as the feasibility of a fission weapon had been in 1942; it would take "fantastic good luck" for a deuterium-burning device to be ready within three years. Just a realistic appraisal of the physical size of a deliverable thermonuclear weapon would probably not be available for at least another year, further complicating the selection of a delivery vehicle.

The costs of this program would be considerable: LASL asked to withhold 400 to 500 kilograms (880 to 1,100 lbs.) of fissionable materials from stockpile in order to conduct experiments. (This material would still be available immediately for fabrication into weapons, if necessary.) Manufacture of tritium instead of plutonium would also cost the equivalent of several hundred more kilograms of fissionable material.

The lab also noted that it was abandoning research and leadership in important areas of basic research except for those immediately applicable to thermonuclear weapon design. This could mean that by 1960, LASL might not be in a position to put to immediate use information that might have been developed had it not been preoccupied nearly full time with an all-out program for the Super.

²⁶⁴Letter dated March 10, 1950, from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1950, pp. 1, 2, 3.

²⁶⁵According to Berend Derk Bruins in his 1981 Columbia University PhD thesis, U.S. NAVAL BOMBARDMENT MISSILES, 1940-1958: A STUDY OF THE WEAPONS INNOVATION PROCESS (p. 508), the gun-type TX-11 penetration bomb was studied for the application of thermonuclear boosting to raise its yield.

LASL noted that a number of important physicists would not work on the Super for moral reasons and that their decisions were having a significant influence on younger brilliant scientists. Furthermore, misleading public statements, many made by high government officials to intimidate the Soviets, had left the impression that all that remained to be done on the H-bomb was a relatively simple engineering job, which scientists would not do.

Unfortunately, to inform these physicists of the true state of the Super program would be a violation of the Atomic Energy Act, and the lab feared that the U.S. was either losing or had already lost the atomic armaments race with Russia.²⁶⁶ Any day now, stratospheric winds might carry evidence of hostile foreign thermonuclear experimentation.

To many, it appeared that unwise policies of secrecy were impeding recruitment. Practically all aspects of the thermonuclear problem as then known were already in the public domain, and further attempts to restrict or recover such information were foolish. On the other hand, other matters, such as an actual feasible device design, were presently unknown and were therefore presumably still secret from everyone.²⁶⁷

At the end of March, Norris Bradbury cabled Air Force Brigadier General James McCormack, Jr., then Director of Military Application of the AEC, the AEC division directing weapons development, and urged that some of the secrecy and misinformation surrounding the Super program be dispelled:

Believe strongest thing which could be done (to obtain theoreticians to work at LASL) would be to find some way to inform scientists generally of the real state of affairs in the H-bomb field and of the need for scientific rather than engineering assistance at this time. We recognize high-level difficulties in this but believe that combined efforts of GAC and AEC might convince appropriate authorities that an authoritative AEC statement on these matters would do much more long-range good than harm and would probably tell Uncle Joe (Stalin)

²⁶⁶Notes of a Briefing Held at the Los Alamos Scientific Laboratory on February 23, 1950, pp. 4, 5, 6, 8-11, 13, 14.

²⁶⁷Minutes of the Twentieth Meeting of the General Advisory Committee to the AEC, March 31-April 1, 1950, p. 8.

very little that he does not now know or suspect.²⁶⁸

Despite Bradbury's plea, no such statement was ever issued by the AEC, and speculation and misinformation continued to characterize most public discussion, knowledge, and debate about the H-bomb.

In conclusion, the revised LASL Technical Program plan contained the following paragraphs regarding the thermonuclear program and its effects on the concurrent development of new fission weapons:

The program proposed above appears to the Los Alamos Laboratory to be almost the maximum possible effort in the direction of understanding and attaining thermonuclear reactions and at the same time to maintain progress in the field of fission weapons sufficient to play a real role in the short range position of this country.

It must be clearly understood, however, that much less than the maximum progress which could be made in the fission weapon field will be made under these circumstances, and that this sacrifice would be made in the attempt to ascertain the thermonuclear weapon possibility at the earliest possible date.

It must also be clear that the general nature and philosophy of the thermonuclear program will differ from those hitherto employed by Los Alamos in its study of fission weapons. Because of experimental and theoretical necessity and in an attempt to gamble on the chance of maximum progress, (the 1951) tests at Eniwetok, involving the expenditure of sizable amounts of fissionable material, will take the place of part of the heretofore extensive model testing and detailed theoretical calculation.

This more empirical approach can, with great good luck, materially shorten a

²⁶⁸"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 (1989), p. 314.

development period; on the other hand, the chance of failure in such tests will be appreciably higher than under the old philosophy.²⁶⁹

The new technical program began well. The Theoretical Division physics staff at Los Alamos, which supported both the fission and fusion weapons programs, grew from about 20 persons in summer 1949 to 38 by January 1951.²⁷⁰ Not only did enthusiasm return to many who had become skeptical, but between 300 and 400 new scientists were hired to participate in the revived program -- a great change from the small handful who had worked part time on the Super before the President's decision in January.

The staff at Los Alamos began working a voluntary six-day work week in March 1950, a schedule they had not been on since the end of World War II. (This work schedule continued for almost the next three years in the fear that the Soviets were not far behind in developing their own thermonuclear weapons).²⁷¹ Although there was still some disbelief in the feasibility of a thermonuclear reaction, the attitude of Los Alamos personnel improved considerably.²⁷² This was only a temporary uplift, however: discouraging events would soon dampen spirits again.

²⁶⁹AF ATOMIC ENERGY PROGRAM, Vol. IV, pp. 195, 196; Los Alamos Scientific Laboratory Technical Program for 1950.

²⁷⁰York, THE ADVISORS, p. 75. At a press conference in September 1954, the lab released figures showing 22 theoreticians in 1949, 35 in 1950, and 45 in 1951. (LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, p. 37.)

²⁷¹Bradbury statement at September 24, 1954 press conference in Los Alamos; Mark interview, LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, p. 36; Work Week for the Los Alamos Scientific Laboratory, Report by the Director of Military Application, USAEC, March 6, 1950; Minutes of 378th Meeting, USAEC, March 7, 1950, AEC 304: Work Week for the Los Alamos Scientific Laboratory; teletype dated March 7, 1950, from Brig. Gen. James McCormack, Jr., Director of Military Application, USAEC, to Carroll Tyler, Manager, Santa Fe Operations Office, USAEC. In a letter dated late May 1950, Dr. Teller expressed his concern to Norris Bradbury that reduction of this schedule to a five-day week in July and August 1950, to accommodate summer vacations, would delay development of a workable model of the Super.

²⁷²AF ATOMIC ENERGY PROGRAM, Vol. IV, pp. 196, 197.

The alleged Soviet threat had not diminished between September 1949 and February 1950: a current CIA estimate concluded that the Soviets might have as many as 10 to 20 atomic bombs by the middle of 1950, 25 to 45 a year later, 45 to 90 by mid-1952, 70 to 135 by mid-1953, and 120 to 200 by mid-1954.²⁷³

Another event with far-reaching consequences also occurred in March. LASL's Dr. Alvin Graves, acting as Scientific Director of Pacific Proving Ground activities, suggested the establishment of a research contract with Dr. E. O. Lawrence's Berkeley Radiation Laboratory, also known as the University of California Radiation Laboratory (UCRL) at Berkeley, to measure soft thermal radiation x-rays from nuclear explosions, from both fission weapons and thermonuclear systems.

It appeared that the proposed work could be carried out under terms of an existing AEC-UCRL contract.²⁷⁴ The involvement of UCRL in these experiments would lead within two and a half years to the institution of a second U.S. nuclear weapons lab.

At the beginning of April, the GAC, in a report to the MLC, stated that it was fully aware of the possible sacrifices in the development of more efficient, more flexible, more deliverable, and more useful fission weapons which the adoption of the current LASL thermonuclear program might entail.²⁷⁵

The GAC, echoing Bradbury's request of a few weeks earlier, also urged greater declassification of some aspects of the Super program:

²⁷³"Soviet Atomic Capabilities," Enclosure A to "Estimate of the Soviet Possession of the Atomic Bomb upon the Security of the United States and upon the Probabilities of Direct Soviet Military Action," ORE 91-49, Central Intelligence Agency, 6 April 1950, p. 11.

²⁷⁴Memorandum dated April 29, 1954 to Chester G. Brinck, Assistant General Counsel, from Leonard A. Jacobvitz, Attorney, Office of Assistant General Counsel, subject: Participation of the University of California Radiation Laboratory in Weapons Development Program.

²⁷⁵Borden-Walker H-bomb chronology, p. 47; letter dated April 1, 1950 to Sumner T. Pike, Acting Chairman, USAEC, from J. R. Oppenheimer, Chairman, GAC; "A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, p. 10.

We share with the AEC a considerable anxiety about the state of the public information on thermonuclear weapons. We agree with the Commission that this can in part be traced to the freedom with which both technical and practical aspects of the problem have been discussed by non-project personnel and to the restrictions which have been established on the official release of information.

We sincerely hope that Commission will be successful in its efforts to have these restrictions on the release of official information liberalized. (Among the information that might be released is an) appropriate rough description of the present state of development of thermonuclear weapons, of the work required to accomplish their development, and of the uncertainties of success inherent in the program.²⁷⁶

Despite the GAC's wishes, very little of this type of information would be officially released in the near future, and rampant public and media speculation would often paint a highly inaccurate picture of the Super program.

The DOD was also aware that some sacrifices in fission weapon development were involved and was prepared to accept them. At the same time, military interest in certain fission weapons might serve to stimulate future work on them. On the other hand, physicist Enrico Fermi had great misgivings as to the wisdom of the scale of the presently-planned thermonuclear program and he felt that the U.S. military position five years hence would be much weaker as a result of it.²⁷⁷

The AEC and DOD recommended on April 4 that President Truman approve the following general principles to guide the U.S. thermonuclear program in the future:

- a. The amounts of tritium to be produced will be determined jointly by the Atomic Energy Commission and the Department of Defense from time to time to meet estimated requirements of the thermonuclear

²⁷⁶Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p. 24.

²⁷⁷Minutes of the Twentieth Meeting of the General Advisory Committee to the AEC, March 31-April 1, 1950, p. 3.

development program rather than to meet a fixed yearly rate.

b. Work on the thermonuclear program will be carried on with the objective of determining the feasibility of a thermonuclear weapon at the earliest practicable date. At the same time, promising developments of fission weapons will be carried forward effectively.²⁷⁸

President Truman approved on April 18 a directive for the AEC's materials production program for 1950; Truman noted that considerable amounts of fissionable material, and 10% of the capacity of the Hanford reactors, would be required for the thermonuclear weapons program.²⁷⁹

Further calculations in the spring of 1950 by Dr. Stanislaw Ulam and Dr. C. J. Everett on the progress of a thermonuclear reaction in a spherical assembly indicated that the proposed Super model would not work, and that energy would be drained off much more rapidly than it was produced, yielding a "fizzle": hydrodynamical disassembly proceeded faster than the build-up and maintenance of thermonuclear reactions.²⁸⁰

²⁷⁸Recommendations of the Special Committee in "Development of Thermonuclear Weapons" in "List of Policies Approved by the President in the Atomic Energy Field on the Recommendation of the National Security Council or the Special Committee of the National Security Council of Atomic Energy," attachment to memorandum dated December 1, 1952, to the Secretary of State, the Secretary of Defense, and the Chairman, Atomic Energy Commission, from James S. Lay, Executive Secretary, National Security Council, Subject: Atomic Energy Policies Approved by the President on Recommendation of the National Security Council or its Special Committee, p. 11.

²⁷⁹"Chronology of Major Decisions Regarding Thermonuclear Program," USAEC, June 14, 1951, p. 3.

²⁸⁰Hewlett and Duncan, p. 440; letter dated May 18, 1950 from John von Neumann to Edward Teller. Some of these findings may be contained in two LASL reports, LA-1076, "Ignition of A Large Mass of Deuterium by a Burning Deuterium-Tritium Mixture, Problem I" by Everett and Ulam, dated March 7, 1950 and LA-1124, "Ignition of A Large Mass of Deuterium by a Burning Deuterium-Tritium Mixture, Problem II" by Ulam, Everett, et. al. dated June 16, 1950. ("Publications of Stanislaw M. Ulam," compiled by Barbara Hendry, Los Alamos Scientific Laboratory, April 1968; Josephine L. Elliott, Joan Houston, and Francoise A. Ulam also co-authored LA-1124.)

An extraordinary and prohibitively expensive amount of tritium would be required to maintain the reaction at a much lower temperature than the proposed D-D burn.²⁸¹

In May, tritium for experiments at Los Alamos and for the GREENHOUSE George test device was to be provided by the AEC reactors at Hanford. Long-range production of tritium in sufficient quantity for the Super required more reactors taking advantage of improved technology; although an early 1950 estimate had indicated a substantially larger tritium requirement than previously believed necessary, the mid-year calculations increasingly pointed to an even larger amount required to ignite the Super.²⁸² By this time, a satisfactory fuel slug of aluminum and lithium had been developed for tritium production.²⁸³

The DuPont Corporation was requested to design, construct, and operate two new heavy water reactors to produce tritium, using deuterium oxide as a moderator (D₂O is a much better neutron moderator than highly-neutron-absorbent graphite).

Negotiations with DuPont began, and on June 12, the AEC asked the company to conduct site surveys and begin planning the construction of both a new reactor installation and a large-scale heavy water production facility²⁸⁴ (deuterium oxide -- heavy water -- was also a source of deuterium gas for liquefaction).

In the middle of June, Ulam, Everett, and three other LASL researchers published a report entitled "Ignition of a Large Mass of Deuterium by a Burning Deuterium-Tritium Mixture, Problem II," a sequel to their March paper on the same subject. This 22-page study investigated the problem from a different point of view and concluded that there still was not a practical and economical means to accomplish ignition.²⁸⁵

²⁸¹Bethe, "Comments on the History of the H-Bomb," p. 47; Mark, LA-5647-MS, p. 8; ADVENTURES OF A MATHEMATICIAN, Stanislaw M. Ulam, Charles Scribner's Sons, New York, New York, 1976, pp. 213 - 219 (revised edition published in 1991 by University of California Press, Berkeley, California, and referred to hereafter as Ulam biography, 1991).

²⁸²Borden-Walker H-bomb chronology, p. 52.

²⁸³Borden-Walker H-Bomb Chronology, p. 56.

²⁸⁴"Chronology of Major Decisions Regarding Thermonuclear Program," USAEC, June 14, 1951, p. 4.

June 1950: Diminishing Prospects For the H-Bomb

The Super program was buffeted by even more problems in mid-1950. The outbreak of the Korean War on June 25 shifted the emphasis of the nation's nuclear weapons development program away from an uncertain thermonuclear weapon to the development and production of small fission weapons for tactical purposes.

The Korean War also caused much fluctuation in the Los Alamos plans to fire a thermonuclear device in the spring of 1951.²⁸⁶ Planning for the thermonuclear test consumed a great deal of time and effort at LASL during the summer of 1950.²⁸⁷

In addition, further manual calculations by Ulam and Everett later in the summer showed more difficulties in the initial D-T reaction which would ignite the Super²⁸⁸ The success of this part of the "classical Super" had previously been taken almost for granted.

These studies showed that a fusion reaction would not proceed in a volume of deuterium once it was ignited: energy loss due to expansion and radiation prevented ignition of the remainder of the mass of unburned fuel by the "hot" reacting region immediately adjacent to it.²⁸⁹ There was little doubt that the reaction would die out long before most of the fuel

²⁸⁵Letter from unknown correspondent dated 8 August 1961 to Senator Clinton P. Anderson, Chairman, Joint Committee on Atomic Energy; LA-1124 by Josephine L. Elliott, Cornelius J. Everett, Joan Houston, and Francoise A. and Stanislaw M. Ulam dated June 16, 1950.

²⁸⁶AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 197. The Korean War also brought Hans Bethe back to thermonuclear weapons work LASL: on March 14, 1950, Bethe had written to Norris Bradbury and stated that because of moral reservations, he could not in good conscience work on the H-bomb; however, he added that in case of war, he would obviously reconsider his position.

²⁸⁷Duncan, "Thermonuclear Weapon Development, January 1950 - May 1951," p. 5.

²⁸⁸Bethe, "Comments on the History of the H-Bomb," p. 47.

²⁸⁹Ulam biography, 1991, p. xii.

was consumed.²⁹⁰ (These calculations were confirmed by computers in early 1951.)²⁹¹

Coming on the heels of the discouraging results in the spring, the calculations were almost the coup de grace for the classical Super.

Dr. Ulam later wrote of this period:

Still Teller kept on hinting that not enough work was being done on his original scheme for the ignition of the "super" ... In collaboration with my friend Everett, one day I decided to try a schematic pilot calculation which could give an order of magnitude, at least, a "ballpark" estimate of the promise of his scheme .. Before we started this calculation of the progress of a thermonuclear reaction (burning in a mass of deuterium or deuterium-tritium mixture), Everett and I had done a lot of work on probability questions connected with the active assemblies of uranium and with neutron multiplications.

We worked out a theory of multiplicative processes, as we called it. (Now the preferred name is "branching processes.") ... Our calculation .. threw grave doubts on the prospects of Edward's original approach to the initial ignition conditions of the "super."

As the results of the von Neumann-Evans calculation on the big electronic Princeton machine slowly started to come in, they confirmed broadly what we had shown. There, in the course of the calculation, in spite of an initial, hopeful-looking "flare up," the whole assembly slowly started to cool down. Every few days Johnny would call in some results. "Icicles are forming," he would say dejectedly.²⁹²

²⁹⁰Hewlett and Duncan, p. 527; Bethe, "Memorandum on the History of the Thermonuclear Program."

²⁹¹Hewlett and Duncan, p. 535.

²⁹²Ulam, ADVENTURES OF A MATHEMATICIAN, pp. 213, 214, 216, 217.

The effects of these disheartening developments were immediate: DuPont was pressing the AEC to tell it whether to design its new reactors for tritium or plutonium production (more plutonium was needed for new, relatively-inefficient low-yield tactical weapons on the drawing boards at Los Alamos). This decision about nuclear materials production depended directly on the fate of the Super.

On July 25, President Truman had declared the DuPont heavy water project to be "of the highest urgency and vital to national security."²⁹³ DuPont thought heavy water reactors were feasible and that a dual-temperature process was the best way to produce heavy water.²⁹⁴ Adding to Truman's concern, about two weeks earlier, intelligence estimates had concluded that the Soviets might already possess at least 10 atomic bombs.²⁹⁵ (By the end of 1950, the Soviet stockpile consisted of only five plutonium-fueled implosion weapons; the U.S. stockpile at that time was nearly 60 times larger and consisted of far more advanced designs.)

The only good news in the earlier July estimate of Soviet capabilities was the understatement that "because our own (thermonuclear) program indicates that some new approaches may be necessary before the thermonuclear is practical, no valid estimate of the Soviet ability to solve the problem nor of the time required for the solution can be made."²⁹⁶

Also during the summer of 1950, Dr. Edward Teller proposed the use of Li^6D in the ALARM CLOCK, in lieu of pure deuterium.²⁹⁷

²⁹³Hewlett and Duncan, pp. 430, 431.

²⁹⁴Hewlett and Duncan, pp. 523, 524.

²⁹⁵"Comment on the Intelligence Report by Committee Staff," undated but ca. July 10, 1950. This report also notes that the Soviets made the decision to embark on their own H-bomb program in late 1949.

²⁹⁶Memorandum dated 10 July 1950 from RAdm. R. H. Hillenkoetter, USN, Director of Central Intelligence, subject: Semi-Annual Estimate of the Soviet Atomic Energy Program (CIA/SCI-2/50), with attachment, Central Intelligence Agency Joint Atomic Energy Intelligence Committee, Status of the Soviet Atomic Energy Program, 4 July 1950, "Facts and Discussion," p. 5.

²⁹⁷Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, p. 12; Teller, COMMENTS ON BETHE'S HISTORY OF THE THERMONUCLEAR PROGRAM, p. 2. The "Alarm Clock" was described at this time as a scheme to obtain a large energy release with relatively cheap materials. (Minutes, Twenty-Third

By this time, both military and civilian organizations were giving considerable thought to the best employment strategies for the new weapon when it was available in quantity. A few high-yield hydrogen bombs were generally believed to be better suited to many Soviet targets than atomic bombs:

It is often pointed out that Russia contains very few large-area targets "tailored" to the H-bomb and that even these consist partly of open, undeveloped territory upon which bomb damage would be wasted. Still, this argument concedes that at least some urban-industrial targets of great size do exist in Russia.

Moreover, many of her smaller cities occupy considerable space, have low population densities per square mile, and are surrounded by outlying installations such as factories and bases important in waging war. A-bombs would be relatively inefficient if used against these targets, whereas one H-bomb would eliminate the entire complex in each case.

It is also pertinent that the H-bomb is capable of creating an immense crater -- which would preclude repairs in the case of railroad marshalling yards, underground plants and the like, and which might be of special interest in attacking oil fields.

The problem of neutron economy (i.e., how much fissionable material for A-bombs is sacrificed by using the neutrons available in reactors to make tritium instead) is directly relevant in this context since it must first be determined how many A-bombs are sacrificed in order to make one H-bomb before deciding what ratio between the two weapons best exploits production capacity in light of the nature and characteristics of the potential targets to be destroyed.

...

Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, October 30, 31, and November 1, 1950, Los Alamos New Mexico, p. 62.)

... General LeMay, testifying before the Joint Committee last fall, indicated that he would desire to carry out an atomic offensive against Russian atomic bases before hitting strategic targets.

Delivery of the Super still posed problems, both with bombing accuracy and the likely loss of bomb carriers:

The matter of deliverability has many aspects -- including the large dimensions expected to characterize an H-bomb [deleted] and the fact that a hydrogen explosion would be so violent as to destroy the delivery aircraft and any crew members aboard.

At the same time, estimates as to H-bomb dimensions indicate that it could be carried in a B-36, and crewless "drone" bombers of the type which have already crossed the Atlantic Ocean by remote control from the "mother" ship are comparatively easy to visualize.

Because delivery of an H-bomb would mean putting the equivalent of "many eggs in one basket," a specially-designed jet-propelled carrier seems to be indicated, and the Air Force is actually working along these lines.

Present war plans contemplate strikes with one A-bomb aboard only one plane in every five comprising the offensive force, so that Soviet interception of any given aircraft would involve only a 20% chance of loss of an atomic weapon. We could doubtless afford to increase such a ratio to one H-bomb carrier in every 10 or 20 or more planes, where only a single ship would be needed to transport so damaging an explosive. ...

So far as aiming accuracy is concerned, I have relatively little confidence in the ability of our plane crews to place an A-bomb consistently close to definite pin-points -- especially at night and in bad weather, against strong opposition.

The virtue of a weapon which could do its assigned job even though it missed its bull's eye by five, 10, or even 15 mile is

obvious. Moreover, we may conceive of heavily-defended targets which could not be approached within (the destructive radius of an A-bomb) without probable destruction of the delivery aircraft; such targets, which could take overriding priority in our war plans, might succumb to the H-bomb alone.²⁹⁸

In preparation for a late September meeting of the General Advisory Commission, Dr. Edward Teller and Dr. John Wheeler prepared at the beginning of August a report on the current status of the thermonuclear work. Their outlook was pessimistic: through the summer, the Theoretical Group at Los Alamos had worked hard and accumulated much new data, but nothing of sufficient promise had developed pointing to a near-term solution to the Super problem.

The report claimed that theoretical analysis was a major bottleneck in the thermonuclear field, and that the number of theoretical physicists at LASL was decreasing, rather than increasing:

... the required scale of effort (for the Super) is clearly much larger than that needed for the first fission weapon. Theoretical analysis is a major bottleneck to faster progress in analyzing thermonuclear weapons, and the bottleneck to the theoretical analysis is the shortage of the right men ...

Approximately a dozen members of the T-Division have been working on all phases of the thermonuclear project. Their effort was divided approximately evenly between the question of [deleted] and the planning of the 1951 test ...

At the present moment, the total number of senior theoretical physicists at Los Alamos is not increasing but decreasing. No

²⁹⁸Memorandum for Mr. Cole dated July 24, 1950 from Bill Borden, subject: The Hydrogen Bomb in Relation to Atomic Bombs, pp. 3, 4, 5. This memo seems to reinforce the claims of some GAC members that the Air Force wanted the H-bomb to compensate for bombing errors. It also highlights the Air Force's "shell game" approach to an attack, with the nuclear weapon aboard only one aircraft in a formation of many.

improvement is foreseeable without a real change in the plan of recruitment.²⁹⁹

In the opinions of Teller and Wheeler, "As of August 1950 it is still impossible to say whether or not any thermonuclear weapon is feasible or economically sensible."³⁰⁰

A few days later, at its 18th meeting, the Family Committee reached a decision on the general nature of the thermonuclear test planned for the spring 1951 test (the GREENHOUSE George shot). The device considered for test was to give evidence of high-energy fusion-spawned neutron production from a tritium-deuterium fuel mixture, the cleanest possibility of proof of reaction.³⁰¹

The Committee also considered alternate thermonuclear test sites, including Churchill, Canada, on the shores of Hudson Bay. This location might be used for a thermonuclear test in the summer of 1952.³⁰²

In a letter at the end of August to the AEC, Norris Bradbury stated (very presciently, in light of what would happen seven months later) that:

We are, of course aware of a general concern in many quarters regarding the progress of the so-called H-bomb. It will be clear ... that concentrated effort in this field during the last eight months has shown that the probability of early, practical success in this endeavor is considerably less than might have been anticipated earlier; and that, indeed, success in the development of a practical "Super" may depend upon entirely new and as yet unforeseen approaches to the matter. It is clearly impossible to make

²⁹⁹Borden-Walker H-bomb chronology, p. 54; LAMD-444, App. 1-A. August 1, 1950, excerpted in "Technical References Relating to Thermonuclear Program," Appendix E to September 12, 1952 Walker memorandum.

³⁰⁰Duncan, "Thermonuclear Weapon Development, January 1950-May 1951," pp. 5, 6. By 1993, tritium cost about \$850,000 per ounce. ("Lab Scientists Play Critical Roles in Historic Fusion Breakthrough," Newsbulletin, Los Alamos National Laboratory, Vol. 13 No. 46, December 17, 1993, p. 1.)

³⁰¹Borden-Walker H-bomb chronology, p. 54.

³⁰²"Notes on Minutes of Meetings of the 'Family Group'," Appendix B to September 12, 1952 Walker memorandum.

a precise evaluation of these probabilities ...³⁰³

On September 12, SAC commander General Curtis LeMay noted that in light of the ongoing Korean War, steps should be taken to ensure the development of a successor to the B-36, then the United States' primary long-range nuclear bomber:

In view of the rapidly deteriorating international situation, it is urged that positive action be taken by Headquarters USAF to insure development by the earliest possible date of a heavy bombardment type aircraft with substantially higher performance characteristics than the B-36.

The high probability of war between the United States and the USSR prior to 1955 rules out all possibility that, with the exception of the B-52, any new development in either heavy bombardment aircraft or long-range missiles will be available to supplant or augment our current B-36 force.³⁰⁴

Late in September, the GAC commented on the detailed report which had been submitted to the committee by Drs. Wheeler and Teller. The report described schemes for the transfer of energy from an atomic bomb to the Super; analyzed problems to be solved and estimated manpower required to solve them; presented calculations showing advantages of the ALARM CLOCK over the Super; and stressed the importance of the forthcoming GREENHOUSE George test.

The GAC noted that there was interference at LASL between the fission and fusion weapon programs, primarily in demands made upon the Theoretical Division and on computer resources allocated between the two programs. The GAC also expressed at this time a desire that promising fission weapons not be unduly delayed by the fusion weapons program.³⁰⁵

³⁰³Letter dated August 29, 1950 from Norris Bradbury to Brig. Gen. James McCormack, Jr., Division of Military Application, USAEC.

³⁰⁴HISTORY OF THE STRATEGIC AIR COMMAND, THE B-52: BACKGROUND AND EARLY DEVELOPMENT, 1946 - 1954, Historical Study No. 60, Historical Division, Office of Information, Headquarters Strategic Air Command, p. 28.

³⁰⁵Borden-Walker H-bomb Chronology, p. 56.

The GAC also stated that "we are happy to note that great progress has been made at Los Alamos in setting up quantitative methods for coping with the difficult problems of the hydrogen bomb; we urge the Commission to support these efforts by making computational machinery available to the laboratory."³⁰⁶

Of the planned thermonuclear test at Eniwetok during the forthcoming Operation GREENHOUSE, the GAC stated:

We wish to make it clear, however, that the test, whether successful or not, is neither a proof firing of a possible thermonuclear weapon nor a test of the feasibility of such a weapon. This test is not addressed to resolving the paramount uncertainties which are decisive in evaluating the feasibility of the Super.³⁰⁷

Near the end of the month, Fermi and Ulam presented further theoretical findings in a 36-page LASL report entitled "Considerations on the Thermonuclear Reactions;" the paper looked at yet another proposed Super model and concluded that it too was without merit.³⁰⁸

In October, research on the "multi-crit" gun-type trigger for the Super was ended, and study concentrated instead on a cylindrical implosion type trigger considered earlier in the year.³⁰⁹ (This cylindrically-imploded trigger became the basis of the "sparkplug" in successful U.S. hydrogen bomb designs.)

At this time, LASL decided to proceed with the test of the BOOSTER at Eniwetok in the spring of 1951, in spite of the discouraging prospects for the full-scale Super.³¹⁰

³⁰⁶Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p. 25.

³⁰⁷Borden-Walker H-bomb Chronology, p. 58.

³⁰⁸Letter from unknown correspondent dated 8 August 1961 to Senator Clinton P. Anderson, Chairman, Joint Committee on Atomic Energy; LA-1124 by Enrico Fermi, Miriam Planck, and Stanislaw M. Ulam dated September 26, 1950.

³⁰⁹Letter dated 3 October 1950 from Sumner T. Pike, Acting Commissioner, U. S. Atomic Energy Commission, to Senator Brien McMahon, Chairman, Joint Committee on Atomic Energy, Congress of the United States.

President Truman approved on October 9 an expanded program for production of fissionable materials; this expansion would be met by construction of three additional heavy water reactors at the DuPont site in South Carolina as well as additional gaseous diffusion facilities to be built in Ohio.³¹¹

Late in October, the designs of the experimental thermonuclear test devices (the BOOSTER and the CYLINDER) slated for detonation the following spring at Eniwetok were "frozen" by the Family Committee at LASL.³¹² For most of the fall, computing resources at LASL were devoted almost entirely to detailed iteration of calculations concerning the test devices. However, some theoretical work continued on the ALARM CLOCK.³¹³

October 1950: The GAC Meets

At the end of October, the General Advisory Committee met at Los Alamos. Dr. J. Carson Mark, head of LASL's T Division, summarized the recent Ulam-Everett calculations and the ENIAC verification.³¹⁴ Dr. John von Neumann concluded from them that a thermonuclear reaction was possible, but not by any easy method. Edward Teller admitted that the practicality of the Super depended upon the amount of tritium that might be required and that the trend was unfavorable.

At the GAC meeting, Dr. James Tuck reviewed data on D-D and D-T cross-sections; Carson Mark reviewed calculations on D-D and D-T burning; John Wheeler reviewed the crucial fission-to-D-T-link; and Dr. Teller reviewed expectations of Super feasibility.

Accurate measurement of D-D and D-T cross-sections was complicated by the fact that they had to be done within a difficult range of energies, energies not easily obtained outside of a full-scale thermonuclear reaction. Extrapolation

³¹⁰Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, p. 5.

³¹¹"Chronology of Major Decisions Regarding Thermonuclear Program," USAEC, June 14, 1951, p. 4.

³¹²"Notes on Minutes of Meetings of the 'Family Group'," Appendix B to September 12, 1952 Walker memorandum.

³¹³Borden-Walker H-bomb chronology, p. 57.

³¹⁴Duncan, "Thermonuclear Weapon Development, January 1950-May 1951," p. 6.

of existing measurements to low energies was quite unreliable.

Previously-measured cross-sections for D-D interactions might be in error by anywhere from 35% to 100%; D-T cross-sections were more reliable, ranging from 3.4 to 6.9 barns with the best present guess at 5.4 barns. A new experiment on D-T cross-section using a gas target was now being started; it would cover an energy range of 20 to 120 thousand electron volts (KeV), and results were expected by January of 1951.

After Carson Mark reviewed the Ulam-Everett and ENIAC calculations on the burning of deuterium and tritium in the Super, Dr. Libby asked how long a theoretical effort would be required to resolve current questions about propagation and burning if all physical data were known (which they were not now). Edward Teller said that time to collect this data would be measured "in years."

John Wheeler described the schemes proposed for coupling a fission explosion to subsequent burning stages in the Super (at this time, a D-T burn was to be used to ignite a large D burn). When asked by Willard Libby whether purely theoretical considerations would suffice to decide the feasibility of the Super, Teller said that he believed that the D-T mixture would burn but that there were uncertainties about the best arrangement.

Before embarking on a list of uncertainties currently facing the quest for the Super, Teller discussed the effect of compression of the thermonuclear reactants on reaction behavior. He then went on to list and describe what he perceived to be three classes of uncertainties: reducible uncertainties, those perhaps reducible, and irreducible ones. Among the reducible uncertainties were the cross-sections of D-D and D-T reactions within relevant energy ranges.

The most irreducible uncertainty was Taylor instability within the thermonuclear reactants. This instability arose when low-density material accelerated high-density material. No uniform acceleration occurred; instead, the low-density material penetrated into the high-density material, with the result that "hot" regions acted like low-density regions, and "cold" regions like high-density regions.

Teller had no new ideas, but he believed fervently that the quest for the Super would be successful. He urged that more theoretical work be done, and that LASL try to hire more theoretical physicists. He also argued that more detailed calculations should be carried out to determine how much tritium might be required for any particular Super design. Teller also noted that the family of proposed thermonuclear devices was not limitless: so far, since 1942, although a

number of schemes had been examined, only the Super and the ALARM CLOCK had survived scrutiny.

A year after the GAC's recommendation to the AEC to stop development of a hydrogen weapon on technical and moral grounds, little had happened to prove them wrong on a purely technical basis.

When Dr. Willard Libby suggested that too much weight was being given to the significance of calculations done to date, none of which had yet been checked by experiment, Drs. Oppenheimer, Isidor Rabi, and Lee DuBridge countered with the argument that while this was true, and while unknown phenomena could still change the picture, the most probable direction of change in the Super's prospects would make success even more doubtful.

As far as LASL's workload and resources were concerned, Bradbury expected the lab's staff to grow from 2,400 to 2,750 and that the six-day work-week would continue past the spring 1951 Eniwetok tests. The director noted that merely increasing headcount did not guarantee the best results and that individuals of special abilities and judgment were also required, especially among theoretical physicists.

At this GAC meeting, Dr. J. Robert Oppenheimer restated his belief that the military wanted high-yield weapons partially to compensate for poor bombing accuracy.

Oppenheimer also noted that the increasing importance of rapid improvement in fission weapon yield, to obtain a corresponding increase in return from the fissionable materials likely to be available within the next year or two, suggested that LASL should place more emphasis on this situation, especially in regard to the attainment of high yields from small amounts of materials (e.g., the so-called "fractional crit" bombs). Only through the near-term efforts of LASL would an increase in stockpile size be realized, since any increase in the production of enriched uranium, plutonium, and thorium would not be forthcoming for some time.

When Bradbury replied that stockpile directives received by LASL from the AEC had not reflected this sort of thinking, Oppenheimer stated that LASL developments could provide important guidance for the formulation of AEC stockpile directives, and that in many cases, the laboratory could hardly expect detailed weapon specifications based on anticipated military requirements, which were often at best a very difficult matter to analyze and specify on a thoroughly quantitative basis.

When the subject arose of reducing LASL's six-day work week to the traditional five days, Oppenheimer said that it would

be difficult to revert to a five-day week as long as LASL admitted that there was competition among the lab's resources between fission and fusion weapon design. He also noted that many lab personnel had become dependent upon the extra take-home pay that a six-day week generated.

Another point discussed at this GAC meeting was the statuses of present and planned cryogenic facilities to liquefy hydrogen isotopes. There was already a small-scale operational liquid gas plant at LASL; another larger-scale facility was to be located not far from Los Alamos and operated by an outside contractor, possibly the Bureau of Standards.

Although the latter facility was originally conceived to solve dewar problems for Super components, it now appeared that the new laboratory might concentrate on the design and fabrication of transport dewars so that LASL and test sites could be supplied with liquid deuterium and tritium.³¹⁵

In a letter to AEC Chairman Gordon Dean, the GAC summarized its findings as to the status of the thermonuclear program at this time, noting that the forthcoming CYLINDER test, while potentially very important and useful, was not a breakthrough:

(a) At the time the General Advisory Committee considered this program at its Seventeenth meeting (at the end of October 1949), it had before it a relatively specific proposal for a super weapon. The proposal involved the ignition of a tritium-deuterium mixture and the use of a fusing (sic) and booster train of such material to ignite deuterium in which the main reaction would propagate. ...

(It) was indicated that theoretical study of this model would be required. This study is not complete, but intensive and highly competent theoretical work has been carried out on the model which gives a substantially altered picture of its properties.

³¹⁵Hewlett and Duncan, pp. 529, 530; Minutes of the Twenty-Third Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, October 30, 31, November 1, 1950, pp. 9, 12, 13, 14, 17, 32, 37, 38, 43, 44, 48, 49, 50, 52, 63, 64.

One such set of calculations has to do with whether or not the detonation will proceed in deuterium. The recent calculations made are inadequate to decide the question. Nevertheless, in the opinion of knowledgeable experts, the calculations intensify the doubts that this propagation will occur.

The second set of calculations has to do with the amounts of tritium required to ignite the deuterium, if such ignition can take place. Here, too, the adequacy of the calculations leaves much to be desired. They all indicate, however, that quantities of tritium of the order of 100 or so grams will be quite inadequate and suggest a lower limit for this model in the range of 3 to 5 kilograms. (Author's note: emphasis added.)

It is clearly of importance that these calculations be refined and that the experimental data incorporated in them be improved. We believe that within a year very substantial progress will have been made. The comments above thus constitute an interim report.

(b) The outcome of the calculations above has, of course, directed the attention of the (Los Alamos Scientific) Laboratory to other thermonuclear models. For example, some attention has been directed toward the possibility of imploding parts of the super bomb in order (to) greatly reduce the requirements (for) tritium. (Author's note: emphasis added.) New attention has been given to a new model of the [deleted] These proposals have not been formulated in enough detail to warrant any assessment of feasibility or of value. They will require, and they will receive, further analysis.

(c) We have discussed the proposed test programs incorporated in (the LASL technical program for calendar year 1951), and have examined with particular care the CYLINDER shot which, if no unforeseen developments occur, will be part of the test program for the spring of 1951. New and elaborate instrumentation forms an essential part of this test.

If the tests and the instrumentation are reasonably successful, radically new information will be obtained. This information bears on the non-equilibrium burning of tritium-deuterium mixtures and on the phenomena associated with the flow of radiation from fission weapons into materials of varying density, and will be relevant to many thermonuclear models.

We wish to make it clear, however, that the test, whether successful or not, is neither a proof firing of a possible thermonuclear weapon nor a test of the feasibility of such a weapon. The test is not addressed to resolving the paramount uncertainties which are decisive in evaluating the feasibility of the super.³¹⁶

The estimation by the GAC of a "lower limit" of three to five kilograms of tritium required to ignite the Super was an order of magnitude greater than its estimate of "several hundred grams" of tritium a year earlier. (Even as late as the summer of 1952, the nation's entire tritium stockpile would be measured in grams, not kilograms.³¹⁷)

Such a requirement would reduce plutonium production by 210 to 350 kilograms to generate the tritium needed by each hydrogen weapon. This was a substantial penalty to pay in terms of the number of fission bombs that could be fabricated from this much plutonium.

Even if the AEC's desire for production of a kilogram of tritium per year, as stated in the spring of 1950, were achieved, it would take at least three to five years to produce enough tritium for just one Super. This was clearly unacceptable.

At a joint Atomic Energy Commission - Military Liaison Committee meeting at Los Alamos on November 14, 1950, lab director Norris Bradbury stated that the Super, as then

³¹⁶Letter dated November 1, 1950 from J. R. Oppenheimer, Chairman, General Advisory Committee to the U.S. Atomic Energy Commission, to Gordon Dean, Chairman, USAEC. The reference to "imploding parts of the super bomb" is quite intriguing, considering what was to happen about three months later. It is not clear from this reference what the implosion mechanism was to have been, either high explosives, or some product of a fission explosion.

³¹⁷September 12, 1952 Walker memorandum, p. 20.

envisioned, could probably still be built and fired successfully, in spite of current difficulties. He added that the cost of the bomb in terms of the amount of tritium required for ignition would probably make it economically unfeasible. The thermonuclear "burning" prospects were worse, and the possibility of a D-T reaction propagating in the deuterium seemed less likely.

There did not appear to be any new or novel design approach to solve the propagation problem. In the spring, it had looked like a test of the Super might be possible in early 1953; now it looked more like early 1954 at the soonest. In the meantime, LASL could only hope that the Russians would not discover any "new or novel" design approaches of their own.

With respect to the spring 1951 tests, Bradbury stated that a test of an experimental thermonuclear device called the CYLINDER was contemplated; if the CYLINDER failed to reach its predicted yield, then a test of the BOOSTER was likely. The BOOSTER was applicable primarily to fission weapons, and was of only incidental benefit to fusion problems.

If the test of the CYLINDER were successful, it might be possible to conduct another thermonuclear test within a year. If the CYLINDER failed, then further fusion experiments might be delayed at least until the fall of 1953. In any event, a successful test of the CYLINDER was required if the thermonuclear program was to proceed. Bradbury thought the CYLINDER had only about a 50% chance of success.³¹⁸

The LASL Technical Program for 1951-1952

Three days later, Bradbury submitted LASL's Technical Program for Calendar Year 1951 and Fiscal Year 1952 (a draft of which the GAC had seen in its meeting at the end of October). It included several points concerning the Super project:

Cryogenics -- Research in this field will be primarily devoted to studying the fundamental physical properties and constants of tritium, He³, and deuterium, the handling of these elements, and the properties of materials of specific interest at cryogenic temperatures.

Thermonuclear Weapons -- The
concentrated research and

³¹⁸Draft Memorandum dated November 17, 1950, to the Chairman, AEC; Subject: Notes on the AEC-MLC-LASL Conference on Tuesday, November 14, 1950, pp. 2, 3.

investigation in this field over the past year has shown that the probability of early, practical success along the lines originally conceived is considerably less than might have been anticipated earlier. Furthermore, practical success along those lines, if it can be attained at all without new and presently unforeseen conceptions, must be regarded as more distant.

Nevertheless, the importance of arriving at firm conclusions regarding the application of these or other potential thermonuclear techniques to military use make it imperative that vigorous work be continued in this field. It is proposed that the work of the Los Alamos Scientific Laboratory on thermonuclear weapons emphasize the following problems:

1. The determination of the specific conditions which are required for the initiation and propagation of a thermonuclear reaction in a mass of deuterium and the efficiency of the "burn-up" therein.
2. The determination of the practical amounts of tritium which are required to achieve the conditions (if any) under which reasonable fractions of a deuterium mass will ignite.
3. A study of the devices by which mixtures of tritium and deuterium may be brought to the conditions required by 1 and 2 immediately above. Two specific devices of this sort will be under active development during at least the early part of this program. These devices are:
 - a. The CYLINDER
[deleted]
 - b. The BOOSTER
[deleted]

Clearly, of these two devices, (a) represents a more radical as well as a more informative step, although at the same time more subject to unforeseen difficulties. Both of these devices will be brought to the most complete stage of development which the Laboratory is able to accomplish prior to the spring 1951 test program. Following the test, there will be detailed studies of the experimental results obtained.

The results of these studies may indicate the necessity, the desirability, or the futility of further experimental attempts to ignite deuterium and tritium, but the status of this program is such that no clear forecast can be made at this time concerning future devices or experiments in this field.

4. Although maximum effort in this field will be devoted to understanding the problems and practicality of the so-called SUPER as outlined in 1, 2, and 3 immediately above, nevertheless, the other possible applications of thermonucleonics to weapon use will be actively explored as appropriate in the light of their ultimate probable success and the other preoccupations of the Laboratory.

Among such applications, two which will receive attention are:

- a. The use of D-T to "boost" the low yield weapons of low multiplication rate. The BOOSTER described above is an illustration of this general technique, but the device itself is designed for another objective and is only

distantly related to the specific problem of stockpile improvement.

It is improbable that work in this field will progress fast enough to make a test of a device of this nature which might have weapon possibilities (if any such test is ever appropriate) before the spring of 1952.

- b. Devices of the ALARM CLOCK character which have been noted in earlier programs but have so far presented no outstanding advantages. Recent suggestions, however, involving the use of Li^6 in such devices have indicated the advisability of further exploratory work along these lines. Such work, for the period of this program, will probably be largely theoretical and no specific nuclear test proposal can be made at this time.³¹⁹

³¹⁹Letter dated November 17, 1950, from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Santa Fe Operations Office, U.S. Atomic Energy Commission, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1951 and Fiscal Year 1952, pp. 2, 5, 6, 7. In a memorandum dated October 11, 1950 to N. E. Bradbury from Edward Teller, Subject: Comments on the Laboratory Program, Teller took issue with a few statements in this plan. First, he suggested that Bradbury's sentence that "Furthermore, practical success along those lines, if it can be attained at all without new and presently unforeseen conceptions, must be regarded as more distant," be replaced by the statement "Furthermore, practical success of the burning of deuterium, if it can be attained at all, must be regarded as more distant. Insufficiently explored possibilities or completely unforeseen schemes might lead to

The Technical Program memo closed with a few remarks about the division of manpower and research between fission weapon development and thermonuclear research:

The extensive effort placed on thermonuclear reactions during the last year has been productive to the extent that the difficulties inherent in their application are now much clearer and their ultimate practical success much less certain.

The nature of the gains possible for fission weapons, while smaller in magnitude than that originally anticipated for fusion weapons, is nevertheless balanced by a relatively higher certainty and immediacy. Accordingly, in the opinion of this Laboratory, it would be unwise to give overriding emphasis to thermonuclear research and development in the areas where these conflict with fission weapon development.

To the greatest extent practicable both lines of research will be continued as far as possible without serious interference and at the highest levels of achievement. There may, however, occur from time to time occasions in which a choice must be made between two objectives for first attack.

an earlier success." (Teller was eventually right about this.) Teller argued that one (deleted) device did give some promise of relatively early success and that the design concept was not based on conceptions which were then unforeseen.

Teller also recommended that the sentence "Such work, for the period of this program, will probably be largely theoretical and no specific nuclear test proposal can be made at this time" be deleted, on the grounds that it precluded the possibility of the test of a lithiated ALARM CLOCK to a greater extent than seemed justifiable. Teller also suggested that Bradbury add that further thermonuclear test shots aimed at large-scale explosions may be planned for the spring of 1952 or later, and that data decisive to such test plans would be available in the spring of 1951 (Teller was anticipating success of one or more small-scale thermonuclear experiments during Operation GREENHOUSE).

The Los Alamos Scientific Laboratory will endeavor to minimize such occasions, but when they occur, it is now our opinion that the course should prevail which promises the greatest gain considering its probable time of accomplishment.³²⁰

Between 1946 and 1949, the work of the theoretical division at Los Alamos had been divided almost evenly between fission and thermonuclear weapon design.³²¹ To Edward Teller, Bradbury's attitude that, in a conflict between technically-promising near-term fission weapon improvements and less-certain long-term thermonuclear weapon development, the greatest effort should be concentrated on fission weapons was tantamount to a declaration of war on the Super program.

The 1951-1952 LASL technical program intensified a long period of increasingly-ill feeling between Teller and Bradbury which culminated in the establishment (largely on account of military and political lobbying efforts by Teller) of a second nuclear weapons lab at Livermore, California in the fall of 1952.

LASL's technical program for 1951 and FY 1952 provided for the testing of atomic devices in the spring of 1952 as well as the fall of 1951. The program for full-scale tests in the spring of 1952 included, tentatively at least, for the detonation of a very high-yield device in the Pacific, probably at Eniwetok.³²²

Late in the month, Chinese troops entered the Korean fighting in large numbers, and pushed embattled and outnumbered United Nations forces back down the peninsula. This was a major escalation of the war which significantly raised tensions both in the Far East and in Europe, where a massive Soviet army might be poised to strike westward. The prospect of general war, either in Europe or Asia, added impetus to the development of tactical weapons at LASL, and put even more pressure on the lab and the conflicting allocation of resources between its fission and fusion weapon programs.

³²⁰Los Alamos Scientific Laboratory Technical Program for Calendar Year 1951 and Fiscal Year 1952, p. 13.

³²¹Bethe, "Comments on the History of the H-Bomb," p. 46. Bradbury was strongly opposed to the establishment of "a second Los Alamos."

³²²"History of Operation IVY," Maj. F. E. Moore, Jr. and Lt. H. G. Bechanan, Joint Task Force 132, undated.

At the end of November, in a meeting between the AEC and the Joint Committee on Atomic Energy, AEC Chairman Gordon Dean re-emphasized the cost of the Super in terms of fission weapons. Dean stated that although there was at this time not as much uncertainty that a hydrogen bomb could be developed (an opinion that was much more optimistic than present circumstances warranted), the cost of each hydrogen weapon might be from 100 to 150 fission bombs, in terms of lost plutonium production from reactors diverted to tritium production.³²³ This was the number of atomic weapons that could be made from 210 to 350 kilograms of plutonium; each weapon core also contained a similar amount of uranium.

A report prepared at this time by LASL noted that since the beginning of the year, an average of two "topnotch," five senior, and four and a half junior theoretical physicists had participated on the LASL thermonuclear program. During this period, LASL had considered a number of aspects of the program, including initiation of reactions in a D-T mixture via the BOOSTER, the multi-crit gun, or by non-fission means, including implosive jets or electrical means.

The next step in the Super, the so-called "link" which included calculations dealing with actual schemes for transmitting the energy obtained from the D-T reaction to a large mass of deuterium, had also been studied.

Ancillary effects were another aspect that had been reviewed. Included among these were the inverse Compton effect (loss of energy from the D-T and D-D reactions to radiation), compression of the main deuterium charge, and wall effects, i.e., the boundary turbulence effects on the thermonuclear reaction of the walls of the casing around the fuel mass.³²⁴

A month later, at the end of December, a special DOD planning committee chaired by Dr. J. Robert Oppenheimer presented its own assessment of the current status and prospects of the Super program and its likely impact on the national nuclear weapons stockpile, and -- echoing one of the findings of the GAC more than a year earlier -- urged emphasis on fission weapon development instead:

The last two years have witnessed important changes in our knowledge of thermonuclear

³²³Borden-Walker H-bomb chronology, p. 58; "A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, p. 11.

³²⁴Enclosure (Appendix B), "Thermonuclear Effort" to letter from Dr. Frederick de Hoffmann to Gordon Dean, dated November 27, 1950.

weapons. In part as a result of the Presidential decision of January 27, 1950 to pursue the development of thermonuclear weapons with high urgency, new, still provisional, but very much improved calculations have been made of probable characteristics of the super (sic) bomb.

As a result of these calculations, it now appears fairly certain that the minimum amounts of tritium required, according to present models, to initiate the main charge of a deuterium super bomb (if this initiation is at all possible) will be of the order of [deleted] of tritium.

The production of this amount of tritium will divert plutonium production probably equivalent to [deleted] fission weapons; this makes the cost in fissionable material of carrying through the development along present lines extremely high.

A second result of the calculations is to cast serious doubt on whether or not deuterium can, in fact, maintain an explosive detonation, and to suggest that even if this turns out to be possible the efficiency of the reaction will be rather low, so that energy releases per ton of material will fall far below those initially expected.

It should be stressed that the calculations hitherto made are far from adequate for firm conclusions; that in some cases the basic nuclear data, and specifically the D-D cross sections, are subject to substantial uncertainties; and finally that the calculations apply to a specific model of a thermonuclear weapon and in their nature are highly idealized, leaving out many factors, most of which are believed to be adverse and some of which, it may be anticipated, can be favorable.

It should be further stressed that the adverse results of these calculations have, of course, given rise to the consideration of alternate schemes. At the present time none of these proposals appears manifestly feasible or attractive.

As far as can now be determined, the most important element in present activities directed toward the feasibility of thermonuclear weapons lies in further analysis and calculation. Should additional study lead to promising new ideas for thermonuclear weapons, it is likely that these will be fully established only with the explosion of a full scale bomb. Nor can one hope to achieve confidence that no promising model has been overlooked without protracted and difficult study.

We conclude that the determination of the feasibility of thermonuclear weapons is an important, but very definitely a long range undertaking, and that in the light of the high promise of fission weapon development, and the urgency of rapid increase in our capacity in this field, it should be so recognized.

In fact, we believe that only a timely recognition of the long range character of the thermonuclear program will tend to make available for the basic studies of the fission weapon program the resources of (the) Los Alamos Laboratory.³²⁵

At one of the regularly-scheduled status review meetings at LASL during this period, physicist George Gamow graphically demonstrated the state of the hydrogen bomb design program by placing a can of cigarette lighter fluid, a small ball of cotton, and a piece of wood on a table. He squirted some lighter fluid on the cotton, placed the cotton next to the block of wood, and lit it with a match. The cotton flared up briefly, then died out. He passed the piece of wood around the assembled group of scientists. It wasn't even warm -- and it was petrified wood.³²⁶

Despite this gloomy outlook for success, the Air Force continued planning for H-bomb delivery. The Glenn Martin glide bomb project was canceled at the end of 1950, and in lieu of

³²⁵"Military Objectives in the Use of Atomic Energy," 29 December 1950, pp. 5, 6. The panel also included Dr. Robert F. Bacher, Dr. Luis Alvarez, Dr. Charles C. Lauritzen, Dr. Mervin J. Kelly, Prof. Walter G. Whitman, Maj. Gen. K. D. Nichols, RAdm. W. S. Parsons, Maj. Gen. R. C. Wilson, and Brig. Gen. James McCormack, Jr.

³²⁶McPhee, p. 90.

this, the Air Materiel Command engineering laboratories at Wright-Patterson AFB near Dayton, Ohio undertook a program to develop a drogue parachute to slow the bomb's rate of fall to allow the delivery aircraft time to escape blast and thermal effects from the explosion. Because of the probable size and weight of the Super, only B-36, B-52, and B-47 aircraft were considered as likely carriers.³²⁷

On December 28, the AEC approved an increase in the number of heavy water reactors to be built in South Carolina from three to five, and also approved the construction of plutonium-producing piles.³²⁸ The DuPont reactors were to be dual-purpose, and capable of producing either tritium or plutonium. The piles were deliberately designed to be slightly inefficient in order to preserve this option. The reactors were not expected to be loaded for tritium production until the fall of 1953 at the earliest.³²⁹

By the end of 1950, Teller was desperate for a solution to the Super ignition problem (and it was a super ignition problem). He proposed a number of complicated schemes to save the original concept, but none of them showed any promise. Nevertheless, he urged that the resources of Los Alamos be put at his disposal for another year or more after the upcoming GREENHOUSE tests, at which time there should then be another test of some sort of thermonuclear device.³³⁰

Unsuspected by Teller, and unforeseeable by anyone at the time, a coup de' theatre was not far away. A radically new approach, although based in part on work of the preceding nine years, was just around the corner: Bradbury's prophetic remarks of late August were about to become reality.

By mid-January 1951, LASL had completed calculations on the thermonuclear device slated for test in the spring. While these computations confirmed expectations for the first step in the operation of the "runaway Super" might be achieved, the available evidence continued to indicate that the last and most vital step, the ignition of a large mass of

³²⁷Alling, pp. 112, 113.

³²⁸"Chronology of Major Decisions Regarding Thermonuclear Program," USAEC, June 14, 1951, p. 4.

³²⁹Memorandum for Mr. Cole dated March 7, 1951 from Bill Borden, subject: Your inquiries relating to the hydrogen program.

³³⁰Bethe, "Comments on the History of the H-Bomb," p. 48. It can be said of this period that Teller was "desperately seeking fusion."

deuterium, would require almost prohibitively expensive amounts of tritium.

A decision on whether or not to test the BOOSTER at the same time was deferred.³³¹

Late in January, the AEC recommended to the MLC that in light of diminished prospects for early success in developing a hydrogen weapon, a second charge of enriched uranium (to produce tritium) should not be loaded into "H" pile at Hanford. To do so would interrupt the current production of tritium, and cost both U-235 and lost plutonium production.³³² Three days later, the MLC concurred in this opinion, and suggested that a joint AEC-DOD letter be written to President Truman advising him of the current prospects for a hydrogen bomb.³³³

Early in February, at a closed JCAE meeting, AEC Chairman Gordon reported that the Hanford "H" pile would not be reloaded with enriched uranium for further tritium production. Thus far, 284 kilograms of bomb-grade U-235 had been used for tritium production since the end of May 1950; of this quantity, about 56 kilograms had been consumed. The remaining 228 kilograms were unusable for weapons until it was chemically separated at a new Arco, Idaho plant which was now under construction and would not be completed until well into 1952. The use of "H" pile for tritium production had to date caused about a 25% loss of plutonium output since May.³³⁴

At this time, Edward Teller took exception to the estimate of three to five kilograms of tritium required for the Super, arguing that it was misleading because it referred only to the initial approach toward the Super, based upon the multi-crit gun initiator or the CYLINDER. Teller also stated that, in his opinion, there was at this time "no conflict in Los Alamos priorities between the hydrogen program and other programs sufficient to interfere with the progress of the former."³³⁵

³³¹Borden-Walker H-bomb chronology, pp. 61, 62.

³³²Letter dated January 22, 1951 to Robert LeBaron, Chairman, MLC, from M. W. Boyer, General Manager, USAEC.

³³³Memorandum for the General Manager, Atomic Energy Commission, dated January 25, 1951 from Robert LeBaron, Chairman, MLC, subject: Tritium Production Program.

³³⁴Memorandum for Mr. Cole dated March 7, 1951 from Bill Borden, subject: Your inquiries relating to the hydrogen program.

The Application of Radiation Implosion

The Super program at Los Alamos was in very bad shape at the beginning of 1951. The calculations and setbacks of the preceding year had damaged morale and almost ended Edward Teller's credibility. Extensive computations and numerical modeling indicated that all configurations of fission and fusion material studied to date would disperse before any significant thermonuclear burning occurred and that they would require prohibitive amounts of tritium merely to start a thermonuclear reaction in the deuterium fuel.³³⁶

Teller himself had been besieged by bad news and unfortunate developments that rivaled the afflictions of Job. Nonetheless, he had not lost his faith in thermonuclear weapons, no matter how craftily nature had hidden its secrets from him and his colleagues in New Mexico.

Teller's ordeal would soon be ended, and the fate of humanity put into the balance like never before in history. A new conjunction of ideas that together promised thermonuclear plenty were born from, of all things, the same fission weapons program that Teller so detested for its diversion of the time and energy of the staff of Los Alamos' Theoretical Division.

During the summer of 1950, some calculations by Ulam and Enrico Fermi led Ulam to conclude that if the fusion cross-sections of deuterium and tritium could somehow be increased, thermonuclear weapons might be possible. The Ulam-Fermi work had shown that, given current estimates of D and T cross-sections (which were too low, as James Tuck would show a year later), a sustainable fusion reaction in liquid deuterium could not be maintained within the current geometry:³³⁷

My calculations with Everett concerned the first phase of the explosion, the problem of initial ignition. An important part of the story has been overlooked in the official accounts and concerns some quite fundamental work that Fermi and I did following the first calculation of the progress of the (thermonuclear) reaction, its propagation and explosion.

³³⁵Memorandum for the Files dated February 9, 1951 from Bill Borden, subject: Conversation with Dr. Edward Teller.

³³⁶Ulam, "Thermonuclear Devices," p. 597.

³³⁷Ulam biography, 1991, p. xii.

In numerous joint discussions we outlined the possibilities of propagation, assuming that some way or another (perhaps by the expenditure of large amounts of tritium) the initial ignition could be achieved. There again we had to use guesses in place of the enormously difficult detailed calculations that would have required computers even faster than those in existence. We did this again in time-step stages with intuitive estimates and marvelous simplifications introduced by Fermi.

The numerical work was done on desk computers ...

A joint report was written by Fermi and myself. Enrico exercised very great caution in its conclusions. In fact, one conclusion stating the unpromising nature of the reaction as it was planned contained this sentence: "If the cross-sections for the nuclear reactions could somehow be two or three times larger than what was measured and assumed, the reaction could behave more successfully."

I believe this work with Fermi to have been even more important than the calculations made with Everett. It turned out to be basic to the technology of thermonuclear explosions...³³⁸

In a 1966 treatise, Ulam provides another account of his work in 1950 with Everett and Fermi:

³³⁸Ulam, ADVENTURES OF A MATHEMATICIAN, pp. 217-219. The "desk computers" Ulam mentions were probably Marchant or Monroe "Green Monster" electromechanical push-button-actuated desk-top calculators, little more than souped-up adding machines. Operators of these machines were called "programmers." (In a letter to the author dated 28 November 1991, Francoise Ulam remembers operating a Marchant machine; she claims that operators were called "data analysts.") The report by Ulam and Fermi was probably "Considerations on the Thermonuclear Reactions," LA-1158, published at LASL in 1950. Recent computer simulations of the Ulam-Fermi calculations, using the knowledge gained from 40 years of thermonuclear weapons development and testing, have confirmed the marginal character of a self-sustaining reaction in the 1950 design. (Ulam biography, 1991, p. xiii.)

Quite independently, however, (I), in collaboration with C. J. Everett, had undertaken the calculation of the ignition process and the following course of the thermonuclear reaction by using numerous simplifications and guesses as to the values of certain multidimensional integrals defining the distribution of neutrons and the products of reaction during the changing geometry of the mass of active material...

The results of this work ... showed a very "weak" process. The mechanism of ignition, as considered in the existing schema, was submarginal and led to disappointingly weak initial conditions and, after a time, to decreasing rates of reaction.

Shortly after that time, a calculation was similarly performed by Fermi and (myself); we considered the "next" problem: Assuming that in some way an ignition of the mass of deuterium -- perhaps supplemented by quantities of tritium -- could be made satisfactorily, how would the subsequent reaction proceed in a volume of deuterium?

Again, the calculation had to rely during its course on intuitive estimates of the properties of distribution, e.g., the neutron density, neutrons of at least two different classes in energy being products of the thermonuclear reactions; the distribution of gamma rays, i.e., the photons constituting the radiation field, etc.

The result of this work was, again, quite negative in the sense of showing that, most likely, even if the initial reactions were established, as in the scheme proposed, it would not continue and go anywhere near to completion or even sizable burning of the remaining material.

Shortly afterwards, the calculations which were proceeding in the meantime on the electronic computing machines, and in which some of our intuitive arguments were replaced by a full numerical work, were completed; and the results confirmed (our hand calculations).

(These calculations) indicate(d) how sensitive the whole complicated arrangement constituting the "thermonuclear device" was to the interacting processes determining the burning of the deuterium. In fact, the reports by Fermi and (myself) indicated that, if the cross sections for the D+D reactions were between two to three times greater than they actually are, i.e., the measured ones, the process, as proposed during the war, would indeed go.

In a sense, therefore, the thermonuclear device, i.e., the proposed H-bomb, was not a trivial object to construct (even) after the fission bomb (was) available.³³⁹

Ulam realized from his work with Fermi that the success of thermonuclear systems was directly dependent upon the reaction rate of the fuels. That reaction rate in turn depended upon an uncontrollable and unchanging physical parameter, the fusion cross-section for deuterium. However, at some later point, Ulam further surmised that the reaction rate might also depend upon a controllable physical parameter, namely, the density of the fusion fuel mixture (nuclear and chemical reactions always proceed more rapidly in a dense medium).

The latter deduction led Ulam to consider some means of compressing the fusion fuel before its ignition.³⁴⁰ Some thought had been given to imploding parts of the Super as early as the fall of 1950. By compressing the fuel, the fusion cross-sections would be significantly and effectively increased, by crowding the nuclei closer together before they began interacting and acquiring kinetic energy. (As explained in Volume I, the fusion reaction rate was eventually discovered to be proportional to the square of the fuel density.)

If the volume of the fusion fuel were reduced, and its density simultaneously increased, energy loss by radiation would also be reduced, making more energy available for the reacting nuclei and making fusion more likely. This situation was analogous to the practice of increasing the density of fission fuels to reduce the mean free path of neutrons

³³⁹Ulam, "Thermonuclear Devices," pp. 596, 597.

³⁴⁰Undated letter from Lee Barker to author, received in August 1988.

between collisions with nuclei and to increase the fission rate.

In December 1950, quite apart from the struggling thermonuclear program, Ulam had proposed a radically new fission bomb design, using a focused shock wave from a small exploding atomic device to mechanically compress a physically separate and larger second fissile core.³⁴¹ Ulam called this technique of focusing shock waves "hydrodynamic lensing."³⁴²

Among other things at the time, Ulam was engaged in LASL's program to mathematically analyze the process of high explosive implosion to reduce the size, increase the yield, and improve the efficiency of fission bombs. There was a limit to how effectively conventional chemical high explosives could compress solid metals, and implosion via a nuclear shock wave could, at least theoretically, provide a profoundly higher density.³⁴³

Ulam's new bomb design was a result of this fission weapon improvement program. The strong shaped implosive shock wave generated by the explosion of the first core and transmitted mainly via neutrons would compress the second core much more effectively than conventional high explosives, resulting in a higher yield for a given amount of fissile material. Ulam called the scheme a "heterocatalytic detonation."³⁴⁴ (It is

³⁴¹Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, p. 7; "Thermonuclear Weapons. The United States Project," subsection of "Development of Nuclear Weapons," in "War, Technology of," article in ENCYCLOPEDIA BRITANNICA, February 1990, Stan Norris and Tom Cochran, Natural Resources Defense Council. Francis Duncan credits Ulam with a (similar?) idea in February 1950 to initiate a thermonuclear reaction ("Thermonuclear Weapon Development, January 1950-May 1951," p. 8.) and states that he brought this idea to the attention of Darol Froman at that time. This, however, is probably a typographical error; Duncan probably meant February 1951, rather than 1950. In a public address at the Massachusetts Institute of Technology on November 14, 1990, Dr. Hans Bethe stated that Ulam's invention was originally intended to be applied to fission bombs to enhance their yields.

³⁴²Rhodes, DARK SUN, p. 462.

³⁴³Hirsch and Mathews, pp. 6, 23.

³⁴⁴"Heterocatalytic" means "a different form of catalysis": in a classical chemical reaction, a catalyst modifies or speeds the rate of reaction and is not consumed in the process; in Ulam's design, the catalytic first fission stage would be

possible that Ulam suggested an elliptical shock wave chamber; see comments on focusing in Volume I.)³⁴⁵

Since there did not seem to be an immediate military requirement for Ulam's new design, its development remained only theoretical until the spring of 1953, when it became the basis of the first incarnation of the TX-15 enhanced fission ZOMBIE bomb (see Volume VI; it is also possible that Ulam's idea was the basis of the LITTLE EDWARD semi-thermonuclear design in the spring of 1952).

By late February 1951, Ulam had occasion to study his concept of "heterocatalytic detonation" anew while examining the early stages of fission bomb detonation and he now considered its possible application to the intractable and seemingly-insurmountable Super ignition problem which had stymied his colleagues' efforts so far. Could this be a workable technique to compress fusion fuels in a thermonuclear device?³⁴⁶

Ulam's wife Francoise, who also worked at LASL, recalls that one day at this time, her husband stated that he thought he had found a way to make the Super work:

The technical and political debates were raging when Stan, mulling over the problems, suddenly came upon a new and totally intriguing new approach. Engraved on my memory is the day when I found him at noon staring intensely out of a window in our living room with a very strange expression on his face. Peering unseeing

vaporized. The term "staging" to describe separately-centered multi-element thermonuclear systems did not come into widespread use immediately at Los Alamos.

³⁴⁵According to Edward Teller in Blumberg and Panos, Ulam proposed a hydrogen bomb configuration with deuterium arranged in a star-like structure around the fission trigger (p. 124). In Winterberg, THE PHYSICAL PRINCIPLES OF THERMONUCLEAR EXPLOSIVE DEVICES, pp. 23-40, the author proposes a number of focussed shock wave thermonuclear ignition systems; one of them, the so-called polyhedron configuration described and depicted on pp. 23-25, bears an inverse resemblance to the Ulam configuration identified by Blumberg and Panos in that it shows a number of fission explosives grouped around a central fusion fuel charge.

³⁴⁶Letter dated February 23, 1951 from Stanislaw Ulam to Dr. John von Neumann; Ulam biography, 1991, p. xiv; memorandum dated May 3, 1954 to Brig. Gen. K. E. Fields, AEC DMA, from N. E. Bradbury, Subject: Thermonuclear History, p. 2.

into the garden, he said: "I found a way to make it work." "Make what work?" I asked. "The Super" he replied. "It is a totally different scheme, and it will change the course of history."

I, who had rejoiced that the "Super" had not seemed feasible, was appalled by this news, and anxiously asked him what he planned to do. He replied that he "would have to tell Edward." Fearing that Teller might pounce on him again, I ventured that maybe he ought to test his idea on (J. Carson) Mark or (Norris) Bradbury first. He did, but went to Teller the next day just the same.³⁴⁷

At his wife's urging, and because of his strained relationship at this time with Teller, Ulam first wrote to

³⁴⁷Ulam biography, 1991, p. 311; letter dated August 10, 1988 to John Manley from Francoise Ulam; review by Priscilla Johnson McMillan of "Edward Teller: Giant of the Golden Age of Physics," in SCIENTIFIC AMERICAN, Volume 262 No. 5, May 1990, p. 132; "The Master Spy Who Failed," Philip Elmer-Dewitt, TIME, January 15, 1990. Other persons have different recollections of Ulam's attitude: in a 1957 interview, Dr. Robert Oppenheimer stated that when Ulam had his idea in 1951, he (Ulam) was not thinking about the H-bomb and was not interested in thermonuclear weapons and took a dim view of the project. (Interview of Dr. J. Robert Oppenheimer by Warner R. Schilling on June 11, 1957 at the Institute for Advanced Studies, transcript p. 14.) In a letter to the author dated 28 November 1991, Francoise Ulam states that "Stan felt all along that theoretical investigations of the feasibility of a Super should be pursued, but that he was not particularly eager to see it developed. He felt that there were more than enough new and more powerful A-bombs." Priscilla McMillan states that Ulam sometimes rationalized the hydrogen bomb and said it kept the peace after World War II. (Letters from McMillan to author dated July 27 and September 13, 1990.)

Ulam, like many of his colleagues at Los Alamos had mixed feelings about his work: while he was fascinated by the challenge and discovery of the physics of it, he may have been occasionally appalled and troubled by the consequences of its likely military applications. (Ulam biography, 1991, pp. xv, xvi.) He was, however, apparently proud, to at least a limited extent, of his contribution to the birth of the American hydrogen bomb: in ADVENTURES OF A MATHEMATICIAN, pp. 222-223, he shows few moral doubts about doing theoretical work that might end in horrible new weapons.

John von Neumann, then approached Carson Mark and Norris Bradbury with his ideas and finally took his thoughts to Edward Teller, who quickly realized their utility.³⁴⁸ If a thermonuclear fuel capsule were substituted for the second fissile core, the design might be applicable to the Super.

Teller had not been immediately receptive to the concept of compression. Teller originally believed that competing physical processes would obviate any advantages to be gained by compressing the main fuel charge; in addition, only limited means -- possibly high explosives -- were then available to accomplish the compression. However, by the spring of 1951, he had realized that a sufficiently high compression would overcome these competing processes:

Now look -- there are two processes on which we concentrated, and they were obviously the important processes, under our work on it. And they were the reaction between two nuclei, that produced energy, and the emission of radiation that was supposed to lose energy.

And a third one -- the increase of energy of these low-frequency quanta. Now every one of these processes varied with the square of density. If you compress the thing, then in one unit's volume, each of the three important processes increased by the same factor, you know. Therefore, their balance could not change. Therefore, compression was useless.

Now when -- (later) -- it seemed clear that we are in trouble -- then I wanted very badly to find a way out. And it occurred to me that an unprecedentedly-strong compression will just not allow much energy to go into radiation. Therefore, something had to be wrong with my argument and then, you know, within minutes, I must be what was wrong.

We took into account the emission of radiation -- I did not know what I did take into account -- the absorption of radiation -- in hydrogen is in fact an exceedingly feeble process.

³⁴⁸Hewlett and Duncan, pp. 536, 537.

It was rightly neglected but emission occurs when an electron and a nucleus collide. Absorption does not occur when a light quantum and a nucleus, or a light quantum and electron collide, it occurs when a light quantum finds an electron and a nucleus together. It's the three-body process, and it does not go with the square of the density, it goes with the cube of the density.

And at very high compressions, this very feeble process becomes important. By using compression, if applied much more strongly than ever before, then (this problem is overwhelmed).³⁴⁹

Teller now took Ulam's idea one step further, making the concept more convenient and universal, and suggested using focused x-radiation from the fission stage as a means to compress a separate thermonuclear fuel stage.³⁵⁰ This was a logical outgrowth of the radiation transport aspect of the Operation GREENHOUSE George shot test device -- the CYLINDER -- and Teller's application of radiation to Ulam's ideas may have been a result of work by Teller on the design of the George test device.³⁵¹

X-rays traveled at the speed of light, much faster than any form of shock wave, and might be employed to compress the fusionable fuel much more quickly and more densely than Ulam's proposed mechanical shock wave. Teller knew at the time that the primary product of a fission explosion was "soft" relatively long-wavelength x-rays.

Because this radiation traveled at the speed of light, much more rapidly than any particulate matter from the explosion, including fission products, bomb debris, and even neutrons

³⁴⁹Transcript of interview of Dr. Edward Teller, conducted at Los Alamos National Laboratory on June 7, 1993.

³⁵⁰AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 212. This is probably the first officially-released U.S. government linkage of Dr. Teller's name with the concept of radiation implosion; it occurs in a passage describing what useful U.S. thermonuclear weapons data, if any, Dr. Klaus Fuchs may have given the Soviets by the time he left the U.S. in 1946.

³⁵¹Hirsch and Mathews, p. 6; memorandum dated May 3, 1954 to Brig. Gen. K. E. Fields, AEC DMA, from N. E. Bradbury, Subject: Thermonuclear History, p. 2; Rhodes, DARK SUN, p. 469.

and other subatomic particles, its work could be done well before the shock wave and particles radiating from the expanding fission "trigger" blew the fusion fuel assembly apart.

Teller noted the large amounts of x-radiation emanating from the fission "trigger," and the very slow motion of the expanding mass of fission material relative to the flow of radiation. He also took note of the recoil forces generated by the x-rays as they deposited their energy in the surrounding materials, particularly the casing around the fusion fuel capsule. As early as 1942, LASL weaponeers had discovered that the temperature of an exploding atomic bomb was so high that radiation pressure was much greater than any material pressure.³⁵²

In addition, Teller knew that the relatively massless x-rays penetrated materials much more easily than heavy neutrons and lost much less of their energy along the way. He wondered if these x-rays could be used to initiate a fusion reaction: could x-radiation, rather than neutrons, be used as the primary means of energy transport from the fission "trigger" to the fusion fuel mass by generating a symmetrically-shaped compressive plasma to significantly increase the density of the fuel? This was the concept of radiation implosion, first brought up for discussion at Los Alamos in April 1946 and apparently soon forgotten afterwards.

The use of radiation as an implosive circumferential piston also meant that a standard implosion weapon -- perhaps tailored to enhance its x-ray output -- could be used as the trigger, rather than a bulky, heavy, and highly-inefficient gun-type device. By early 1951, the fission devices then on the drawing boards at LASL were much more "transparent" to x-rays than the World War II models, primarily due to the reduction in quantity and masses of the tampers and high explosives surrounding the fissile cores of the latter.

With this reduction in the number and volume of interfering and absorptive atoms and subatomic particles, more x-rays, of many energy levels, could now find their way out of the exploding core and be usefully employed for compression or other purposes.

Ulam had also discovered another key element of the solution -- a technique of staging to generate high yields by kiloton detonations followed by fractional-megaton detonations followed in turn by megaton explosions. This method obviated the apparent impossibility of initiating megaton-yield

³⁵²Transcript of public address by Dr. Hans Bethe at the Massachusetts Institute of Technology on November 14, 1990.

explosions directly from even large kiloton-yield triggers, another blind alley that had been pursued fruitlessly until now.

In his autobiography, Ulam describes both his original idea and Teller's generalization of it, showing the synergy of the two scientists:

Perhaps the change came with a proposal I contributed. I thought of a way to modify the whole approach by injecting a repetition of certain arrangements. Unfortunately, the idea or set of ideas involved is still classified and cannot be discussed here...

Shortly after responding (to a memo from Darol K. Froman querying lab staff members about the Super program) I thought of an iterative scheme. After I put my thoughts in order and made a semi-concrete sketch, I went to Carson Mark to discuss it. Mark, who was by then the head of the theoretical division, was already in charge of the very extensive theoretical work supporting Teller's and Wheeler's special groups.

The same afternoon I went to see Norris Bradbury and mentioned this scheme. He quickly grasped its possibilities and at once showed great interest in pursuing it. The next morning, I spoke to Teller. I don't think he had any real animosity toward me for the negative results of the work (I had done) with Everett (that was) so damaging to his plans, but our relationship seemed definitely strained.

At once Edward took up my suggestions, hesitantly at first, but enthusiastically after a few hours. He had seen not only the novel elements, but had found a parallel version, an alternative to what I had said, perhaps more convenient and generalized.

From then on pessimism gave way to hope. In the following days I saw Edward several times. We discussed the problem for about half an hour each time. I wrote a first sketch of the proposal. Teller made some changes and additions, and we wrote a joint report quickly. It contained the first engineering sketches of the new possibilities of starting thermonuclear explosions.

We wrote about two parallel schemes based on these principles. The report became the fundamental basis for the design of the first successful thermonuclear reactions and the test in the Pacific called "Mike."³⁵³

³⁵³Ulam, ADVENTURES OF A MATHEMATICIAN, pp. 219, 220. Norris Bradbury was director of the Los Alamos lab in 1951; "Wheeler" was Dr. John A. Wheeler, who in July 1951 established and directed at the Institute for Advanced Studies at Princeton University a group known as Project Matterhorn B to engage in studies of specific thermonuclear weapon designs. ("In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 No. 2 (1989), p. 321.) It is indeed ironic that the H-bomb was born in the U.S. amidst the frustration and personal ill feelings between Teller and Ulam. The "iterative scheme" Ulam mentions is an indirect reference to staging; this is as much as he was allowed to say about it until after the PROGRESSIVE case when the "three concepts" cited in note 75 in Volume I were finally declassified in September 1980. The report that Teller and Ulam co-authored is LAMS-1225, dated March 9, 1951 (Hewlett and Duncan, pp. 537, 657). The official AEC history downplays Ulam's contributions to the so-called "New Super" concept. According to Hewlett and Duncan, p. 537:

..Ulam continued, "Had the following couple of thoughts (ideas) about bombs..." He needed only a few sentences to sketch a scheme which could be applied to several members of the thermonuclear family, even the Super. He had mentioned his idea to Teller: "Edward is full of enthusiasm about these possibilities; this is perhaps an indication they will not work."

Teller was indeed enthusiastic. He listened to Ulam describe a particular approach to apply his idea. Teller's mind raced over the possibilities. He rejected Ulam's approach as posing enormous technical difficulties. He had a scheme of his own, based partly on the nuclear mechanics which were to be used at the GREENHOUSE test of thermonuclear principles. In March, Teller and Ulam completed a joint

In a 1962 letter to then-AEC Chairman Glenn Seaborg, Ulam offered more details on the genesis of his ideas and how they were received initially by Edward Teller:

In addition to the work of mine ... proving that the old Super scheme was unpromising, the history of the "new" ideas leading to the present class of designs is as follows: One day early in January 1951 it occurred to me that one should employ an implosion of the main body of the device and thus obtain very high compressions of the thermonuclear part, which then might be made to give a considerable energy yield.

I mentioned this possibility, with a sketch of a scheme how to construct it, to Dr. Bradbury one morning. The next day I communicated it to Edward (Teller), who by that time was convinced that the old scheme might not work. For the first half hour or so during our conversation, he did not want to accept this new possibility but in the subsequent discussion took to it very eagerly.

In subsequent discussions, we devised certain arrangements which appeared in a report written jointly by us (LAMS-1225). There are two ways to pursue this original suggestion of what is called now a "two-stage" device: one by means of shock wave implosion which I proposed to Edward when I told him about the two-stage possibilities and the other, radiation implosion of the second stage, which (was suggested by) Edward.³⁵⁴

report in which each presented his own scheme to achieve the conditions which Ulam had suggested.

³⁵⁴Letter dated March 16, 1962 to Dr. Glenn Seaborg, Chairman, USAEC, from S. M. Ulam. This letter was occasioned by the publication of one of Teller's books, THE LEGACY OF HIROSHIMA (Doubleday, Garden City, New York, 1962), and the entirely self-serving description therein of the origins of the "three concepts" which mentioned Ulam only to the extent of his work in destroying Teller's original ideas; Teller also accused Norris Bradbury of dragging his feet on hydrogen bomb development. In later years, Teller has often either minimized or denied Ulam's crucial contributions to the

J. Carson Mark also remembers the Ulam-Teller interaction:

Ulam came into my office one afternoon and drew some pictures on the board. He said these were more interesting than what we were doing. (In our discussion, Ulam gave me no indication that he was thinking about the fusion bomb.) He went through the same exercise the next day with Teller, who instantly saw their relevance to the fusion bomb.³⁵⁵

Norris Bradbury, one of the first persons Ulam approached with his ideas, recollected the birth of the "three concepts" thusly:

Now the particular case of what, let us say, has been called the new idea, widely attributed to and sometimes correctly to Mr. Teller (sic), that appeared in 1951. Actually, what are the facts? The basic problem that faced us in the fission field and thermonuclear field has been the subject of thought by people ever since the first fission bomb was successfully detonated.

The underlying idea, the underlying problem -- I can't discuss what it is for it's classified -- the underlying attack dates back to 1945. Ulam was working directly on the problem of how to make a

American H-bomb program; in Blumberg and Panos' 1990 hagiography, Teller claimed that "Ulam contributed nothing" (pp. 124-126).

³⁵⁵WAR AND PEACE IN THE NUCLEAR AGE, John Newhouse, Alfred A. Knopf, New York, 1989, p. 79; letter from Mark to the author dated 18 September 1979 (revising third sentence in his Newhouse quote). Priscilla McMillan believes that Ulam spoke rather elliptically to Carson Mark and Norris Bradbury; this may explain Mark's revised quote. Newhouse originally quotes Mark as saying that Ulam was not thinking about the hydrogen bomb and did not make any connection between his ideas and the bomb. Ulam's wife Francoise later stated that "the postwar H-bomb was essentially Teller's, except that Stan had to show him how to go about it" (Ulam biography, 1991, p. 310) and "Los Alamos had to build it without (Teller.)" ("Stan Ulam - A Few Personal Remarks," address by Francoise Ulam to the Ulam Quarterly Inaugural Conference on February 26, 1990.)

thermonuclear reaction, and he had an idea. It turned out that his idea was discussed; that idea bifurcated or trifurcated and had various directions in which it might go. And indeed the direction in which it might go that was suggested by Dr. Teller turned out to be the most profitable direction to explore.³⁵⁶

Yet another account of the application of radiation implosion is provided by Hans Bethe, who stressed the serendipitous nature of the new insight and warned of the risk of expecting similar scientific breakthroughs on a regular basis:

Between January and May 1951, the "new concept" was developed. (This I shall call Method D.)...

In January, Teller obviously did not know how to save the thermonuclear program. On March 9, 1951, according to Bradbury's press statement, Teller and Ulam published a paper which contained one-half of the new concept. As Bradbury has pointed out, Ulam as well as Teller should be given credit for this. Ulam, by the way, made his discovery while studying some aspects of fission weapons. This shows once more how the important ideas may not come from a straightforward attack on the main problem.
....

It is difficult to describe to a non-scientist the novelty of the new concept. It was entirely unexpected from the previous development. It was also not anticipated by Teller, as witness his despair immediately preceding the new concept. I believe that this very despair stimulated him to an invention that he might not have made under calmer conditions.

The new concept was to me, who had been rather closely associated with the program, about as surprising as the discovery of fission had been to physicists in 1939. ... The fact was that the new concept had

³⁵⁶Transcript of Press Conference by Norris Bradbury, et. al. at Los Alamos, New Mexico, September 24, 1954.

created an entirely new technical situation. Such miracles incidentally do happen occasionally in scientific history but it would be folly to count on their occurrence.

One of the dangerous consequences of the H-bomb history may well be that government administrators, and perhaps some scientists too, will imagine that similar miracles should be expected in other developments.³⁵⁷

Another account of the symbiosis of ideas is contained in an early 1953 congressional H-bomb chronology:

Dr. Ulam and Dr. Teller, working at Los Alamos, jointly submitted a report which proposed the use of a conventional A-bomb to compress materials to high densities. [deleted] Teller modified (Ulam's proposal) to using radiation, and this was the origin of "radiation implosion" ... when Teller recognized the importance of high densities ... he suggested that radiation implosion might be capable of achieving the required extreme densities over large volumes. This led to our present concept of thermonuclear reactions.³⁵⁸

Ulam and Teller jointly submitted their two separate hydrogen bomb design schemes -- Ulam's using a shock wave from a conventional atomic bomb to mechanically compress a deuterium-tritium thermonuclear fuel mixture to an extremely high density, and Teller suggesting x-rays to accomplish the same purpose in a large volume of fuel -- in a short report, entitled "On Heterocatalytic Detonations I: Hydrodynamic Lenses and Radiation Mirrors," published at Los Alamos in early March 1951.

³⁵⁷Bethe, "Comments on the History of the H-Bomb," pp. 48, 49. Dr. Bethe's commentary was written in 1954. Since then, four other countries have developed deliverable thermonuclear weapons, so perhaps the Teller-Ulam inventions were not so miraculous after all. Obviously, radiation implosion was a physical principle that transcended political and geographical boundaries.

³⁵⁸Borden-Walker H-bomb chronology, p. 62. Ellipses in original.

In this paper, Teller and Ulam presented the seminal "three concepts" that are the basis of every thermonuclear weapon in the American stockpile.³⁵⁹ In one bold step, a radical departure from previous thinking and a major scientific breakthrough, the physicist and the mathematician uncovered the long-sought secret to the limitless power of explosive, uncontrolled hydrogen fusion.

The monograph described, in 20 pages of text, tables and four illustrations, the principles of staged explosions, pre-ignition fuel compression, and radiation channeling (or "ducting") and implosion. The thesis emphasized the utility of "lenses" to focus and direct radiation from the primary onto the fuel mass in the secondary, identified in the report as "auxiliary" and "principal" bomb, respectively, and the practicality of using the device casing as a "radiation mirror" to "reflect" x-rays into the secondary.³⁶⁰ (See Volume

³⁵⁹In a letter dated February 26, 1958 from R. F. Taschek to Professor Warren C. Johnson nominating Teller and Ulam for the AEC's Fermi Award, Taschek states that "LAMS-1225 contains the nucleus for all useful further development of today's thermo-nuclear (sic) bomb complex and must be accounted jointly as a major achievement of Teller and Ulam."

³⁶⁰The comments in the report about "radiation mirrors" "reflecting" x-rays are virtually identical to Howard Morland's description of the workings of a hypothetical H-bomb in the November 1979 issue of THE PROGRESSIVE. Indeed, in an affidavit filed during the case, Robert N. Thorn, then acting director of the Los Alamos Scientific Laboratory, stated:

I have read the Article, "How a Hydrogen Bomb Works" (sic), written by Howard Morland, and am familiar with its contents. It is perhaps as suggestive of the process used in thermonuclear weapons as the original outline on the subject by Teller and Ulam. The original conceptualization of this process led rapidly to successful efforts by the United States in this area. (Affidavit of Robert N. Thorn, filed March 12, 1979 in Civil Action 79-C-98 in the United States District Court for the Western District of Wisconsin.)

In a letter to Dr. Hugh DeWitt dated July 25, 1990, Priscilla McMillan quotes Robert Serber as saying that he (Serber) was not surprised at the idea of using radiation to achieve

I for a description of radiation-focusing lenses and the use of the weapon casing to absorb and re-radiate x-rays.)

The report repeatedly stressed the need for high thermonuclear fuel densities:

In this discussion the following general scheme is considered. By an explosion of [deleted] auxiliary fission bombs, one hopes to establish conditions for the explosion of a "principal" bomb.... We propose to discuss certain general features of such an arrangement. The main purpose of the "auxiliary" system is to induce very high compressions in the principal assembly...

The scheme then depends on concentrating, as much as possible, the energy released by the explosion of a fission bomb in the mass of the principal assembly and doing it so as to achieve a high compression in this mass.³⁶¹

A thermonuclear device using the new "three concepts" -- separate stages, pre-ignition compression, and radiation ducting -- could reasonably be expected to yield energy 500 to 600 times greater than that of a conventional atomic bomb using the same amount of fissionable materials.³⁶²

The major thesis of the report was the idea of using high fuel densities, resulting from compression via a plasma generated by the x-rays, in lieu of high fuel temperatures,

compression, but that he was surprised by the means Ulam suggested, namely, "reflection" of x-rays from the casing interior (the radiation is "reflected" only in the sense that it is absorbed by and re-emitted from an intermediary substance).

³⁶¹"On Heterocatalytic Detonations I: Hydrodynamic Lenses and Radiation Mirrors," LAMS-1225, E. Teller and S. Ulam, Los Alamos Scientific Laboratory of the University of California, March 9, 1951, pp. 3, 5. Emphasis in original. On August 12, 1957, Ulam and Teller jointly filed a 54-page Top Secret patent application entitled "Thermonuclear Device." ("Publications of Stanislaw M. Ulam," LA-3293-MS, compiled by Barbara Hendry, Los Alamos Scientific Laboratory of the University of California, April 1968, pp. 2, 8. LA-3293-MS notes that the patent was not expected to be published in the near future.)

³⁶²Borden-Walker H-bomb chronology, p. 62.

i.e., high particulate kinetic energies in the fusion fuel mass induced by neutrons from the fission trigger. A year later, Dr. Hans Bethe stated:

...the H-bomb designs for which we now expect success are almost exactly the opposite of those proposed in 1946.

...As you are undoubtedly aware, the "super bomb" envisaged in 1946 was based on a self-propagating nuclear reaction in deuterium. However, calculations in 1950 showed that this reaction probably cannot take place at all, or if it can, that it is highly uneconomical and impractical...

In 1951, Teller discovered an entirely new approach to thermonuclear reactions. I believe that among all scientists in the United States, he was the only one who could have made this discovery, due to his ingenuity and his persistent belief in thermonuclear reactions, in spite of the severe setback implied by the 1950 calculations.

Even with Teller, the discovery was largely accidental. The new approach used high densities of deuterium rather than high temperatures and was based on two separate discoveries, (a) that high densities would be useful and (b) that they could be achieved by a radiation implosion. Whether this approach will actually be successful, only the test late this year can show.³⁶³

When Teller met with Dr. Herbert York at Eniwetok later in the spring of 1951, Teller described the new concept to York, who stated that "it was immediately evident to me that it

³⁶³Letter dated 23 May 1952, from Dr. Hans Bethe, Los Alamos Scientific Laboratory, to Gordon E. Dean, Chairman, U.S. Atomic Energy Commission, Washington, D.C. The self-propagating deuterium burn, if it could be started, would be uneconomical and impractical because of the need for several kilograms of tritium to initiate it. The comment about current (1952) designs being opposite to those proposed in 1946 is a reference to the position of the fissionable "sparkplug" inside the fusion fuel capsule; the fission trigger in 1946 was outside the fusion fuel capsule.

would work, that here was the long-sought solution to the (Super)."³⁶⁴

The new approach was as radical a departure from the "classical Super" design as the implosion-type plutonium-fueled atomic bomb had been from the earlier unworkable plutonium gun-assembly bomb design (see Volume VI). Like the implosion bomb, the "three concepts" depended upon compression to prepare the thermonuclear fuel for detonation. Either consciously or unconsciously, Ulam was following in the footsteps of Seth Neddermeyer, the Cal Tech physicist who had conceived the plutonium implosion idea upon which the wartime FAT MAN had been based.

In early April, Teller suggested an alternate approach, an "equilibrium thermonuclear gadget," in which energy generated by the explosion could theoretically maintain equilibrium with energy, in the form of x-rays and other ionizing radiations and particles, lost to the environment in the vicinity of the explosion.³⁶⁵

In this refinement of his earlier approach, Teller incorporated an internal fusion igniter within the thermonuclear fuel mass itself. By placing the "sparkplug" initiator (as explained in Volume I) in the center of the fusion fuel capsule, and setting off a second fission explosion in the middle of the compressed fuel mass, tritium would be generated inside the bomb and no external supply would be required. The end result would be an outwardly-advancing shock wave further compressing the fusion fuel by its interaction with the inwardly-directed shock wave generated by the x-rays from the primary.

In addition to the outwardly-directed shock wave from the sparkplug explosion, neutrons generated by the fissioning sparkplug could breed tritium via lithium conversion in a solid-fueled device, or bred via neutron-deuterium collisions and unions (equation "M" in "Fusion Physics" in Volume I).

Hans Bethe describes the maturation of the "three concepts:"

Within a month, the very important second half of the new concept occurred to Teller, and was given preliminary checks by de Hoffman (sic). This immediately became the

³⁶⁴York, THE ADVISERS, p. 80.

³⁶⁵"A New Thermonuclear Device," LA-1230, E. Teller, Los Alamos Scientific Laboratory of the University of California, April 4, 1951, p. 3; Borden-Walker H-bomb chronology, p. 62. This design later became known as the "equilibrium Super."

main focus of attention of the
thermonuclear design program.³⁶⁶

The official AEC history corroborates this account:

Some of Teller's anxiety might have stemmed from his realization that he was, at last, upon the right track. His report with Ulam had done little more than to point out possible approaches. Another idea, based upon the first, came to him probably in the latter part of March. De Hoffmann began a mathematical analysis, feeling fortunate, as he worked night and day, that the calculative techniques he had worked out for some of the GREENHOUSE tests were applicable to Teller's latest suggestion. The results looked good. In early April, de Hoffmann signed the report with Teller's name. The approach could have been called the "New Super."³⁶⁷

Teller himself later described the process thusly:

³⁶⁶Bethe, "Comments on the History of H-Bomb," p. 48.

³⁶⁷Hewlett and Duncan, p. 541. The second report is LA-1230; according to Hewlett and Duncan (p. 658), the name "New Super" was their own invention, because (in 1969, at least) the actual name of the approach was still classified. Widespread usage of the term "sparkplug" to describe this igniter in the secondary did not occur until later in the development of thermonuclear test devices. No officially-published account of the discovery of radiation implosion and the "three concepts" uses the term "sparkplug." There is very strong circumstantial evidence which suggests the "sparkplug" was an integral part of the "three concepts:" Howard Morland's reference to it in his book and his description of "sparkplugs" mentioned in the mistakenly-declassified UCRL-4725; Dr. Ray Kidder's third affidavit in the PROGRESSIVE case which stated that Morland's original H-bomb schematic (without "sparkplug") lacked "a vitally important secret of H-bomb design" which "is clearly revealed in UCRL-4725;" a recent issue of LOS ALAMOS SCIENCE referring to the inclusion of a "sparkplug" in the Mike device at Eniwetok in 1952; and the Blumberg and Panos biography of Edward Teller cited in Volume I in which Teller says "we needed a primary, we needed a secondary, we needed a sparkplug". Lacking a "sparkplug," the compressed fusion fuel capsule would be nearly impossible to ignite.

There was no question in my mind how the super (sic) should be made. We needed a primary, we needed a secondary, we needed a sparkplug. The secondary had to have cylindrical symmetry because that was the symmetry natural to the process. The energy transfer had to go by radiation in order to make it as close to simultaneous as possible. All this was clear to me.³⁶⁸

The "new Super" became the basis for a test in November 1952 that yielded several megatons, an explosive force hundreds of times that of the wartime atomic bombs.

In terms of George Gamow's earlier analogy-by-demonstration, the Teller-Ulam configuration essentially pulverized the petrified wood (by compression), soaked the resultant dust with lighter fluid (the tritium produced amidst the fuel by the fissioning "sparkplug"), and then ignited the powder.³⁶⁹

³⁶⁸Interview of Dr. Teller by Dr. Keyworth cited in Blumberg and Panos, p. 130.

³⁶⁹To some extent, the order of names in their invention is implied by the term radiation implosion itself: Teller suggested radiation and Ulam suggested implosion. It might more properly be called the "Ulam-Teller" configuration: Stan Ulam had the original ideas and Edward Teller carried them to fruition, adding ideas of his own along the way. Dr. George Kistiakowsky, an early nuclear weapons explosives pioneer, stated in "A is for Atom, B is for Bomb," a 1970s Public Broadcasting System documentary on Edward Teller, that Teller should be called the "mother of the H-bomb," rather than the more common appellation of "father of the H-bomb." (Other sources claim that either Dr. Hans Bethe or George Gamow coined the former title for Teller; according to David Hawkins, wartime LASL historian, a variation of this characterization was that although Teller claimed paternity, Ulam had slept with the mother nine months earlier.) According to Kistiakowsky, Ulam "impregnated" Teller with the idea and Teller nurtured it to maturity.

There had been some personal friction previously between Teller and Ulam; in addition, there may have been a feeling of elitism among the physicists and a bias against the "mere" mathematicians working on the Super. Francoise Ulam states that Teller objected loudly and disbelievingly to the Ulam-Fermi calculations in 1950 and that "what should have been the common examination of difficult problems became an unpleasant confrontation," and that after the seminal discoveries were jointly made, "Teller pushed Stan aside and

The primary fission trigger no longer acted as the igniter for the fusion process -- an approach that had been pursued so diligently and so fruitlessly for many years -- but merely served as a catalyst to prepare the fusion fuel for ignition by the "sparkplug."

Increasing the fuel density had the same effect as adding tritium: raising density reduced heat loss, making ignition possible, while adding tritium reduced the ignition temperature, allowing for burning even with high heat loss. However, using up large amounts of tritium at the expense of many dollars and potential plutonium, i.e., dozens of fission weapons, was clearly uneconomical and impractical.

Why Did It Take So Long?

After more than 40 years, an enduring question about the discovery of radiation implosion is why the Teller-Ulam principles did not surface much earlier than they finally did. Today, the "three concepts" seem obvious (despite DOE assertions in 1979 to the contrary), given present knowledge about the products of a fission explosion and the properties of thermonuclear fuels. However, much of what is known now was either unknown or barely known four decades earlier.

There seem to be three major reasons for the relatively-late discovery of the application of radiation implosion to U.S. thermonuclear weapons. The first, and perhaps most important reason, was the stubborn refusal of some Los Alamos weaponeers, Edward Teller foremost among them, to consider

refused to deal with him any longer; he never met or talked with Stan meaningfully ever again." (Ulam biography, 1991, pp. 310, 311.)

Frederick de Hoffmann, whose name is spelled differently in almost every history of the H-bomb, contributed significantly to the mathematical verification of the final design. In an obituary in the May 28, 1984 issue of NEWSWEEK (Ulam died in Santa Fe, New Mexico, on May 13, 1984, at the age of 75), the magazine states: "Although Edward Teller is often referred to as the 'father' of the H-bomb, it was actually Ulam who solved the problem of how to initiate the fusion of hydrogen atoms and thereby release the bomb's devastating energy." Edward Teller has also never forgiven Stan Ulam for expressing doubts about the thermonuclear project after, in Teller's words, its feasibility was clear to others. (Letter dated March 13, 1990 from Edward Teller to the Editor, New York Times.) Ironically, Teller himself would, on many occasions between February 1951 and November 1952, express grave doubts as to whether the first LASL thermonuclear test device would work as intended. Apparently it was permissible for Teller to doubt, but others could not.

anything other than the "classical Super" design, wherein a burning wave front would (theoretically) self-propagate through a mass of deuterium.

Even after it became abundantly clear that this idea would not work, Teller still clung to it tenaciously, as evidenced both by the comments about thermonuclear weapons in the Los Alamos Technical Program for 1951 and 1952, and Teller's last-ditch suggestions at the end of 1950. As late as August 1952, after the Mike device had been designed, but not yet tested, Teller claimed that the 1946 "classical Super" might still work, with minor modifications.³⁷⁰

This single-minded obsession, coupled with Teller's dominating personality and influence over the entire program, doomed both the consideration and the viability of competing ideas.

In his 1982 article, Dr. Hans Bethe states:

Teller himself was desperate between October 1950 and January 1951. He proposed a number of complicated schemes to save Method A (the "classical Super"), none of which seemed to show much promise. It was evident that he did not know of any solution. ...

After the failure of the major part of his program in 1950 (a reference to the discouraging Fermi-Ulam-Everett calculations), it would have been folly (for) the Los Alamos Laboratory to trust Teller's judgment, at least until he could present a definite idea which showed practical promise.

This attitude was strongly held by most of those on the permanent staff of the Laboratory who were responsible for its operation. As might be expected, the many discussions of aspects of this situation bred considerable emotion.³⁷¹

Stan Ulam also later claimed that Teller's stubbornness delayed the consideration of alternate approaches; Teller

³⁷⁰Teller, COMMENTS ON BETHE'S HISTORY OF THE THERMONUCLEAR PROGRAM, p. 1.

³⁷¹Bethe, "Comments on the History of the H-Bomb," p. 48.

tried to blame this delay in part upon LASL director Dr. Norris Bradbury:

... the impression which is intended to strike the reader is that Dr. Bradbury ... did not intend to proceed as quickly as possible in the planning and in the construction of an H-bomb. This seems to me (to be) utterly incorrect.

In fact, one could take the opposite view. The plans originally pushed by Edward were not very good and would have led to failures which would have postponed the development of (successful) thermonuclear devices and his insistence on salvaging his original schemes cost a lot of time. It seems to me that, apart from a general (sense of) enthusiasm, there was very little on which to base an actual program.³⁷²

Dr. John A. Wheeler, then supporting Los Alamos efforts at Princeton, later stated:

All our ideas about how to create a fusion explosion were based on one premise. Within a fixed framework, we went through unbelievably clever and subtle distortions to try to make it work, and we couldn't. This new idea (the "three concepts") changed the framework.³⁷³

During his loyalty/security clearance hearing in April 1954, Dr. J. Robert Oppenheimer, defending the recommendations of the GAC in October 1949, said of the "classical Super" of 1949:

We had in mind, but I don't think we had clearly enough in mind, that we were talking about a single design which was in its essence frozen, and that the possibility did not occur to us very strongly that there

³⁷²Letter dated March 16, 1962 to Dr. Glenn Seaborg, Chairman, USAEC, from S. M. Ulam.

³⁷³MEN WHO PLAY GOD, Norman Moss, Harper & Row, New York, 1968, p. 52. Emphasis added.

might be quite other ways of going about it.³⁷⁴

In 1954, Oppenheimer stated:

I am inclined to think that if we had before us a program like the one that matured by mid-1951, there would have been little doubt of the need to develop it. I think that if we had such a program in 1945, the development might have been ripe by 1947 or 1948; and if we had it in 1949, we would have been far along by 1951.³⁷⁵

In a recent letter to the author, Dr. J. Carson Mark says of the Super between 1942 and 1951:

The general pattern, with occasional inconsequential modifications, was that proposed in 1942; that worked on at Los Alamos in 1943-45; that considered at the 1946 conference; that given such attention as possible at Los Alamos from 1946-49; that sold to a few pugnacious politicians, administrators, military men, and subsequently to President Truman, in 1949; and that worked on for a year following President Truman's directive. No one knew just what it might take to ignite the main charge; nor whether, if ignited, it would burn effectively.³⁷⁶

In his defense, it can be argued that Teller pursued the "classical Super" instead of a simpler, more physically-feasible design like the ALARM CLOCK because the former offered, in theory, at least, the promise of a much larger yield and potential for future adaptation than the latter.

Second, and perhaps almost as significantly, not much accurate information was available between 1945 and 1950 concerning the fusion cross-sections of deuterium and tritium, which turned out to be the key to Ulam's fuel

³⁷⁴IN THE MATTER OF J. ROBERT OPPENHEIMER, p. 78. Emphasis added.

³⁷⁵1954 J. Robert Oppenheimer memo to self in Box 202 of Oppenheimer Papers in Library of Congress, enclosure to letter dated 13 September 1990 from Priscilla J. McMillan to author.

³⁷⁶Letter dated 18 September 1989 to author from J. Carson Mark, Los Alamos, New Mexico; Ulam biography, 1991, p. xi.

compression proposal. While some wartime measurements of these phenomena had been obtained, they were sketchy and unreliable. Precise values were not available until 1951, after President Truman's January 1950 statement, when a group at Los Alamos led by Dr. James Tuck re-measured D-D, D-T, and D-He³ cross-sections. The cross-sections were discovered to vary widely at low energies, and the results of previous measurements diverged widely.³⁷⁷

Third, sufficiently powerful computers did not exist until after the discovery of the "three concepts." The "number crunching" required to analyze Super behavior was one of the largest mathematical projects of the twentieth century, in terms of both volume and complexity of calculations. Dr. Teller himself noted in September 1947 that it might take as long as two more years of tests, calculations, and experiments until just the feasibility of the Super or the ALARM CLOCK could be adequately determined.

Teller later claimed repeatedly both in public and in private that very little work had been done on the Super at LASL between 1946 and 1951, conveniently overlooking the fact that in the fall of 1947 he had suggested a two-year delay in the project until further calculations and experiments could be completed.

Only in the spring of 1952 did the new MANIAC computer at Los Alamos, joined later in the year by a companion model at Princeton, New Jersey, begin to equal the task; coincidentally, a major theoretical problem with the Mike device became apparent and a redesign of that system followed.

The MANIAC, or Mathematical And Numerical Integrator and Calculator, was a major improvement over the earlier ENIAC. A successor, the MANIAC II, was in service at LASL from 1956 until 1977.³⁷⁸

³⁷⁷"Nuclear Data: The Numbers Needed to Design the Bombs," LOS ALAMOS SCIENCE, Number 7, Winter/Spring, 1983, Los Alamos, New Mexico, p. 121; Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, p. 5; Minutes, Twenty-Third Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, October 30, 31, and November 1, 1950, Los Alamos New Mexico, pp. 37, 38; memorandum for the File dated August 28, 1951 from Ken Mansfield, subject: Conversation with Dr. Teller.

³⁷⁸"Computing and Computers -- Weapons Simulation Leads to the Computer Era," Francis H. Harlow and N. Metropolis, LOS ALAMOS SCIENCE, No. 7, Winter/Spring 1983, Los Alamos National Laboratory, pp. 135, 137. Another interpretation of

Between 1945 and 1951, theoretical understanding and mathematical formulation of the properties of many thermonuclear reactions far outstripped the computational abilities of existing electromechanical devices to verify these phenomena. Even if the new ideas of Teller and Ulam had arisen in 1946, it is quite likely that computer resources and mathematical formulae required to demonstrate the feasibility of the "three concepts" would not have been available until many years later, as was finally the case.

Equally important was the accumulation of knowledge between 1946 and 1951 about the behavior and energy emissions of exploding fission bombs. Neither Ulam's shock wave implosion principle nor Teller's adaptation of radiation implosion could have been successfully theorized without a firm understanding of the results of Operations CROSSROADS and SANDSTONE, and more complete, more accurate mathematical modeling of fission processes. Obtaining this data took time and experimentation.

In addition, small, highly-efficient atomic explosives were required to generate enough radiation to drive the compression of the thermonuclear stage. Large-diameter primaries, surrounded by thick layers of radiation-absorbing high explosives and tampers, would not provide sufficient compression.³⁷⁹ These types of more-efficient weapons, most notably the smaller-diameter MK 5 and MK 7 bombs, were simply not available until 1951.

In 1954, Dr. Hans Bethe summed up LASL growth and learning between 1949 and 1951:

The size of the Los Alamos Laboratory, the experience of the staff, and the sophistication of (its) control over fission bomb design were all enormously greater in 1951 than they had been a couple of years before. In addition, there is the matter of the revolutionary change in computing

the acronym MANIAC was "Metropolis And von Neumann Invent Awful Contraption." (Rhodes, DARK SUN, p. 383.) In an article in the May 26, 1995 issue of the LANL Newsbulletin, Nicholas Metropolis claimed that MANIAC was an acronym for "Multiplicative and Numerical Integrating Automatic Computer." The acronym has also been alleged to stand for "Mathematical Analyzer, Numerator, Integrator, and Calculator" and "Mathematical Analyzer, Numerical Integrator, and Computer." It is possible that the name MANIAC was never intended to be an acronym at all.

³⁷⁹Interview of Hans Bethe by Richard Rhodes, 3 May 1993.

facilities and techniques between 1947 and the present time, which was just beginning to take real effect about the beginning of 1951.³⁸⁰

As Hans Bethe later noted, there were no essential delays in LASL work on the Super after Truman's January 1950 directive, except those caused by the lab's technical ignorance.³⁸¹ Events had proceeded about as fast as they could have.

Compression: The Key To Success

Perhaps the most difficult point to comprehend is why a high fuel density, rather than a high internal temperature, was not seen much earlier as a solution to the Super ignition problem. By 1945, it was widely known that both chemical and nuclear reactions proceeded more rapidly and more completely in dense media.

In addition, the scientists at Los Alamos were well aware that densely-compressed cores were required for high fission weapon efficiencies. Ironically, according to Bethe, it was Teller himself who first suggested that implosion would compress fissile material to a higher-than-normal density within a bomb.³⁸²

As early as 1934, the possible utility of compression was known to astrophysicists and may have been known to the Soviets:

Arthur Eddington -- a classical author on
astrophysical thermonuclear topics --

³⁸⁰Bethe, "Comments on the History of the H-Bomb," p. 50.

³⁸¹"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 No. 2 (1989), p. 333.

³⁸²Bethe, "Comments on the History of the H-Bomb," p. 44; letter dated April 14, 1945 to Dr. H. D. Smyth, Princeton University, from Dr. J. R. Oppenheimer, Director, LASL. According to another source, "Teller and Hans Bethe pointed out that this (implosion) would also have the advantage of compressing the plutonium and increasing the efficiency of the chain reaction." ("The University of California and the Mobilization of Science for National Defense," paper by Robert W. Seidel, Los Alamos National Laboratory, presented during February 1992 in Pleasanton, California at the workshop on "The Decade of Innovation: Los Alamos, Livermore, and National Security Decisionmaking in the 1950s," p. 22.)

wrote in his publications, for example, in the 1934 book "New Pathways in Science," that igniting thermonuclear reactions (he had already foreseen the role played by deuterium, and even tritium, for laboratory experiments and warned of the possibility of creating weapons on this basis) is very sensitive to the degree of compression of thermonuclear fuel.

The German theoreticians Busemann, Bechert, and Guderlei in 1940-42 produced analytical theories of compression of a continuous medium, which were immediately applied by the experimenter Trinks (in Berlin, 1942-43) for compression of deuterium in order to find a way to of making a hydrogen bomb.³⁸³

However, in May 1945, Oppenheimer apparently believed that implosion did not lend itself readily to Super development, although it is not clear from Oppenheimer's comments whether he was referring to implosion of the thermonuclear fuel, or to the inherent unsuitability of an implosive fission device as a trigger for the Super, as it was then designed.³⁸⁴

The principle of radiation implosion was discussed very briefly, and apparently then only in passing as a theoretical possibility, at the April 1946 conference at Los Alamos. In a

³⁸³Additional footnote 14a to "The H-Bomb: Who Really Gave Away the Secret," Daniel Hirsch and William G. Mathews, in Soviet Physics Usp. 34 (5) May 1991 (Uspekhi Fizicheskikh Nauk 161, 153-169), translated from the Russian (after originally appearing in English in the Bulletin of the Atomic Scientists, January/February 1990, pp. 22-30. According to another source, G. Guderlei (or Guderley, in the German) proposed in 1942 the use of a high explosives-driven spherical implosion wave to initiate thermonuclear fusion. Trinks conducted laboratory experiments with one-meter diameter hollow silver spheres enclosing deuterium gas, the sphere being surrounded by layers of high explosives and a tamper, in a configuration very similar to that later used by LASL for the first implosion atomic bomb. These spheres, when compressed, were subject to rippling and non-symmetrical implosion, and Trinks never detected any neutrons from thermonuclear reactions. ("Some Reminiscences About the Origins of Inertial Confinement Fusion," Dr. Friedwardt Winterberg, FUSION, Vol. 3 No. 2, September 1979, pp. 42, 43.)

³⁸⁴Letter dated May 7, 1945 to General Leslie R. Groves, USA, from J. Robert Oppenheimer, Director, LASL.

September 1954 press conference, LASL director Dr. Norris Bradbury stated:

... One of the ideas, that later turned out to have an effective means of accomplishment, was originated in 1946. ... I can remember very vividly in 1946 exploring in great detail this original idea with the members of my technical staff. At the time, we saw absolutely no way to make it into a usable system. ... the basic idea involved in the final successful development arose out of a series of ideas and discussions that (date) back to 1945.³⁸⁵

While much consideration had been given to preventing fuel mass dispersion in early designs of the "classical Super," few of them had included compression of the deuterium to delay its disintegration and improve its "burning."

In August 1952, Dr. Edward Teller claimed that the successful 1951 application of radiation implosion differed from its suggested utilization in 1946 in only two respects: by imploding a larger volume of material, and by achieving greater compression by keeping the imploded material cool as long as possible, which was accomplished by ignition of the central "sparkplug" after the secondary was compressed.³⁸⁶

As early as the spring of 1947, embodied implicitly in the design and operation of the ALARM CLOCK, highly-compressed deuterium was seen as a requisite to successful fusion.³⁸⁷ However, both the ALARM CLOCK and the designs of the BOOSTER and CYLINDER were configured for only small-scale

³⁸⁵Transcript of Press Conference with Norris Bradbury at Los Alamos, New Mexico, September 24, 1954, pp. 5, 6, 18.

³⁸⁶"Comments on Bethe's History of the Thermonuclear Program," Dr. Edward Teller, August 15, 1952, p. 2. Teller also later claimed that if J. Robert Oppenheimer had provided moral support in 1945 for the Super, many scientists would have remained at LASL and the Super would have been ready by 1949 or 1950. ("In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 No. 2 (1989), p. 332, citing Teller's testimony during the Oppenheimer loyalty/security hearings.) This statement by Teller was mostly his opinion and showed little connection to reality.

³⁸⁷LAMS-560, p. 9.

thermonuclear reactions, and not a theoretically-unlimited reaction, which was reserved exclusively to the Super.

Although by 1951 some thought had been given to the idea of thermonuclear ignition by means of ordinary high explosives,³⁸⁸ an idea which seemed to point directly toward compression, not enough experimental work had been done in this area to provide useful data about how fuel compression might improve fusion cross-sections. (This may have been the basis of the comment about "imploding parts of the super bomb" mentioned in the GAC report at the end of October 1950). It was probable that ordinary high explosives would not compress the deuterium sufficiently to either initiate or maintain fusion reactions.³⁸⁹

After World War II ended, a few Los Alamos weaponeers had considered applying the principles of fission weapon design and dynamics to fusion devices:

The mode of burning (hypothesized) for the thermonuclear fuel in this (classical Super) design (if, that is, it would burn at all) would be negatively affected by the presence of any surrounding dense material. If, for example, one attempted to compress the fuel, the material -- such as high explosive -- used for the compression would present the chance of quenching the burning entirely.

People were aware of a different possible mode of burning thermonuclear fuel for which compression would be advantageous -- or perhaps essential -- and examples of this were given serious attention for a year or so (1946-47). But the levels of compression achievable with high explosive -- even in amounts too large to be of practical interest -- were insufficient to lead to any useful result. The "iterative scheme" discussed in the Teller-Ulam report of March 1951 changed all that.³⁹⁰

³⁸⁸"Status of Hydrogen Project," transcript of meeting of the Joint Committee on Atomic Energy, February 21, 1952.

³⁸⁹Interview of Hans Bethe by Richard Rhodes, 3 May 1993. A possible discussion of thermonuclear ignition via high explosives may be contained in LA-560, the July 1946 Tuck-Ulam paper, "Possibility of Initiating a Thermonuclear Reaction by Non-Nuclear Methods."

The JCAE Polls The Scientists

In November 1953, before the tests of the first solid-fueled U.S. thermonuclear devices, and after the test of a limited-yield solid-fueled Soviet device in August 1953, Representative Sterling Cole, then chairman of the JCAE, posed a number of questions to several U.S. nuclear scientists, including Ernest O. Lawrence, Enrico Fermi, Hans Bethe, Norris Bradbury, Edward Teller, John von Neumann, and GAC chairman Isidor Rabi. One of his queries was "How much, if any, do you think our progress would have been forwarded (again measured in months or years) had we undertaken a major thermonuclear effort in 1946 rather than in 1950?"

Von Neumann stated his belief that the thermonuclear program might have been advanced by as much as three years, had it begun at LASL in 1946:

I think that our thermonuclear program could have been started in 1946 just as well as in 1950. Had this happened, there would have been some handicaps which did not operate in 1950, namely:

- A. The AEC laboratories underwent a severe shock at the end of the war by loss of personnel loss of understanding of purpose, etc., at all levels. This condition kept deteriorating for over a year (until) it was gotten under control, and a possible modus operandi established, in the course of 1946.
- B. Some techniques which existed in 1950 were a good deal less available in 1946, in particular, high-speed electronic computing. Nevertheless, it was not wholly unavailable in 1946, and to some degree, alternative methods were possible.

In this context, it should be noted that the thermonuclear program went to the successful "Mike" test in November 1952 in a direction which differed in some very important ways from the original concepts.

This redirection was based on certain quite decisive calculations and theories of E. Teller and S. Ulam. These did not in their

³⁹⁰Letter dated 18 September 1989 to author from Dr. J. Carson Mark, Los Alamos, New Mexico; Ulam biography, 1991, p. xiv.

original form use electronic computings, and they would have been just as possible in 1947 or 1948 as in 1951, when they actually occurred.

On the basis of these, and of some other, quite similar considerations, I would think that if the program had been started in 1946, it would not have been relatively slowed by more than one year against the course it took after 1950.

The developing of a [deleted] would, in fact, not have been slowed at all, starting from 1946 rather than from 1950. Hence, I think that the time that could have been gained is at most one year short of the four years lapsed between 1946 and 1950. In other words, I believe that we would have gained three years.³⁹¹

Dr. von Neumann's comments about computers notwithstanding, he overlooked the fact that there was in 1946 very little impetus to proceed with the Super, even at the highest levels of the U.S. government which would have had to order -- and finance -- such a program. There was little enough appreciation at that time within government circles of the revolutionary nature of the advent of the atomic bomb, let alone the likely consequences and nature of a possible future weapon that could be 1,000 times more destructive. The U.S. government did not begin devoting significant financial support to the Super program until after the Soviet nuclear blast in August 1949.

Von Neumann also completely ignored Teller's single-minded obsession with the "classical Super" which effectively stopped any other approaches until early 1951.

Enrico Fermi stated that "an earlier start of a crash program might have put us ahead in this line by perhaps a year or a year and a half; this is (with) the assumption (that) the effort along more conventional lines of the atomic program had not been correspondingly weakened."³⁹²

³⁹¹Letter dated November 23, 1953 to Honorable Sterling Cole from John von Neumann. Emphasis in original.

³⁹²Letter dated November 24, 1953 to Sterling Cole, Chairman, JCAE, from Enrico Fermi.

Dr. Rabi offered a more measured response to Cole's question, and noted the lack of high-level early postwar enthusiasm for the Super project:

Opinions will differ very greatly on this subject. My own impression is that we could not have undertaken a major thermonuclear program until we had some good ideas as to how to proceed.

In my judgment, these ideas did not appear (until) about early 1950. If these ideas had appeared earlier, we would have had our major thermonuclear program earlier, provided (that) Congress would have been willing to back it up with production facilities for the necessary materials.

One of the most important factors in our advance was the demonstration by Dr. Ulam of Los Alamos that the plans of 1946 were scientifically unsound. When the misconceptions were cleared away, new ideas appeared (chiefly Dr. Teller's) and were immediately recognized as appropriate for a practical solution of the problem. The program then went away at a rapid pace.³⁹³

Edward Teller, who early in the fall of 1947 had suggested a two year delay in U.S. thermonuclear research in order to gather further information, and whose stubborn insistence on the "classical Super" had further delayed the American program, had no doubts whatsoever as to how much time could have been saved had the Los Alamos laboratory taken up his ideas in 1946:

This question I can answer in a straightforward manner. Our work on thermonuclear bombs was practically stopped, on or around September 1, 1945. It was resumed on the second of February 1950. In the intervening years, there was some progress in the theory of thermonuclear reactions. This progress, however, was exceedingly slow.

Furthermore, a great deal of momentum was lost by discontinuing our effort. Thus in

³⁹³Letter dated November 25, 1953 to Honorable Sterling Cole, Chairman, JCAE, from I. I. Rabi, Chairman, GAC.

1950, we lost more time by repeating old calculations and by training new people.

Thus I think it is realistic to say that the decision to discontinue work on thermonuclear reactions in 1945 has retarded our work in this field by four years and five months.³⁹⁴

Teller did not mention to Cole that he had left LASL in 1946 when Norris Bradbury and the AEC could not guarantee a grandiose annual nuclear testing program that Teller wanted, and that Teller had not offered to remain at LASL to continue to try to push the thermonuclear program along between 1946 and 1950.

Teller also failed to mention his original 1947 ALARM CLOCK idea or the great progress that had been made on the designs of the BOOSTER and the CYLINDER between 1945 and 1950, work sufficient to permit tests of the latter devices at GREENHOUSE in the spring of 1951. He also neglected to note that the calculations "repeated" in 1950 pretty well ended any hopes for the "classical Super." Teller was always adept at tailoring history to suit his current political agenda.

Perhaps the best answer to Cole's question was that provided by Bradbury:

In 1946, any major developmental effort in the thermonuclear field would have had to follow along the lines of the so-called classical "super" bomb -- a type of system now believed to be nuclearly inefficient and almost certain to be unrealistic in practical application because of its probable size. Attempts to proceed along these lines would have had an almost insurmountable handicap arising from the absence in this country at that time of practical high-speed electronic computing machinery.

Indeed, the machines essential to the calculation of the weapon systems of today have been in existence for only the last two years, and without them, our present progress would have been impossible.

It should also be pointed out that the two outstanding and primary needs in 1946 were:

³⁹⁴Letter dated December 18, 1953 to Honorable Sterling Cole, Chairman, JCAE, from Edward Teller.

(1) the development of an efficient use for U-235 in atomic weapons, and (2) the general development of more efficient and more versatile atomic weapons. Our present rapid and practical success in the thermonuclear field has been possible in a large measure because of our skill in understanding the behavior of fission weapon systems, and in developing small but nevertheless highly efficient devices of this nature.

Accordingly, had we devoted less effort to the fission field in 1946-1950 and more effort to the thermonuclear, we would probably have attempted to follow an ill-advised line, and might not have developed as rapidly the necessary fission skills to follow the present profitable approach.

It is, of course, a misconception that thermonuclear effort was neglected during this period. Indeed, the first ideas (now abandoned but pointing in the correct direction) for the ALARM CLOCK were discussed and explored in 1947; "boosting," which as a process appears to be essential to the smaller primary bomb field, has been under active development since 1947-48.

The directives of 1950 did little more than formalize for the LASL a trend which was already underway -- a trend which led eventually to the idea of radiation implosion and to our present effectiveness in this area.³⁹⁵

In any event, by the spring of 1951 terrestrial fusion explosions were a distinct possibility. Many large and complex engineering problems remained to be solved, and devices designed and tested, to prove the Teller-Ulam ideas, and to make workable, practical, powerful, and deliverable thermonuclear weapons in the form of aircraft bombs and missile warheads.

The major theoretical points to confirm in tests would be whether x-rays could in fact be directed and channeled within the bomb casing to compress fusion fuels, and if

³⁹⁵Letter dated 2 December 1953 to Honorable W. Sterling Cole, Chairman, JCAE, from Norris Bradbury, Director, LASL. (Bradbury apparently never wrote a short letter.)

multiple nuclear explosions within the same device could be sequentially staged to generate ever-increasing yields. Large-scale radiation implosion could not be tested in a laboratory (although very small-scale radiation implosion is now routinely demonstrated during inertial confinement fusion experiments).

Even with these new ideas, there was no guarantee or certainty of success short of a full-scale test. The concepts that Teller and Ulam had set forth in their March report would require much theoretical effort. Perhaps years might be needed to evaluate the approach. Some small-scale experimental work could provide data, but even so, Ulam foresaw a long future of hard analysis.³⁹⁶

In addition, because radiation implosion was very new and still untested, it might, like other Teller ideas, contain a fatal flaw. If this turned out to be the case, then an alternate method -- perhaps Ulam's shock wave compression technique -- might be required, leading to the development of implosion systems larger and more complex than any built to date.

Plans for testing the new principle were incorporated quickly into Los Alamos's contemporary test program:

Among the tests which seem to be of importance at the present time are those concerned with boosted weapons. Another is connected with the possibility of a heterocatalytic explosion, that is, implosion of a bomb using the energy liberated from another, auxiliary bomb. A third concerns itself with tests on mixing during atomic explosions, which question is of particular importance in connection with the ALARM CLOCK.

...Cryogenic work is of importance only for the thermonuclear program...³⁹⁷

Before a large-scale experiment of the new principles could be conducted, the thermonuclear ideas developed in 1947 and

³⁹⁶Hewlett and Duncan, p. 540.

³⁹⁷Los Alamos Memorandum ADWD-250 dated March 24, 1951, from Edward Teller to Norris E. Bradbury, Subject: Plan for Setting up a Separate Thermonuclear Division. The term "staging" soon replaced the cumbersome expression "heterocatalytic explosion."

1949 were scheduled for tests in the near future at Eniwetok atoll.

Operation GREENHOUSE

Operation GREENHOUSE was the first Pacific Proving Ground weapons test series to feature thermonuclear experiments.³⁹⁸ GREENHOUSE was the fourth series of nuclear tests conducted by the U.S. following World War II.

Operation CROSSROADS, in 1946, was the first test series, consisting of an air and underwater detonation at Bikini atoll in the Marshall Islands using an array of target ships. CROSSROADS was the first large-scale weapons effects test, and did not include explosions of new weapons types; the two CROSSROADS bombs were drawn from the U.S. nuclear stockpile, which in 1946 consisted of only nine implosion-type cores.³⁹⁹

Operation SANDSTONE in 1948 included three tower-mounted devices detonated at Eniwetok atoll; these shots tested levitated composite cores for the new MK 4 weapon. The third atmospheric test series, Operation RANGER, featured five airdropped device detonations in January and February 1951 at the newly-established Nevada Proving Ground (later the Nevada Test Site) north of Las Vegas, Nevada.⁴⁰⁰

The two initial U.S. nuclear test sites were chosen mainly for their isolation and remoteness from inhabited areas. The Nevada Test Site, 65 miles northwest of Las Vegas, is in the middle of what was then the Las Vegas Bombing and Gunnery Range, an Air Force training and practice site. The terrain of the NTS is typical of south-central Nevada, featuring ranges of hills and mountain peaks, and high desert valleys that

³⁹⁸Much of the following GREENHOUSE history, except where noted, is drawn from OPERATION GREENHOUSE 1951, DNA 6034F, prepared by the Defense Nuclear Agency, Washington, D. C., June 1983.

³⁹⁹"U.S. Nuclear Stockpile, 1945 to 1950," David Alan Rosenberg, BULLETIN OF THE ATOMIC SCIENTIST, Vol. 38 No. 5, May 1982, pp. 25-30. For a good account of Operation CROSSROADS, see OPERATION CROSSROADS: THE ATOMIC TESTS AT BIKINI, Jonathan M. Weisgall, Naval Institute Press, Annapolis, Maryland, 1994.

⁴⁰⁰The name "Nevada Proving Ground" had been changed to "Nevada Test Site" by November 1956 (memorandum dated November 23, 1956 to Brig. General Alfred Starbird, USA, Director of Military Application, USAEC, from K. F. Hertford, Manager, Albuquerque Operations, USAEC, subject: Standard Operating Procedures for Exercise Desert Rock VII). Both Nevada site names are used interchangeably.

drain into dry lake beds. The altitude of the NTS varies from 3,075 to 4,050 feet above sea level.⁴⁰¹

The Nevada site, an empty desert, was already owned by the U.S. government and was adjacent to an existing Air Force training field, Indian Springs Air Force Base, which could be expanded to fulfill AEC requirements. The base included an airstrip long enough for takeoffs and landings of medium-sized aircraft.

There was also adequate water available, and the site was conveniently located for both the Los Alamos and Livermore labs. The site was also bounded naturally by mountains with no roads or trails allowing easy access. The only major road close by was Highway 95 seven miles south of the site.⁴⁰²

Far to the west, the Pacific Proving Ground⁴⁰³ comprised the enclosed lagoon and land areas of Eniwetok and Bikini atolls, and the open ocean within three miles of the two atolls. Bikini and Eniwetok atolls in the Marshall Islands were selected as test sites primarily for their proximity to a major military airbase on Kwajalein atoll, and for their remoteness from shipping lanes, airways, and fishing grounds, although itinerant Japanese fishing vessels venturing close to the atolls would be the bane of the AEC for many years to come.

The two atolls were an excellent location that would be available for an indefinite period, in a secure area. The atolls were within 300 miles of Kwajalein, site of a major U.S. military airbase and ship anchorage, and within 2,500 miles of both Guam in the Mariana Islands and Oahu in the Hawaiian chain. Eniwetok atoll encircled a fine deepwater lagoon suitable for anchoring large vessels.

Eniwetok was also eminently appropriate for its relatively isolated location with an empty ocean area downwind, its small native population that would have to be relocated, the low likelihood of contaminating neighboring atolls, ease of

⁴⁰¹"Background Information On Nevada Nuclear Tests," Office of Test Information, Nevada Test Organization, Las Vegas, Nevada, May 1, 1957, p. 28.

⁴⁰²LASL J Division memorandum LAS-J-1609 dated November 22, 1950 entitled "Desirability of an Area in the Las Vegas Bombing Range to Be Used as a Continental Proving Ground for Atomic Weapons."

⁴⁰³The Pacific Proving Ground, or PPG, was also known colloquially as the "picnic grounds." (Anders, FORGING THE ATOMIC SHIELD, p. 257.)

logistic support, and favorable year-round climatic conditions.⁴⁰⁴ Temperatures at Eniwetok generally range between 67⁰ and 97⁰ Fahrenheit.

Figure III-1 shows the major islands of Eniwetok atoll, with the code names assigned to them by the AEC. For the most part, the islands on the northern half of the atoll were given women's names, while those on the southern part of the atoll were named after men.

The names of the northern "feminine" islands were sequenced alphabetically in a clockwise order from the northwest corner of the atoll to the eastern side. The southern "masculine" islands names were sequenced alphabetically clockwise from the eastern side of the atoll to the western side.

The atolls were out of the way of most western Pacific typhoons, and the prevailing northeastern winds would not subject nearby inhabited islands to deposition of radioactive fallout from nuclear tests.⁴⁰⁵

The atolls are about 5,000 statute miles from the U.S. west coast, and 2,700 statute miles southwesterly from the Hawaiian islands.⁴⁰⁶

The Marshall Islands archipelago comprise 34 low-lying atolls and single islands scattered over 500,000 square miles in the southwestern central Pacific. The atolls are arranged roughly in two parallel rows running from the northwest to the southeast.

The westernmost chain is called the Ralik (Sunset) Chain, and Eniwetok is the most northerly of the Ralik Chain atolls. Bikini, in turn, is the northernmost atoll in the Ratak (Sunrise) Chain, the eastern atoll group. There are nearly 1,225 individual islands and islets in the Marshalls, comprising a total of only 70 square miles. Twenty-five of the 34 atolls are inhabited.

⁴⁰⁴Memorandum for the Record dated November 24, 1947 by Capt. J. S. Russell, USN, subject: Conference with Secretary of State, 1000, November 20, 1947.

⁴⁰⁵"Analysis of Sites," Enclosure "D" to JCS SM 1908 - Report by JPTC, October 18, 1947.

⁴⁰⁶Report of the Field Manager, Eniwetok Field Office, to the Manager, Santa Fe Operations, AEC, July 1, 1950 to July 1, 1953, p. 2.

Eniwetok is located at latitude 11°21' N and longitude 162°20' E; Bikini lies 165 miles almost directly to the east, at latitude 11°35'N and longitude 165°20'E. About 47 small islands comprise Eniwetok atoll, which is about 20 miles wide and 30 miles long. The chain of coral islands ringing the 388 square mile lagoon are broken in two locations by deepwater passages from the surrounding ocean into the lagoon.

The largest island at Eniwetok is only 2 1/2 miles long and 1,500 feet wide. Eniwetok island, at the southeastern edge of the lagoon, had been the location of a World War II U.S. bomber base. About 22 miles to the north, at the northern edge of Eniwetok lagoon, lay Engebi island, about 4,500 feet long; Engebi featured a 2,475-foot airstrip first occupied by Japanese and later by U.S. forces. Directly northeast of Eniwetok island was Parry island; during World War II, Parry island had been the site of U.S. naval seaplane and supply bases.⁴⁰⁷

Bikini atoll, also known as Eschholtz atoll, is oval in shape, measuring approximately 24 miles by 14 miles, with only one navigable passage into its lagoon. Bikini is an atoll containing 26 islands surrounding a lagoon, located approximately 11° north of the equator on the northwestern fringe of the Marshalls. All of the Bikini atoll islands comprised a total of only 2.332 square miles; the lagoon covers an area of about 229 square miles.

Figure III-2 shows the major islands of Bikini atoll and their AEC-assigned code names. Starting at the northwest corner of the atoll, the islands were named in a clockwise order after the U.S. military phonetic alphabet in effect in 1946.

Both Bikini and Eniwetok were chosen as test sites because they were far from densely populated areas and the prevailing winds would, it was believed, blow radioactive fallout from the test shots into the vast uninhabited stretches of the central Pacific ocean.⁴⁰⁸

⁴⁰⁷OPERATION SANDSTONE, NUCLEAR EXPLOSIONS 1948, Atomic Weapons Tests, Operation SANDSTONE, 1948, Annex I Part II, Volume I, Report to the Joint Chiefs of Staff, Section VIe, p. 2; "Anatomy of Eniwetok," Capt. Gilbert M. Billings, USAF, Historian, Task Group 7.1, JTF 7, April 1958.

⁴⁰⁸"Bikini Q's and A's," attachment to letter dated August 2, 1968 to Chairman Seaborg, Commissioner Ramey, Commissioner Tape, and Commissioner Johnson thru General Manager from John A. Harris, Director, Division of Public Information, USAEC, subject: Questions and Answers on Bikini.

The highest land on both atolls is only 12 feet above high tide; the lagoon at Bikini is not generally as calm as that at Eniwetok, due in part to the wide passage to the sea that the south end of Bikini atoll, and due also to the length of the lagoon lying parallel to the prevailing winds.⁴⁰⁹

Because of its position in the low latitudes, Eniwetok atoll has a tropical marine climate wherein the average mean daytime temperature is 81⁰ F. Day-to-night temperature variance is between 10 and 12 degrees. Humidity is also high, averaging about 85% year-round. The northern islands are drier and cooler and January through March are the driest months; July through November are the wettest.

The Marshalls are cooled by northeast trade winds, sea breezes that predominate from December to April. The islands have virtually no sources of fresh water.⁴¹⁰

At Eniwetok, Runit island was one of the major shot sites. Runit was approximately two miles long and averaged only 500 feet wide. No point on the island was more than 10 feet above sea level. Coconut trees were once cultivated on the island; in 1948, it was heavily covered by coconut trees. A 700-foot airstrip was constructed on Runit in 1948; this strip was later extended to 1,050 feet.⁴¹¹

The prime attraction of these islands lay in their good physical condition and a combination of location, layout, climate, and isolation. By 1953, the Pacific Proving Ground was a 150 by 350 mile area between latitudes 10⁰15' N and 12⁰45' N and longitudes 160⁰35' E and 166⁰16' E.⁴¹²

⁴⁰⁹Report of the Field Manager, Eniwetok Field Office, to the Manager, Santa Fe Operations, AEC, July 1, 1950 to July 1, 1953, p. 3.

⁴¹⁰OPERATION GREENHOUSE, Joint Task Force Three, pp. 6, 7 (this was an AEC-DOD orientation booklet provided to visitors to the Marshalls); ADVENTURES IN THE PACIFIC, Susanna Margolis, Sierra Club Books, San Francisco, California, 1988, pp. 374, 375.

⁴¹¹OPERATION SANDSTONE, NUCLEAR EXPLOSIONS 1948, Atomic Weapons Tests, Operation SANDSTONE, 1948, Annex I Part II, Volume I, Report to the Joint Chiefs of Staff, Section VIB, Chapter XIII, pp. 61, 62; Section VI-d, Chapter IV, pp. 8, 9.

⁴¹²Report of the Field Manager, Eniwetok Field Office, to the Manager, Santa Fe Operations, AEC, July 1, 1950 to July 1, 1953, p. 2.

Although far more expensive in terms of time, money, and logistics, nuclear tests were conducted in the Pacific to avoid heavy fallout over the continental U.S. and physical damage to nearby towns and cities. In general, nuclear devices fired in the Marshalls were much more powerful and generated far more fallout than devices fired in Nevada; the course of world history turned remote and tiny Pacific atolls into laboratories and proving grounds for some of the most advanced high-energy physics experiments ever conducted:

The Pacific tests of nuclear devices were conducted on coral atolls and in an environment that was altogether oceanic. It is conceivable that circumstances might have been otherwise, but the size and character of the devices and the time of their creation determined the course that events would take. Thus tiny flecks of land in the largest of oceans were drawn into the orbit of nuclear developments
...⁴¹³

Planning for GREENHOUSE

The next U.S. Pacific test series, Operation GREENHOUSE, was conceived even as the SANDSTONE tests were being completed in 1948. The AEC realized that its ambitious and dynamic weapons development program required more tests in the near future, both overseas and at a continental test site. Planning for GREENHOUSE began in the summer of 1948 at the Los Alamos Scientific Laboratory; in July 1948, J Division was established at LASL to assume responsibility for planning and conducting nuclear tests.⁴¹⁴

These test series were quite complex. Each series, or operation, comprised a number of shots, or individual test explosions. Each shot in turn included a number of associated diagnostic and effects programs and projects.

A program was a single series of connected technical measurements, usually concerned with a single phenomenon, such as a program to measure gamma radiation, air pressure versus time, long-range detection, etc. A project was a sub-component of a program, such as a free-air pressure

⁴¹³PROVING GROUND: AN ACCOUNT OF THE RADIOBIOLOGICAL STUDIES IN THE PACIFIC, 1946-1961, Neal O. Hines, University of Washington Press, Seattle, Washington, 1962, p. 20.

⁴¹⁴Memorandum dated September 3, 1948 to Alvin C. Graves from N. E. Bradbury, Subject: J Division Directive #1.

measurement project which might be part of a pressure versus time program.⁴¹⁵

The AEC initiated in the summer of 1948 a three-year research and development program aimed at readying three new types of experimental atomic weapons for full-scale test in 1951.⁴¹⁶

When the Los Alamos lab first considered the 1951 tests, it intended to fire three bombs. In the autumn of 1948 these devices or weapons were in various stages of study and research and represented fundamental elements of the AEC's weapons program.⁴¹⁷ One of these devices, the MK 5, was designed and built to utilize a more advanced implosion design, and the size of the bomb was reduced in dimensions without a comparable loss in efficiency. Another, the MK 6, was the latest FAT MAN weapon, an improved, higher-yield version of the MK 4 tested in 1948.

The first change in the plans of J Division for the new weapons test series resulted from laboratory research during the summer of 1949 and from the detection in September 1949 of the first nuclear explosion in the Soviet Union, which prompted the presidential directive at the end of January 1950 to accelerate the development of a thermonuclear weapon.

In July 1949, a joint civilian-military task force named JTF 3, for Joint Task Force 3, was formed. This organization was composed of representatives of the U.S. Army, Air Force, Navy, Marine Corps, Atomic Energy Commission, LASL, and civilian contractors. (By the time of the tests two years later, JTF 3 included nearly 9,500 men, six ships, 110 small boats, and 113 aircraft; the Korean War, which started in June 1950, put heavy demands on military manpower and equipment at the same time and led to more pressure for a continental test site that would lessen logistical problems and increase security).⁴¹⁸

⁴¹⁵"Definitions," AEC-MLC Meeting, January 23, 1952.

⁴¹⁶Enclosure "Q," Facts Bearing on the Problem, p. 91 in Report by the Joint Proof Test Committee to the Joint Chiefs of Staff on Armed Forces Participation in 1951 Atomic Weapons Proof-Testing, JCS 1998/13, 25 November 1949.

⁴¹⁷Memorandum for Information, Resume of Joint Proof Test Committee Planning Principles for Greenhouse, Director of Military Application, USAEC, November 18, 1949.

⁴¹⁸Letter dated August 4, 1950, from Secretary of Defense Louis A. Johnson to Gordon Dean, Chairman, U.S. Atomic Energy Commission; letter dated 22 August 1950 from Los Alamos Scientific Laboratory Director Norris E. Bradbury to

One of the experiments under study in August 1949 for the 1951 test program was "an experiment in which (a) nuclear explosion is (to be) used to send material down a tube and cause a thermonuclear reaction of small magnitude in deuterium."⁴¹⁹ (This became the GREENHOUSE George shot.)

The primary purpose of the 1951 test program at this time was "to prove, experimentally, new weapon designs and theories developed by the AEC's Los Alamos Scientific Laboratory."⁴²⁰

The name "Operation GREENHOUSE" was approved for the test series in August, and a major construction program began at Eniwetok in mid-1949. Among the many new structures erected was a large hydrogen and nitrogen liquefaction plant on Parry island, for the production of liquid deuterium and nitrogen for thermonuclear experiments.

The purposes of the test series were further defined in September:

The primary purpose of (GREENHOUSE) is to determine the characteristics of three new atomic weapons designed (to) increase effectiveness and efficiency (of use of fissionable materials). The performance of these weapons cannot be predicted by laboratory methods within the accuracy desired. The explosions also offer an

Brigadier General James McCormack, Jr., Director, Division of Military Application, U.S. Atomic Energy Commission, Washington, D. C. In addition to drawing off ships and aircraft that could have been used in the GREENHOUSE and IVY series of tests, troop requirements during the Korean War even dictated the type and firepower of weaponry diverted to Eniwetok to arm a permanent garrison force to protect the atoll from invasion. ("History of Operation IVY," Maj. F. E. Moore, Jr. and Lt. H. G. Bechanan, Joint Task Force 132, undated, p. 22.)

⁴¹⁹"Los Alamos Scientific Laboratory Program," p. 5, Enclosure A to memorandum dated 2 August 1949 to Norris E. Bradbury, Director, LASL, from Alvin C. Graves, J-Division Leader, LASL, Subject: 1951 Test Program. Rhodes, in DARK SUN (p. 457), quotes one LASL physicist as saying that "Teller's idea was that we needed to put this DT where we could see it, not in the middle of an explosion, so we'll pipe some radiation out to it."

⁴²⁰"Statement of Policies," attachment to Memorandum for Record dated 18 August 1949 from Col. Paul T. Preuss, USAF, Division of Military Application, USAEC.

opportunity to the military services to carry out research programs which could not otherwise be attained.⁴²¹

The test series was expected to include three shots, with the possibility of a fourth shot if a repeat test were necessary. Because of the nature of the models to be tested, greater emphasis would be placed on the detection of 14 MeV neutrons and measurements of alpha over a wider range than in previous tests.⁴²²

By the end of 1949, the target date for the first shot was early March 1951, with subsequent shots at two-week intervals.⁴²³ LASL still expected to fire three shots: a smaller and lighter implosion bomb (the TX-5), a uranium hydride-fueled implosion weapon, and an implosion device of the "booster" type.⁴²⁴ These three shots were named Dog, Easy, and George. At least one of the shots would also be a full-scale test of the MK 4 HE assembly.⁴²⁵

By mid-January 1950, LASL had re-run the calculations on the thermonuclear device to be tested during GREENHOUSE. While these calculations confirmed expectations that the test of the CYLINDER would succeed, and that at least a limited thermonuclear reaction would occur, the available evidence

⁴²¹Memorandum for General Landry dated 2 September 1949 from Col. E. J. Hale, HQ USAF, Subject: Assignment of Lt. General Quesada to the Joint Technical Planning Committee.

⁴²²Minutes of the Sixteenth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, September 22-23, 1949, p. 9.

⁴²³Memorandum for Information, Resume of Joint Proof Test Committee Planning Principles for Greenhouse, Director of Military Application, USAEC, November 18, 1949.

⁴²⁴Letter dated December 9, 1949 from Norris E. Bradbury, Director, Los Alamos Scientific Laboratory, to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, Subject: Los Alamos Scientific Laboratory Technical Program for Calendar Year 1950, pp. 1, 2, 3; Appendix to Enclosure "F," Memorandum for the Commander, Joint Task Force Three, p. 57 in Report by the Joint Proof Test Committee to the Joint Chiefs of Staff on Armed Forces Participation in 1951 Atomic Weapons Proof-Testing, JCS 1998/13, 25 November 1949; Borden-Walker H-Bomb Chronology, p. 21.

⁴²⁵Shelton, p. 4-4.

continued to indicate that the vital next step of igniting a mass of deuterium would fail.⁴²⁶

In February 1950, LASL made a decision to work for a four shot, rather than a three shot, program. So much progress had been made toward the realization of a fusion "gadget" that it seemed possible to include a thermonuclear device among the shots (the BOOSTER test shot was tentatively added to the GREENHOUSE agenda in the summer of 1950).⁴²⁷

In January and February, LASL had begun planning the design of a new cryogenic facility at Eniwetok and had started purchasing equipment for it. This gas liquefaction plant was to support both the BOOSTER experiment and any other tests that might require the use of cryogenic fusion fuels. Because of slow delivery of highly specialized equipment, a full year was required to specify, purchase, ship, construct, and install the components of the low-temperature facility. (Only by the end of 1950, when construction at Eniwetok was well underway, was sufficient information available to estimate cryogenic production requirements with reasonable certainty.)⁴²⁸

By April, J Division had revamped its plans for GREENHOUSE. In June, the shot days for the operation were tentatively scheduled and designated as Dog, Easy, George, and Item in order of time sequence of firing, regardless of the exact order in which the nuclear devices were to be fired. Soon after, calendar dates were chosen for the shot dates, and locations assigned. Dog would be fired on March 1, 1951 on Engebi island at Eniwetok; Easy on March 15 on Bogallua; George on March 29 on Eberiru; and Item on April 12 on Runit.

Although GREENHOUSE had several objectives, the test program had three aspects that were paramount. These were summed up in a letter dated July 31, 1950 to Secretary of Defense Louis A. Johnson from AEC Chairman Gordon Dean:

- (1) The results of the planned shots (will) have an immediate bearing on the quality of the U.S. stockpile of atomic bombs.
- (2) The test program (is) planned to develop an important body of information regarding the blast and radiological effects of atomic weapons.

⁴²⁶Borden-Walker H-Bomb Chronology, p. 61.

⁴²⁷AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 283.

⁴²⁸WT-50, p. 2.

- (3) The results of two of the shots (are) expected to make a direct and significant contribution to the currently-inadequate understanding of the technical and economical feasibility of a thermonuclear weapon.⁴²⁹

In addition, two new advanced weapon designs were to be proof-tested; if testing were delayed, it could have significant effects on the American nuclear weapons stockpile. In April 1949, the GAC had reviewed LASL studies that showed that the new "Fox" composite core would provide a more effective use of fissionable material and would reduce the likelihood of predetonation, especially when compared to stockpiled SANDSTONE X-Ray bomb cores:⁴³⁰

With regard to the bearing of the planned tests on the United States stockpile, two of the several shots under consideration for GREENHOUSE are fission bombs which, if proven successful, will be introduced into stockpile at the earliest possible date.

Both of these bombs will have new nuclear configurations which, in combination with newly designed non-nuclear components, will mark a considerable extension of the known weapons art. One of these bombs, having about the same size and weight as the standard bomb now being stockpiled, is expected to give an energy release at least twice that of the present design with the same amount of nuclear material. (Author's note: a reference to the MK 6.)

The other bomb will be substantially smaller than the present design, less than half as heavy, and is expected to give an energy release about 50% more than the present design with the same amount of nuclear material. (Author's note: a reference to the TX-5.) The primary importance of this latter bomb is that it may be carried by smaller aircraft than are required to carry the present stockpile weapon, and for the same reason will be

⁴²⁹Letter dated 31 July 1950 from USAEC Chairman Gordon Dean to Secretary of Defense Louis A. Johnson.

⁴³⁰Letter dated April 7, 1949 to D. E. Lilienthal, Chairman, USAEC, from J. R. Oppenheimer, Chairman, GAC.

better adapted for use in the most promising guided missiles now under development. It should therefore increase the flexibility of use of atomic weapons, as in tactical operations in the field.

The alternative to testing these weapons would necessarily be either to stockpile presumably improved but untested weapons or else to continue more conservatively to stockpile weapons believed far less effective than the new available designs.

We wish to emphasize further that the fissionable core planned for use in the test of the smaller bomb mentioned above is now being manufactured for use in the present standard nuclear configuration and non-nuclear assembly. This core is fabricated from uranium and plutonium in proportions different from any design of core that has been proof-fired. It is being manufactured for stockpile because it affords a significant increase in the effectiveness of utilization of uranium and plutonium in the percentages of production of these materials at Oak Ridge and Hanford.

In approving this core for manufacture, the Department of Defense pointed out the desirability of detonating it in a test before its number exceeds 50% of the total weapons in stockpile. This time will actually occur before the spring of 1951.⁴³¹

A new type of composite core, the Type F or "Fox" core,⁴³² was slated for test early in 1951. Calculations had shown that the Fox core would provide a significant increase in total stockpile yield; with addition of only a small percentage of additional fissionable material, the yield of the Fox core could be raised even further.

⁴³¹July 31, 1950 Dean letter to Johnson (emphasis in original); Minutes of the Fourteenth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission June 2-4, 1949, p. 16.

⁴³²Letter dated January 22, 1952 to Senator Brien McMahon, Chairman, JCAE, from Thomas E. Murray, Commissioner, USAEC.

By August 1950, there was a military requirement that a certain percentage of the national stockpile consist of these higher-yield cores.⁴³³

Late in August, when requirements for men, ships, aircraft, and materiel to fight the Korean War which had begun two months earlier began to endanger LASL's test plans for Eniwetok, the laboratory further altered its test program.⁴³⁴ The lab noted that:

... the nuclear test program proposed for Operation GREENHOUSE has not only an immediate bearing on the stockpile position of the United States, but is also expected to make a direct contribution to the rapid understanding of the technical and economic feasibility of a thermonuclear weapon program.⁴³⁵

The test program proposed by Los Alamos must always be based on considerations of maximum technical advance as well as on the somewhat more mundane considerations of immediate stockpile relevance. The best test program, presumably, capitalizes as much as possible on both of these premises.

It has been noted time and again both that the national stockpile should not be based upon untested weapon techniques, and that maximum progress along weapon development lines is only possible when careful physical experimentation can be done in connection with full-scale atomic detonations.

At the present time, we find the country faced with a real probability, even if of

⁴³³Letter dated August 21, 1950, from Sumner T. Pike, Acting Chairman, U.S. Atomic Energy Commission, Washington, D.C., to Lt. General Elwood R. Quesada, USAF, Commander, Joint Task Force Three, Washington, D.C.

⁴³⁴Memorandum dated July 20, 1950 to Carroll L. Tyler, Manager, Santa Fe Operations Office, USAEC, from Brig. Gen. James McCormack, Jr., Director, Division of Military Application, USAEC, Subject: GREENHOUSE.

⁴³⁵Memorandum dated 21 July 1950 for Brig. General James McCormack, Jr., through Carroll L. Tyler, from N. E. Bradbury, Director, LASL, Subject: GREENHOUSE.

uncertain magnitude, that Eniwetok will not be available for atomic weapon testing in the spring of 1951. To the Los Alamos Scientific Laboratory, it appears almost fantastic that, precisely at a time in international relations when the most rapid progress should be made in this field, that such a denouement can be contemplated without the gravest concern.

We recognize the good intentions of the Chiefs of Staff, but we have no illusions as to the priority which they have granted and will have to continue to grant to "police actions" in Korea and potentially elsewhere. In plain fact, only the USSR can really determine whether or not we can test at Eniwetok in 1951 or any other time in the near future.

It has been apparent that the problems of support by the Joint Chiefs would be somewhat easier if the needs of the Laboratory at Eniwetok could be reduced or minimized. Although it could be argued that our test program should be enlarged and expanded in view of the deterioration of the international situation; nevertheless, the tendency so far has been of a different nature.

In any event, we have agreed on a reduction in the number of test sites to be prepared at this time to three, only one of which will be capable of full thermonuclear instrumentation.⁴³⁶

There was also some concern about conducting nuclear tests at a time of open hostilities, a situation that had not arisen since 1945, which resulted in extensive White House-AEC-DOD discussions regarding the logistic and political feasibility and timeliness of GREENHOUSE. (These deliberations culminated in a approval of the series by President Truman on February 13, 1951.)⁴³⁷

⁴³⁶Letter dated 22 August 1950 from Norris Bradbury, Director, LASL, to Brig. Gen. James McCormack, Jr., Director of Military Application, USAEC. In July, the JCS wanted to divert ships needed for GREENHOUSE to Korea to help with a blockade; by September 1950, the JCS had abandoned this plan. (Anders, FORGING THE ATOMIC SHIELD, pp. 65, 70.)

Plans to study the practicality of initiating fusion by firing a tritium-deuterium fueled gun-type device were dropped; a thermonuclear cylindrically-imploded device called the CYLINDER, which seemed to offer greater flexibility, adaptability, and economy of fissionable material than a gun-type approach to the problem, was substituted.⁴³⁸ The CYLINDER was designed mainly as a test mechanism to transfer radiation energy from a fission device to ignite a small amount of a liquid D-T mixture.⁴³⁹

The final design of the CYLINDER had evolved essentially between March and July 1950; by the summer of 1950, all fundamental cryogenic features of the device were incorporated.⁴⁴⁰

Shots scheduled to test the TX-5 and the "How Prime" or "How Double Prime" devices (the high-yield TX-6) were retained. These were tests mainly of stockpile interest, and required shooting at the earliest possible time in 1951 under conditions of great reliability and accuracy. Depending upon the results, decisions would be made to retrofit new cores to the entire national nuclear weapons stockpile.

A test of the TX-5 with its 92-point ignition system could, in principle, be delayed until the fall of 1952 if no considerations other than its scheduled stockpiling early in 1952 were involved. However, the general behavior of a 92-point system was of great interest and applicability to both larger- and smaller-diameter implosion systems than the TX-5. Accordingly, a careful test of the TX-5 was highly desirable for the spring 1951 test series.

In addition, the laboratory began to plan for a continental U.S. test site, which eventually became the Nevada Proving Ground. The feasibility of an air drop of the "How" weapon in

⁴³⁷Memorandum dated September 3, 1954 to W. B. McCool, Acting Secretary, USAEC, from Harold D. Anamosa, Documentation Secretary, Subject: Mr. Strauss' Memorandum Regarding Presidential Directive to Hold No Further Tests; memorandum for Mr. Strauss dated September 7, 1954 from W. B. McCool, Assistant Secretary, USAEC.

⁴³⁸Letter dated 5 October 1950 to Honorable Brien McMahon, Chairman, Joint Congressional Committee on Atomic Energy, from Sumner T. Pike, Acting Chairman, USAEC. The CYLINDER may also have been known as the SILEX device. (WT-102, pp. 32, 34.)

⁴³⁹Hirsch and Mathews, "Fuchs and Fallout," pp. 5, 23.

⁴⁴⁰WT-50, pp. 30-33.

MK 4 geometry off the west coast of the U.S. was also examined: the Air Force also wanted to test the new Fox core in an airburst.⁴⁴¹

In early September, the Joint Chiefs of Staff directed Joint Task Force 3 to proceed with Operation GREENHOUSE without compromising weapon development.⁴⁴² The JCS approved a revised plan for GREENHOUSE that called for a delay of 45 days in the first shot, from 1 March 1951 to on or about 15 April 1951. The Korean War was causing delays in the shipment of materiel to the Marshall islands.⁴⁴³

LASL held high hopes for the new Fox core, in particular, that it might lead to a 30 to 50 percent increase in stockpile size:

After expressing these views, the General called attention to some new thinking that has begun to emerge from Los Alamos only very recently, and which is not reflected as yet in any formal documents from the Laboratory (sic). It has long been known that by varying the amount of fissionable material in the core of an implosion weapon, the energy release might be varied over quite a wide range.

[deleted]

This increase in numbers should be 30% or more, and could be substantially more than 50%, depending on the military choices which may be made as to numbers of weapons desired in various ranges of energy release.

It is important to note that these new developments, if successful, may be used in the test now planned for the spring of

⁴⁴¹Letter dated 22 August 1950 from Norris Bradbury, Director, LASL, to Brig. Gen. James McCormack, Jr., Director of Military Application, USAEC.

⁴⁴²WT-48(EX), p. 76.

⁴⁴³History of Air Task Group 3.4, Provisional, 1 November 1950 - 31 January 1951, p. 2; attachment dated September 12, 1950 to letter of same date to William L. Borden, Executive Director, JCAE, from Brig. Gen. James McCormack, Jr., Director of Military Application, USAEC, p. 5.

1951, and making these improvements available for stockpile may not require manufacturing new non-nuclear assemblies, which is a time-consuming task. If the developments are completely successful, all that will be required will be to refabricate the pits and cores and to substitute the new designs in the existing high explosives assemblies in stockpile.

The tie-up and burn-up of U-235 in the H-bomb tritium program combines with aggravated world conditions to highlight the advantage of quickly gaining greater numbers of weapons from a given amount of (fissionable) material.⁴⁴⁴

In the early fall, it appeared that many of the problems of dewar design (dewars, large insulated bottles, held liquid gases) and telemetry components and filling apparatus for the BOOSTER depended upon their physical relation to each other in the shot cab. Detailed mockups of the BOOSTER cab and all of its equipment were constructed at the lab; this mockup resolved many design problems and assisted operational plans.⁴⁴⁵

Planning for GREENHOUSE was also hindered at this time by competition between the fission and fusion weapons programs for the relatively-limited computing facilities at LASL and elsewhere in the U.S.⁴⁴⁶ Both projects enjoyed almost the same priority; the Korean War had emphasized the need for small tactical fission bombs at the same time that the U.S. government's fear of a Soviet invasion and general nuclear war in Europe reinforced development of thermonuclear weapons.

Given this reasoning, it appeared that work had to move ahead on all possible technologies, regardless of conflicts over resources:

General McCormack climaxed his remarks by indicating that, with more strategic

⁴⁴⁴Attachment dated September 12, 1950 to letter of same date to William L. Borden, Executive Director, JCAE, from Brig. Gen. James McCormack, Jr., Director of Military Application, USAEC, pp. 2, 3. Between 1951 and 1952, the U.S. nuclear weapons stockpile almost doubled, from 438 to 832 weapons.

⁴⁴⁵WT-50, p. 2.

⁴⁴⁶Letter dated September 13, 1950 from J. Robert Oppenheimer, Chairman, GAC, to Gordon Dean, Chairman, USAEC.

weapons using less (fissionable) material per unit in prospect, and with promising tactical weapons under development -- all considered against the background of world conditions -- a question arises as to how these projects fit in relative priority with the hydrogen program.

There is no really serious conflict between the programs at the moment, but it seems clear that there is a possibility of a conflict in the future, perhaps by sometime next year. A year ago the Los Alamos Laboratory (sic) was estimating that continued work on fission weapons might increase the effectiveness of material available by a factor of, say, two; and this was a prime reason for desiring to transfer major efforts to the thermonuclear field.

Now that the calculations of a year ago appear to have underestimated the potential of fission weapons, the question of priorities becomes more pointed. It now seems, for instance, that by use of a not-unreasonable amount of fissionable material in a new design of fission weapon, energy releases may be very high indeed. Perhaps the amount of material contained [deleted]

The trend of Los Alamos thinking on the H-bomb is toward the conclusion that this device may require [deleted] tritium and that, due to the not entirely foreseen difficulty of propagating the reaction in light elements, [deleted]

If such thinking is corroborated by further study, the price of a Super bomb might be, [deleted] The General stated, however, that the investigation to date tends to confirm, from the theoretical point of view, the feasibility of initiating a thermonuclear reaction, although it is still impossible to say whether or not a thermonuclear weapon, as such, is feasible; further, that Los Alamos thinking may swing back to a lower estimate of the tritium required; and that whatever the outcome, the program must be pushed ahead.

We must determine as best we can the facts regarding the Super and what we might be

able to accomplish, so that we may know what Russia may accomplish.

At this time, the AEC's Director of Military Application was quite fearful of an imminent general atomic war between the U.S. and the USSR. This was a recurring nightmare of many U.S. government officials for the next few years, and was one of the many driving forces behind the development of American hydrogen weapons:

Threaded throughout the foregoing discussion were references to Russia's weapon policy and the danger that she might strenuously seek to increase her number of bombs by using less (fissionable) material per unit. This, the General felt, would be a logical step for Russia to take, [deleted] he hardly sees how the United States could pass through a war begun after 1954 without absorbing ruinous atomic punishment. [deleted]

If military incentive to delay war diminishes, and if political incentive to begin war increases, then there is reason to worry about an earlier date for a war than we might have predicted previously. This reasoning serves to solidify the General's belief that we must make every effort to enhance our own weapons position during 1951 and 1952.⁴⁴⁷

In mid-October, LASL advised the AEC that a "multi-crit" gun device might not be ready for firing in the second GREENHOUSE shot; the laboratory expected to make a final decision in January 1951 as to whether the device would be fired. By the latter part of the month, GREENHOUSE included five shots at Eniwetok atoll: Dog on April 14, 1951; Easy on May 5; George on May 19; Item on June 2; and King, an airdrop of a "half crit" test weapon from a B-50D over Bogallua island, on June 16.

Item was to be fired only if Easy failed; if Easy succeeded, then it was to be repeated as a full-scale test of the complete weapon system, and not just its nuclear components. King was to be detonated at an altitude of 4,000 feet with a bombing (release) altitude of 35,000 to 40,000 feet.

⁴⁴⁷Attachment dated September 12, 1950 to letter of same date to William L. Borden, Executive Director, JCAE, from Brig. Gen. James McCormack, Jr., Director of Military Application, USAEC, pp. 5-9.

Rehearsals for King might include the dropping of either inert concrete-filled "pumpkins" (dummy FAT MAN bombs) or surplus high-explosive-filled MK 4 cases that would be exploded in the air over Bogallua.⁴⁴⁸ The BOOSTER was to be fired only if the CYLINDER were not successful, a probability then considered quite small.⁴⁴⁹

There was some debate at this time over which version of the "How" weapon to fire during GREENHOUSE. By this time, two competing types, the "How Prime" and the "How Double Prime" were available for test. The "How Double Prime" was favored because of the preponderance of Fox cores in the stockpile and the likely higher-yield of the "Double Prime."⁴⁵⁰

Dates and locations of tests fluctuated for the next several months. Construction of test effects structures for shot Easy began in 1950; by year's end, over 75% of the buildings were complete.

By November, the MLC viewed the forthcoming Eniwetok tests as a demonstration of the greatly widened range of military choices which now confronted the Defense Department: between small weapons containing relatively little fissionable material and capable of being carried by fighter aircraft, on the one hand, and on the other, large weapons containing much fissionable material and generating perhaps 16 times as much energy release as the small weapons.⁴⁵¹

In November and December, DOD was giving a good deal of thought to the air defense of Eniwetok. Before this, it had not been believed necessary to provide air cover; military ground personnel on the islands and naval surface forces had been presumed adequate. After SANDSTONE ended, a 74-man

⁴⁴⁸HISTORY OF OPERATION GREENHOUSE, 1948-1951, Lt. General Elwood R. Quesada, Joint Task Force Three, WT-48 (EX), undated, pp. 75, 76, 77; Headquarters, Task Group 3.1, LASL memorandum dated 23 October 1950 from Chief of Staff, TG 3.1, to Distribution, Subject: Notes on Meeting at Los Alamos Scientific Laboratory, 18 October 1950. The "half crit" weapon is probably one of the "fractional crit" weapons tested during the RANGER series of nuclear detonations in Nevada in early 1951.

⁴⁴⁹Memorandum for the Military Liaison Committee dated November 8, 1950 from Gordon Dean, Chairman, USAEC, Subject: Revisions to GREENHOUSE Schedule.

⁴⁵⁰Los Alamos Laboratory Program for Calendar Year 1951 and Fiscal Year 1952, November 17, 1950.

⁴⁵¹Memorandum for the Files dated November 28, 1950 from Bill Borden.

garrison occupied Eniwetok in what was probably some of the least-desirable and loneliest military duty in the world.⁴⁵²

However, the Chinese invasion of Korea in November and a concurrent buildup of Soviet forces in the Far East raised the possibility that there might be airborne interference with the tests, either by aircraft and/or paratroops. The Air Force took steps to obtain aircraft for defense, and six F-80C fighter aircraft were assigned to Eniwetok.⁴⁵³

The GREENHOUSE schedule approved on December 7 included the Dog shot on April 7 at Runit; Easy at Engebi on April 18; George on May 9 at Eberiru; the King airdrop on May 16 over Bogallua; and Item on June 2 at Engebi.⁴⁵⁴ Dog shot was to use a tower-mounted TX-5; the King airdrop was to be a MK 6 test; and the Item shot was to be the BOOSTER detonation. The George shot would be the CYLINDER; the BOOSTER was not to be fired if results of the CYLINDER test were sufficiently conclusive.⁴⁵⁵

Near the middle of December 1950, the AEC informed the JCAE that "it should be recognized that the test at Eniwetok in the Spring (sic) of 1951 will necessarily be an intermediate step, perhaps only the first of several intermediate steps, in the development of a hydrogen bomb, and may well not provide a clear-cut answer to the question of whether or not it is possible to construct a hydrogen bomb."⁴⁵⁶ This statement contrasted sharply with Teller's conversation in mid-November 1949 with Brien McMahon and Robert LeBaron that test of the BOOSTER would prove the feasibility of the Super and accomplish most of its engineering development at the same time.

⁴⁵²Atomic Energy Commission, Responsibilities Associated with the Maintenance of Eniwetok Proving Ground and Conduct of Future Test Operations, Report by the Director of Military Application, USAEC, attachment to AEC 334/10, August 17, 1951.

⁴⁵³History of Air Task Group 3.4, Provisional, 1 November 1950 - 31 January 1951, p. 6.

⁴⁵⁴History - Task Group 3.1, 1 November 1950 - 31 January 1951, Capt. Albert S. Knauf, USAF, p. 2.

⁴⁵⁵Progress Report to the Joint Committee on Atomic Energy, June through November 1950, Part III, Weapons, December 15, 1950, p. 15.

⁴⁵⁶U.S. Atomic Energy Commission, Progress Report to the Joint Committee on Atomic Energy, June through November 1950, Part III, Weapons, December 15, 1950, p. 3.

By the beginning of January 1951, the construction of a gas liquefaction plant at Eniwetok was well along. The facility was located on the lagoon side of Parry island just south of the island's airstrip and occupied an area of 310 by 280 feet.

Three principal units comprised the cryogenic plant. A liquid nitrogen plant rated at 50 liters/hour supplied nitrogen which was used as a precoolant by the liquid hydrogen plant. The main liquid hydrogen unit was acquired from Navy surplus; when the AEC bought it, it was standing idle on the grounds of the Aerojet Engineering Corporation at Azusa, California, where it had been built under Navy auspices to assist the company in the development of liquid fueled rocket engines. The basic design of the hydrogen liquefier was the work of Dr. Herrick L. Johnson at Ohio State University. The plant was rated at 30 liters/hour.

For the next two years, until the construction of a new National Bureau of Standards liquefier at Boulder, Colorado, this used Navy equipment would be the largest hydrogen liquefier in the world. This hydrogen liquefier assumed a new and major importance for the AEC's H-bomb program between 1950 and 1953, after the Navy had terminated its hydrogen-fueled rocket program. Much of the liquid hydrogen used in early large-scale cryogenic experiments which led to the engineering of the cooling system around the Mike device tested during Operation IVY in November 1952 would come from this Aerojet liquefier.

The third cryogenic unit at Parry island was a Collins helium cryostat which could be converted into a hydrogen liquefier rated at a capacity of two liter per hour. This unit was considered an emergency standby in case the main hydrogen plant failed completely. The helium cryostat was manufactured by Arthur D. Little, Inc. of Cambridge, Massachusetts.⁴⁵⁷

Operation RANGER: Prelude to GREENHOUSE

One event that had major effects on the planning for weapons to be tested during GREENHOUSE was Operation RANGER conducted at the Nevada Proving Ground in early 1951.⁴⁵⁸

⁴⁵⁷WT-50, pp. 2, 5, 26.

⁴⁵⁸RANGER was originally named Operation FAUST. (Operation GREENHOUSE, Nuclear Explosions 1951, Scientific Director's Report, Annex 9.1, Staff Reports; Annex 9.4, Documentary Photography, WT-98 (EX), Los Alamos Scientific Laboratory, Los Alamos, New Mexico, p. 36.; letter dated January 16, 1951 to John C. Clarke, LASL, from E. F. Cox, Manager, Weapons Effect Department re Request for Weapons Effects Experiments Observers at Faust.) This initial name may have been

In November 1950, LASL discovered that insufficient data was available to determine satisfactory design criteria for nuclear devices to be tested during Operation GREENHOUSE. Recent theoretical hydrodynamic studies and small-scale experimental results had given rise to concerns about the reliability of yield estimates and yield reproducibility of the more advanced hydrodynamic models destined for eventual stockpiling and test during GREENHOUSE.

Variations in core compression might significantly affect predicted GREENHOUSE yields. Some type of test was deemed advisable to determine stockpile confidence, and to better define the choice of models for GREENHOUSE shots and help interpret results of the shots.

LASL decided that a series of small nuclear tests was required to gather data related to the variation of yield with compression of the fissile pit.⁴⁵⁹ These tests would involve explosions on the order of a few kilotons, perhaps less than one kiloton in some instances. The experiments would produce early information on the anticipated performance of the pits, cores, and high explosive ignition systems to be used during GREENHOUSE, and would also clarify some important uncertainties about expected yields vital to GREENHOUSE planning.⁴⁶⁰

Following meetings of the LASL Fission Weapons Committee on December 6 and 11, the laboratory recommended, and the AEC agreed, that five airburst tests should be staged within two months, starting on or about January 25, 1951.⁴⁶¹ The shots would be staged at the AEC's new continental test site in

indicative of the price the scientists were willing to pay to learn the secrets of the atom.

⁴⁵⁹Operation RANGER, Shots ABLE, BAKER, EASY, BAKER-2, FOX, 25 January - 6 February 1951, DNA 6022F, Defense Nuclear Agency, Washington, D. C., 1982, p. 18; Los Alamos Laboratory Program for Calendar Year 1951 and Fiscal Year 1952, November 17, 1950; Shelton, p. 4-9; U.S. Atomic Energy Commission, Quarterly Progress Report July - September 1949, Report to the Joint Committee on Atomic Energy, November 4, 1949, p. 7.

⁴⁶⁰Memorandum for the Chairman, Military Liaison Committee, from Gordon Dean, Chairman, USAEC, Subject: Possibility of Pre-GREENHOUSE Low Yield Atomic Tests, USAEC, Washington, D. C., December 20, 1950; WT-98 (EX), p. 36.

⁴⁶¹Furman, p. 571; OPERATION RANGER, Volume 1, Report of the Deputy Test Director, WT-206(EX), Los Alamos Scientific Laboratory, University of California, Nevada Test Site, January-February 1951, p. 18.

Nevada. In connection with Operation SANDSTONE, plans for a continental test site had been made as early as the spring of 1947; a number of sites were surveyed as candidates, including locations on the coast of North Carolina.⁴⁶² The Las Vegas Bombing and Gunnery Range was finally chosen as a continental U.S. nuclear test site.

The stated purpose of RANGER was as follows:

During the past year or two, improvements in fission weapons has been achieved by more advanced hydrodynamics [deleted] this has resulted in calculated yields that a year ago were quite unexpected, and, as is natural when theory goes so far beyond experiment, there has been a serious question of the accuracy of the calculations.

The resolution of these questions has been given urgency by the desirability of having accurate yield predictions before Operation GREENHOUSE and the desirability in general of being sure of what one is doing in fission weapon design.

The purpose of Operation RANGER is to settle, by a series of radical nuclear explosions, the present questions concerning the theory of the implosion method. The questions are primarily two [deleted]

[deleted]

(There are) four models available for the tests ... The so-called "Model A" involves [deleted] It is believed that an explosion of the size calculated can be achieved only if the densities are approximately as calculated and the cross sections employed are approximately correct.

Model B consists of [deleted]

[deleted]

⁴⁶²Letter dated 19 April 1947 to Dr. Joseph O. Hirschfelder, Dept. of Chemistry, University of Wisconsin, from N. E. Bradbury, Director, LASL; letter dated June 9, 1948 to RAdm. W. S. Parsons, USN, from N. E. Bradbury, Director, LASL.

The tentative schedule of shots prepared by the Fission Weapon Committee is: A, B, B, E, F. As shots are made, this may have to be altered in view of the results in order to provide the necessary information in the most economical manner.⁴⁶³

All that was made public about RANGER's purpose was that it was "based on the urgent necessity for certain laboratory data essential to continued progress in weapons development."⁴⁶⁴

The first plans for RANGER included possible compression and initiation timing tests of five different combinations of pits in various high explosive geometries. There were by this time six different pits designs, designated Types A, B, C, D, E, and F, and at least four HE geometries, the MKs 4, 5, 6, and 7.

Shot "A" would be fired first; if it were a dud, it would be fired again. If still a dud, the Type D pit would probably not be stockpiled. If the yield and alpha of Shot "A" were such that the indicated compressions were within 25% of those calculated, two successive "B" shots would be fired next.

This was the first time that two identical bomb models would be detonated in a test series. While the CROSSROADS Able and Baker tests both used Model 1561 MK 3A bombs, they were not identical, and CROSSROADS was a weapons effects series, not a weapons development test series.

If the yields and alphas of the two "B" shots were not reasonably close to each other, or if both were consistently very low, the logical conclusion would be that asymmetric implosions leading to timing problems with the new TOM initiator were too great for stockpiling purposes, and the test operation would be ended.

If the two "B" tests agreed fairly well and indicated correct timing within about 0.2 microseconds and compressions within about 25% of those calculated, the "E" shot would be fired next. Regardless of the results of "E," the "F" shot would follow next.⁴⁶⁵ All shots would use Type D pits; at least two

⁴⁶³WT-202, p. 13.

⁴⁶⁴Letter dated January 29, 1951 to Honorable Millard Caldwell, Federal Civil Defense Agency, from Gordon Dean, Chairman, USAEC.

⁴⁶⁵WT-206(EX), p. 20.

would use the new 92-point HE implosion system. RANGER would also include the first tests of the new TOM initiator.⁴⁶⁶

The major technical purposes of the tests were fourfold and affected development of future MK 5 and 6 weapons, as well as the whole GREENHOUSE test program and the future composition of the U.S. nuclear weapons stockpile. The tests were expected to produce a number of advantages and to provide a sounder experimental basis for GREENHOUSE:⁴⁶⁷

1. An early experimental determination of the efficiency of the Type D pit in MK 4 HE prior to large scale stockpile reconversion. It is expected that the results of the tests can be interpreted to give at least a very strong indication of the performance of this system. The tests may even be conclusive on this point.
2. The attainment of a sound experimental basis for designing the stockpile model of the [MK 5 weapon] in time for its test at GREENHOUSE. The February Mercury (Nevada) tests, as envisaged at Los Alamos, hold possibilities ranging all the way from a conclusion that [deleted] geometry because of poor high explosive behavior or other factors, to the complete proof test for stockpile.

The latter possibility would be the case only if the proposed [Fox core] is tested and found to give a yield appropriate for military requirements, but, in any case, favorable results of the proposed tests will form a sound basis upon which to design a stockpile model. If so, there appears to be just sufficient time to construct such a model for proof test at GREENHOUSE.

At the same time, tooling-up for the production fabrication of the model could be accomplished so that large scale production could begin immediately following a successful GREENHOUSE test.

⁴⁶⁶Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 5.

⁴⁶⁷Minutes of AEC Meeting 513, January 2, 1951.

Such tooling-up, based upon our present knowledge and in view of known uncertainties, cannot be undertaken with much assurance that the behavior of the model will conform closely with the desired characteristics.

3. The provision of a sound experimental basis for obtaining the maximum value from Operation GREENHOUSE with a minimum expenditure of fissionable material. This objective includes two somewhat different considerations.

In the first place, we are most anxious to provide a test model for Engebi island which will not only have a high probability of yielding the pressures desired for the several million dollar effects program, but which will also be of value in determining the nature of the stockpile in the near future and in supplying basic information pertinent to future implosion weapon development.

Without the Site Mercury tests it may even be necessary to use a Type C pit in order to give high assurance of providing a yield within the specified limits. With the aid of the proposed February tests, it may be found advisable not only to test an advanced hydrodynamic model, but possibly also to gain information on some important side question such as, for example, the effect of reduction of plutonium content in composites below the [deleted] now used.

In the second place, the results of the February tests will be of enormous aid in the basic understanding of GREENHOUSE observations, especially if the behavior of the GREENHOUSE models differ appreciably from prediction. Such understanding can point the way toward improvement.

In addition, it is conceivable, for example, that the performance of the Type D pit in MK 4 HE is either so poor on the average or so variable, shot to shot, that a single test of say "How" or a [deleted] would be essentially meaningless. If this unfortunate supposition should be true, not only would the material in a GREENHOUSE shot not preceded by the proposed February tests be

wasted, but we might easily fail to recognize that the shot was not representative of the model and therefore base the design of a large part of the stockpile upon a misconception.

Although the probability that the designs are this bad may not be very great, the stakes are so high it seems important to be quite confident (that) we are on the right track.

4. An experimental determination and, where appropriate, comparison with theoretical prediction, of certain basic physical quantities directly affecting yields. The quantities of concern are the [alpha, core compression, and transit time] from other statistical factors affecting the time of initiation, and the effect of composition upon yield. ... ⁴⁶⁸

By the end of December, a series of drop tests of up to five devices was contemplated for the first part of February 1951 in Nevada. The shots were to use TOM initiators, and include yields of 1.5, 2, 9, and 35 to 40 kilotons; the latter shot would be a test of the "Fox" core in a Type D pit. The different test devices were designated A, B, E, and F; the recommended order of firing, assuming success of all models, was to be A, B₁, E, B₂, and F. The b shot was to be repeated to verify yield reproducibility.⁴⁶⁹

The F, or "Fox" shot, was to give information on the effect of composition in a well-determined hydrodynamic system with accurately-known initiation time, and to give direct information on the performance of a "fractional crit" model similar to a design that was a candidate for stockpiling.

The choice of weapon models for firing during GREENHOUSE depended to a critical extent on a number of theoretical calculations, and on estimates of the effect of expected variations in the behavior of materials inside the bombs during explosion. Minute variations in behavior could have very great effects on the performance of the weapons; it was not possible by numerical modeling at LASL to make an experimental check on all possible behavior variations, so

⁴⁶⁸Atomic Energy Commission, Pre-GREENHOUSE Test Program, Memorandum by the Director of Military Application, USAEC, December 29, 1950.

⁴⁶⁹WT-206(EX), p. 21.

full-scale tests were required. The ultimate design of the "fractional crit" cores to be tested during GREENHOUSE would depend upon the results of Operation RANGER.⁴⁷⁰

A second technical area where additional information was urgently required was the effect of new bomb designs on the U.S. stockpile. During the course of developmental work in 1949 and 1950, the TX-5 had been designed, which was both substantially smaller and lighter, and more efficient in its use of fissionable materials, than the MK 4, which made up the bulk of the national nuclear weapons stockpile at the beginning of 1951. The TX-5 was slated for testing during GREENHOUSE.

It had also become apparent by this time that the Type D pit for the TX-5 was also applicable to the MK 4. If it were adaptable to the MK 4, then its substitution into the stockpile would be a relatively simple way of improving at an early date the U.S. potential in atomic weapons, mainly by improving both the yield and the nuclear efficiency of existing MK 4 weapons.

Total stockpile yield could potentially double without adding more fissionable material to it, by increasing the yield of stockpiled cores and by using substantially less nuclear material in each core to obtain the same yield. Vital data on the application of the Type D pit to the stockpile could be secured by exploding a few devices containing relatively small amounts of fissionable material.⁴⁷¹

If the "F", or "Fox" shot, were successful, it could be regarded as a complete proof-test for stockpiling and save about 3 1/2 months in its testing and stockpiling.⁴⁷² Favorable results of this shot would form a sound basis upon which to design a stockpile model. All RANGER shots would use the Type D pit; only the cores would vary in their composition (in this respect, the RANGER tests were similar in their controlled nature to the SANDSTONE shots in 1948; see Volume II).⁴⁷³

⁴⁷⁰WT-98, p. 36.

⁴⁷¹Memorandum dated January 4, 1951, for the Chairman, Military Liaison Committee, from Gordon Dean, Chairman, U.S. Atomic Energy Commission, Subject: Pre-GREENHOUSE Test Program (RANGER).

⁴⁷²Excerpt from Gordon Dean office diary dated January 8, 1951.

⁴⁷³Statement by Gordon Dean, January 9, 1951, used as background to a briefing to a subcommittee of the National

Nominal height-of-burst for the A, B, and E shots was set at 1,050 feet; the F shot, because of its substantially higher expected yield, was set at 1,500 feet. Existing ARCHIE radar fuses were modified for the tests.⁴⁷⁴

Operation RANGER, as finally conducted, was a series of five airdrop tests of the so-called "fractional crit" weapons so-named because they required only a fraction of the critical mass used in Manhattan Project weapons. Although conceived during World War II, test devices using the new cores had not been designed until 1948.⁴⁷⁵

During an 11-day period between January 27 and February 6, 1951, five devices were dropped from a B-50D at altitudes between 19,700 feet and 29,700 feet; the bombs were detonated at heights between 1,100 and 1,400 feet over the desert.⁴⁷⁶ Two of the tests yielded one kiloton each; two others yielded 8 kilotons each; and the last test yielded 22 kilotons.

The final firing sequence (by model designation) was A, B₁, E, B₂, F; the B model was tested twice to establish yield reproducibility and prove that the shots were representative of the pit model. The RANGER shots were named Able, Baker-1, Easy, Baker-2, and Fox. Shot Able tested compression of an oralloy core; Baker-1 and Baker-2 tested initiation timing effect, and Easy tested initiation timing and plutonium behavior.⁴⁷⁷

Security Council on approval of the RANGER program. The "Fox" shot was also known as the "Freddy" shot so that the name of the test would not be identical to the name of the core being tested.

⁴⁷⁴Memorandum dated January 13, 1951 to Don Cotter, J-8, from Gaelen Felt, J6, Subject: Fusing Heights for Operation RANGER.

⁴⁷⁵"Comments on the History of the H-Bomb," Bethe, pp. 44-45; Report of the Manager, Santa Fe Operations, U. S. Atomic Energy Commission, July 1947 to July 1950, Book Two, p. 16.

⁴⁷⁶NWD 86-2 (Rev. 2C), "Known U.S. Nuclear Tests, July 1945 to 31 December 1988," Robert S. Norris, Thomas B. Cochran, and William M. Arkin, Natural Resources Defense Council, Washington, D. C., January 1989, pp. 17, 18.

⁴⁷⁷Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 7. The RANGER Able test core proved to be quite popular; it was fired five times during RANGER, twice during Operation BUSTER-JANGLE, at least twice during Operation TUMBLER-

The 22 kiloton F (Fox) pit shot was a proof-test of an early pit-HE assembly for a future stockpile bomb. The test device used a composite or alloy-plutonium "How Double Prime" core that was close to planned production size.⁴⁷⁸

All of the test devices except the F shot met their predicted yields. The F shot had been predicted to yield 34 kilotons: its failure to attain its predicted yield was ascribed variously to either implosion asymmetry or imperfect initiation timing; the bomb was slightly predetonated.

RANGER results were very rewarding. Besides providing design data for the GREENHOUSE bomb pits, RANGER proved that low-yield devices making efficient use of critical material could be designed and built.⁴⁷⁹ All of the shots except for the Fox test used MK 4 HE with Type D pits; the Fox shot used a MK 6 How Double Prime HE assembly, the same geometry to be used in the forthcoming GREENHOUSE Dog shot.

The efficiency of the Type D pit had proven satisfactory in a MK 4 HE geometry; in addition, because of the success of the RANGER Fox shot, LASL was in a position to design for stockpiling a new composite pit yielding from 30 to 40 kilotons in a MK 4 HE assembly, if the DOD continued to require such a weapon. As a general result, the tests had confirmed and refined LASL's ability to predict the yield from weapon designs up to about 50 kilotons.

Immediately after the conclusion of RANGER, a decision was made to incorporate the Type D pit into stockpiled weapons, and all bombs were remodeled by the end of the following July.⁴⁸⁰

In addition, Operation RANGER showed that the basic LASL program proposed for Operation GREENHOUSE did not require major changes, but had been underscored by RANGER results and

SNAPPER in 1952, and once during Operation TEAPOT in 1955. (Notes on J-7 Meeting, 11 June 1951, Discussion of Experiments for JANGLE.)

⁴⁷⁸Background statement by Gordon Dean, Chairman, USAEC, to National Security Council, January 9, 1951; Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 7; "Handbook for Operation RANGER and a Summary of Test Results," WT-202.

⁴⁷⁹WT-206(EX), p. 67.

⁴⁸⁰Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 14.

should go on with an enhanced probability that its goals would be met.⁴⁸¹ RANGER had also proved that variable yields could be obtained from differing critical masses in MK 5 and MK 6 HE assemblies (GREENHOUSE would provide further confirmation of this).

Another RANGER experiment had ramifications for the thermonuclear program. Three samples of U-238 metal, each approximately one kilogram (2.2 lbs.), were put on posts near surface zero to measure the fissions induced from both gamma rays and neutrons emanating from the airbursts.

This experiment was intended to answer a militarily-significant question: How closely together in time and space could two atomic bombs be detonated without the gamma rays and neutrons from the first detonation causing serious predetonation of the second?

A "Pajarito" assembly for testing criticality was used, involving a mockup of a bomb with a sphere of plastic material simulating HE. (the name "Pajarito" was a reference to the site near Los Alamos where critical and subcritical assemblies were tested remotely following the accidental deaths of two physicists in August 1945 and May 1946; some assemblies simulated stockpiled weapons.

The "Pajarito" assembly for RANGER contained all nuclear components, with counters in the pit to record the rate of induced nuclear reactions versus time.⁴⁸²

While this experiment was primarily an early test of "fratricide," the effects of nuclear warheads exploding near each other, the effects of various radiations on relatively-large pieces U-238 was also useful in the design of the SAUSAGE thermonuclear device to be tested in the fall of 1952 (the device used a mantle of U-238 around its fusion fuel chamber).

Midway through RANGER, the National Security Council outlined GREENHOUSE plans and objectives for President Truman at the end of January 1951:

As you have been advised previously, the Atomic Energy Commission proposes to conduct full-scale weapon tests at the

⁴⁸¹Minutes of 525th AEC Meeting, 10:30 AM, Wednesday, February 14, 1951; Atomic Energy Commission - Military Liaison Committee, Minutes of the Fifty-Fourth Conference, 3:40 PM, Wednesday, February 14, 1951.

⁴⁸²WT-202, p. 41.

Eniwetok Proving Ground in the Spring of 1951 (Operation GREENHOUSE).

Final determination of the weapon models and nuclear devices actually to be detonated during GREENHOUSE will depend in part on the outcome of the tests to be conducted at the Las Vegas site during February (Operation RANGER), and also on the course of events during GREENHOUSE.

As now foreseen, the GREENHOUSE program will include from 3 to 5 detonations during the period April 1, 1951 - June 3, 1951. The principal objectives of GREENHOUSE are the following:

- a. Proof-testing of nuclear components and associated weapon designs which may permit the attainment of much greater yields from a given amount of fissionable material, or, conversely, a given yield with the expenditure of less material.
- b. Examination of atomic weapons effects on buildings, aircraft structures, various materials, and on animals.
- c. Gaining information concerning the feasibility of the thermonuclear weapon.

The weapons models and nuclear devices under consideration for this firing program are:

- a. A weapon approximately $\frac{3}{4}$ the size and $\frac{1}{3}$ the weight of the present stockpile model, yet expected to give an energy release exceeding that of the present standard. (Author's note: this is a reference to the MK 5; the standard stockpile model at this time was the MK 4.)
- b. A weapon model of approximately the same size

and weight of the present standard model, expected to be comparable in energy release while using substantially less fissionable material. The design of this model, and the question whether it need be fired, will depend on the RANGER results. (Author's note: a reference to the device fired in the Fox shot of Operation RANGER; it was predicted to yield 34 kilotons but yielded only 22 kilotons; kilotons; it was not re-fired during GREENHOUSE.)

- c. A weapon model as in (b) above, but using the present standard nuclear core, expected to give an energy release from this core of 2 or 3 times that of the present standard model. (Author's note: a reference to the MK 6 weapon.)
- d. A large fission device (the Cylinder) for determining experimentally one aspect of the feasibility of a thermo-nuclear weapon, that is, whether the most favored current idea for initiating a thermonuclear reaction will work in fact. This experiment will not determine whether a thermonuclear reaction, once initiated, would continue to propagate itself.
- e. A fission device (the "Booster"), less ambitious than the Cylinder in its prospects for obtaining needed experimental data, and by the same token, more likely to succeed in its

relatively limited goals.
This shot would be fired in
the event either of
experimental failure in
firing the Cylinder or if
scientific developments in
the meanwhile seem to make
firing of the Cylinder less
desirable than now appears
to be the case.⁴⁸³

After RANGER, there was some reshuffling of the test devices slated for GREENHOUSE; late in February, the shot schedule included a test of the MK 6 prototype (then known as the "How Double Prime") on Runit; the TX-5 on Engebi; and the CYLINDER on Eberiru.

The TX-5 would be activated by an URCHIN initiator instead of a TOM initiator if the first shot gave an unexpectedly high yield (the MK 6 shot would use a TOM initiator). The BOOSTER would be fired only if the CYLINDER failed conspicuously; the probability of this happening was considered to be remote. No air drops were planned at this time; all tests would be tower shots.⁴⁸⁴

Figure III-3 shows a 300-foot high shot tower used during a Nevada test series in 1952. The towers for the GREENHOUSE Dog and Easy shots were identical to this Nevada tower.

The Type F "fractional crit" pit test was eliminated from the program in March 1951. This was probably the B-50D airdrop planned for the five-shot program; the device was to have been airdropped in a test named "King," a designation later assigned to a high-yield fission device airdrop test during Operation IVY.⁴⁸⁵

⁴⁸³Memorandum to the President, January 31, 1951, from S. Everett Gleason, Acting Executive Secretary, Executive Office of the President, National Security Council. Emphasis in original.

⁴⁸⁴Atomic Energy Commission, Revision of GREENHOUSE Shot Schedule, Report by the Director of Military Application, USAEC, undated, but circa February 12, 1951.

⁴⁸⁵WT-98 (EX), p. 49; memorandum dated March 9, 1951 for the Military Liaison Committee, from Gordon Dean, Chairman, USAEC, Subject: Revision of GREENHOUSE Shot Schedule. "Operation GREENHOUSE Nuclear Explosions 1951, Scientific Director's Report on Atomic Weapon Tests at Eniwetok 1951, Annex 9.2, Sandia Corporation Proving Ground Group, Part II - Mechanical Assembly," WT-73, Robert A. Knapp and Task Unit 3.1.4.1.2 Personnel, Sandia Corporation Proving Ground

Soon afterwards, firing of the BOOSTER was made part of the schedule, independent of the behavior of the CYLINDER. Testing of the BOOSTER was believed beneficial to the design and proof-test of the ALARM CLOCK, prospects for which had become more favorable since formulation of the Teller-Ulam principles. The BOOSTER test would also help reduce data ambiguities resulting from the vaporization of the complex and massive cryogenic installation surrounding the CYLINDER.⁴⁸⁶

The TX-5 test (the Easy shot) was to have a dual purpose: to test a weapon which represented as nearly as possible the proposed stockpile model, and to generate a blast pressure consistent with the stress limits of the test structures erected for the military effects program. In order to meet these requirements, an appropriate core or core-initiator combination had to be selected to yields between 40 and 60 KT. The Fox core in the MK 6 prototype, when used with a TOM initiator, was expected to give more than 60 KT.⁴⁸⁷ For this reason, a less-efficient URCHIN initiator was finally used in the TX-5 test.

Despite plans for the forthcoming test series, and in spite of the appearance of the promising but as yet unverified Teller-Ulam concepts, the AEC, echoing Bradbury's comments of November 1950, reported in mid-March to the President that prospects for a successful thermonuclear weapon were still remote, and that production of tritium for the program was having a major impact on plutonium production:

In accordance with your directive of January 31, 1950 ... an extensive effort has been devoted at Los Alamos to a program

Ordnance Department, Sandia Corporation, Albuquerque, New Mexico, August 1951, notes on p. 38 that several Type 43D HE practice weapons and four inert ballistic MK 4 bombs were returned to Sandia Corporation from Port Chicago, California after the airdrop (King shot) was canceled.

⁴⁸⁶Minutes of AEC Meeting No. 531, March 6, 1951; Minutes of AEC Meeting No. 552, April 26, 1951, Subject: Fourth GREENHOUSE Shot; Firing of "Booster" Shot, Minutes of 57th AEC-MLC Meeting, April 26, 1951; teletype dated April 26, 1951, to Lt. General Elwood Quesada, Commander, JTF 3, from Gordon Dean, USAEC, Washington, D.C.; letter dated May 15, 1951 from Gordon Dean, Chairman, USAEC, to James S. Lay, Jr., Executive Secretary, National Security Council.

⁴⁸⁷Memorandum for the Chairman, Atomic Energy Commission dated March 14, 1951 from Robert LeBaron, Chairman, MLC, Subject: Revision of GREENHOUSE Shot Schedule.

for the determination of the feasibility of thermonuclear weapons. The results have been that the inherent difficulties are clearer, the time to carry out the program is believed to be longer, and ultimate practical success is less certain than a year ago. The crucial question is how much tritium is needed in order to ignite a mass of liquid deuterium. [deleted] As it now stands today, we do not have the answer.

Thermonuclear Experiment at Eniwetok

The plans for Eniwetok this spring include an experimental test of the reaction of small quantities of tritium and deuterium under conditions of temperature and pressure produced by an atomic explosion. This experiment should supply information of value for further work, but it will not determine the feasibility of thermonuclear weapons. The present status of the program is such that it is not yet possible to make definite plans for additional thermonuclear tests in the future.

Production of Tritium

... the existing reactors at Hanford have been used to produce tritium in an effort to meet a goal of 500 grams a year, which was then estimated to be the rate required by the thermonuclear program. One reactor has been charged with enriched uranium for this purpose. This involves an investment of weapon-level uranium-235 and a decrease in the output of plutonium and results in a significant reduction in the number of atomic weapons that would otherwise be in the stockpile. (Author's note: as mentioned above, in terms of neutrons required for its creation, one kilogram of tritium was equivalent to 70 kilograms of plutonium.)
...

We propose to follow the general principle that tritium be produced to meet definite requirements of the thermonuclear program rather than to meet a fixed goal of 500 grams per year.

The five new reactors to be built and operated by the DuPont Company at Savannah

River site in South Carolina will each have a capacity of [deleted] using natural uranium as feed and producing plutonium at the same time. The reactors are designed for flexibility and can efficiently produce either plutonium and tritium or plutonium alone, as our needs dictate.

More plutonium could be produced if tritium were not required; on the other hand, more tritium could be produced at the expense of plutonium production by using enriched uranium as feed instead of natural uranium.

The report also commented on the status of fission weapons development, and -- again echoing Bradbury's comments five months earlier -- the most desirable relationship between the fission and fusion weapons programs:

During the past year, while work has been greatly intensified on the thermonuclear program, much effort has also been devoted to improving fission weapons. One new design (Author's note: the Fox core in the Type D pit) will enable yields comparable to those from existing weapons to be obtained from nuclear cores containing considerably less fissionable material, so that the total number of weapons which can be made from a given quantity of fissionable material can be increased significantly.

This advance has been demonstrated in the recent tests in Nevada and will be utilized in the current production of weapons. The same new design of weapons, using existing nuclear cores, is expected to give yields approximately double those now available; this will be tested at Eniwetok this spring. Another goal is to develop weapons of smaller size and weight; one design (Author's note: the TX-5) will be tested at Eniwetok this spring and other designs intended particularly for tactical employment are being developed.

Further gains in these various directions are likely. It is therefore important to continue a vigorous program of fission weapon development simultaneously with the thermonuclear program. We propose to continue work in the thermonuclear field with a view to determining the feasibility

of the weapon at the earliest practicable date, and to do so in such a manner that promising developments in the fission weapon field may be carried forward effectively.

Since we are now convinced that determination of the feasibility of the thermonuclear weapon, as well as the quantity of tritium required for experimental work and production, are now much less certain than a year ago, we do not consider it advisable to establish and maintain a fixed production rate for tritium at this time.⁴⁸⁸

A few weeks later, on April 6, the AEC and DOD recommended to President Truman that while "work on the thermonuclear program will be carried on with the objective of determining the feasibility of a thermonuclear weapon at the earliest possible date," promising developments on fission weapons would also be carried forward expeditiously.⁴⁸⁹

Operation GREENHOUSE: Operational Phase

Operation GREENHOUSE finally included four tower shots (200' and 300' high) in April and May 1951 on the northeast islands of Eniwetok atoll. The tower cab floors measured 20'x29', with an additional 7'x9' extension on one cab. Within the cab, each explosive device was enclosed by a protective exterior case for attaching or supporting the device and auxiliary components. Each tower could safely carry several hundred tons. The devices tested included the TX-5 (Easy shot), the "How Double Prime" MK 6 prototype (Dog shot), the BOOSTER (Item shot), and the CYLINDER (George shot).⁴⁹⁰

Before movement of the test devices to the Pacific, a 25-foot high mockup shot tower had been erected at Tech Area 33 (TA-

⁴⁸⁸Draft Report to the President on Status of Thermonuclear Program, March 8, 1951, attachment to Memorandum dated March 19, 1951 for the Chairman, Military Liaison Committee, from Gordon Dean, Chairman, USAEC, Subject: Draft Report to the President on the Thermonuclear Program. Emphasis in original. The final version, little changed from the draft cited, was signed by Dean and Gen. George C. Marshall, Secretary of Defense, and presented to Truman in early April 1951.

⁴⁸⁹"Chronology of Major Decisions Regarding Thermonuclear Program," USAEC, June 14, 1951, p. 5.

⁴⁹⁰WT-73, p. 4.

33) at LASL. Here personnel from the lab's W Division practiced device assembly and core insertion, using inert units, simulated capsules, and tools and equipment designed for the GREENHOUSE operation.⁴⁹¹

All of the GREENHOUSE shots were weapons development tests; shot Easy also included extensive experiments on the effects of nuclear explosions on military materiel and industrial structures.⁴⁹² In addition, shots George and Item were related to the development of thermonuclear weapons: George was a test of the tritium-deuterium "ignition" technique for the "classical Super," and Item was a test of thermonuclear boosting of a standard fission weapon. Shots Dog and Easy were tests of devices related the development of the MK 6 and 5 weapons, respectively.

Elaborate instrumentation to measure and record device performance was installed for each shot. In some cases, special instruments set close to surface zero measured events that took place in a fraction of a microsecond and transmitted the data to safer locations where it could be recorded before the detecting instruments were vaporized or before electromagnetic signals from the explosion obscured the diagnostic signals. The description of one GREENHOUSE diagnostic experiment noted that its sensing equipment "had to be designed in such a manner that it would continue to function while blowing up."⁴⁹³

Data was transmitted by cables and radio links, and recorded by high-speed oscilloscopes, magnetic tapes, photographic plates and other means.

Some instruments depended on laboratory analyses of radioactive samples flown from Eniwetok to continental U.S. sites within 30 to 40 hours after each test detonation. Other measurements involved the use of high-speed cameras operating at speeds of a million frames per second.

More complicated instrumentation included mass spectrographs, beta particle spectrographs, photocells, photomultipliers,

⁴⁹¹WT-102, p. 45.

⁴⁹²One of the structures erected for destruction during GREENHOUSE was full-scale model of a Soviet apartment house. (Furman, p. 582.)

⁴⁹³Operation GREENHOUSE Nuclear Explosions 1951, Scientific Director's Report, Annex 1.8 Measurement of X-Rays, Part II Design Considerations and Data Analysis, H. Bradner, H. F. York and Co-Workers, University of California Radiation Laboratory, Berkeley, California, September 1951, p. 7.

and ion chambers. The complexity of these instruments dictated that devices be detonated from fixed towers, rather than in airdrops of less-certain air-zero location.

After the extensive construction program had been completed and personnel assembled and test operations rehearsed, actual test operations began. These were divided into three overlapping phases.

The first phase began three days before each planned shot hour and lasted for 48 hours. During this period, aerial and surface surveillance of a "Danger Area" security zone around Eniwetok was intensified and task force personnel were withdrawn from camps near the shot islands to safer locations further away at the southern end of the atoll.

The second phase began at midnight the day before the shot and continued through shot day. During this phase, the test device was assembled and moved from a workshop aboard the USS CURTISS (AV-4), a converted Navy seaplane tender first used for this purpose at Operation SANDSTONE, and hoisted to the cab atop the shot tower.⁴⁹⁴ Shops 5 and 6 aboard the CURTISS were used as general repair and storage areas, and for assembly operations. The two shops were air-conditioned and offered very satisfactory and secure working quarters.⁴⁹⁵

All of the test devices except the Item (device the BOOSTER) were shipped from the naval weapons station at Port Chicago near San Francisco to Eniwetok aboard the CURTISS in late February 1951. The CURTISS had loaded devices aboard at Port Chicago on February 19 and 20; she sailed two days later. For security purposes, she did not stop at Hawaii.⁴⁹⁶

The CURTISS was heavily escorted by protective ships and aircraft; there was a submarine scare the day before she reached Eniwetok. Deceptive courses were followed and the ships zigzagged, sailing well south of established steamship lanes in the central Pacific. Strict radio silence was

⁴⁹⁴Besides her claim to fame as a nuclear test support ship, the CURTISS was also the duty post of Lt. Henry Fonda, a naval air combat intelligence officer, between the spring of 1944 and the end of World War II. ("Henry Fonda and the U.S. Navy," Capt. Alexander G. Monroe, USNR, NAVAL HISTORY, Vol. 5 No. 1, Spring 1991, pp. 77-79.) This may be where Lt. Fonda gained some of the experience so useful to him later in his role in "Mr. Roberts."

⁴⁹⁵WT-102, pp. 1, 29.

⁴⁹⁶WT-102, p. 49.

observed; at night, all ships were blacked out with no lights showing.⁴⁹⁷

The CURTISS arrived at Eniwetok atoll and anchored off Parry island on March 8.

The Item device was flown out from Travis Air Force Base near Sacramento, California, in mid-May.⁴⁹⁸ Nuclear cores for the bombs were airlifted to Eniwetok during March.⁴⁹⁹

Experimental teams made final instrument adjustments and calibrations, and animals for effects tests were placed at their exposure stations. This phase ended with device detonation.

The third phase, which lasted several days after the shot, began with the first efforts to recover recorded data following the explosion. During this phase, the predominant activities were shot site surveys and radiochemical and radiobiological sample recovery. Radiochemical data was used to determine device efficiency and fuel reactions; radiobiological data was used to determine human dosimetry and short- and long-term radiation effects on the local environment.

Each of the units detonated during GREENHOUSE was an implosion-type device. The Dog and Easy tests were conducted mainly to verify certain theoretical calculations used to design the bomb capsules and hydrodynamic systems.

Before GREENHOUSE, the designs of capsules and pits for the MK 5 and 6 weapons had undergone laboratory and field testing to various degrees at LASL. During GREENHOUSE, some of the remaining nuclear components of these two bombs that had not been tested during Operation RANGER received their first actual full-scale field test. The results of GREENHOUSE ultimately demonstrated the suitability of the MK 5 and 6 bombs for stockpiling.

⁴⁹⁷Report dated 16 June 1951 to Commander Joint Task Force Three, from Commander Task Group 3.3, Subject: Report of Participation in Operation GREENHOUSE, pp. A-3, A-4; Commander Task Group 3.3, History, 15 November 1950 to 31 March 1951, pp. 8, 9.

⁴⁹⁸WT-48 (EX), pp. 82, 84; Bethe, "Comments on the History of the H-Bomb," p. 47.

⁴⁹⁹WT-98 (EX), p. 36.

The George and Item shots were to test the feasibility of large-scale thermonuclear reactions which could not be conducted safely at LASL.⁵⁰⁰

Although all four shots were weapon development tests, the objectives, associated experiments, and necessary instrumentation for each were different. Shot Easy, the major effects test shot, required a device with a predictable yield with little or no variation. The number and sizes of the test structures caused the Easy device to be positioned at the western edge of Engebi, the largest of the northeastern islands at Eniwetok, and required shot Dog to be fired on Runit island, 11 1/2 miles away, so as not to disturb or damage the Easy structures (Easy was fired after Dog).

On March 24, the LASL weapons assembly group conducted a practice nuclear loading of the "How Double Prime" and loaded the live components on April 7.⁵⁰¹

Shot Dog, the first test in the series, was fired at 0634 hours local time, 30 minutes before dawn, on April 8, 1951 (April 7 in the U.S. on the eastern side of the International Dateline) atop a 300' high tower at the north end of Runit island. **Figure III-4** shows a replica of the Dog device being hoisted up the shot tower.

Shot times for atmospheric tests in Nevada and the Pacific were usually a half hour or hour before dawn to provide a compromise between minimum light background at ground or sea level for light-sensitive experiments and sufficient daylight at higher altitudes for drone aircraft operations.

In the Pacific, a device could be fired a few minutes before dawn in complete darkness and within 10 to 15 minutes there would sufficient daylight for helicopters and aircraft to fly. At altitudes above 20,000 feet, there was daylight approximately 10 minutes before dawn at sea level.⁵⁰²

⁵⁰⁰WT-102, p. 6.

⁵⁰¹WT-102, p. 49.

⁵⁰²O'Keefe, p. 139; letter dated January 13, 1954 to Robert M. Underhill, Secretary/Treasurer, Board of Regents, University of California, from Henry R. Hoyt, Assistant Director for Administration, LASL; OPERATION SANDSTONE, NUCLEAR EXPLOSIONS 1948, Atomic Weapons Tests, Operation SANDSTONE, 1948, Annex I Part II, Volume I, Report to the Joint Chiefs of Staff, Section VIII, p. 41.

Dog was a test of a stockpiled MK 6 weapon with a "How Double Prime" (How) composite uranium-plutonium core.⁵⁰³ Yield was 82 kilotons, a new record for unboosted fission weapons (the old record was the 49 KT SANDSTONE Yoke shot of April 30, 1948).⁵⁰⁴ According to one account, at zero hour, "the predawn darkness was rent with a brilliance greater than noon and the islands of the atoll shook with the force of the blast."⁵⁰⁵ The explosion lifted 250,000 tons of soil to an altitude of approximately 35,000 feet.⁵⁰⁶

Stockpiling of the MK 6 on an emergency basis was underway at the time of the test;⁵⁰⁷ incorporation of the MK 6 HE assembly and a new pit would more than double the energy yield of the major portion of the national atomic weapons stockpile.

The test program for shot Easy included 69 experiments or projects conducted by several separate task units within the task force. These repeated most, if not all, of the tests conducted during shot Dog, but in addition, the heavily-instrumented experiments of the structural response program were included.

These tests involved full-size and scaled-down military and industrial structures built at various ranges from ground zero, the 300' high shot tower on Engebi island. Test structures were also located on the nearby island of Mujinkarikku. Another island, at greater range to the west,

⁵⁰³"How Double Prime" design and assembly during GREENHOUSE are detailed in WT-52, "Operation GREENHOUSE, Sandia Corporation Proving Ground Group, Part III -- Fuzing and Firing Activities," and WT-73, "Operation GREENHOUSE, Sandia Corporation Proving Ground Group, Part II, Mechanical Assembly;" Operation GREENHOUSE Nuclear Explosions 1951, Scientific Director's Report, Annex 1.7 Radiochemical Yield and Efficiency Measurements, Roderick W. Spence and Jere D. Knight, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, September 1954, p. 15.

⁵⁰⁴OPERATION GREENHOUSE, ANNEX 6.4, FALLOUT PHENOMENOLOGY, C. E. Adams, F. R. Holden, N. R. Wallace, U.S. Naval Radiological Defense Laboratory, San Francisco, California, WT-4, August 1951.

⁵⁰⁵WT-48 (EX), p. 114.

⁵⁰⁶WT-48 (EX), p. 118.

⁵⁰⁷Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 8.

was used for low-level blast testing of instrumented sections of various types of aircraft.

Shot Easy was detonated at 0627 hours on April 21; yield was 47 kilotons. The TX-5 used an URCHIN-type initiator rather than the newer TOM initiator in order to keep the yield within the planned range of values for the associated structures test program.⁵⁰⁸ Easy included a 45" diameter case with a "Fox" composite core in a Type "D" pit (the type of pit also used on the Dog shot); total weight of the unit was reportedly 3,080 lbs.⁵⁰⁹

Measurements of yield, alpha, and transit time for both shots Dog and Easy were good and indicated that the models behaved essentially as predicted. The TX-5 performed satisfactorily and demonstrated its suitability for stockpiling.⁵¹⁰

The George Shot

The CYLINDER device tested during the George shot⁵¹¹ was not an actual weapon. The design concept was instead based upon a configuration that produced the D-T reaction in such a way as to permit a maximum number of recorded measurements. The final test device was the culmination of nearly two years of intensive research and development at Los Alamos. Following the directives of the Family Committee, LASL's Theoretical Division, GMX, CMR, and W Divisions collaborated to build the gadget.

⁵⁰⁸Memorandum dated May 8, 1951 for Dr. Richard W. Dodson, Executive Secretary, General Advisory Committee, from Capt. A. McB. Jackson, USN, Acting Director of Military Application, USAEC, Subject: Preliminary Results of First and Second GREENHOUSE Shots; Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 8.

⁵⁰⁹Memorandum for the President dated 27 April 1951 from Brig. Gen. B. B. Landry, USAF; Operation GREENHOUSE Nuclear Explosions 1951, Scientific Director's Report, Annex 1.7 Radiochemical Yield and Efficiency Measurements, Roderick W. Spence and Jere D. Knight, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, September 1954, p. 18; Shelton, p. 4-26. The official weight of the Easy device was 2,700 lbs.

⁵¹⁰Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 8.

⁵¹¹Anders, FORGING THE ATOMIC SHIELD, p. 143.

Because the bomb was to be fired from a stationary position atop a tower, it was possible to reduce the device's weight and componentry to a minimum. The unit was stripped of its nonessential elements, and a lightweight three-section protective cover can encased the unit and also acted as a security cover to conceal its inner parts.

To compensate for the absence of a heavy ballistic case, support beams had to be installed to carry the weight of the device while being shipped and while in the shot tower. An opening in the cover casing allowed removal of trap-door HE charges and insertion of the core; this opening also allowed access to several detonators. The X-unit was covered by one of the three cylindrical sections of the unit's casing.⁵¹² The bomb used an oralloy core.⁵¹³

The George shot was detonated atop a 200 foot high tower on Eberiru island at the eastern side of Eniwetok atoll at 0930 hours on the morning of May 9, 1951 (local time). The yield was 225 kilotons, the largest nuclear explosion to date (this record would be dwarfed within a year and a half by the world's first full-scale thermonuclear explosion).⁵¹⁴

The official GREENHOUSE history states:

Shot GEORGE was the climax of Operation GREENHOUSE.

Never before had there been an explosion of such magnitude. It was known that if the device functioned in the test according to theory, the bombs dropped on Japan in 1945 would be regarded as small by comparison, and even Shot DOG, until that time the most powerful detonation ever made, was sure to be placed a poor second in the total kilotons of released energy.⁵¹⁵

⁵¹²WT-73, p. 7.

⁵¹³Operation GREENHOUSE Nuclear Explosions 1951, Scientific Director's Report, Annex 1.7 Radiochemical Yield and Efficiency Measurements, Roderick W. Spence and Jere D. Knight, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, September 1954, p. 20.

⁵¹⁴A telex message dated May 10, 1951 from Jane H. Hall to Paul Fine and J. Robert Oppenheimer, quoting Alvin C. Graves, stated that George "yield ...was much greater than How Double Prime, perhaps by (a) factor between two and three. The interesting mixture certainly reacted well ..."

Figure III-5 shows the George shot cloud with a "skirt" of cold air around its stem.

Only a small fraction of George's power came from tritium-deuterium fusion. The best-detailed, but not necessarily most accurate, glimpse into the design of the George device comes from a recent article (this description of the George device is uncorroborated):

The Greenhouse experiment, which took place in 1951 on Eniwetok atoll in the Pacific, was supposed to create the first man-made thermonuclear reaction, using the energy of a 500-kiloton (sic) atomic bomb to ignite a fraction of an ounce of deuterium and tritium placed in a small adjoining chamber. The project was more of a public-relations stunt than a genuine experiment, because everyone knew beforehand that it was pretty certain to work; using a huge atomic bomb to ignite the little vial of deuterium and tritium was like using a blast furnace to light a match....

...As the house theorist, I applied the methods of the branch of physics known as radiative transfer theory to compute the temperature inside the vessel of deuterium and tritium, using measurements on the amount of radiation coming from the outside.⁵¹⁶

The CYLINDER (the device detonated in the GREENHOUSE George shot) had a "complex physical installation;" the liquid deuterium in the secondary was refrigerated, requiring a large cryostat.⁵¹⁷

⁵¹⁵WT-48(EX), p. 135.

⁵¹⁶"Why Strategic Superiority Matters," Robert Jastrow, COMMENTARY Magazine, March 1983, Dartmouth College, Hanover, New Hampshire, p. 27. According to Dr. Ted Taylor, "George had a fission bomb, a rather large one." ("An NHP Interview with Theodore Taylor," conducted at The University of Maryland, College Park, MD, 12 April 1989, Nuclear History Program Oral History Transcript #1, p. 41.) The George fission trigger was much smaller than 500 KT (see following note).

⁵¹⁷Teletype dated April 26, 1951 from Gordon Dean to General Elwood Quesada, Commander Joint Task Force Three.

The George shot included a very small scale test of radiation implosion (this was not the primary purpose of the shot, which was to test D-T fusion).⁵¹⁸ The test was successful:

⁵¹⁸"Thermonuclear Weapon Development, January 1950-May 1951," pp. 1, 8. Although the public literature about the early days of the U.S. thermonuclear weapons program contains many statements that the success of the George shot was a direct result of the Teller-Ulam principles, or that George led to the discovery of radiation implosion, all of these claims are mistaken. The full theoretical prediction of the behavior of the George device was completed in January 1951, before the principle of radiation implosion was applied to the Super. (Bethe, "Comments on the History of the H-Bomb," p. 47.) The experiment may have been altered after April 1951 to include an engineering configuration suitable to test radiation implosion. According to Bethe's letter to the author dated 11 September 1989, George "was designed to find out whether compression by a fission explosion was possible, and whether it would then cause a thermonuclear reaction;" Mark, in his letter to the author dated 18 September 1989, acknowledges Bethe's interpretation of the purpose of the George shot, then adds:

In my view the George shot was designed primarily to try to use energy from a fission explosion to heat a mass of deuterium-tritium mixture to the point at which it would be expected to ignite and proceed with a thermonuclear burning, and to observe what we could of such a process.

Certainly, some compression was associated with such a pattern of behavior, but that was a rather secondary matter. The experiment was conducted on May 8, 1951, and was fully successful. In particular, it confirmed to us that the methods we were using to calculate the elements of the process were sufficiently accurate to provide an adequate description of the actual events.

Of course, the objectives of the George experiment (laid down more than a year in advance), all the details of its design, and all the elements of the observational program were necessarily in final form considerably before the February-March 1951 developments. In fact, by that time the device was fully fabricated (or almost

high-energy neutrons radiating from thermonuclear reactions left their traces in sample materials and instruments near the explosion.⁵¹⁹ The blast dug a crater 10 feet deep and 1,140 feet in diameter.⁵²⁰

The official GREENHOUSE history described George shot in apocalyptic terms:

Detonation occurred promptly at 0930 and there followed the greatest spectacle within recorded history. The white day became dark by comparison with the brilliant light radiating from Eberiru island. From the

completely fabricated) and parts (or all) of it were on their way to Eniwetok.

The George shot, then, was an item of the program of the Classical Super. Nevertheless, the results of George were of very real value because of the confirmation obtained that the methods we were in a position to apply could be relied on wherever similar processes might be involved in the new program.

Dr. Bethe later reviewed Marks' comments about George, and concurred with them fully in a note from Bethe to the author dated 26 September 1989. During a September 1954 press conference, LASL director Dr. Norris Bradbury stated that "we conducted the GREENHOUSE experiments and these were phenomenally successful. They gave important information which we could use in the exploitation of the Ulam-Teller (sic) idea. Very important data." (Transcript of Press Conference with Norris Bradbury at Los Alamos, New Mexico, September 24, 1954, p. 12.)

The total fission-fusion yield of the George device was only 225 kilotons; if a 500-kiloton trigger was used, it was less than 50% efficient. It is more likely that a much smaller trigger was used, probably on the order of 160 to 200 KT. Stockpiled MK 6 bombs at this time could accept 160 KT cores.

⁵¹⁹O'Keefe, p. 149.

⁵²⁰Crater size was originally estimated at 30 feet in depth and 500 feet in radius. There was considerable flooding in the crater after the test; there apparently was an underground connection with the sea as the water level in the crater rose and fell with local tides. (Letter dated 2 October 1952 to Major Norair M. Lulejian, USAF, HQ Air Research and Development Command, from William R. Kennedy, Nuclear Field Test Section, Health Division, USAEC.)

ashes of the exploded device, the mushroom catapulted upward to an altitude of 70,000 feet or more, far beyond the reach of tropical thunderheads. Boiling and seething as it rose, the atomic cloud formed three ice caps in quick succession, each of which was left behind as a shining, crystal bell along the rope-like column. When the spectacle was at its maximum, there was suddenly visible the translucent blue-violet nimbus of ionization.⁵²¹

To a large extent, George was a scientifically-irrelevant event: the "classical Super" had been made obsolete by the discovery of radiation implosion, and tritium-deuterium fusion caused by indirect exposure to a fission explosion was no longer a part of the "new Super" design.⁵²² This reaction would henceforth be initiated by the "sparkplug" in direct contact with the fusion fuel.

While the George shot successfully demonstrated radiation implosion, it was on only a very small scale, did not include a large quantity of deuterium by itself without tritium to lower its ignition temperature (the CYLINDER was "rich in tritium"), and certainly did not indicate even remotely that radiation implosion would work in large solid-fueled thermonuclear devices.⁵²³

⁵²¹WT-48 (EX), p. 135.

⁵²²In his letter to the author dated 11 September 1989, Bethe claims that in January 1951, if Ulam's ideas had not been conceived soon afterwards, Teller would not have known what to do next even if the George test were successful. A later report to the Joint Committee on Atomic Energy said of the CYLINDER: "Some design such as this would probably be required to act as a fuze (sic) for the ignition of a large quantity of deuterium; for this reason, the CYLINDER may be regarded as a step in the direction of a true thermonuclear weapon, or hydrogen bomb. The important theoretical question of whether a large mass of deuterium will burn has not yet been answered, however." (Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 12.)

⁵²³Hirsch and Mathews, p. 13; Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 9. Bethe, in "Memorandum on the History of the Thermonuclear Program," states that the CYLINDER included transfer of energy by radiation from the fission bomb to both heat and compress the T-D mixture in an adjacent vessel.

Despite a successful thermonuclear burn, total George yield was lower than predicted and the "runaway burn" expected from the test did not happen, which indicated that even this part of the "classical Super" design might not have worked as expected, in addition to the low probability of self-propagating thermonuclear reactions in deuterium.⁵²⁴

The success of the CYLINDER did not guarantee the propagation of a deuterium-deuterium reaction, nor did it assure the maintenance of the deuterium fuel mass long enough for an adequate burn.

The final design of the CYLINDER was selected from three competing configurations. One promising device had been suggested by Edward Teller in the fall of 1949; the final design of the CYLINDER just inadvertently incorporated energy conduction by radiation from a fission trigger. This choice led to the first small-scale experimental confirmation of the Teller-Ulam principles.⁵²⁵

Dr. Hans Bethe later stated that the "GREENHOUSE (George) test had a vague connection with radiation implosion, but one

⁵²⁴AF ATOMIC ENERGY PROGRAM, Vol. IV, pp. 198, 201. NUCLEAR WEAPONS DATABOOK, Vol. II, states (p. 13): "While yield from the ignition of the D-T mixture far exceeded expectations, it contributed only a small amount to the 225 Kt yield of shot George." This is confirmed by "Thermonuclear Weapon Development, January 1950-May 1951" (p. 8) which states:

The George shot resolved several uncertainties. The tritium-deuterium mixture [deleted] had been compressed; a fusion reaction had occurred; the percentage of tritium and deuterium burned exceeded expectations.

The latter source does not indicate how well George's actual yield compared with its predicted value; it is probable that predicted yield covered a wide range, given the novelty and unpredictability of the experiment. A September 28, 1951 Joint Committee on Atomic Energy paper entitled "Some Comments on the Hydrogen Bomb Program," states that "The hydrogen experiment at Eniwetok last Spring was successful beyond expectations." In a February 26, 1958 letter to GAC Chairman Warren C. Johnson from R. F. Taschek, Taschek refers to "the somewhat unexpected success of the GREENHOUSE George shot."

⁵²⁵Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, pp. 6, 7.

that nobody, not even Teller, could have foreseen or did foresee when that test was planned."⁵²⁶

Even if the "new Super" had not been invented, George was irrelevant because the calculations by Ulam and Everett in the spring of 1950 showed that the classical Super would not work.⁵²⁷ All the George shot did was prove experimentally that a large fission explosion could generate enough heat and pressure to fuse liquid deuterium to tritium.⁵²⁸ This was hardly a surprise to anyone working on the Super at Los Alamos in 1951.

The Item Shot

The final test in Operation Greenhouse, the Item shot, was a test of the BOOSTER, Edward Teller's 1947 idea to improve the yield of fission weapons. The BOOSTER used liquid hydrogen isotopes in a "booster ball" charged by a cryogenic pump at the base of the 300 foot high shot tower on Engebi island.⁵²⁹

Because of the delicate nature of the BOOSTER capsule, final assembly of the capsule was completed at LASL before the BOOSTER was flown to Eniwetok. Since the capsule did not contain either plutonium or oralloy, assembly at LASL was considered safe. A copper pit liner encased within the capsule was tested for leaks with various inert gases under high pressure several times between May 2 and May 9.

The LASL weapons assembly team conducted practice loadings of BOOSTER nuclear components on May 17 and May 20; live components were loaded on May 22.⁵³⁰

Item was detonated at 0617 hours on the morning of May 25, 1951. The yield was 45.5 kilotons⁵³¹ (unboosted yield probably would have been around 30 kilotons). Like the George shot, the device used an all-oralloy core.⁵³²

⁵²⁶Bethe, "Comments on the History of the H-Bomb," p. 49.

⁵²⁷Bethe, "Comments on the History of the H-Bomb," p. 47.

⁵²⁸Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, pp. 13, 63.

⁵²⁹WT-98 (EX), p. 53.

⁵³⁰WT-102, pp. 32, 49.

⁵³¹NWD-2 (NRDC), p. 13.

As in the George shot, thermonuclear reactions were evidenced by high-energy neutrons radiating from the explosion. The test results from the BOOSTER shot helped to reduce some of the ambiguity surrounding the results of the George shot. These ambiguities were caused by the design and structure of the CYLINDER; the tower which supported the George shot device also carried 283 tons of diagnostic equipment.⁵³³

All of this material made it difficult for remote instruments to "look through" to the interior of the explosion; in addition, the vaporization of such a large mass of items of varying densities led to unusual diagnostic measurements of questionable reliability.

Ten years later, the results of the Item and George tests were summarized by the AEC:

Operation GREENHOUSE of 1951 provided the first tests in the development of thermonuclear weapons. The GEORGE (sic) shot was an experiment to determine if the fusion reaction could occur in a proper configuration from the initial energy release of a fission device.

GEORGE was not only a success in that it achieved a thermonuclear energy release, but the idea of radiation implosion and two-stage systems grew rather naturally and unexpectedly out of these results.

When GEORGE performed so well, it dispelled any doubt (and there was some)

⁵³²Operation GREENHOUSE Nuclear Explosions 1951, Scientific Director's Report, Annex 1.7 Radiochemical Yield and Efficiency Measurements, Roderick W. Spence and Jere D. Knight, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, September 1954, p. 22.

⁵³³Teletype dated April 26, 1951, from Gordon Dean, USAEC Chairman, to General Elwood Quesada, Commander, Joint Task Force Three; FORGING THE ATOMIC SHIELD: Excerpts from the Office Diary of Gordon E. Dean, edited by Roger M. Anders, University of North Carolina Press, Chapel Hill, North Carolina, 1987, p. 144. The towers holding both the BOOSTER and the CYLINDER had to be redesigned in the spring of 1950 to carry all the extra weight of special diagnostic equipment. (U.S. Atomic Energy Commission, Progress Report to the Joint Committee on Atomic Energy, January through May 1950, May 31, 1950, p. 5.)

as to the success of the ITEM (sic) shot. ITEM, however, was still very important because it demonstrated the first boosting of a fission bomb (using D-T gas placed at the center), and laid the foundation for most of the fission weapons now in stockpile.⁵³⁴

An important part of every nuclear test series was the rollup phase, during which dangerous or radioactive debris or craters were collected or filled in, and test facilities put into "standby" or "mothballed" status until the next series, which might not occur for the next year or two. Prevailing weather conditions in the central Pacific around the Marshall Islands required that storage and preservation of delicate laboratory and scientific equipment be exercised carefully.

Relative humidities as high as 85%, combined with salt-laden air and outdoor temperatures between 65⁰ and 90,⁰ presented ideal conditions for metal corrosion and decay of organic materials. Violent seasonal storms and typhoons, often coupled with high waves that washed completely over low-lying atoll islands, frequently destroyed and uprooted buildings, test towers, buried cables, and underwater installations.

All buildings, and especially the cryogenic facilities, had to be sealed and dehumidified after being filled with as much material as possible that might otherwise deteriorate outdoors. Expensive machine tools were coated with cosmoline, a sprayed-on preservative grease.

Immovable outdoor equipment adjacent to the buildings was enclosed with temporary, weather-resistant plywood coverings "cocooned" in a thin plastic foam coating derived from materials originally used to protect aircraft outdoors. A small hoist motor was immersed almost completely in waste machine oil in a 50-gallon drum which was then sealed and stored.⁵³⁵

⁵³⁴Letter (with attachment) dated November 6, 1961 to Mr. James T. Ramey, Executive Director, JCAE, from Brig. Gen. A. W. Betts, USA, Director of Military Application, USAEC, re Unexpected Results from Previous Weapons Tests. General Betts' claim that the ideas of two-stage weapons and radiation implosion followed from the George shot is incorrect: these ideas predated GREENHOUSE.

⁵³⁵WT-50, pp. 131-148.

Results of GREENHOUSE

At the end of May 1951, the AEC reported to Congress on the results of RANGER and GREENHOUSE:

Weapons tests were successfully held at Las Vegas (January 27 and 28, February 1, 2, and 6) and at Eniwetok (April 8 and 21, May 9 and 25). Both operations provided us greater reassurance of the direction weapon production and development is expected to take over the next few years.

In brief, RANGER (Las Vegas) demonstrated the practicability of "fractional crit" weapons and the new Type D pit, the effect of which will be to increase greatly the potential effectiveness of the stockpile of fissionable material.

GREENHOUSE (Eniwetok) not only further confirmed the effectiveness of the Type D pit and a new non-nuclear assembly for fission weapons, but it also provided a demonstration of the deuterium-tritium reaction in experimental devices. This marks a major step toward the development of thermonuclear weapons and provides valuable data in the continuing effort to develop a hydrogen bomb, but it does not yet indicate that such a weapon is feasible.⁵³⁶

The report also noted that the fission weapons stockpile had been affected in two ways by the thermonuclear research program. Substantial amounts of U-235 had been diverted to tritium production and reactors loaded with U-235 to make tritium sacrificed plutonium production.

Some of the best news from GREENHOUSE was that most the developments tested had been conceived after the summer of 1947, when Klaus Fuchs left Los Alamos, and hence were presumably unknown to the Soviets:

As we know, the results of GREENHOUSE have already produced a spurt in ideas for new developments in the nuclear weapon field. If we are ahead of the Russians in know-

⁵³⁶United States Atomic Energy Commission, Draft Foreword by the Commission to the Progress Report to the Joint Committee (on Atomic Energy), December 1950 through May 1951, May 29, 1951, pp. 1, 2.

how it is because of these developments,
which are, for the most part, post-
Fuchsian.⁵³⁷

As Operation GREENHOUSE ended, planning and construction for the hydrogen bomb program continued. A tentative schedule of devices for Operation IVY, the next Pacific test series, was prepared; it included a very high yield implosion weapon, an implosion device of the smallest design of military interest, a large thermonuclear device, and a multi-crit gun-type weapon.⁵³⁸

At the beginning of June 1951, Edward Teller and Frederick de Hoffmann published at LASL a report entitled "Effectiveness of Lithium-6 in an Equilibrium Super;" the paper examined the theoretical utility of a "dry" solid thermonuclear fuel in Teller's March design.⁵³⁹ Teller had been energized by GREENHOUSE; in the middle of May he wrote "it is almost needless to say that the results of the recent tests have filled many people here (including myself) with great enthusiasm; it is now my conviction that the thermonuclear program is past its ignition point."⁵⁴⁰

The uncertainty in the H-bomb's characteristics became even more evident at a conference on June 14 attended by representatives of the Air Force, the Sandia Corporation, and the Los Alamos laboratory. The Los Alamos scientists indicated that it was their intention to build a Super small enough to carry in conventional Air Force aircraft, but when they offered bomb size estimates, they thought in general terms of from 35,000 to 80,000 lbs. in weight, about 20 feet in length, and approximately 9 feet in diameter. This was a far cry from the 10,000 lb. H-bomb envisioned in the Air Force's early planning days of 1949.

⁵³⁷Letter dated 16 May 1951 to Dr. Norris E. Bradbury, LASL, from Frederick Reines. In this letter, Reines complains about documentary film-making during GREENHOUSE and elsewhere by the Air Force's Lookout Mountain Lab, which he felt was endangering national security.

⁵³⁸Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 10.

⁵³⁹Borden-Walker H-bomb chronology, p. 64; LAMS-1263 cited in "Technical References Relating to Thermonuclear Program," Appendix E to September 12, 1952 Walker memorandum.

⁵⁴⁰Letter dated May 17, 1951 to Dr. H. D. Smyth, USAEC, from Edward Teller.

LASL's assistant technical associate director, Dr. Harold M. Agnew, pointed out that the larger the bomb the better, because it could then contain a greater proportion of high explosives and less of the costly fissionable material. (See Volume II for comments on high explosive quantities versus fissionable material efficiency.) For this reason, B-36 and B-52 aircraft looked very appealing to the scientists inasmuch as they could lift heavier loads, had larger bomb bay volumes, and allowed much greater thermonuclear packaging freedom than did the smaller B-47.

Even more attractive was a proposed large cargo aircraft, the new Douglas C-124, which would probably carry anything the scientists devised. When confronted with Air Force comments that such an aircraft would be extremely vulnerable, the scientists suggested that only one aircraft in an entire formation of identical airplanes carry the H-bomb, forcing the enemy to play the role of a small town "rube" in a rural carnival shell and pea game and try to guess which plane to attack. This suggestion paralleled earlier USAF atomic strike plans which envisioned a single atomic bomb carried by one airplane in a five-bomber formation.

During the conference, studies by the Massachusetts Institute of Technology and the University of California were examined which showed how dangerous it was for a manned aircraft to deliver thermonuclear weapons. When told of USAF Air Materiel Command drogue parachute studies, the Los Alamos scientists objected because they regarded such a drop method would make the slowly-falling bomb too vulnerable to air and ground interception and attack.

A BRASS RING report concluded that a drone B-47 could probably carry up to 50,000 lbs., but because of its bomb bay size, might be limited to a bomb 25 feet long and five feet in diameter. Whether or not the B-47 could even perform the mission depended entirely on the final H-bomb configuration. Los Alamos and Sandia personnel promised they would furnish more exact measurements within six months.⁵⁴¹

The analysis of data from devices tested during GREENHOUSE was given high priority at Los Alamos. These results were of crucial importance, regardless of the approach finally chosen for a thermonuclear weapon. Despite the unfavorable calculations of Ulam, Everett, and the ENIAC, the Super was still in the running; indeed, some recent data showed its chances to be slightly improved. If this trend continued, the Super might be tested in the spring of 1954.

⁵⁴¹Trester, pp. 58, 59, 60.

The new Teller-Ulam ideas also appeared promising, but because their origin was so recent, there had been no time for close and critical study. Two mathematical analysis groups at Los Alamos were to undertake their study as a main task. It was too soon to establish a test schedule for a device based on the new ideas, but if a general feasibility study were finished by October 1951 and showed promise, perhaps a test of the Teller-Ulam principles could be held in the spring of 1953.

LASL would not carry all the thermonuclear approaches through to the testing stage. As soon as one became less attractive than another, it would be dropped until eventually the effort would narrow to a single design.⁵⁴² LASL would not attempt another test similar to the GREENHOUSE George shot to acquire further data on thermonuclear phenomena. To do so would detract from the effort to test a full-scale device.

Edward Teller's suggestion of using lithium-6 in the ALARM CLOCK in the summer of 1950, when coupled with radiation implosion, showed up well in theoretical calculations and gave promise for a practical thermonuclear weapon. However, the ALARM CLOCK was prone to problems with Taylor instabilities, caused by the intermingling of materials of widely-differing densities within the device.⁵⁴³ This concern with instability was such that the CYLINDER had been deliberately designed to minimize the unknown effects of this factor.⁵⁴⁴

⁵⁴²This approach caused AEC Commissioner Thomas E. Murray to ask, during the June 1951 thermonuclear program review at the Institute for Advanced Studies in Princeton, New Jersey, if limiting work to only one design was a violation of President Truman's directive of January 31, 1950 which ordered the AEC to "continue its work on all forms of atomic weapons." (IN THE MATTER OF J. ROBERT OPPENHEIMER, p. 84.) In August 1951, Edward Teller claimed that this one-design-at-a-time approach dictated by Norris Bradbury might be used as an excuse to drop the H-bomb program altogether, and that he had gotten Gordon Dean to overturn Bradbury's directive. (Memorandum for the File dated August 28, 1951 from Ken Mansfield, subject: Conversation with Dr. Teller.)

⁵⁴³Bethe, MEMORANDUM ON THE HISTORY OF THE THERMONUCLEAR PROGRAM, pp. 12, 13, 15. The Taylor instability was discovered by British mathematician Geoffrey I. Taylor; the theory stated that when a light material pushes against a heavier one, the interface between them will be unstable. (Szasz, BRITISH SCIENTISTS AND THE MANHATTAN PROJECT: THE LOS ALAMOS YEARS, p. 22.)

⁵⁴⁴Blumberg and Panos, pp. 127, 128.

While the thermonuclear reaction in any design would be concluded in a matter of microseconds, the effect of any instability was nonetheless consequential.

The Princeton Conference, June 1951

By the latter part of June, following a milestone thermonuclear program review meeting at the Institute for Advanced Studies at Princeton University, LASL scientists had analyzed and absorbed some of the results of GREENHOUSE, and were making new plans based on them. The meeting was conducted to discuss the status of H-bomb materials production and three major thermonuclear device models.

Among meeting attendees were all five AEC commissioners; five of nine members of the GAC; J. Carson Mark, Hans Bethe, Norris Bradbury, Darol Froman, Edward Teller, and others from LASL; and Drs. J. Robert Oppenheimer, Enrico Fermi, and John Wheeler.⁵⁴⁵

⁵⁴⁵"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 No. 2 (1989), p. 232. The Princeton meeting on June 16 and 17, 1951 has been documented at varying lengths in almost every popular history of U.S. thermonuclear weapons development. A recurring and still-unresolved conflict about the meeting is whether it was convened specifically to review the new Teller-Ulam ideas (as Hans Bethe claims in "Comments on the History of the H-Bomb") or whether it was called to review the entire thermonuclear program at Los Alamos.

The latter interpretation is Edward Teller's recollection; after William J. Broad wrote an article entitled "Rewriting the History of the H-Bomb," in SCIENCE magazine, Vol. 218, 19 November 1982, Teller claimed in a subsequent letter, printed in the December 24, 1982 issue of SCIENCE, that when he asked to speak at the June 16 session, Norris Bradbury denied him the opportunity. In a rebuttal letter printed in the same issue, Bethe stated that the meeting was called to discuss "Method D," the concepts of staging and radiation implosion; in "Comments on the History of the H-Bomb," Bethe states (p. 53) that Teller was allowed ample opportunity to speak and that Teller grew impatient only after the Los Alamos laboratory proposed a slower thermonuclear development schedule than Teller was willing to accept.

In his article "The Hydrogen Bomb: The Work of Many People" (originally published in THE LEGACY OF HIROSHIMA, by Teller and Allen Brown, Doubleday & Co., 1962, and most recently reprinted in BETTER A SHIELD THAN A SWORD, The Free Press,

The findings of the meeting were:

I. General

The recent Eniwetok experiments in the thermonuclear field have been conspicuously successful, not only in the extraordinary number of pertinent observations and records which were made, but in demonstrating the practical fact that a thermonuclear reaction involving tritium and deuterium can be made to take place out of equilibrium with radiation, both external to and at the center of a fission reaction (Author's note: references to the George and Item shots, respectively).

Moreover, the experiments have demonstrated that thermonuclear reactions do not present any unforeseen phenomena and that the prediction of the behavior of thermonuclear systems on a theoretical basis can henceforth be carried out with great confidence. Finally, the experiments have demonstrated a new and interesting technique of radiation engineering, i.e.

New York, 1987), Teller also claims he was almost denied an opportunity to speak at the Princeton meeting. According to Hewlett and Duncan (p. 543), the agenda for the meeting included prepared remarks by J. Carson Mark on the GREENHOUSE results; comments by Darol Froman on Los Alamos lab plans for the future, including a proposed reorganization to emphasize thermonuclear research; and a few remarks by Dr. Norris Bradbury on lab philosophy and division of effort between fission and fusion weapon development. Bradbury did not include Teller in the list of laboratory spokesmen so that the physicist could express his own views freely (so say Hewlett and Duncan; Teller may have felt differently about being excluded from the list of designated speakers).

Given the enthusiasm of most of the Los Alamos T-Division staff for the new Teller-Ulam ideas, it is unlikely that they were downplayed at Princeton, although there may have been some lingering doubts about their utility: Teller had previously come up with many impractical and unfeasible thermonuclear weapon design ideas. As mentioned in Hewlett and Duncan (pp. 542-545), it is quite likely that Teller was extremely impatient at this time with any design approaches but his own. This memo shows that there were at least three separate thermonuclear concepts being considered in the early summer of 1951.

"radiation implosion" (Author's note: on a small scale, in the George shot) which has promise of great interest, and they have demonstrated that the technique of "boosting" fission weapons is one which merits further exploration.

Three general thermonuclear systems having a potentially "open-ended" nature (extremely large yields from inexpensive materials) are currently being considered. The degree of understanding of these systems is not now in such a state that one can describe with certainty either their relative or absolute expectations. However, it is the general objective of the Los Alamos Scientific Laboratory during the next four to six months, if possible, to determine the relative priority which should be accorded the study of these systems, and, thereafter, to put primary emphasis upon the development of a device of demonstrable significance along the lines of the best system.

II. Specific Thermonuclear Systems

The three lines upon which theoretical research in this field will be primarily concentrated are:

(A) Runaway Super

The prospects of the runaway super have slightly improved since the report presented to the General Advisory Committee in the fall of 1950 in view of the larger T-D cross sections experimentally observed. Furthermore, recent calculations have indicated that the inverse Compton effect is not as large as had been believed and will interfere less with the progress of the reaction.

It is still an open question whether it will be possible or economically appropriate to make a runaway super. No really significant progress in this matter can be expected prior to a full-scale Maniac calculation. This calculation is being prepared by John von Neumann and members of the staff of T-Division. No significant expansion of this group is

believed necessary except possibly to handle the work on the Maniac.

....

...It is apparent from the above that the theoretical effort on the runaway super will be continued at approximately its present level.

(B) Alarm Clock

Calculations in this field have so far been carried out by manual methods but have demonstrated that a compressed system of this nature has a significant probability of serving as the basis for a practicable device although Taylor instability and mixing are more likely to affect the functioning of a system of this nature than that of others being considered. (The uncompressed system seems to offer little encouragement.) A group headed by Nordheim will continue to work on this problem, including the effects of mixing.

The first reasonably complete machine calculations on an idealized spherical model are hoped to be complete during the fall of 1951 and should assist in determining the status of this system with respect to the other open-ended thermonuclear systems.

(C) Equilibrium Super

The potential promise of this system appears at this time to be as high as any yet proposed. On the other hand, it has had the least study and the possibility still exists that a decisive flaw may be found. The Theoretical Division at Los Alamos, supplemented by the Wheeler group and other outside assistance, will devote extensive theoretical effort to this system during the remainder of 1951.

The two fundamental questions to which the major effort will be devoted are:

- (1) The nature and extent of the burning to be expected in deuterium (and its modifications with Li^6 and/or H^3) under the relevant conditions for this system; and

- (2) The nature and extent of the compression which may be obtained by the technique of radiation implosion under various conditions. It may be noted that information of this sort will also be important for the Alarm Clock.

III. Boosted Systems

....General exploration of the field of boosted implosion weapons will continue. The gains to be expected in this field are limited but might ultimately turn out to be of interest. For the present, however, the demands for smaller and more versatile implosion weapons require developmental priority.

IV. Relevant Programmatic Activity

- (A) Detailed analysis of the results of the Cylinder and Booster shots. (Rough draft of report ... scheduled for August 1951.)
- (B) Cross section measurements involving D, T, and Li^6 . (Work currently in progress at Los Alamos Scientific Laboratory on T and D to be completed during summer, 1951, with the D-D cross sections to follow. Program being established at California Institute of Technology to become effective during 1952.)
- (C) Lithium Hydride (Deuteride and Tritide) studies. (To be undertaken at the Los Alamos Scientific Laboratory by CMR-Division with the objective of being able, in the case of the deuteride and hydride, to make pieces of considerable size by mid-1952. In the case of the tritide, to explore the fabrication and stability problems of this compound in a similar time scale.)
- (D) Opacities. (Work to be continued at Argonne at present level, 2-3 professional people.)
- (E) Boulder cryogenic facility. (Preliminary work to be devoted to problems of storing and transporting large amounts of liquid hydrogen. Ultimately may become involved in large dewar problems for thermonuclear device requiring these.)
- (F) Military worth. (A group of 3-6 people at

Los Alamos Scientific Laboratory who will be concerned with the economics, deliverability, and other considerations relevant to the military worth of thermonuclear and fission devices during the course of their development with the objective of not only assisting Los Alamos Scientific Laboratory thinking along these lines, but the preparation of relevant information for the military. This part of the program is, of course, not specifically thermonuclear in character, and may be staffed not only at Los Alamos but elsewhere.)

- (G) Equation of state of liquid hydrogen.
(Studies by GMX-Division at fairly low priority with the ultimate objective of being able to supply information of this nature during 1952-53 if required.)
- (H) High explosive compression large systems.
(No current experimental work contemplated until requirements become clearer.)

V. Considerations Relevant to Tests of
Thermonuclear Devices

- (A) It is presently intended that a device be proposed for test only if such a test is required for a significant increase in the understanding of thermonuclear systems, or if such a test is capable of leading in some foreseeable way to a device having weapon utility.
- (B) While any test contemplated will include as much diagnostic information as may conveniently be included, or as may be required for the success of the test, every effort will be made in the future to avoid experiments having the complexity of some of those employed at Greenhouse. Wherever possible, it will be planned that some macroscopic or readily accessible effect (such as yield) will be capable of revealing the significant information desired.
- (C) A time of six months to a year or more is likely to be required between the date on which it may be possible to describe a device fully enough to warrant a decision to test and the date on which an actual

model may be ready to test. Considering the present state of theoretical knowledge, it is felt to be improbable that a test of any thermonuclear device which meets the above conditions can be ready before (the) fall (of) 1952. Even to meet such a date it will be necessary to begin to focus effort on the test device sometime late in 1951, with a consequent low priority on studies of other systems.

- (D) Should the device selected for test require large amounts (kilograms) of Li^6 or lithium hydrides (deuteride or tritide), then its test date will also depend upon the success of production facilities in meeting production rates of the order of several kilograms per day. Steps are understood to be underway to meet such a rate at the earliest practicable date.
- (E) Should the device selected for test be along the lines of the classical Super, it seems improbable that the test could be conducted earlier than 1954. Such an estimate is implied by the fact that consideration of the classical Super is more difficult than either of the other systems and the basis for a decision to test seems unlikely to be available in time to make an earlier test possible, even if full attention should now be devoted to this system.
- (F) No new decisions concerning tritium production at Hanford are believed to be required at this time. Presently planned production will lead to an amount of approximately [deleted] by the latter part of 1952. This amount seems to be reasonably related to any test program which might be conceived in the time scale described above; and, in the event tritium were not required for open-ended thermonuclear tests, such an amount is still of interest in connection with boosted weapons. Clearly, however, this question should be re-examined whenever production decisions at Hanford so require.
- (G) In no sense should the above test dates be regarded as more than conceptual and as a guide to fiscal and administrative planning. Clearly, unforeseen difficulties can enormously prolong the intervening period;

unforeseen simplifications might shorten it. However, it is hoped that in October or November of this year sufficient theoretical progress will have been made to warrant a thorough review of the situation and any significant steps which should then be taken.⁵⁴⁶

Operations IVY and CASTLE would soon destroy some of the lab's "high confidence" in predicting theoretical behavior of thermonuclear systems: predicted yields of experimental devices in both test series would be seriously underestimated more than once.

The Princeton conference also included proposals to conduct tests on several key points of the new bomb designs and to undertake further tests on each of the separate steps before full-scale integrated tests were made. The first approach was referred to as the "shoeshine" approach to make three tests together at one time. The latter approach, to conduct lower-

⁵⁴⁶LOS ALAMOS SCIENTIFIC LABORATORY THERMONUCLEAR PROGRAM, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, 22 June 1951. All emphasis in original. "Mixing" is the (undesirable) intermingling of non-nuclear materials with nuclear fuels (see Glossary for full definition).

"Nordheim" was Lothar W. Nordheim, a LASL consultant who worked at the lab full-time between September 1950 and September 1952. The "Wheeler group" was the Matterhorn Project at Princeton University mentioned in a preceding footnote. The "Cylinder" is a reference to the GREENHOUSE George device also named in an AEC teletype dated April 26, 1951 from Gordon Dean to General Elwood Quesada and in a letter dated June 22, 1951 from Robert LeBaron, Chairman of the Military Liaison Committee, to the AEC, to Gordon Dean, AEC Chairman.

"Argonne" is the Argonne National Laboratory near Chicago, Illinois. The "Boulder cryogenic facility" was a new National Bureau of Standards (NBS) plant under construction at Boulder, Colorado, in 1951. Dewars are essentially large thermos bottles (insulated containers with silvered glass double walls with an evacuated space between the walls) for storing and transporting large amounts of liquefied gases; they are named after their inventor, Sir James Dewar, a late-nineteenth century Scottish chemist and physicist who was the first person to liquefy hydrogen, and who also produced large quantities of liquid oxygen. The reference to "high explosive compression of large systems" may be a reference to using HE for thermonuclear ignition.

level unit tests, was referred to as the "components testing" program.⁵⁴⁷

The use of Li^6D was also discussed, along with means of producing it in large quantities. At least one thermonuclear design approach involving "in situ" generation of tritium (probably Teller's "Equilibrium Super") was considered promising enough to warrant immediate production of Li^6 .⁵⁴⁸

During the Princeton meeting, a strong difference of opinion surfaced over how soon a full-scale thermonuclear test could be conducted. LASL was not prepared to consider any date earlier than the spring of 1953 -- nearly two years hence -- while Edward Teller campaigned for the spring of 1952.⁵⁴⁹

Preparations for Enriched Lithium Production

Also in June, clearing and grading were underway and foundations were being poured at a 240,000-acre site close to the Savannah River near Aiken, South Carolina for the DuPont tritium reactors and heavy water plant. This location had been chosen because the chemical composition of the river water was good, and the mild climate offered a long construction period during the year.

More than 25,000 DuPont workers and contractors were building five heavy water reactors for tritium production, and six dual-temperature deuterium oxide distillation towers.⁵⁵⁰ In spite of all this progress, heavy water would still be in critically short supply until January 1952, when six more dual-temperature production units would go into operation at an AEC facility in Dana, Indiana. **Figure III-6** shows the heavy water plants eventually built at the AEC's Savannah River facility.

A pilot plant at Dana had already supplied valuable corrosion data for the larger, permanent units being built at the billion-dollar Savannah River site. Tritium would also be in short supply for at least fifteen months until the reactors were completed.⁵⁵¹

⁵⁴⁷Borden-Walker H-bomb chronology, p. 65.

⁵⁴⁸Borden-Walker H-bomb chronology, p. 65.

⁵⁴⁹Memorandum for the File dated January 13, 1953 from John Walker.

⁵⁵⁰Hewlett and Duncan, pp. 531, 544.

During the summer of 1951, the AEC decided to start producing lithium-6 in large quantities.⁵⁵² In August, funds were approved for construction of a pilot plant to be in production by autumn 1952. Analysis of the results of Operation IVY and a review of the thermonuclear weapon development program led to the conclusion that the design approach used in the SAUSAGE device, namely the inclusion of a "sparkplug," which generated tritium in situ in a weapon was promising enough to require immediate production of large quantities of separated Li-6.⁵⁵³

Early in July, the first heavy water unit at the AEC's Dana plant was 67% complete.⁵⁵⁴ By early August, a pilot plant was being built at Oak Ridge to separate lithium-6 from lithium-7.⁵⁵⁵

In an executive session held on August 23, the Atomic Energy Commission requested its General Manager to prepare a study of the likely future work load at LASL and Sandia and of possible ways in which the AEC's nuclear weapons research and development program could be expanded most effectively.⁵⁵⁶

By the middle of the summer, DOD had an extensive "shopping list" of new atomic weapons for a wide spectrum of potential applications; coping with this "wish list" added further burdens to the overworked staff at LASL:

The military services are currently considering new types of atomic weapons for

⁵⁵¹Letter dated June 22, 1951, from Robert LeBaron, Chairman, Military Liaison Committee to the USAEC, to Gordon Dean, Chairman, USAEC.

⁵⁵²Teller, COMMENTS ON BETHE'S HISTORY OF THE THERMONUCLEAR PROGRAM, p. 2.

⁵⁵³Letter dated August 27, 1951 to Robert LeBaron, Chairman, MLC, from Gordon Dean, Chairman, USAEC.

⁵⁵⁴Memorandum to the File dated July 11, 1951 from Wayne P. Brobeck, subject: Meeting with Walter Williams, AEC, on Friday, July 6, p. 2.

⁵⁵⁵Memorandum for the File dated August 10, 1951 from Walter P. Brobeck, JCAE, subject: Isotope Separation Group.

⁵⁵⁶Memorandum dated April 29, 1954 to Chester G. Brinck, Assistant General Counsel, from Leonard A. Jacobvitz, Attorney, Office of Assistant General Counsel, subject: Participation of the University of California Radiation Laboratory in Weapons Development Program.

which there may be a requirement in the near future. Study is being devoted to new applications of current weapons which will require increased ordnance effort and nuclear improvements on weapons already in the program. Following is a tabulation of some of the potential weapon requirements which are under consideration by the Department of Defense.

The weapons listed are in various stages of study as pertains to the establishment of operational requirement and technical feasibility. Although these projects are not to be considered as a formal requirement at this time, it is felt that this information may be of use to the Atomic Energy Commission.

- a. Improved 60-inch (diameter) implosion weapon.
- b. Improvement of gun-type weapons to give better nuclear efficiency.
- c. Atomic weapons for air defense.
- d. Warheads for rockets of both surface-to-surface and air-to-surface types.
- e. Weapons of one to 10 KT yield for tactical employment.
- f. Atomic depth charges for air, surface, and subsurface launching against submarines.
- g. Atomic warheads for torpedoes for anti-submarine and anti-shipping use.
- h. Air-delivered implosion-type weapons (to use as mines) in harbors.
- i. 16-inch atomic projectile for precision shore bombardment.
- j. Warhead for small guided missile for use against guided missile-firing

submarines.⁵⁵⁷

As to what the Soviets were up to on the other side of the Iron Curtain, a late July Central Intelligence Agency (CIA) estimate of the status of Soviet thermonuclear weaponry noted a "lack of specific evidence of Soviet activity in this field," and stated:

There is no specific evidence that the Soviet atomic energy program is or is not being directed toward the production of thermonuclear weapons. It is known that (Klaus) Fuchs, in answer to a question of "super" weapons in the spring of 1947, told the Soviets what the general thinking on this problem was at the time of his departure from Los Alamos in 1946. Undoubtedly, the possibilities of developing such weapons are being extensively investigated in Soviet laboratories.

Considering the most recent American technical information, there is considerable uncertainty as to whether or not it is physically possible to construct a thermonuclear weapon that will function satisfactorily. Even if one can be built, it seems likely at the moment that it may require the sacrifice of a disproportionately large quantity of fissionable material usable in ordinary fission weapons.⁵⁵⁸

Since at least the spring of 1951, the AEC had in place a long-range fission detection project to detect Soviet nuclear atmospheric tests. Another part of this project measured atmospheric content of radioactive krypton-85, a by-product effluent of the production of plutonium.⁵⁵⁹ In March 1951, the AEC had estimated that the Soviets had produced only about

⁵⁵⁷Memorandum for the Chairman, Atomic Energy Commission, dated July 18, 1951 from Col. H. McK. Roper, USA, Executive Secretary, MLC, Subject: Military Guidance in the Atomic Weapon Field, p. 3.

⁵⁵⁸"Status of the Soviet Atomic Energy Program," CIA/SI 113-51, 28 July 1951, Central Intelligence Agency, p. 12.

⁵⁵⁹"Krypton-85 Method of Concealing," paper dated June 4, 1952 prepared by the JCAE staff.

20% as much plutonium as had been manufactured by the U.S. at that time.⁵⁶⁰

The AEC proposed installing chemical "scrubbers" in the effluent stacks of the Hanford and Savannah River isotope separation facilities to help conceal the extent of U.S. generation of radioactive atmospheric krypton-85. The Soviets, of course, could easily do likewise to conceal the extent of their plutonium production.

Later in the spring, there was intelligence evidence that, besides plutonium-generating reactors, gaseous diffusion plants and electromagnetic uranium isotope separators were being built by the Soviets in the Ural Mountains, with assistance from captured German scientists. The diffusion plant was expected to supply uranium enriched to one to five percent in the U-235 isotope, which would be fed to an electromagnetic stage like the AEC's Y-12 beta units; the U-235 would then be enriched to bomb-grade.⁵⁶¹

By the end of August, the AEC expected that an American thermonuclear device would not be ready for testing until the winter of 1952-53.⁵⁶² Late in the month, AEC Chairman Gordon Dean wrote to MLC Chairman Robert LeBaron and advised him that the AEC had authorized the test and construction of a plant capable of producing one pound (about a half kilogram) per day of lithium enriched to 95% in the Li-6 isotope.⁵⁶³ This followed a recommendation of the June conference at Princeton, which had stated that the approach to weapon design involving a "sparkplug" and in situ tritium generation within the weapon merited quantity production of lithium-6.⁵⁶⁴

The "sparkplug" for the forthcoming test device was to be boosted with tritium; this would "cost," in terms of lost

⁵⁶⁰Minutes of the Twenty-Fifth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, March 15, 16, and 17, 1951, p. 13.

⁵⁶¹Minutes of the Twenty-Sixth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, May 8, 9, and 10, 1951, p. 2.

⁵⁶²Memorandum dated 18 December 1952 to Commander, JTF 132, from Commander, TG 132.1, Subject: Submission of Report of the Commander, Task Group 132.1 on Operation IVY.

⁵⁶³"Comments Concerning the H-Bomb Development," Department of Defense, Military Liaison Committee to the Atomic Energy Commission, 21 April 1954.

⁵⁶⁴Borden-Walker H-Bomb Chronology, p. 65.

plutonium production, the equivalent of only two to four atomic bombs. The device was expected to have a yield on the order of three to 10 megatons.

At this time, Edward Teller thought that deliverable hydrogen weapons might be as big as nine feet in diameter, and weigh from 10 to 20 tons. He stated that some thought had been given to using the Air Force's new four-engine Douglas C-124 transport as a delivery vehicle, but that it would be highly vulnerable to interception by enemy fighters. He believed that more suitable bombers would be developed, and that future technical improvements would, almost without question, considerably reduce the size of thermonuclear weapons from the dimensions now contemplated for a full-scale test device.⁵⁶⁵

Teller and the Second Weapons Laboratory

Edward Teller continued his relentless lobbying for a second weapons lab in the fall of 1951. He had started this campaign in the spring, not long after he and Ulam had devised the first technically feasible thermonuclear weapon design principles.

As early as April 1951, just a few weeks after jointly writing with Stan Ulam the seminal LAMS-1225 report, Teller had sent to AEC Chairman Gordon Dean a two-page memo with a 12-page attachment laying out his plans for a second weapons lab at Boulder, Colorado specializing in thermonuclear research:

Following our conversation, I have given thought to the alternatives which present themselves concerning the future of the thermonuclear program.

A detailed plan for a new site would enable one to judge more realistically the advantages and disadvantages of a new location. The past two weeks have been too short to formulate such a plan, but I have tried to arrive at an outline of manpower and space requirements as well as some estimates of the cost of principal equipment.

⁵⁶⁵Memorandum for the File dated August 28, 1951 from Ken Mansfield, subject: Conversation with Dr. Teller. In this memo, Teller mistakenly names the much-smaller twin-engine Fairchild C-123 as the proposed USAF bomb carrier.

I am attaching this outline for your use, but would like to emphasize that it is submitted only in order to put discussions on a more concrete basis and not as a definite proposal to the (Atomic Energy) Commission.

A wide choice for the location of a site presents itself. I have singled out Boulder but have briefly discussed some other places to show (their) relative merits.

While I am sending you this outline in order to complete the picture, I am fully aware that in order to avoid delay and duplication it might be of considerable advantage to keep the thermonuclear program at Los Alamos. If anyone hopes to achieve practical results within an extremely short time span -- such as a year -- then a delay would be particularly serious. Recent theoretical considerations offer some prospect of a fairly simple thermonuclear system. On the other hand, Los Alamos carries a heavy burden in the fission weapon development program and it is not evident whether a more prolonged and ambitious (thermonuclear) program could be carried out at Los Alamos.

Los Alamos is in my opinion the best scientific laboratory of any government department. I believe nevertheless that the following changes would have to be introduced in order to pursue the thermonuclear program in an effective way:

- (a) Concentrate responsibility for the scientific administration of the thermonuclear work in a single individual who actively heads the program and participates in its exploration as a full-time job.
- (b) Induce a considerable number of scientists, including some now outside the AEC, to spend

full time on thermonuclear
questions at Los Alamos.

These changes would be effective only if they bore the fullest support of the laboratory administration.

A new site would automatically meet conditions (a) and (b) above. In addition, considerable impetus would be given to the recruiting program. The drive and enthusiasm in a project with a single but large goal was shown in the early days of Los Alamos.

A new site should operate in such a spirit and I believe that it is important that the project be kept relatively small. The top scientific staff might amount to not more than 50 people. Frequent discussions and daily contact on the single subject of thermonuclear work would distinguish such a site from the compartmentalization of ideas (not dictated by security) now so prevalent.

...

I sincerely hope that a vigorous program will be planned and that I may contribute to the defense work by participation in the field of Atomic Energy (sic) rather than by helping other branches of defense.

Teller appended to this memo a lengthy attachment in which he detailed his estimates of the scientific manpower and equipment and fields of study for the new lab. He stated that the goal of the new site would be "to investigate theoretically all avenues of thermonuclear work; to prepare for early intermediate tests (about nine to 24 months hence); to design and execute a full-scale demonstration of a thermonuclear explosion; and to design (but probably not to build) thermonuclear weapons."

To achieve these expansive objectives, Teller proposed a lab at Boulder, Colorado, to operate in conjunction with the new National Bureau of Standards (NBS) cryogenics lab, which was to begin operations at Boulder early in 1952. Teller also suggested that a second MANIAC computer be built at Boulder by early 1952, to supplement computations being conducted by Project Matterhorn at Princeton, New Jersey.

Teller noted that liquid hydrogen facilities were essential for the oldest type of Super work, and that the NBS facility at Boulder could provide all the necessary technical support required by the adjacent thermonuclear lab.

The greatest delay in establishing a new lab would likely be caused by time required to set up a new explosives testing site. Teller noted that HE lenses required by the new lab could be fabricated at LASL, and might subsequently be modified by an HE machining facility in Boulder. An HE magazine sufficient to hold as much as 25 tons of explosives would also be required.

Teller reviewed alternate possible sites for the new lab, including Princeton, Chicago, Brookhaven National Laboratory (on Long Island, New York), and Tonopah, Nevada. All of these alternates were judged unsuitable for several reasons, including their distance from the NTS or their vulnerability to a surprise Soviet bomber attack.

Teller concluded by stating that the total manpower requirements for a Boulder site would include 50 senior scientists, 82 junior scientists, and 228 technicians. He proposed an optimistic schedule for the establishment of the new lab, and concluded that it could be in "fairly routine operation" by the spring or summer of 1952.⁵⁶⁶

In August, Teller had accused LASL of becoming "less and less adventurous scientifically" and now regarding its main mission as protecting itself from outside criticism by embarking only upon projects that were certain to be successful. He accused LASL of "rapidly taking on all the features of a monopolistic and secret bureaucracy at its worst."

Teller parenthetically noted that the New Mexico lab's record of consistently-successful field tests were not so much a measure of innovative risk-taking as they were of a determination to proof-test a gadget only after enough time and effort had been expended on it to make its success virtually assured.

Teller blamed LASL's too-conservative attitude upon the lack of publicity and competition in the hydrogen weapons arena, and suggested that the only ways to accelerate progress on the Super program would be to either fire Norris Bradbury and Marshall Holloway, set up a separate and enlarged thermonuclear research division at LASL, set up a second independent nuclear weapons lab, or amend the Atomic Energy

⁵⁶⁶Letter and attachment dated April 20, 1951 to Chairman Gordon E. Dean, USAEC, from Edward Teller.

Act to allow allied countries, especially Great Britain, to assist with the program.⁵⁶⁷

Teller wrote to AEC Chairman Gordon Dean on September 11, 1951 and proposed again that a new lab be located in Colorado, now possibly adjacent to the new Rocky Flats plutonium fabrication plant being run by the Dow Chemical Company. Teller's planned "Rocky Mountain Laboratory" was to be entirely administratively independent of Los Alamos and even of the AEC's Santa Fe Operations Office (Dean was very unlikely to agree to the latter condition).

The laboratory was to conduct both theoretical and experimental work; the experimental research was to be dedicated toward the pursuit of new ideas, rather than be committed entirely to one or more developmental programs. In this respect, the "Rocky Mountain" lab would be oriented far more towards research and development than LASL. In fact, Teller proposed that the new lab would not even have an explosives research division at the outset.

To further theoretical research, the lab would be equipped with complete up-to-date computing facilities.

Teller did not plan to "raid" LASL for scientific personnel; he suggested that only a "small number" of "scientific people intimately concerned with the thermonuclear program or particularly suited to the work to be performed at the new laboratory might be transferred from LASL." These persons would presumably be selected from the few LASL scientists still on friendly terms with Teller.

Teller also made it clear that he did not intend for either LASL or the "Rocky Mountain" lab to be limited to either fission or fusion weapon research. Although the main purpose of the new lab was to "(speed) weapon development," Teller stated in an attached appendix of technical details that "the following is an outline for a proposed thermonuclear laboratory."

In this appendix, Teller delineated the overall scope-of-work for the new lab:

To investigate theoretically all avenues of unconventional nuclear weapon design and in particular thermonuclear work; to prepare for and execute or help to execute weapon

⁵⁶⁷Memorandum for the File dated August 28, 1951 from Ken Mansfield, subject: Conversation with Dr. Teller.

tests; to design (but probably not to build) thermonuclear weapons.

As much experimental and developmental work as possible is to be farmed out to existing AEC installations or to outside organizations. This policy will help to keep the place small, utilize existing AEC laboratories to the fullest extent, and allow the new site to concentrate on essentials.

The new lab was to contain facilities and manpower dedicated to experimental physics, electronics, chemistry, metallurgy (with all uranium and plutonium fabrication to be done by the Dow Chemical Company), theoretical physics, computing, engineering (for test devices only), field testing, and machine shops.

Cryogenic work would be farmed out to the NBS Boulder lab and there would be no need for an independent explosives research and development group. The new lab would comprise a total of nearly 300 people at a cost well in excess of a million dollars.⁵⁶⁸

Planning for the First Large-Scale Thermonuclear Test

In mid-September, Dr. Norris Bradbury reported to a joint AEC-MLC conference and outlined the current status of the thermonuclear program and the various methods which might be used to apply fusion reactions to weapons. The principal part of Dr. Bradbury's discussion dealt with a proposed plan and schedule for the actual design and possible testing, sometime before the end of 1952, of a certain type device based on principles successfully demonstrated during GREENHOUSE.

Although Bradbury indicated that the whole program in regard to this device was very tentative and optimistic nature, he felt it necessary to proceed with the design and test of such a device on an accelerated schedule because the principles involved looked more promising than any others considered to date.

Bradbury emphasized that the proposed design, the probable yield, and the proposed completion date for the construction and testing of such a device were all a matter of guesswork at the moment and that the ensuing months would probably see

⁵⁶⁸Letter dated September 11, 1951 to Chairman Gordon E. Dean, USAEC, from Edward Teller, and attached appendix entitled "The Rocky Mountain Laboratory." Emphasis added. This September 1951 paper appears to be a rewrite of Teller's April 1951 proposal.

many schedule changes. Even with these reservations, he offered a figure of twelve months in which to design and proof-test the device; in short, the system should be increasingly ready for test between July and September 1952, or very soon thereafter.⁵⁶⁹

Bradbury stated that theoretical progress on the "classical Super" was awaiting completion of calculations on computers that were not yet available. He reviewed at length the Los Alamos Laboratory's thinking as to new "Super" devices, and he outlined the physical characteristics of two equilibrium devices around which the laboratory was now concentrating its thinking.

Stressing the advanced state of thermonuclear design as opposed to its status just six months earlier, Bradbury emphasized that virtually every component of the equilibrium device they envisaged testing in 1952 revolved around engineering, development, and logistical problems, rather than theoretical difficulties.

Bradbury stated that neither tritium nor lithium-6 were needed for the test, and that all materials required for the operation were either in production or contracted for and on schedule. Tritium was expected to be necessary only for the "booster" program to enhance fission weapon yields.⁵⁷⁰

The test device itself was expected to be a 30-to-50 ton basilisk about six feet in diameter, and about twenty feet long with an outer case of steel or other ferrous metal one foot thick to provide the longest possible disassembly (i.e., containment) time. Fabrication of a smaller-scale "breadboard" model posed some difficulty because no heavy elements were allowed to be present in the hydrogen and deuterium during the functioning of the device.

If design of the full-scale device were "frozen" in mid-November 1951, a test could be conducted sometime in the last half of 1952. Since LASL was not entirely qualified to conduct the engineering design of the device, the AEC had approached the Arthur D. Little Company, American Car & Foundry, and the Chrysler Corporation to design and manufacture device components. A. D. Little specialized in cryogenics and ACF had built both large conventional and atomic bomb ballistic casings.

⁵⁶⁹Moore and Bechanan, p. 24.

⁵⁷⁰Notes on Informal Conference of the AEC and MLC, September 13, 1951.

An important feature at this time in the design of the secondary of the test device was the use of a U-235 tamper.⁵⁷¹ This would give the highest possible yield-to-weight and yield-to-volume, since U-235 was fissionable over a much greater range of neutron energies than was U-238.

As a technical matter, it was necessary to decide whether to study each of the separate steps in an "Equilibrium Super" and/or to undertake a test which combined several steps. In addition, there was the technical question of the earliest possible date for the next thermonuclear test.

While a majority of LASL scientists approved a test at the beginning of November 1952, a minority (led by Teller) believed that a date four to six months earlier was feasible. LASL finally decided to test on October 1, 1952, and on September 17, 1951 responsibility for construction of the test device and meeting the November test date was assigned to Dr. Marshall Holloway.⁵⁷²

A simple diagnostic test program was proposed, including measurement of alpha, neutron intensity, various forms of radiation, and percentage of thermonuclear burning. In addition, the motion of the outer case would be photographed during a period of several microseconds. Bradbury anticipated that the next thermonuclear test, following the large-scale "equilibrium" experiment, would use lithium hydride in place of liquid hydrogen and lithium-6 deuteride instead of liquid deuterium.⁵⁷³

Later in September, LASL recommended the production of large quantities of lithium-6 on a daily basis starting in mid-September 1953, two years in the future. Previously, the expectation had been for production of several kilograms of lithium-6 by the fall of 1952 and one kilogram per day thereafter, but it was now thought that a lower production rate might result in a more efficient production plant.

⁵⁷¹"Chronology of Major Decisions Regarding Thermonuclear Program," USAEC, June 14, 1951, p. 5. This document states: "Theoretical work has continued on [deleted] a specialized high-yield thermonuclear device which would involve a large mass of uranium-235 [deleted]."

⁵⁷²Borden-Walker H-bomb chronology, pp. 65, 66.

⁵⁷³USAEC Memorandum for the Record by Col. Richard T. Coiner, Jr., USAF, Deputy Director of Military Application, Subject: Dr. Bradbury's Presentation of Tentative Laboratory Conclusions Regarding Thermonucleonics.

LASL also recommended that there be no further loading of enriched uranium at Hanford for tritium production. (The initial loading was expected to be processed by July 1952.)⁵⁷⁴

Near the end of the month, the JCAE met with the commissioners of the AEC to discuss the U.S. thermonuclear weapons program. It looked now like the earliest possible date for a test would be July 1952, and it certainly could be held by December 1952. The JCAE was told that the AEC had the alternatives of proceeding step-by-step (the so-called components testing program) or of combining several technical experiments into one full-scale test.

A JCAE staff paper discussed during this meeting stated that qualified witnesses disagreed on the question of whether the AEC had dedicated enough effort to the hydrogen project. In response to this comment, AEC Chairman Dean asserted that progress on the H-bomb had to be broken down into two periods: from late 1949, when the program was stepped up, until June 1951, and the second period from June 1951 to the present.

As to the first period, Dean said that things went slowly because of discouraging calculations which indicated that very large amounts of tritium -- on the order of kilograms -- would be required by a Super, and because of the promising development of a new fission weapon pit (possibly the Fox pit) which would greatly increase the yield of atomic bombs.

As to the second period, Dean noted that when the "three concepts" were proposed, "the atmosphere changed, I would say, even on the part of the GAC, and the people who had been reluctant to (support) the Super weapon became rather interested, and some rather enthusiastic."

Regarding current proposals for a second U.S. nuclear weapons laboratory, Dean said that the AEC had not made a decision one way or the other. Dean also noted that AEC commissioner Thomas Murray was an enthusiastic and outspoken advocate of a "thermonuclear (weapons) laboratory."⁵⁷⁵

The JCAE surveyed several persons to get their comments on the idea of a second weapons lab; the consensus was that LASL was becoming increasingly over-burdened, conservative, and bureaucratic, and that the competition that might be provided

⁵⁷⁴Borden-Walker H-bomb chronology, p. 66.

⁵⁷⁵Borden-Walker H-Bomb Chronology, pp. 66, 67; "A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, pp. 12, 13.

by a separate laboratory would spur LASL onto better achievements. The new lab should also not limit itself solely to thermonuclear weapons research, but "might perform weapons work all across the board."⁵⁷⁶

At the end of September, the AEC's Oak Ridge facility was ordered to produce lithium deuteride in large volumes as rapidly as possible.⁵⁷⁷ At this time, the consensus of the AEC was that the next series of thermonuclear tests would not be held before the winter of 1952-1953.⁵⁷⁸

At the beginning of October, LASL began negotiations with ACF to start engineering design of the equilibrium test device;⁵⁷⁹ later in the month, ACF began work on the device, including fabrication of its "less unique parts." The system would be ready for test in a year.⁵⁸⁰

At this time, Enrico Fermi believed that the "classical Super" was not regarded as "a serious proposition" by those familiar with its problems. He also stated that calculations on it should be continued, but he added that there were enormous and very lengthy theoretical problems to be solved before LASL could even understand the theoretical operation of such a design.

It looked to Fermi now as if Taylor instability would not be as big a problem as thought earlier,⁵⁸¹ and that while the current test device would have a diameter of five or six feet, he was unable to make any prediction concerning its length or weight. (Fermi may have been unaware of the current LASL specifications of the proposed test device.)

Fermi also expressed very real concern over whether or not a deliverable weapon could be made from this design. He also believed that at least two years would elapse between the time of a successful H-bomb test and a the deployment of a

⁵⁷⁶"Some Comments on the Hydrogen Bomb Program," prepared by the (JCAE) Staff, September 28, 1951.

⁵⁷⁷Hewlett and Holl, p. 163.

⁵⁷⁸Moore and Bechanan, p. 24.

⁵⁷⁹Minutes of the Sixtieth Conference of the Atomic Energy Commission and the Military Liaison Committee, October 5, 1951; Borden-Walker H-bomb chronology, p. 67.

⁵⁸⁰Letter dated October 23, 1951 from Sumner T. Pike, Acting Chairman, USAEC, to Honorable Robert LeBaron, Chairman, MLC.

⁵⁸¹Memorandum for the File dated August 28, 1951 from Ken Mansfield, subject: Conversation with Dr. Teller.

deliverable bomb. In this estimate, the Italian physicist was not far off the mark: deliverable weapons would be available within 15 months of the first large-scale test of the "three concepts."

On the subject of a second lab, Fermi thought that it would, in the short run, damage LASL because there were not, in his opinion, enough sufficiently-qualified scientists in the U.S. to staff two facilities. In the long run, however, he believed that a competing lab might well pay considerable dividends and help offset the "increasing rigidities and bureaucratization" of LASL.⁵⁸²

Establishment of the Panda Committee

Also starting in October 1951, a new theoretical physics group was established at LASL to carry out work on large-scale thermonuclear explosions. The Family Committee was disbanded at LASL and replaced by a new committee, the "Theoretical Megaton Group," or TMG.

This group, known outside the lab as the "Panda Committee," was to continue thermonuclear weapons work; at its first meeting on October 5, the committee noted that its charter referred specifically to the test of a large-scale thermonuclear explosion and not the specific design of a weapon model, although a weapon would of course be the result of the committee's labors. The committee met on a weekly basis up to the fall 1952 test.

The committee was divided into five separate and coordinated subgroups: Dr. J. Carson Mark was to lead theoretical work; device engineering would be the responsibility of ACF; LASL's J Division was to handle the forthcoming Eniwetok test; and LASL's CMR, GMX, and P (Physics) divisions would also be involved.

Overall direction and coordination of the TMG was the responsibility of Dr. Marshall G. Holloway, then Division Leader of W Division, who was relieved of his duties at the latter and reassigned to head the newly-formed committee.⁵⁸³ The TMG's first assignment was to design, fabricate, and test a thermonuclear device in the fall of 1952, possibly as early as October 1, 1952.

⁵⁸²Memorandum for the File dated October 2, 1951 from J. Kenneth Mansfield, subject: Conversation with Dr. Enrico Fermi on Monday, September 24, 1951.

⁵⁸³"Notes on Meetings of the Theoretical Megaton Group," Appendix C to September 12, 1952 Walker memorandum; Moore and Bechanan, p. 54.

One production problem was that of inducing various shops throughout the U.S. to manufacture one or two strange or odd pieces about which they could not be given complete information for security reasons. This practice paralleled wartime work on the first U.S. atomic bombs.⁵⁸⁴

Also, by late 1951, successful thermonuclear weapon delivery appeared much more feasible than just a year earlier. The general size and weight of a militarily-practical weapon had become much clearer. An estimate in early October stated that a deliverable hydrogen bomb would be enclosed in a case approximately 20 feet long and six feet in diameter. Its weight would be approximately 50,000 lbs., within the lifting capabilities of the B-36 and the newly-developed B-52.⁵⁸⁵

If the test scheduled for late 1952 was successful, LASL could probably have one or two custom-made thermonuclear bombs ready for aerial delivery by the fall of 1953.

Thinking of B-36 or B-52 aircraft as carriers (the new XB-52 was nearing completion and rollout at the Boeing plant in Seattle, Washington at this time), LASL feared that information then on hand indicated that the Air Force could not safely deliver a bomb of more than 15 megatons explosive force.

On the brighter side, the bombs conceived at the time would be constructed with thick casings absolutely immune to antiaircraft fire or shell fragments, reopening the way for drogue parachute delivery.

LASL also suggested increasing the reflectivity of exposed delivery aircraft surfaces to lessen the effects of the bomb's heat radiation. (The Air Force eventually painted the undersides of the wings, tails, and fuselages of B-47, B-52, B-58, and B-36 aircraft with reflective white heat-resistant paint after weapons tests revealed the vulnerability of these bombers to thermal effects of nuclear explosions.)⁵⁸⁶

Opposition to the Second Weapons Laboratory

One consideration in the future of LASL, even as successful thermonuclear weapons appeared increasingly likely, was the rising chorus of voices, orchestrated mainly by Edward

⁵⁸⁴Moore and Bechanan, p. 54.

⁵⁸⁵Letter from Harold Agnew to Commanding General, Wright Air Development Center (WADC), 3 October 1951; Trester, p. 60.

⁵⁸⁶Trester, p. 60.

Teller, asking that a second U.S. nuclear weapons laboratory be established to "increase the rate of development of atomic weapons in the United States."

LASL director Norris Bradbury did not view favorably the possibility of a second lab, which would only draw resources and future work from Los Alamos, and he was quick to strongly defend LASL's past and likely future accomplishments and to argue against the creation of a competing weapons lab:

Attention is currently being directed to the question of increasing the rate of development of atomic weapons in the United States, and you have asked for the comments of the Los Alamos Scientific Laboratory on this important subject.

It must first be clearly understood that the phrase "development of atomic weapons" means different things in different circumstances. In one context, this phrase may mean such things as the development of new techniques of attaining a nuclear explosion (e.g., thermonuclear explosions or "boosted" fission weapons), development of nuclear explosions in different geometries, increase in the efficiency of utilization of fissionable material under specific geometrical circumstances, and related developments which lie almost wholly in the nuclear, hydrodynamic, and explosive fields.

In quite another context, the development of atomic weapons may mean developments in the application of existing types of atomic warheads to various conditions of military use. Under these circumstances, one may be discussing the development of atomic mines, of atomic warheads for rockets and guided missiles, of one or another specialized adaptation of the "gun" type nuclear assembly.

These two differing interpretations may have a technical interrelation since the characteristics required by an application may interact with the basic warhead to a serious degree. Nevertheless, it is essential to distinguish between the two basic goals since the technique of enhancing progress toward one may not be the same as towards the other.

It is appropriate to discuss next the possible extent of the unexplored areas in the two basic fields above. With respect to the development of fission weapons, there is, of course, a natural limit set by 100% consumption of the fissionable material in any given device. (Even this limit is probably theoretically unattainable.)

Since the current efficiencies of fission weapons run from possibly [deleted] one may expect that future research and development can lead to perhaps [deleted] as many weapons of a given characteristic from a given mass of fissionable material at the high end of the spectrum.

The field at the lower end of the spectrum extends farther but is intrinsically more difficult and perhaps more limited, but conceivably could permit as much as possibly a factor of [deleted] to be attained in relative effectiveness.

It is important, however, to note that the foreseeable prospects in fission weapon efficiency are not spectacular. Moreover, precisely this sort of gain has been attained through the efforts of the Los Alamos Scientific Laboratory since 1945, and it is not unreasonable to suppose that similar gains could be anticipated over a comparable period in the future even under present circumstances.

The exploration of the thermonuclear field is, of course, just beginning. However, even at this time, one may feel fairly confident that fusion weapons will be of importance and interest only if their yield is very large -- certainly in the megaton region. Moreover, one may expect that they will be inherently large and bulky devices of considerable weight.

In this respect, they will compete with the potential development over the next years of fission weapons of perhaps one megaton yield in "conventional" geometries and weights.

The combination of fission and fusion devices (boosted weapons, [deleted] and the

like) is also just beginning, although the efficacy of simple boosting is fairly well understood and in many cases requires only application to a specific device and comparison of the practical problems presented with those presented by boosting [deleted]

More spectacular developments such as autocatalysis (at present no more than a name) and the [deleted] as well as thermonuclear devices present fields for exploration beyond the areas currently seen -- but are now of such unknown fertility that any real prognosis is impossible except that efficiencies beyond 100% cannot be achieved.

In the general field of the application of atomic warheads, the development is necessarily paced by the ingenuity of military thinking concerning the effective use of atomic weapons in warfare.

The early strategic employment of free-fall atomic bombs against large city areas caused a preoccupation following the war with this concept which has only recently been forced to compete with ideas of tactical use.

In general, one tends to be struck by the fact that concepts, plans, and techniques for the strategic use of free-fall bombs are reasonably well understood if not actually developed.

In contrast, the tactical use is largely a name coupled with some sort of intuitive feeling that there must be uses of this sort, and that the actual techniques of employment and their relative worth and effectiveness are almost entirely guess-work at this time.

An unfortunate consequence of this situation has been the fact that the Department of Defense, being uncertain as to just which tactical employment (or employments) should have priority, has requested the development of practically every type of application conceivable.

This, in turn, leads to the result that none of the application techniques proceed as fast as their particular proponents would wish or as the gravity of the situation may demand.

To clarify the distinction between the two interpretations of weapon development, it is instructive to consider the informal list of desired developments presented to the Atomic Energy Commission by the Military Liaison Committee in a memorandum dated July 18, 1951. This list contains the following items:

- "a. Improved 60-inch implosion weapon.
- b. Improvement of gun-type weapons to give better nuclear efficiency.
- c. Atomic weapons for air defense.
- d. Warheads for rockets of both surface-to-surface and air-to-surface type.
- e. Weapons of one to 10 KT yield for tactical employment.
- f. Atomic depth charges for air, surface, and subsurface launching against submarines.
- g. Atomic warheads for torpedoes for anti-submarine and anti-shipping use.
- h. Air-delivered, implosion-type weapons for mining action in harbors.
- i. 16-inch atomic projectile for precision shore bombardment.
- j. Warhead for small guided missile for use against guided missile-firing

submarines.

Since this list may be regarded as the best evidence of what is actually desired most rapidly, it is instructive to consider it item by item.

Item (a) clearly falls in the category of basic warhead research and development, although it is understood that the main preoccupation of the Department of Defense with this request lies in ballistics, fuzing, overall weight, automatic nuclear insertion, and related details of ordnance efficiency.

Item (b) is presumably thought [deleted] Moreover, since the improvement of efficiency in gun weapons is almost invariably accompanied by an increase in yield and not by an increase in (the) number (of stockpiled weapons), there must clearly be further exploration of the urgency of this request examined in the light of application to which it is intended to put the weapon.

Item (c) and item (d) are clear examples of pure application of atomic warheads to one vehicle or another. Their only interrelation with atomic warhead development lies in the specialized environmental conditions which the vehicle may impose and which may require study of the effect of the necessary mechanical engineering upon nuclear efficiency. (Author's note: the latter is a possible reference to the nucleonic and hydrodynamic effects of warhead support structures and delivery vehicle materials and configuration.)

Item (e), in spite of its general vagueness, probably implies fundamental warhead development and undoubtedly falls partly at least into a basic research category.

The remaining items, in their entirety, are examples of warhead application to specialized use, and their interrelation with fundamental development is the same as that described for (c) and (d).

As noted above, there is a relation between basic atomic warhead development and atomic warhead application. This lies almost entirely in the effect of the environment upon specific warhead components. In the case of current implosion weapons, practically the entire problem of the [deleted] Naturally, this does not suggest that the importance of an application may not well be worth such a loss.

Although the method employed to increase the rate of development of atomic weapons will necessarily depend upon the definition of the goals to be achieved, nevertheless one suggestion is occasionally made which is, perhaps, independent of this consideration. This suggestion makes reference to the business world and argues that "competition" should be introduced into research and development to prevent the evil effects of "monopoly." We wish to take strong exception to this principle for the following reasons:

- (a) In no research institution known to us is competition used as an incentive. It is effective, perhaps, in the commercial field where a cheaper or better product results in more sales and the inventor of this product can be directly rewarded by royalties or a commission.
- (b) In research, competition is wasteful of research talent. True competition would mean the insulation of one group from another resulting in duplication and overlap -- as well as the failure to cross-fertilize with new ideas and developments. The latter has been a basic concept of scientific research since its inception. If the "competitive" groups are to "keep in touch," i.e., collaborate, then the best collaboration exists when the distance is smaller --

i.e., when they are both
part of one group.

- (c) The location, staffing, and cost of a "competitive" organization to either Los Alamos or Sandia would present formidable problems. We compete for technical personnel now with every phase of academic, industrial, and government organizations. Were one thinking of a parallel Los Alamos, would one duplicate "S" site -- or would Los Alamos become a service organization to a new enterprise?
- (d) In commercial industry, the stimulus of added financial reward provides the basis for competition. In atomic weapon development, the stimulus to succeed and to advance arises from the unique responsibility of the laboratories engaged therein to the United States. Were this responsibility to be shared, the task loses much of its driving purpose; responsibility is shrugged off; blame is easily shifted. This unique sense of individual responsibility to the country on a matter of great importance should be regarded with the utmost seriousness. It is a strong factor in recruitment; it is a factor in morale and enthusiasm; it is the factor that brings from a laboratory the suggestion of a six-day week and fourteen atomic tests in one year.

It is believed to be a fundamental axiom that if the rate of atomic weapon development by the Atomic Energy Commission

is to be increased, this increase must take the form of narrowing the responsibilities of existing institutions, and then supplementing these institutions by others with equally limited responsibilities. Moreover, anything which is to be done rapidly must be done largely within the framework of institutions currently engaged to some extent in the program.

As a specific example of this, one may regard the entire problem of application of gun-type fission weapons to various uses as primarily a problem of mechanical engineering. It is a problem requiring ordnance equipment, and a problem largely in fact carried out by one or another agency of the Department of Defense. However, the primary responsibility for accomplishment and all the preoccupations that this implies is not with the Department of Defense but with Los Alamos.

It is possible and perhaps feasible to consider transferring the entire responsibility for this type of development to the Department of Defense and to reduce the responsibility of the Los Alamos Scientific Laboratory to that of consultation on request.

Since most of the applications of atomic weapons are going to involve primarily engineering problems whose exact extent will require extensive experimentation by the Department of Defense, it is tempting to suggest methods of operation which transfer the entire responsibility for these developments to the Department of Defense and utilize Los Alamos and Sandia as expert consultants.

This may involve some legal question of responsibility under the Atomic Energy Act which might have to be interpreted through a delegation by the Atomic Energy Commission to the Department of Defense of the actual developmental responsibilities while the Atomic Energy Commission retains the legal and fiscal responsibility.

An alternative proposal, which may be freer of legal questions, would be to regard Los

Alamos as responsible for the entire field of basic atomic warhead research and development and for consultation in all fields of application. The Sandia Corporation could then be responsible for all practical application of atomic weapons to whatever use. They would, of course, be free (and probably should) to make extensive use of the Department of Defense facilities through direct arrangements elsewhere and through absorption of Department of Defense personnel at Sandia.

To make such a proposal effective in accelerating the rate of atomic weapon development, it would probably be necessary to free the Sandia Corporation from all production responsibilities. These would then have to be taken up by another agency which had production as its sole objective. Such a division between development and production will slow down the immediate arrival of new weapons in the stockpile, but, in the long range, may make a better and wider stockpile.

All proposals of this sort have to take into consideration the precise relationship between the application and the basic development. To relieve the situation at Los Alamos in any real way, it would be necessary to specify that the procedure, for example in a rocket warhead development, would be as follows: [deleted] expected, if any.

This might well involve actual tests which Los Alamos would conduct in its capacity as "expert" to the extent it believed appropriate. Clearly, any good ideas forthcoming either as a result of these tests or from background of experience would be fed to the designers as early as possible in the design development.

However, it would not be a Los Alamos responsibility to come up with a design to fit this environment. Rather it would be a responsibility elsewhere to make the best engineering design compatible with the environment and compatible with nuclear efficiency as expounded by Los Alamos.

As a further example of this trend, it is assumed that Project Apple will ultimately relieve Los Alamos of engineering responsibilities in connection with nuclear components of free-fall weapons as well as assembly and inspection responsibilities.⁵⁸⁷

Just two days later, Bradbury sent a second letter to the AEC covering much of the same ground and making many similar arguments against a second U.S. nuclear weapons lab.

Bradbury claimed that creation of a "second Los Alamos" would not "show much accomplishment within two to four years" and would "present serious difficulties of staffing, would result in a considerable duplication of effort and expenditure (of funds), and would consume a rather considerable amount of time and effort between the two installations in the essential endeavor of keeping in touch with each other, no matter how much care we put upon program determination and division of effort (between the two labs)."

Bradbury also stated at that time, a year before IVY, that although setting up a second lab to accelerate development of thermonuclear weapons would probably only delay the date at which the next significant large-scale fusion experiment could be conducted, a second lab might speed up conversion of an experimental device to a militarily-useful weapon.

The LASL director also argued that the establishment of a new lab to develop implosion weapons would require extensive new facilities to fabricate high explosive charges, requiring personnel and time. The transfer of information alone from LASL would, Bradbury stated, slow down rather than expedite

⁵⁸⁷Letter dated 9 October 1951 to Col. Kenneth E. Fields, Director of Military Application, USAEC, from Norris Bradbury, Director, LASL. All emphasis in original. "S", or Sawmill site, was the location of much of LASL high-explosives implosion testing; after the new Livermore lab was set up, it eventually had its own location nearby, Site 300, for this type of testing. Project Apple was the new Rocky Flats, Colorado weapon pit fabrication facility opened on March 31, 1951; it was originally operated for the AEC by the Dow Chemical Company and its mission was to produce nuclear weapon components, including fabrication, final acceptance inspection of all capsule components, assembly of all new capsules, and limited training of military personnel. Many of these production-related functions were originally performed by LASL. (Report of the Manager, Santa Fe Operations, U.S. Atomic Energy Commission, July 1950 to January 1954, pp. 51, 52.)

the rate of improvement of fission weapons:

From time to time we are aware of a rather thinly-veiled criticism to the effect that the rate of progress in atomic weapons research and development of the Los Alamos Scientific Laboratory is inadequate to the national need.

We are obviously not in a position to contest the validity of this criticism other than to invite attention to the somewhat ironic fact that every current weapon development has arisen out of the suggestion (and in many cases, the urging) of this laboratory.

One possible way, of course, to speed up eventually the rate of weapon research and development is to establish additional laboratory facilities, presumably at some new site. This is the suggestion, occasionally made, to set up a "new Los Alamos."

We regard this proposal as one which could hardly show much accomplishment within two to four years, would present serious difficulties of staffing, would result in considerable duplication of effort and expenditure, and which would consume a rather considerable amount of time and effort between the two installations in the essential endeavor of keeping in touch with each other, no matter how much care was put upon program determination and division of effort.

This proposal (for a second lab) has been made seriously in connection with the thermonuclear program. It is dubious if its implementation at this time could do other than delay the date at which the next significant thermonuclear experiment might be made. However, careful planning might, through the establishment of an appropriate laboratory (or through its attachment to an existing installation) speed up the conversion of an explosive thermonuclear experiment to a militarily-useful weapon.

The establishment of a new laboratory to participate in the development of implosion type weapons seems to be somewhat more

questionable in view of the extensive facilities for explosive fabrication which would be required and the time which it would take to get them and to staff the new laboratory.

Clearly, to withdraw strength from Los Alamos for this purpose, to withdraw (explosive) charges from S-Site, and even the extensive consultation and transfer of information which would be required, would for some time slow down rather than expedite the overall rate of progress in (atomic weapons).

....

In the thermonuclear field, it is highly improbable that a significant test will be appropriate before May 1952. It is perhaps most probable that such a test will occur either in the latter part of that year or the early part of the following year.

The status of this field continues to be one of enormous theoretical interest but whose overall practicality cannot as yet be precisely stated. As noted earlier, it is extremely dubious if extensive administrative changes will do other than slow down this effort.

With regard to the ease of staffing additional (weapons development) effort elsewhere than at Los Alamos, it may be noted that Professor John A. Wheeler at Princeton, with essentially carte blanche and the attractions of an academic atmosphere, has been able to obtain the equivalent services of only eight full-time theoreticians after an extensive campaign of recruitment involving more than 115 (attempts). The tour of duty of many of those whom he did obtain is less than a year. The Los Alamos Scientific Laboratory Theoretical Division now has a staff of 109.

We believe it quite appropriate to examine from time to time both the level of accomplishment and the basic philosophy of the Los Alamos Scientific Laboratory. If we were to criticize ourselves, such criticism would primarily direct itself toward the fact that we have shown too much

preoccupation with the development, engineering, and production of specific weapons and correspondingly to little attention to the investigation of basic and underlying principles and phenomena involved in all atomic weapons.

It is, of course, necessary to point out that this preoccupation has resulted in the more rapid appearance of a given weapon in the stockpile -- but at a sacrifice in longer-range progress. It is, therefore, our intention, unless specifically directed otherwise, to take every opportunity to increase the fundamental and exploratory aspects of laboratory operations.

Perhaps our greatest hope along this line lies in the rapid completion and activation of Project Apple; our other lies in the gradual elimination from the laboratory program of specific developmental projects paced by military time scales.⁵⁸⁸

The AEC, for the most part, shared Bradbury's opinion at this time and was opposed to establishment of a second U.S. nuclear weapons research and development laboratory. The AEC argued that "competition" would not be healthy to weapons engineering, and that LASL's research and development efforts would be better served by moving production work out of the New Mexico lab:

The Joint Committee on Atomic Energy has recently made public reference to the possible desirability of the establishment of a second weapons laboratory similar to the Los Alamos Laboratory. What is the preliminary thinking of the (Atomic Energy) Commission with respect to this suggestion?

Preliminary thinking is that the establishment of a second weapons laboratory similar to the Los Alamos Laboratory is not desirable in the sense in which it is suggested, namely, to create competition between two independent groups in the field of research on atomic weapons.

⁵⁸⁸Letter dated 11 October 1951 to Col. Kenneth E. Fields, Director of Military Application, USAEC, from Norris Bradbury.

The Los Alamos Laboratory is the repository of the national "know-how" in the nuclear aspects of atomic weapons research. If a "second," or duplicate, laboratory were to be established, the Los Alamos group would have to be used fully to bring the second laboratory to the present level of accomplishment and ability with the first.

Assuming that this were done, maximum progress would then be achieved if the two groups worked closely together, with maximum coordination of their efforts, rather than independently, which is the idea behind the "competition" concept. About the only advantage foreseen in the two laboratories (idea) is duality of facilities in event of catastrophe, which is not the primary reason for this suggestion.

A more reasonable approach appears to be to transfer more of the workload in the production and production engineering fields from Los Alamos to other groups to permit expansion of the Laboratory's research and development activities.

Considerable additional progress in this direction will be possible as new facilities presently under construction are placed in operation, particularly Apple and Pantex. The entire problem is under active consideration. Firm recommendations are to be submitted to the Commission at an early date.⁵⁸⁹

The JCAE stated to Congress its desire that a second lab be created because of "the continuing necessity of an imaginative approach and a willingness to assume risks, including the risk of test failures, if the stakes appear to be sufficiently high."⁵⁹⁰

⁵⁸⁹"Budget Bureau Questions - 1953 Budget, Program 3000 - Weapons (and related Construction), Products, Storage and Surveillance," undated, but ca. early to mid-1951, p. 8. Pantex is the U.S. nuclear weapons final assembly plant, opened in October 1951.

⁵⁹⁰Chronology of Second Laboratory Events, 6/15/52 draft, p. 2.

At this time, the majority of the members of the GAC was opposed to the creation of a second lab:

We have been led to consider whether the creation of another laboratory, with responsibility coextensive either with the whole or with a part of present Los Alamos responsibility, would make for the progress of the work. Indeed, we are persuaded that added facilities for weapons improvement are both desirable and practical, and that they can help to free (LASL) from certain routine obligations which have in the past burdened its personnel and facilities.

But with this one exception, to which we shall return below, we are convinced that the establishment of another weapons laboratory similar in nature to Los Alamos is neither necessary, nor in any real sense feasible.

... the new laboratory would appear to be hardly feasible unless there were in fact a great many outstanding and specially-qualified scientists who could be persuaded to work there, and yet could not be persuaded to contribute to Los Alamos. It is our view that no such situation exists.

Circumstances in the future may, of course, call for a reevaluation; but we are convinced that at the present time, the establishment of a new "Los Alamos" or a "thermonuclear laboratory" would tend to impair the effectiveness of Los Alamos, and would not succeed in creating a new facility of comparable promise of achievement.

.. What we would recommend is that insofar as possible, and as rapidly as possible, relatively routine weapons development be transferred to another site, or sites, and that this be done in addition to the steps already undertaken to relieve Los Alamos of responsibilities for (weapons) production.

Specifically, we believe that such work in the development of gun-type weapons could equally well be done outside Los Alamos --

in fact (and this is an essential point) -
- without responsibility on the part of
Los Alamos except as an expert consultant.
We believe that similar arguments may well
apply to the development of more rugged
implosion weapons, such as are now desired
by the (armed services).

We believe that in time any laboratory
which undertakes development activities
for weapons improvement will need to have
some facilities, now unique to Los Alamos,
such as critical assembly sites, and
implosion study sites, in order that it
may itself be able to answer questions
about the performance of the warheads
without referring them to Los Alamos; but
whereas the responsibility for development
should be transferred with as little delay
as possible, the construction of the
special facilities should, as we think,
occur only as the need for them becomes
apparent.

...

(Note: Dr. Libby does not concur with the
views expressed in this section, which are
otherwise strongly held and
unanimous...) ⁵⁹¹

GAC member Willard F. Libby believed that with LASL's current responsibilities for current and future U.S. fission weapons program, the New Mexico lab could not devote sufficient manpower, funds, or attention to thermonuclear weapons development.

Libby favored a new lab for the latter purpose, and stated that "one would hope that every effort would be made to convince Dr. Bradbury and the Los Alamos Laboratory that the creation of the second (lab) was intended to clear the decks for (LASL's) work on fission weapon development, and was not intended as a criticism of their past efforts." ⁵⁹² Libby's comment notwithstanding, the creation of a second lab might also appear as a vote of "no confidence" in LASL's abilities.

⁵⁹¹Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, pp. 29, 30.

⁵⁹²Letter dated October 13, 1951 to Gordon Dean, Chairman, USAEC, from W. F. Libby, Member, GAC.

By mid-October, the GAC viewed favorably the proposed Teller-Ulam test device, calling it "a certainly interesting, possibly encouraging line of attack" that could well lead to a thermonuclear test within a year.⁵⁹³

In the meantime, the Soviet Union continued to make progress with its own nuclear weapons program. Its third fission bomb test, fired on October 18, used a device with a composite uranium-plutonium core; the shot yielded about 50 kilotons. U.S. intelligence sources estimated that the core of the Soviet device used about seven kilograms of plutonium and three and a half kilograms of uranium-235.⁵⁹⁴

This test was comparable in its yield and use of fissionable materials to one of the early SANDSTONE shots by the U.S. in 1948, just about three and a half years earlier. The second Soviet fission device test had been conducted just three and a half weeks earlier, on September 24. The nuclear arms race was accelerating.

The JCAE issued a report to Congress on October 19, calling for "early consideration (of) the possibility of a second weapons development laboratory to supplement Los Alamos at least in certain engineering phases."⁵⁹⁵

⁵⁹³"In Any Light: Scientists and the Decision to Build the Superbomb, 1942-1954," Peter Galison and Barton Bernstein, HISTORICAL STUDIES IN THE PHYSICAL AND BIOLOGICAL SCIENCES, Vol. 19 No. 2 (1989), p. 323.

⁵⁹⁴NUCLEAR WEAPONS DATABOOK, Vol. IV, p. 336; "Status of the Soviet Atomic Energy Program," CIA/SI 118-51, 6 March 1952, Central Intelligence Agency Office of Scientific Intelligence, p. 14. This third Soviet test device, which yielded about 50 KT, may have weighed only about 5,000 lbs. with a diameter considerably less than 60 inches; its design was apparent by 1949. (Translation of article by Yuri Smirnov, Russian Research Center (Kurchatov Institute), IZVESTIA, #265, December 8, 1992, p. 3.) Unlike each of the first U.S., Soviet, British, and French nuclear devices, the first Chinese fission test in October 1964 that yielded 20 KT used uranium implosion, rather than plutonium implosion. This was not because the Chinese were testing advanced technology; at the time, in a very rough analogy to the U.S. situation in 1948, the Chinese had very little uranium and no plutonium. Their first device using plutonium was not fired until their eighth test in late December 1968. (NUCLEAR WEAPONS DATABOOK Working Paper NWD 91-1, CHINESE NUCLEAR WEAPONS: A CURRENT AND HISTORICAL OVERVIEW, Richard W. Fieldhouse, Natural Resources Defense Council, Washington, D.C., March 1991, pp. 6, 49, 50; USAEC Press Release dated October 21, 1964.)

On October 23, the AEC advised the MLC that an improvement on the recent estimate of the winter of 1952-53 as a test date had occurred and that the general plan of the AEC at this time called for a thermonuclear test to be conducted in the fall of 1952. The revision was a result of the informal briefing which Norris Bradbury had given to the joint AEC-MLC meeting in mid-September.

Also at this time, Edward Teller stated his opinion, in a letter to AEC Chairman Gordon Dean, that the current design of the ALARM CLOCK was:

... as hopeful as the SAUSAGE and may be in the long run much more usable because of the earlier possibilities of delivery. In fact, the ALARM CLOCK might easily have become the first practical thermonuclear weapon except for the time that was wasted in getting to you the request for production of Li-6.⁵⁹⁶

In its eventual incarnation as the TX-14, the ALARM CLOCK would, in fact, become the first deliverable U.S. thermonuclear bomb. Teller also complained in this letter of "local," i.e., Los Alamos, objections to the development of any thermonuclear weapon design other than the "Runaway Super."

Designing the First Large-Scale Thermonuclear Test Device

By the end of October, work on the thermonuclear test device at LASL had reached to a point where a number of activities were underway, including theoretical studies required to establish the basic design of the test device, as well as to understand and predict its behavior; the physical engineering design and construction of the device itself, including many of its less-unique parts; procurement, installation, and operation of liquid hydrogen plants to meet the needs of the test; design and construction of dewars (insulated containers) for storage and transportation of liquid hydrogen; and planning and organizing the lengthy Eniwetok test program.⁵⁹⁷

⁵⁹⁵"A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, p. 13.

⁵⁹⁶Letter dated October 23, 1951 to AEC Chairman Gordon Dean from Edward Teller.

⁵⁹⁷Moore and Bechanan, p. 24.

Theoretical studies were being conducted at LASL, the National Bureau of Standards (NBS), and by Project Matterhorn B at Princeton University, under the direction of Dr. John Wheeler. One of the chief study objectives of Matterhorn was the propagation of "flames" in thermonuclear fuels.

Physical device design and construction were assigned to ACF, which established a special office in Albuquerque where the majority of the Panda group was located.⁵⁹⁸

Procurement, installation, and operation of liquid hydrogen plants was assigned to a number of contractors: the NBS installed and operated a 200 liter/hour hydrogen liquefier at its new cryogenics laboratory in Boulder, Colorado, and carried out the testing and development of transport dewars as well as the dewar of the device itself, i.e., the liquid hydrogen circulation and containment system in the device secondary. **Figure III-7** shows the liquefier building, the first building erected at the Boulder site. This plant was completed in 1951 to house large-capacity nitrogen and hydrogen liquefiers for use by the NBS Cryogenic Engineering Laboratory.

A 200 liter/hour hydrogen liquefier, identical to the one at Boulder, was to be erected at Parry island at Eniwetok atoll.⁵⁹⁹

The NBS and the AEC operated a Cryogenics Engineering Laboratory at Boulder that produced liquid hydrogen, nitrogen, and deuterium, and conducted basic engineering research studies of the characteristics of various materials at cryogenic, i.e., extremely low, temperatures.⁶⁰⁰

⁵⁹⁸Borden-Walker H-Bomb Chronology, p. 64; "A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, p. 12. Matterhorn had started work on May 26, 1951. According to an interview of physicist Carl Haussmann in "Preparing for the 21st Century: 40 Years of Excellence (UCRL-AR-108618, University of California/U.S. Department of Energy, 1992, p. 28), Matterhorn was responsible for the design calculations of the Mike device secondary stage.

⁵⁹⁹"Budget Bureau Questions - 1953 Budget, Program 3000 - Weapons (and related Construction), Products, Storage and Surveillance," undated, but ca. early to mid-1951, p. 9.

⁶⁰⁰Report of the Manager, Santa Fe Operations, USAEC, July 1950 to January 1954, 1 January 1954, pp. 46, 47; "NBS-AEC Cryogenic Engineering Laboratory," Summary Technical Report 1736, National Bureau of Standards, U.S. Department of Commerce, Washington D.C., August 1953.

In 1950, the AEC had asked the NBS to design and construct a hydrogen liquefier several times larger than any then in existence. The resulting 300 liter/hour hydrogen liquefier and a research laboratory were commissioned at Boulder in 1952 under a very tight time schedule, and the principal NBS staff that contributed to its design and operation received a Department of Commerce Exceptional Service Award.⁶⁰¹

Ohio State University agreed to install a second 200 liter/hour liquefier on Parry island at Eniwetok atoll and to rebuild a small existing Parry island plant built for the George and Item shots of Operation GREENHOUSE to bring it up to its rated output. Transport dewars were designed and fabricated by the Arthur D. Little Company of Cambridge, Massachusetts, under contract with LASL, and under the technical direction of the NBS.

The transport and storage dewars were many times larger than any hydrogen dewars ever built. In addition to being heavily insulated, the dewars also required constant refrigeration to remove heat generated by the spontaneous change of state of the liquid hydrogen.

Because of this and other technical problems, it appeared doubtful that adequate dewars would be available to transport large quantities of liquid hydrogen from the NBS cryogenics laboratory to Eniwetok. LASL requested the installation of a second 200 liter/hour liquefier on a U.S. Navy escort carrier; this was superseded by a decision to build a liquefier on Parry island.

The transport dewars and the dewar in the device itself would be tested at Boulder, mainly because the NBS lab would be the first plant to have large quantities of liquid hydrogen. This testing would include operational shakedown of the assembly, filling, and check-out of the device.

ACF had been furnished with a schematic of the thermonuclear test device, which would not represent the final design, and had started preliminary design work. There were many peculiar design problems with which ACF had to be familiar. In addition, feedback from ACF to LASL's T Division was expected to bring to light certain practical design limitations or

⁶⁰¹"Cryogenic Engineering at NBS," TECHNICAL NEWS, U.S. Department of Commerce, National Bureau of Standards, STR-3070, September 1964, p. 3; "The World of Cryogenics. III. Cryogenics at the National Bureau of Standards Boulder Laboratories," B. W. Birmingham, CRYOGENICS, Vol. 5 No. 3, June 1965, p. 121; THE COLDER THE BETTER, David Wilson, Atheneum, New York, 1980, p. 115.

construction problems which could affect the basic design. LASL felt that the sooner it came to grips with design and construction problems, the better.

At LASL, T Division expected to have a reasonable guess soon as to the basic schematic of the test device, and to be able to foresee what theoretical work was required to reach a final basic schematic which it hoped to have ready by January 1952 (there were at least two competing versions of the same radiation implosion design at this time).

American Car and Foundry expected to have ready by the beginning of December a schedule of tasks and a budget. All aspects of the Panda Program were to be detailed and organized by the end of January 1952. LASL's newly-formed J Division, established to conduct Operation GREENHOUSE, would perform the test at Eniwetok.⁶⁰²

In addition to the thermonuclear device, Norris Bradbury instructed J-Division, the laboratory's nuclear test division, to plan for an air-drop of a high-yield (500 KT) fission weapon as a second event in the forthcoming test series.⁶⁰³ While this high-yield device might act as a "backup" if the thermonuclear test failed, it had not been developed specifically for that purpose and had military utility of its own.

By mid-November 1951, successful delivery of a high-yield thermonuclear device was still believed to require an unmanned drone:

The prospect of successful development of a thermonuclear weapon have very greatly improved in the last year. Several proposed models are now under study, all with anticipated yields of several megatons. Those models have undergone enough examination to eliminate any obvious reasons for failure, and some tentative designs of working systems have been made. It is now proposed to test fire one of these designs late in 1952.

⁶⁰²Atomic Energy Commission, Development of Large-Scale Thermonuclear Device for Testing at Eniwetok in the Fall of 1952, Note by the Director of Military Application, December 3, 1951.

⁶⁰³Memorandum dated 18 December 1952 to Commander, JTF 132, from Commander, TG 132.1, Subject: Submission of Report of the Commander, Task Group 132.1 on Operation IVY.

It is not possible now to evaluate either the economy or the deliverability of these proposed weapons. They all involve the use of fairly large amounts of fissionable material, and they are all of such weight and size that they would require a carrier probably of the size of the B-36 to deliver by air. The proposed explosion is so large that a piloted carrier is unlikely.⁶⁰⁴

Also at this time, the TMG held a general seminar and review of the theoretical status of the test device. The objective of the proposed 1952 experiment was to test the design of a weapon giving the largest yield possible with delivery from an existing bomber or one that would be available in the near future (within a year or two).⁶⁰⁵

At the end of November, construction of the new DuPont heavy water facility in South Carolina was well along. Footings for one of the new reactors had been poured, and the first of the towers for the hydrogen-distillation plant had been erected. The hydrogen-distillation process, although costly and dangerous, was selected as the quickest method of producing heavy water for use as a moderator in the production reactors.

At the beginning of December, the LASL thermonuclear device scheduled for test did not bear even the remotest resemblance to a military weapon in terms of either its deliverability or maintainability. It was then estimated to be about 15 feet in height, weighing between 50 and 75 tons, with a yield estimated between one and 15 megatons.⁶⁰⁶

⁶⁰⁴Borden-Walker H-bomb chronology, p. 69.

⁶⁰⁵"Notes on Meetings of the Theoretical Megaton Group," Appendix C to September 12, 1952 Walker memorandum.

⁶⁰⁶Facts Bearing on the Problem and Discussion, Enclosure A to Report by the Commander Joint Task Force 132 to the Joint Chiefs of Staff on Armed Forces Participation in the 1952 Nuclear and Thermonuclear Experimental Device Tests, 1 December 1951. The high end of this yield range was apparently not high enough for at least one AEC commissioner: Thomas E. Murray complained (in a December 12, 1951 letter to GAC Chairman Dr. J. Robert Oppenheimer that advocated the establishment of a separate thermonuclear weapons development laboratory) that the forthcoming fall 1952 test was a "closed end limited size thermonuclear reaction," and not a test of an "open ended" thermonuclear weapon of theoretically limitless

It also had to be kept in an upright (vertical) position. In addition, due to the use of liquid deuterium, extensive cryogenic support equipment, including pumps, compressors, and liquefiers, was required to keep the fuel ready for detonation. The technical and logistical problems posed by the preparation and handling of large quantities of liquid deuterium and hydrogen were unprecedented.

The size and shape of the device precluded its transportation by any existing or planned aircraft (the B-36 could carry a maximum bombload of only 42 tons); however, it might be transportable aboard a ship. Nonetheless, the AEC's Director of Military Application inquired as to the MLC's interest in a replica of the device for use as a weapon.⁶⁰⁷

The test device would not use lithium-6, although it was clear that subsequent weapons suitable for delivery by existing aircraft would have to use a compact "dry" fuel. A pilot plant to separate lithium-6 was in operation at Oak Ridge and small quantities of the isotope were being produced. A plant capable of producing kilogram quantities was to be constructed at Oak Ridge and to be in operation by January 1953.⁶⁰⁸

Teller Pursues the Second Weapons Laboratory

Early in December, Edward Teller sent to AEC commissioner Lewis Strauss a brief to be submitted to the GAC that presented Teller's chief arguments for a second weapons laboratory. Although speaking highly of LASL on one hand, Teller nonetheless pressed his agenda for faster development of thermonuclear weapons and was critical of both the forthcoming IVY test and what he saw as LASL's less-than-zealous commitment to the thermonuclear program.

He also criticized LASL's bypassing of "components testing" and thought that LASL might be taking too large a step at once. This was especially ironic, considering that Teller had

explosive power. (Dr. Murray may have had Edward Teller's early 40 MT aspirations in mind.)

⁶⁰⁷Memorandum for the Executive Secretary, Military Liaison Committee, dated December 4, 1951 from Col. K. E. Fields, Director of Military Application, USAEC, Subject: Thermonuclear Device for Test at Eniwetok, Fall of 1952; Atomic Energy Commission, Progress Report to the Joint Committee, June through November 1951, Part III, Weapons, December 17, 1951, p. 3.

⁶⁰⁸Atomic Energy Commission, Progress Report to the Joint Committee, June through November 1951, Part III, Weapons, December 17, 1951, p. 8.

previously constantly complained about LASL's alleged "conservatism" and avoidance of risk in its test programs.

Teller also reiterated many of his plans from his August 1951 letter to Gordon Dean and closed by arguing that the pace of recent developments was evidence of past failure by LASL to pursue those ideas sooner. By saying the latter, Teller completely disregarded the groundwork that was necessary to have brought those ideas to the point where they were now:

Statement to the General Advisory Committee
on Need of Second Weapons Laboratory

In the postwar years, much progress has been made at Los Alamos on the development of fission weapons. In the last two years, Los Alamos has undertaken an additional difficult task: the active exploration of the field of thermonuclear weapons. It is generally recognized that the work on fission weapons in Los Alamos has been of the highest quality. It is remarkable that so much progress could be made at Los Alamos on the thermonuclear program considering that Los Alamos used a small fraction of its manpower on this additional job.

It seems unavoidable that a laboratory with fixed and definite tasks should develop into an organization efficiently performing these tasks and less efficient in exploring new fields.

I have a great deal of admiration for the vigorous way in which the Los Alamos laboratory is attacking the problem of thermonuclear weapons at the present time. However, this effort is necessarily limited in manpower. What is more important, it has become strictly limited to the immediate objectives and little thought is given to developments beyond the next thermonuclear test.

Furthermore, this test is undertaken without intermediate experimental exploration of important and unknown processes such as large-scale radiation implosion. This way of proceeding is best illustrated by imagining that the wartime implosion program would have been

undertaken without any experimentation such as RaLa and pin techniques.

The present procedure adopted by Los Alamos is necessary in view of the limited resources of the laboratory. I have great hopes (but no feeling of certainty) that this procedure will lead to early success. It may, however, end in failure and, even in case of success, it cannot lead to sufficient exploitation of a new and unknown field.

The difficult task at Los Alamos could be faced with greater hope if there were a considerable influx of excellent scientists into Los Alamos. Such an influx, however, did not take place. The senior scientific staff at Los Alamos is, at present, weaker than it was at the outbreak of the Korean war.

During the early months of the Korean war, many of our best scientists were willing to make considerable sacrifices in order to be of help in the rearmament effort and Los Alamos could have secured their services. This did not happen. Following the great success of the thermonuclear tests at Eniwetok, excellent scientists could have been attracted to Los Alamos. This, again, did not happen.

Having watched the operation of recruiting at Los Alamos and having helped in this recruiting effort, I feel certain that the reason for this double failure is the fact that Los Alamos is not willing to offer adequate responsibilities to outsiders.

In connection with the recruiting efforts at Los Alamos, I have come to the conviction that a considerable number of excellent scientists would be available for work at a new defense laboratory. Attractions of a new site would include a flexible program, full responsibility for such a program, freedom from old commitments, and an understanding that most of the personnel are at the laboratory on a temporary basis and will return to academic activities after a number of years.

I should envisage the second laboratory as an establishment of considerably smaller size than Los Alamos. I think an organization of approximately 300 technical employees could accomplish a great deal and one should attempt to hold the size of the laboratory to approximately this size.

I do not believe that thermonuclear developments should be pursued only at the new laboratory. In fact, I firmly believe that thermonuclear efforts at Los Alamos should continue.

I also do not believe that the new laboratory should be limited to thermonuclear work. Rather, it should be encouraged to plan and explore all kinds of new developments in the field of bomb physics. Among the tasks one may envisage:

Exploration of large-scale radiation implosion techniques; this will require tests involving nuclear explosions which, however, generally could be kept to small explosions (to be definite, smaller than 50 kilotons);

Further development of thermonuclear bombs with the aim of simplifying delivery, decreasing the size, and increasing the economy of future thermonuclear weapons;

Development of low temperature techniques in order to explore whether or not transportation, logistics, and delivery problems involving liquid hydrogen can be so simplified as to use liquid deuterium as a permanently useful component of nuclear weapons. (This should be dependent upon theoretical work which is to decide whether addition of liquid deuterium to future thermonuclear designs can usefully decrease the size and weight or increase the economy of the weapon.);

Further development of boosted weapons;

Theoretical and experimental exploration of the implosion of one bomb by another, as applied to cases in which thermonuclear reactions are not involved or involved only in an auxiliary manner;

Theoretical and possibly experimental research on the explosion of reactors, in particular, of water boilers. (This development has some interest for the reactor safety question. In addition, it may be significant in constructing a "suitcase" weapon of limited power. Finally, an exploding water boiler might be an appropriate object to be used in imploding a second weapon.);

Investigation of the effects of atomic bombs, including those effects which do not appear to have direct and pressing military significance.

A few of these developments are likely to yield results in a relatively short period, such as a year. Some of the items mentioned above are not likely to be explored by Los Alamos in the near future. Others will be studied at Los Alamos but will not, in my opinion, be studied sufficiently.

In order to limit the size of the new laboratory, one should try to get along with a minimum of high explosive techniques. The new laboratory should rely to a maximum extent on the excellent facilities at Los Alamos for developing high explosive techniques and, in particular, high explosive lenses. The effort made at Los Alamos in this direction seems to be both admirable and completely satisfactory.

Furthermore, one should try to avoid establishing a new test division and serious thought should be given to the

possibility of the new laboratory and Los Alamos making use of a common test division.

In summary, I believe that, in spite of recent progress, the field of atomic bombs is still insufficiently exploited. In fact, the very rapidity of recent progress is evidence of potentialities which have been neglected for years and which will not be fully exploited unless a new laboratory is established.⁶⁰⁹

When he appeared personally before the GAC a few days later, Teller again criticized what he saw as LASL's conservative approach to thermonuclear design, saying that this approach was neither suited nor able to explore fully what he saw to be the very great possibilities of thermonuclear weapons. He especially singled out LASL's plans for the forthcoming Eniwetok test; Teller believed that the present device design was subject to serious technical uncertainty and that it required intermediate experimental and component studies before the IVY shot was conducted.

Teller compared the present situation in terms of a hypothetical delay in implosion fission weapon development which might have happened if LASL had, in 1943, decided to proceed along that line with only a theoretical research program and a minimum of intermediate experimental testing. He felt that the thermonuclear program at LASL was now falling into this trap.

Teller stated that among physicists, there was little interest in the thermonuclear program because the country was not a "hot" war (American soldiers under fire in Korea probably would have disputed the latter point). He thought that this situation could be reversed if a new, more-flexible nuclear weapons lab were established where a "forward-looking" advanced thermonuclear weapons development group could flourish.

While hydrogen weapons would be the main effort of the new lab -- including theoretical and experimental work on the "runaway Super" (which Teller almost alone still thought could be made to work), more effective use of lithium-6,

⁶⁰⁹Letter dated December 7, 1951 to Admiral Lewis Strauss from Edward Teller with attached statement to GAC. "RaLa" is a reference to the radium-lanthanum technique of testing implosion efficiency. The "water boiler" was a reference to a wartime water-moderated, uranyl sulfate-fueled research reactor at LASL.

further development of boosted weapons, heterocatalytic implosions, and non-military uses of thermonuclear energy. (LASL was already working on many of these subjects.)

Teller also thought that time was of the essence: the Russians might already be far ahead of the U.S. in H-bomb development. He said that the new lab should be established before IVY, arguing that if the LASL thermonuclear test were successful, then there would be a complacent feeling that the new lab was unnecessary.

On the other hand, if LASL were not successful in the test, then although a new lab might be needed, recruiting would be difficult. Teller did not explain why bold and adventurous young physicists would be deterred by a failed LASL test, instead of being energized and galvanized to do better and show that they were superior to LASL.

When Teller had approached Bradbury with the suggestion that a highly-skilled group of theoreticians, experimentalists, and engineers be assembled at LASL to design thermonuclear test devices to the blueprint stage, at which point the test device would be completed and fired by other Los Alamos divisions, Bradbury had declined the idea. Bradbury told Teller that such a group would become an elite division within LASL and would tend to draw the lab's best staff members away from other deserving projects. Teller admitted that Bradbury's objections were valid.

Dr. Teller also admitted that even if a new lab were established, it would have to depend upon LASL's GMX and J divisions for high explosives and field testing. He thought that the new lab could have a device ready for field test within a year.⁶¹⁰

On the face of it, what Teller was asking for was incredible. Here was a physicist who had probably delayed the development of new approaches to successful H-bomb design. On the one hand, he was complaining about LASL's "conservatism" while on the other, he was claiming that Bradbury was taking too big a risk with the forthcoming thermonuclear test by not conducting components testing.

Now that a successful approach was apparently in hand, Teller had the unmitigated gall to blame LASL for what he saw as foot-dragging, and to ask for his own private weapons laboratory from which the solution to all thermonuclear ills would undoubtedly soon spring full-blown.

⁶¹⁰Minutes of the Twenty-Eighth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, December 12, 13, and 14, 1951, pp. 10-14.

Results of IVY would soon show that LASL's approach to full-scale testing of radiation implosion was correct and fruitful, Teller's doubts notwithstanding. Given some of Teller's own unsuccessful attempts later to leap into the unknown after the second lab was established, his criticism of LASL's risk-taking during IVY was hard to understand.

In a later session of the same GAC meeting, Dr. Oppenheimer asked Darol Froman about the desirability of setting up a development group for thermonuclear work at LASL along the line proposed by Teller, and what he thought about the idea of a second weapons lab.

Froman replied that there would be organizational difficulties in the first course of action, and that he was concerned about competition for key staff members, especially those already working on thermonuclear weapons at LASL.

In a later session of this meeting of the GAC, Oppenheimer reviewed the questions about LASL raised by Teller. The main problems appeared to be threefold:

- (1) Los Alamos worked from one test series to the other. This was wasteful, and in some ways, a frustrating way to operate the lab. Some lines of thermonuclear and implosion weapon development, not scheduled for either of the 1952 test series in Nevada and the Pacific, needed longer-range thinking, and more thought than they were currently receiving.

- (2) The LASL organization was almost unchanged from 1943, and reflected an obsolete conception of the lab's function. Something a little bit deeper and more long-range was apparently required. The GAC was in full agreement on this point.

- (3) It was important to create a situation wherein Edward Teller's contributions to the H-bomb program could be maximized. This would not happen if Teller were either preoccupied with the administration of a new laboratory, or if his relations with those at LASL working on the thermonuclear program continued to be discordant.

Two solutions to this situation appeared to be obvious. First, a new laboratory, with the range of interests and quality of personnel advocated by Teller, might be established rapidly.

Second, since LASL had already started a reorganization and redirection of its efforts, the old lab could be left to sort out the problems cited earlier by itself.

The GAC suggested an intermediate solution, namely, that at LASL, a special division or section with an explicit directive to work on problems of greater variety and longer range than those to which LASL was currently committed.

This advanced development division (to use Teller's own words) would be protected from raids by other LASL divisions seeking temporary test program assistance. The proposed division would need either its own facilities, or even better, a guaranteed commitment of a certain portion of the lab's facilities. It would have to be directed by a fabulous person who would be acceptable and helpful to both Teller and Bradbury, who were very strongly politically opposed to each other by now.⁶¹¹

On other matters, Darol Froman noted that LASL planned two initiator timing shots for the spring of 1952, and a third in the fall. (These became the XR1, XR2, and XR3 tests during Operations TUMBLER-SNAPPER and UPSHOT-KNOTHOLE in 1952 and 1953.)

The GAC also reviewed current intelligence findings about the two most recent Soviet nuclear tests, Joe-2 (on September 24, 1951) and Joe-3 (on October 18, 1951). The airborne bomb debris from Joe-3 was glassy and composed mainly of aluminum oxide; it revealed that the shot was an airdrop, and, on the basis of fission product ratios, the bomb appeared to have employed a composite pit of either U-235 or U-233 with plutonium.⁶¹² There was no evidence of 14 MeV neutrons, which meant that it was not a thermonuclear test.

Debris from Joe-2 revealed that it was either a surface or underground burst.

Another point discussed at the 28th GAC meeting was the fact that the continued classification of bomb sizes, shapes, and

⁶¹¹Minutes of the Twenty-Eighth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, December 12, 13, and 14, 1951, pp. 27, 28.

⁶¹²A paper dated 21 February 1952, prepared by the staff of the JCAE, claims that "the Russian 'two year' composite core is more efficient than our 'three year' composite -- and it may be as good as our best composite to date." The "two year" and "three year" comments are references to the fact that while the U.S. did not test a composite core between 1945 and 1948, the Russians tested a similar core within just two years, 1949 to 1951.

moments of inertia were hindering good matches of bombs and delivery aircraft.⁶¹³

Later in December, the AEC's General Manager's study requested in August had been completed and was circulating for review. One of the conclusions of the report was that the establishment of a second laboratory similar to LASL was now neither desirable nor necessary.⁶¹⁴

The LASL 1952-1953 Technical Program

The LASL technical program for calendar year 1952 and fiscal year 1953, delayed and not submitted to the AEC until late December 1951, reflected the lab's new knowledge in the thermonuclear field; especially noteworthy was LASL's continuing hope for HE-initiated thermonuclear explosions:

IV. Thermonuclear Weapon Research

This general field will include the research and developmental effort in the general field of large explosions in which the major source of energy is primarily due to the establishment of an effective thermonuclear reaction.

- A. Non-equilibrium Reactions - To the extent permitted by other preoccupations in this field, attention will be given primarily to the theoretical computation of the burning process in liquid deuterium.
- B. Equilibrium Reactions [deleted] - To the extent permitted by other preoccupations in this field, attention will be given to systems involving Li^6 , deuterium, and U^{238} , and to techniques of high explosive or radiation implosion methods of [deleted]
- C. [deleted] The results of such a test

⁶¹³Minutes of the Twenty-Eighth Meeting of the General Advisory Committee to the U.S. Atomic Energy Commission, December 12, 13, and 14, 1951, pp. 16, 17, 22.

⁶¹⁴Memorandum dated April 29, 1954 to Chester G. Brinck, Assistant General Counsel, from Leonard A. Jacobvitz, Attorney, Office of Assistant General Counsel, subject: Participation of the University of California Radiation Laboratory in Weapons Development Program.

will, it may be hoped, indicate the characteristics of similar or related systems of greater ordnance simplicity.⁶¹⁵

Thermonuclear test plans at this time included Operation IVY at Eniwetok in the fall of 1952, during which a program of possibly two experimental detonations (item "C" in the list above, and a high-yield 500 KT fission device) would be conducted, if deemed appropriate and desirable at the time.

There also existed the possibility that the experimental behavior of the IVY thermonuclear device, coupled with a probable delay in lithium-6 production, might make necessary a test of an equilibrium Super (item "B" in the list above) at Eniwetok sometime in 1953.⁶¹⁶

The new year of 1952 began much more auspiciously than 1951. Early in January, AEC Chairman Gordon Dean stated in a letter to Senator Brien McMahon, chairman of the Joint Committee on Atomic Energy:

I sincerely believe the Commission is doing everything that can be done to develop a thermonuclear weapon at the earliest date possible. In the face of many difficult engineering problems, we are planning to test a thermonuclear device in late 1952 (Operation IVY).

Because of its size, weight, and complexities, the device to be tested is probably not usable as a weapon, although in other respects -- yield in the megaton region and principle of reaction -- it represents the first full-scale thermonuclear explosion to be attempted.

⁶¹⁵Los Alamos Scientific Laboratory Technical Program for Calendar Year 1952 and Fiscal Year 1953, p. 4. Item "C" includes a description of the principles tested in the IVY Mike device, including the use of a U-238 fusion tamper around the secondary, making the SAUSAGE a fission-fusion-fission system.

⁶¹⁶Los Alamos Scientific Laboratory Technical Program for Calendar Year 1952 and Fiscal Year 1953, p. 6; Memorandum for the Files dated January 10, 1952 from John S. Walker, subject: AEC Letter of January 9, 1952 re Thermonuclear Developments and a Second Laboratory.

We expect to learn much from this operation that will be of great value in our thermonuclear work. In addition to this effort, we are taking steps, such as production of Li⁶, to insure that development of subsequent thermonuclear devices more nearly approaching a practical weapon may follow as quickly as possible.⁶¹⁷

The major part of Dean's letter to the chairman was a statement which echoed many of Norris Bradbury's arguments of the preceding October and stated that the AEC believed that it was neither feasible nor constructive for a second U.S. nuclear weapons laboratory to be established, even though this was a pet project of Edward Teller and some DOD officials and AEC commissioners who all felt that LASL had been too deliberate in its pursuit of the hydrogen bomb.

Dean argued that "in the field of atomic weapons research, competition in the same sense as it is applied to research in industry would not provide an equivalent stimulus directed towards new and improved weapons." Time would prove Dean and the AEC wrong on this point.

The chairman also claimed that "creation of dual laboratories such as Los Alamos would dilute the scientific talent and introduce difficult problems of coordination. Further, because of disassociation of talent and effort between the two laboratories, the rate of progress would be reduced." Dean and the AEC were also wrong about this: the eventual creation of a second lab at Livermore, California would lead to an "explosion" of advanced and successful warhead designs.

Another argument against the second lab made by the AEC was that "the division of talent between Los Alamos and a competing laboratory would at this time retard rather than accelerate the development program. Scientists of the caliber necessary to man and administer another laboratory for the prosecution of development programs similar in scope to Los Alamos are limited in number."

⁶¹⁷Letter dated January 9, 1952 from AEC Chairman Gordon Dean to Honorable Brien McMahon, Chairman, Joint Committee on Atomic Energy, Congress of the United States. Later in 1952, the GAC reported to President Truman that at least one thermonuclear test device looked most promising and that even if this particular design didn't turn out well, the LASL scientists were sure of finding one that would. (Interview with J. Robert Oppenheimer, by Warner R. Schilling, 11-12 June 1957, p. 14.)

This was true to some extent; however, Livermore did not have problems recruiting talented and enthusiastic staff members. The second lab's major early problems would result from the pursuit of old, discarded ideas and from trying to move too far too fast with both advanced fission and fusion weapon designs in an attempt to prove that its worth was comparable to LASL.

To support his argument, Dean cited the views of the GAC in a letter to the AEC dated October 13, 1951; the letter had been written by GAC Chairman J. Robert Oppenheimer, wartime LASL director, opponent of Edward Teller, and a staunch adversary of a second weapons lab:

The review of weapons work at Los Alamos has served again to emphasize the very great scope, variety, difficulty, and importance of warhead development, and to raise the question of optimal facilities and personnel for this vital part of the Commission's program. We have thus been led to consider whether the creation of another laboratory, with responsibility coextensive either with the whole or with a part of present Los Alamos responsibility, would make for the progress of the work.

Indeed, we are persuaded that added facilities for weapons improvement are both desirable and practical, and that they can help to free the (Los Alamos) laboratory from certain routine obligations which have in the past burdened its personnel and facilities. But with this one exception, to which we shall return below, we are convinced that the establishment of another weapons laboratory similar in nature to Los Alamos is neither necessary, nor in any real sense, feasible.

It appears to us (to be) unnecessary because of the rapid progress that Los Alamos has made and is likely to make in the future; thus, it seems to us reasonable to expect, though, of course, by no means certain, that within the next year or so a solution to the major thermonuclear problem may have been found.

It is equally reasonable, and rather more certain, that there will be still further achievements in improving implosion assemblies, as Los Alamos becomes free to undertake radical work in this field.

The new laboratory seems to us unnecessary because the overwhelming proportion of scientists experienced in, qualified for, and interested in development work on weapons are now in fact one way or another available to Los Alamos.

The new laboratory would appear to be hardly feasible unless there were in fact a great many outstanding and specially qualified scientists who could be persuaded to work there, and yet could not be persuaded to contribute to Los Alamos. It is our view that no such situation exists.

Circumstances in the future may, of course, call for re-evaluation; but we are convinced that at the present time the establishment of a new "Los Alamos" or a "thermonuclear laboratory" would tend to impair the effectiveness of Los Alamos and would not succeed in creating a new facility of comparable promise of achievement.⁶¹⁸

Dean also cited plans to relieve LASL of production, modification, and inspection work it was now performing on nuclear weapons parts; within a year, a new production facility at Rocky Flats, Colorado would take over much of this work. In addition, all high explosives fabrication, including some now being done at LASL, would be taken over by plants in Burlington, Iowa; Amarillo, Texas (the Pantex facility); and a third plant now being planned (this became the proposed Spoon River, Illinois plant).

Despite this opposition by the GAC and the AEC, a second U.S. nuclear weapons development laboratory would be created before the end of 1952.

The Soviets eventually also founded two nuclear weapons design labs, at Arzamas-16 and Chelyabinsk-70. In the U.S., a second lab was established in part allegedly to mitigate the effects of J. Robert Oppenheimer's lack of enthusiasm for H-bomb development; in the Soviet Union, a second lab was established in part due to anti-Semitism and in part to provide a more conservative, less-adventurous research facility.

⁶¹⁸Letter dated January 9, 1952 from AEC Chairman Gordon Dean to Honorable Brien McMahon, Chairman, Joint Committee on Atomic Energy, Congress of the United States.

In both countries, identical arguments were put forth that while creation of an additional lab would draw away talent and resources from an established facility, it would nonetheless provide in the long term a competitive atmosphere to advance the state-of-the-art of nuclear weaponry.⁶¹⁹

By the middle of January 1952, four possible device models that had been in competition for testing during the proposed Mike shot of Operation IVY were reduced to two. The cryogenic hydrogen and deuterium would be maintained by a relatively standard refrigeration unit contained in the device.⁶²⁰

The basic design of the final proposed LASL thermonuclear test device was frozen on January 18. Consideration was given to testing this device in a solid-fueled "dry" form; however, sufficient quantities of lithium-6 would not be ready in time; the behavior of lithium in thermonuclear reactions had not been studied adequately either theoretically or experimentally; and a liquid-fueled test using deuterium would give "purer" reactions and results, bypassing many unknown lithium-deuterium and lithium-tritium interactions:⁶²¹

In the cryogenic design you had nothing but deuterium -- essentially an infinite medium of deuterons. With lithium deuteride, there were as many lithium atoms as there were deuterons, but we were interested in the deuteron reaction, $D + D$. There wasn't a prominent reaction of deuterium with lithium. The lithium was inert in that respect. In fact, it was a diluent, and certainly a complication.

The cryogenic (design) avoided that complication. It introduced some physical complications in construction, in handling, but those one knew how to deal with. They had nothing to do with any of the thermonuclear behavior. ...

The description of the (thermonuclear) burning process in pure deuterium is much

⁶¹⁹Zaloga, pp. 103-105.

⁶²⁰Memorandum for Record from Colonel Campbell, Subject: Notes on Meeting, January 15, 1952.

⁶²¹Borden-Walker H-bomb chronology, p. 70; Memorandum dated 18 December 1952 to Commander, JTF 132, from Commander, TG 132.1, Subject: Submission of Report of the Commander, Task Group 132.1 on Operation IVY.

simpler than the description of the burning process with either Li-6 or normal (unenriched) lithium deuteride. The description of the compression of liquid deuterium is simpler than of (its) compounds, the deuterides. To avoid (having to deal with) lithium seemed like a virtue. Every departure from the simplest picture seemed like something to avoid.⁶²²

At the end of January, LASL issued a technical report entitled "Interim Report on the 'Alarm Clock.'" This optimistic paper considered, among other models, a thermonuclear weapon smaller than the 60" diameter stockpiled MK 6 fission bomb.⁶²³ The field of fusion weapon development was expected to be as varied in development and as broad in prospect as the fission weapon field. Present prospects appeared to range widely, from devices deriving their yield primarily from thermonuclear reactions, through mixed fission-fusion devices, to fusion-boosted fission weapons.⁶²⁴

Early in February, Operation IVY, now planned for the fall of 1952, was to include tests of both a large thermonuclear device and a 400 to 600 kiloton yield fission device.⁶²⁵ The second shot would add to the knowledge of both the effects of high-yield weapons and of certain factors involved in the design of high-efficiency bombs.⁶²⁶

At this time, the JCAE was still lobbying the AEC to expand thermonuclear research and production of fusionable materials. This might also include the establishment of a

⁶²²Rhodes, DARK SUN, pp. 483-484. In reality, there are at least three separate interactions of lithium with deuterium. See Equations A, E, and G in "Fusion Physics" in "Weapons Physics" in Volume I.

⁶²³Borden-Walker H-bomb chronology, p. 70.

⁶²⁴Letter dated February 11, 1952 to Gordon Dean, Chairman, USAEC, from William L. Borden, Chairman, Joint Committee on Atomic Energy.

⁶²⁵Memorandum dated 12 February 1952 from the Military Liaison Committee for the Director, Division of Military Application, USAEC, Subject: Operation IVY.

⁶²⁶Memorandum dated 18 December 1952 to Commander, JTF 132, from Commander, TG 132.1, Subject: Submission of Report of the Commander, Task Group 132.1 on Operation IVY.

second nuclear weapons laboratory, if that appeared necessary (the JCAE strongly supported the idea of a second lab):

The Joint Committee is most appreciative of your letters of January 9 and 10, 1952 dealing with questions of the thermonuclear program and of a supplemental laboratory effort. The concern created by these questions is very deep. Because all phases of thermonuclear development will undoubtedly continue among the most compelling of issues for the foreseeable future, I am raising questions suggested by recent progress in this field in four general areas which will be of interest to the Joint Committee in connection with our meeting with you next Thursday morning, February 14.

1. I wonder whether it can now be anticipated that the fusion field, generally speaking, may prove to be as varied in development and as broad in prospect as the fission field. Present prospects, for example, appear to range widely, from devices deriving their yield primarily from thermonuclear reactions, through mixed fission-fusion devices, to fusion-boosted fission weapons. This raises the question whether we are approaching not only achievements of a new dimension, but also requirements for development of major and new dimensions.

Furthermore, the question may also be posed whether it is not time to determine whether the scope and scale of our thermonuclear development, particularly the development and actual production of weapons, is not largely dependent upon the effort that is expended. I am deeply convinced that United States development -- to the maximum extent, in maximum scope, and within the shortest period of time -- is not only of vital but unique importance.

I remain apprehensive that the Soviet Union will be first to produce a large-scale thermonuclear reaction or first to produce usable and deliverable thermonuclear weapons.

2. I think two specific problems, one of manpower, and a second of materials, are of special interest to the Joint Committee. The test operation planned for next fall (sic) is most encouraging and will be watched with the

closest possible attention and given any help within our power.

However, the mere presence of this one development in the thermonuclear field necessarily impacts not only our fission weapon program but also upon parallel fusion development of great promise. It would appear, for example, that a shortage of theoretical physicists will retard development of "solid" models of the Super, which, according to my information, promise much easier modification to a deliverable weapon.

(This shortage) will affect not only the date of a 1953 test but (will) also limit the number of tests which can be undertaken. Although the manpower problem, particularly in such special fields, is a recurring one, it impresses me as being of critical magnitude if we are to achieve super weapons in the shortest period of time.

It is the earnest hope, I am sure, of every member of the Joint Committee, that enthusiasm generated by the thermonuclear race for discovery can attract men reluctant to join the program in the past; and that every inducement of salary and organizational arrangement be explored.

Analogous points are suggested by the expediting of new materials in fusion development and, in particular, by the degree that tests of a "solid" thermonuclear device looking to a deliverable weapon will have been delayed by lack of lithium-6. Of course, within the risk that lithium-6 will continue to be essential, the problem is not only for a supply ample to a full test program at the earliest date now possible, but also to anticipate stockpiling requirements for thermonuclear weapons.

The JCAE closed its comments by reiterating its support of a "supplemental" weapons lab, noting that new fission weapons developments alone could keep LASL busy for years. LASL should be relieved of its engineering and production work as soon as possible; however, even resources made available by these changes could be consumed by fission weapons research.

Adding fusion weapon work to LASL's existing load might only, in the JCAE's opinion, make the laboratory insufficient to conduct either fission or fusion weapon research

adequately.⁶²⁷ LASL director Norris Bradbury would later take strong exception to this point of view.

In mid-February, LASL reviewed nuclear weapon test schedules for 1952 and 1953. The next post-IVY series of thermonuclear tests, scheduled for 1953, was highly contingent upon production of a sufficiently-large quantity of lithium-6. During late 1951, the AEC had authorized construction of a lithium-6 plant at Oak Ridge, and the Commission decided at the end of February to proceed with the project.

The MLC directed the AEC to place urgency and emphasis on lithium production so that enough would be stockpiled to incorporate into weapons as soon as a technically-feasible, aircraft-deliverable solid-fueled design had been successfully tested.⁶²⁸ The MLC specifically recommended that construction of a lithium-6 separation plant should be started promptly with the goal of producing two pounds (about a kilogram) of highly-enriched lithium-6 per day starting in April 1953.⁶²⁹

Of several promising hydrogen bomb models considered by LASL, work in late February had narrowed to the "Equilibrium Super," to be tested at the beginning of November in a device called the SAUSAGE.⁶³⁰

At this time, the GAC was still opposed to the creation of a second U.S. nuclear weapons laboratory:

We continue to believe that the creation of a second independent laboratory would have damaging effects upon Los Alamos and would, at least for many years, have no compensating advantages. We are strengthened in this belief by the fact

⁶²⁷Letter dated February 11, 1952 to Gordon Dean, Chairman, USAEC, from Brien McMahon, Chairman, JCAE. Regarding the alleged shortage of theoretical physicists available to the AEC at this time, it is unlikely that even if a concerted recruitment campaign had begun immediately, enough new scientists would have received their "Q" security clearances in time to have any effect upon plans for Operation CASTLE, which would include the first test of solid-fueled hydrogen devices.

⁶²⁸Borden-Walker H-bomb chronology, p. 73.

⁶²⁹"Comments Concerning the H-Bomb Development," Department of Defense, Military Liaison Committee to the Atomic Energy Commission, 21 April 1954.

⁶³⁰Borden-Walker H-bomb chronology, p. 74.

that, as anticipated, thermonuclear and fission weapons have become more and more closely interrelated in design and planning, and that a separation of functions based on the distinction between these two types of weapons would hardly appear to be technically meaningful.⁶³¹

Construction of an enlarged plant to produce lithium-6 for use in thermonuclear weapons was begun at Oak Ridge in March and was scheduled for completion in February 1953 to permit weapon tests later that year (the CASTLE series, which was finally conducted in the spring of 1954).⁶³² The DOD did not believe that lithium-6 would be needed in quantities greater than that required for 10 thermonuclear weapons per year.

This might change, of course, once the armed forces decided just how many hydrogen weapons were to be stockpiled, a decision which was still in the future. DOD thought that only a few strategic Soviet targets merited thermonuclear destruction -- most could be adequately targeted by fission weapons -- and that tactical uses for the hydrogen bomb would be very limited.⁶³³

The hydrogen bomb was still perceived as more an instrument of psychological warfare than as an instrument of military warfare; mostly for this reason, it was vitally important that the U.S. obtain it before the Soviets did.

The only likely tactical employment of thermonuclear weapons might be their use against widely-dispersed troop and armor formations, groups which, while sufficiently dispersed to require the use of many precisely-delivered low-yield atomic weapons to destroy them, would still be vulnerable to the wide-area blast and thermal effects of a single hydrogen warhead.

This tactical use of the hydrogen bomb was postulated partly on possible Soviet war plans for the conquest of Europe.

⁶³¹Excerpts from GAC Meetings, compiled by McKay Donkin, August 18, 1953, p. 31; "A Chronology of the Thermonuclear Weapon Program to November 1952," draft dated August 14, 1953, p. 14.

⁶³²Progress Report to the Joint Committee on Atomic Energy, December 1951 through May 1952, USAEC, Washington, D.C., June 16, 1952, p. 8.

⁶³³Memorandum to the File dated February 26, 1952 from J. K. Mansfield, subject: Military Requirements for H-Weapons.

Conventional military wisdom at this time held that the Soviets would divert large numbers of Red Army divisions for thrusts against Turkey, Scandinavia, and the Balkan states while simultaneously striking westward across the European plain to the English Channel. On the other hand, it was also possible that the Soviets might just as likely concentrate all their forces into one massive blow against the center of Europe. In either case, the use of thermonuclear weapons might be the only way to stop the attack.⁶³⁴

Late in February, the JCAE complained again that LASL was not doing enough to accelerate the hydrogen bomb program:

The Committee staff believes that the present hydrogen program is too little and too late in the following respects:

(1) Of the five promising types of H-bomb, only one is being vigorously developed today.

(2) There is today little or no effort to reduce the time lag between a successful experimental test and the achievement of a practical weapon -- years may elapse between the two.

(3) The sole material now deemed crucial (lithium-6) is to be produced in limited test quantities only. Again, years may elapse between a successful experimental test and stockpiling of H-bombs in quantity.

(4) Not nearly enough scientific manpower is being applied to the hydrogen program. Example: Of 12,000 PhD physicists today working in the United States, less than one in 200 is working full time upon H-bomb problems.

(5) In a word, although the situation has gradually improved since 1949, there is still no real sense of urgency about the hydrogen program.⁶³⁵

⁶³⁴Memorandum to the Files dated March 13, 1952 from J. Kenneth Mansfield, JCAE, subject: Bernard Brodie on the Hydrogen Bomb Program.

⁶³⁵"The H-Bomb Program and the Second Laboratory Proposal -- Staff Comments," paper dated February 21, 1952 prepared by JCAE staff.

At the time these comments were made, the authors of the paper quoted were assuming that the Soviets had been working full-tilt on their own H-bomb, and that a serious effort toward progress had been underway in the U.S. only for about two years. Both assumptions were mistaken.

In addition, any impetus for a "crash" program had to emanate from the highest levels of the government, not from LASL or the AEC. If Congress, the Department of Defense, and President Truman had all been committed to such a program in 1950, then perhaps LASL might have had the resources it would have required to concurrently advance the state-of-the-art of both fission and fusion weaponry.

At a JCAE meeting on February 21, chaired by Senator Brien McMahon, and attended by Gordon Dean, Dean and AEC commissioner physicist Dr. Henry DeWolf Smyth vigorously defended LASL's postwar thermonuclear weapons research program. Smyth noted that if the Soviets had in fact embarked upon a hydrogen bomb program, they would have simultaneously had to embark on a program to improve their fission weapons, given the extremely necessary requirement for an efficient trigger or primary for any thermonuclear weapon.

Of the 1951 Soviet test of a composite core weapon, Smyth said:

As to the composite weapon, the composite weapon is extremely valuable economically to us because we are able to produce U-235 relatively more cheaply than plutonium. It does not seem to me be an essential (ingredient of) a weapons stockpile. That is, I think the composition which you use in the nuclear core is largely a matter of economics over a very wide range of yields.

This I believe I have discussed in here before when we have talked about (the) plutonium (to) uranium-235 ratio.

As to the composite nature of the last Russian shot and (its) efficiency -- I think it is true that it may be as good as our best composite to date, but I do not think the intelligence information is reliable enough to say that is as good or more efficient than any of our other bombs. You just do not know that accurately.

I think all you can say from the intelligence information is that they apparently did have a composite shot and

that it was reasonably efficient. We don't know the yield, and I don't think we know with any great accuracy (its) efficiency.

When questioned about the possible participation in the Soviet hydrogen weapon program by a Russian cryogenics expert (Petr Kapitsa), and the results that might have had, Smyth noted that:

Cryogenics is a very difficult element in getting the thing to work even experimentally, and much more so if you are going to have to use it for a deliverable weapon. But it is a technical problem rather than a problem of a nuclear kind. That is, it is an accessory problem.

It might have had some influence in this sense: that if the (Soviet) cryogenic people said, "Sure, we can build a hydrogen liquefier that you can cart along with this thing in a plane," that might have given them a little encouragement. But it doesn't seem to me that it would have been a major factor.

Smyth also noted that the expected yield of the thermonuclear device to be tested during the forthcoming Fall 1952 operation (Operation IVY) was between one and 15 megatons. He also stated that up until the spring of 1951, the only thought given to compressing fusionable isotopes was by means of high explosives only, and that the resultant ALARM CLOCK design would be prohibitively large and heavy.

At this meeting, Colonel Kenneth Fields, then Director of the AEC Division of Military Application, noted that the "real" design and engineering on the IVY test device was expected to commence in the summer of 1952.

Later during the meeting, Dean stated that while the Fall 1952 test device was expected to weigh around 50 tons, the ALARM CLOCK (the TX-14), as then envisioned, was expected to be deliverable in existing aircraft and weigh around 10 tons and be about five feet in diameter.

Dean also noted that the 1952 test device would require "a specially fabricated core," a "very special fission bomb" that contained explosives with no heavy elements in them (presumably to maximize x-ray output). The uranium castings from which the fusion tamper would be constructed "were the biggest that have ever been made."

When asked about the relative efficiency or advantage of liquid-fueled thermonuclear weapons versus solid-fueled bombs, Smyth said:

Well, the enormous advantage of the solid is in terms of -- once you get it fabricated -- in terms of deliverability, you don't have to have a hydrogen liquefier along with it. But there are complications in getting the right solid material with the right elements in it to make a solid one go.

In other words, solid is what you want, but we aren't able to make it just yet.

So far, about \$30 million had been spent on research and development of the forthcoming thermonuclear test device, and another \$30 million was expected to be spent testing it at Eniwetok. The lithium-6 production program for solid-fueled weapons might involve yet another \$30 million.⁶³⁶

Early in March, the CIA revised slightly its estimate of Soviet thermonuclear weapons research:

There is no evidence that the Soviets have tested thermonuclear weapons or that their program is being directed toward this end. However, if the new approaches to the thermonuclear weapon problem currently being investigated in (the U.S.) are proven feasible, the capability on the part of the USSR to produce such weapons ... must be recognized.⁶³⁷

Near the end of the month, in a misleading memorandum that read like it could have been written by Edward Teller himself in support of a second nuclear weapons laboratory, the Secretaries of the Army, Navy, and Air Force urged the Secretary of Defense to issue a directive to the AEC to speed up work on the hydrogen bomb, and in particular, to establish a "second thermonuclear weapons laboratory:"

⁶³⁶"Status of Hydrogen Project," transcript of meeting of the Joint Committee on Atomic Energy, February 21, 1952.

⁶³⁷Scientific Intelligence Report, Joint Atomic Energy Intelligence Committee, Status of the Soviet Atomic Energy Program, CIA/SI 118-51, Central Intelligence Agency, Office of Scientific Intelligence, 6 March 1952, p. 2.

On 23 November 1949 the Joint Chiefs of Staff sent a memorandum to the Secretary of Defense in which they stated, among other things, that the possession of thermonuclear weapons by the USSR without such possession by the U.S. would be intolerable and recommended that the Atomic Energy Commission take steps to determine the feasibility of a thermonuclear weapon.

The President on January 31, 1950 wrote a letter to the Chairman of the Atomic Energy Commission directing the Commission to proceed to determine the technical feasibility of the thermonuclear weapon, the scale and rate of effort to be determined jointly by the Atomic Energy Commission and the Department of Defense, and directing that the necessary ordnance program and (delivery system) program be undertaken concurrently.

Recent developments in experimentation by our own scientists have shown that there is a serious possibility that the Russians will be successful in developing a thermonuclear weapon. We are informed by competent authorities, including those at Los Alamos, that Klaus Fuchs possessed and may be presumed to have transmitted to the Russians a full understanding of the Los Alamos thermonuclear weapon feasibility report of April 1946.

As you know, this report contained all of the essential ideas which led to the GREENHOUSE George shot in May 1951. The George shot in turn demonstrated the principle of radiation implosion which greatly increased the probability of a practical and economical thermonuclear weapon and thus precipitated our current redirected development program.

Therefore we must assume, for planning purposes, that the Russians have at their disposal all the essential ideas which led to our present thermonuclear program.

In this connection, there is persuasive evidence that the Russians are producing both enriched uranium and plutonium. This is a new development, since the first two

Russian (nuclear tests) contained plutonium only, while the third was a composite of plutonium and uranium.

The fact that the Russians have developed production of enriched uranium means that they are faced with no major production problems in developing thermonuclear weapons, since we are informed that production of lithium-6 and deuterium is simple and cheap by comparison.

In contrast to this Russian activity, the U.S. thermonuclear effort was small between the years 1946 and 1950. Action was resumed only after the first detection of the Russian fission bomb explosion. We have been informed that accordingly it is wholly possible that the Russians may be abreast of, or even ahead of, us in the development of thermonuclear weapons.

For these reasons, we recommend that you take up with the Special Committee of the National Security Council designated by the President to advise him on atomic matters the desirability of a directive to the Atomic Energy Commission to intensify its efforts in the development of thermonuclear weapons.

Specifically, we recommend that you seek the support of the Special Committee for the rapid development of a second thermonuclear weapons laboratory. We do not suggest that this be the limit of the intensification of our thermonuclear program, but rather that it be a first step.

Our broader suggestion is that the prestige and authority of the National Security Council and the President be placed (behind) a vigorous directive for the speeding up of the thermonuclear program.⁶³⁸

⁶³⁸Memorandum by the Acting Secretary of the Army (Archibald S. Alexander), the Acting Secretary of the Navy (Francis P. Whitehair), and the Secretary of the Air Force (Thomas K. Finletter) dated 27 March 1952 to Secretary of Defense Lovett. Emphasis in original.

There were a number of errors and omissions in this memo. First, the exact current state of the Soviet H-bomb program was quite unknown to the U.S. at this time. There was no evidence to support the contention that the Russians were, in fact, either "abreast of, or even ahead of" the U.S. The Soviets were in fact a long way behind the U.S., and had not yet discovered the "three concepts."

Second, the April 1946 report cited contained only speculative calculations and theory about the "classical Super," an idea which was known by scientists at LASL to be unworkable by the spring of 1951, a year before this memo was written. If by early 1952 the Soviets were pursuing the "classical Super," it would only enhance the national security of the U.S., drain the Soviet economy, and slow down the entire Soviet nuclear weapons program.

Third, the assertion that the "U.S. thermonuclear effort was small between 1946 and 1950" was technically true, but it failed to note that this was primarily due to a reduced national allocation of money and manpower to the program, resources which could only have been provided by a politically-motivated Congress and the executive branch of government.

This statement also omitted that fact that even if funding and manpower allocations had both been high between 1946 and 1950, technical constraints -- chiefly in the development of high-speed electronic computers -- would have limited and slowed the U.S. hydrogen bomb program. Even Teller himself in September 1947 had called for a two-year delay in the thermonuclear program so that necessary calculations and tests could be completed.

Finally, the memo failed to state that by this time (spring 1952) the U.S. thermonuclear program, and preparations for the first full-scale test of a workable weapon design, were proceeding about as fast as they possibly could, given the resources then available to LASL and DOD.

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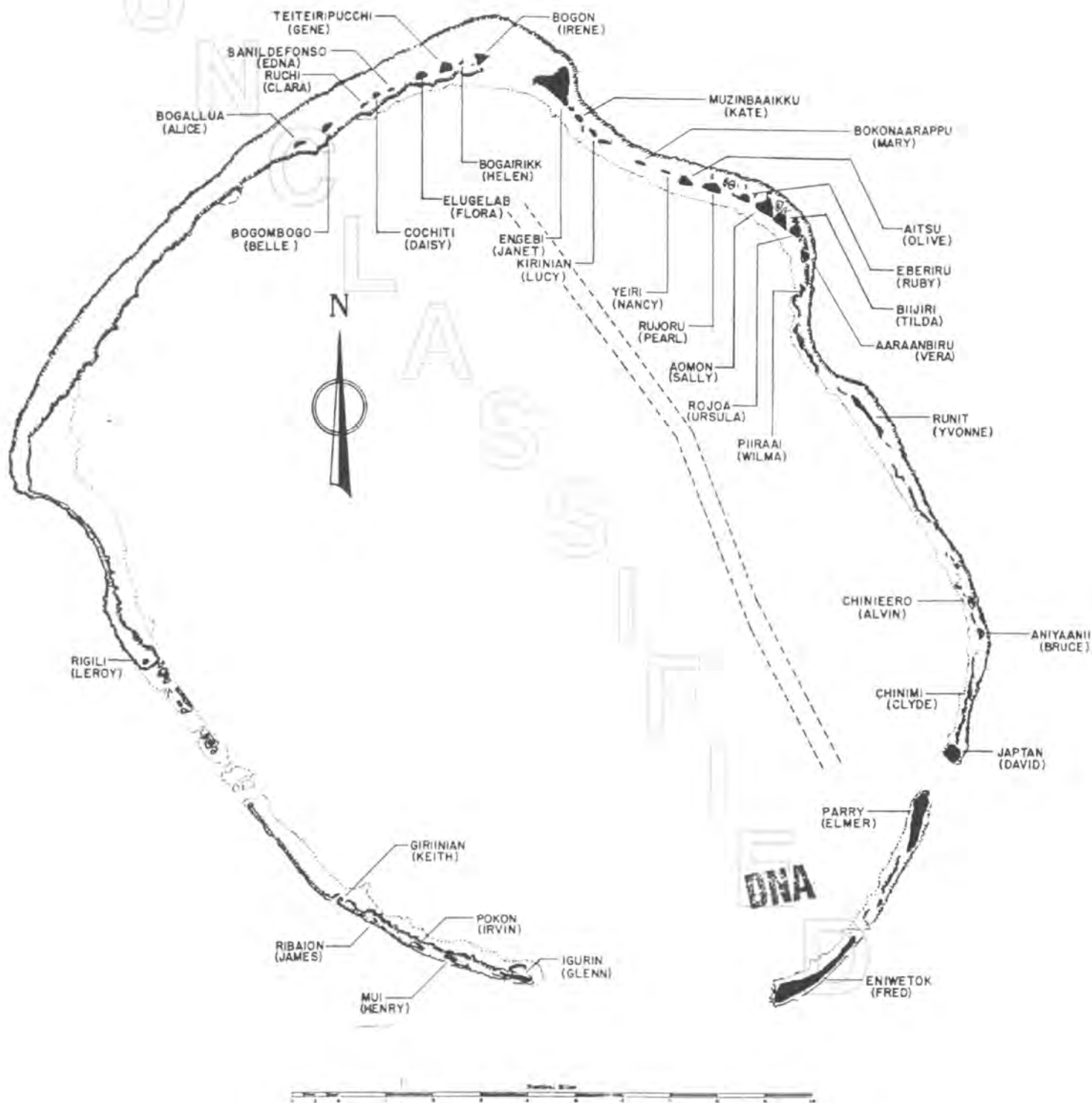
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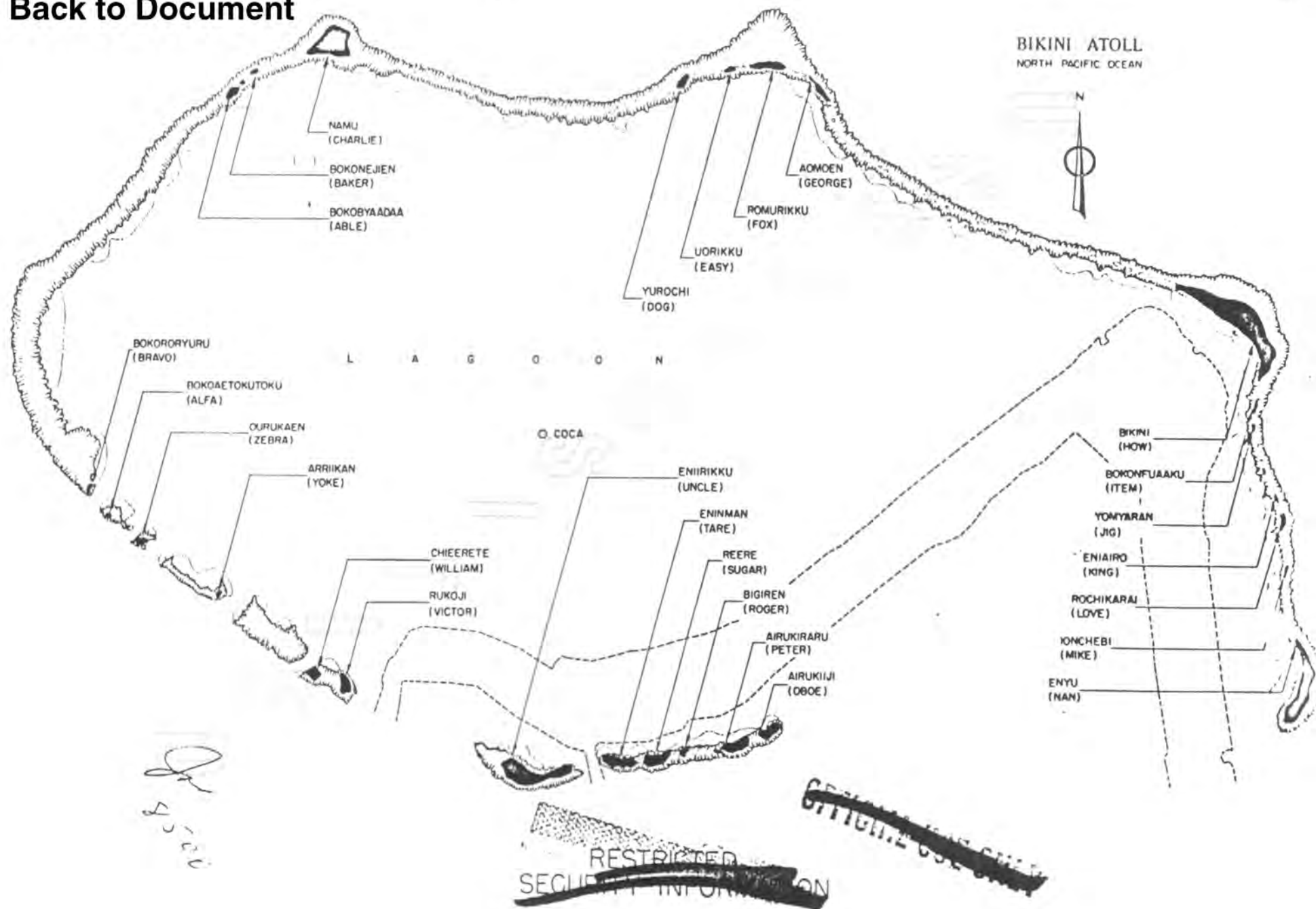
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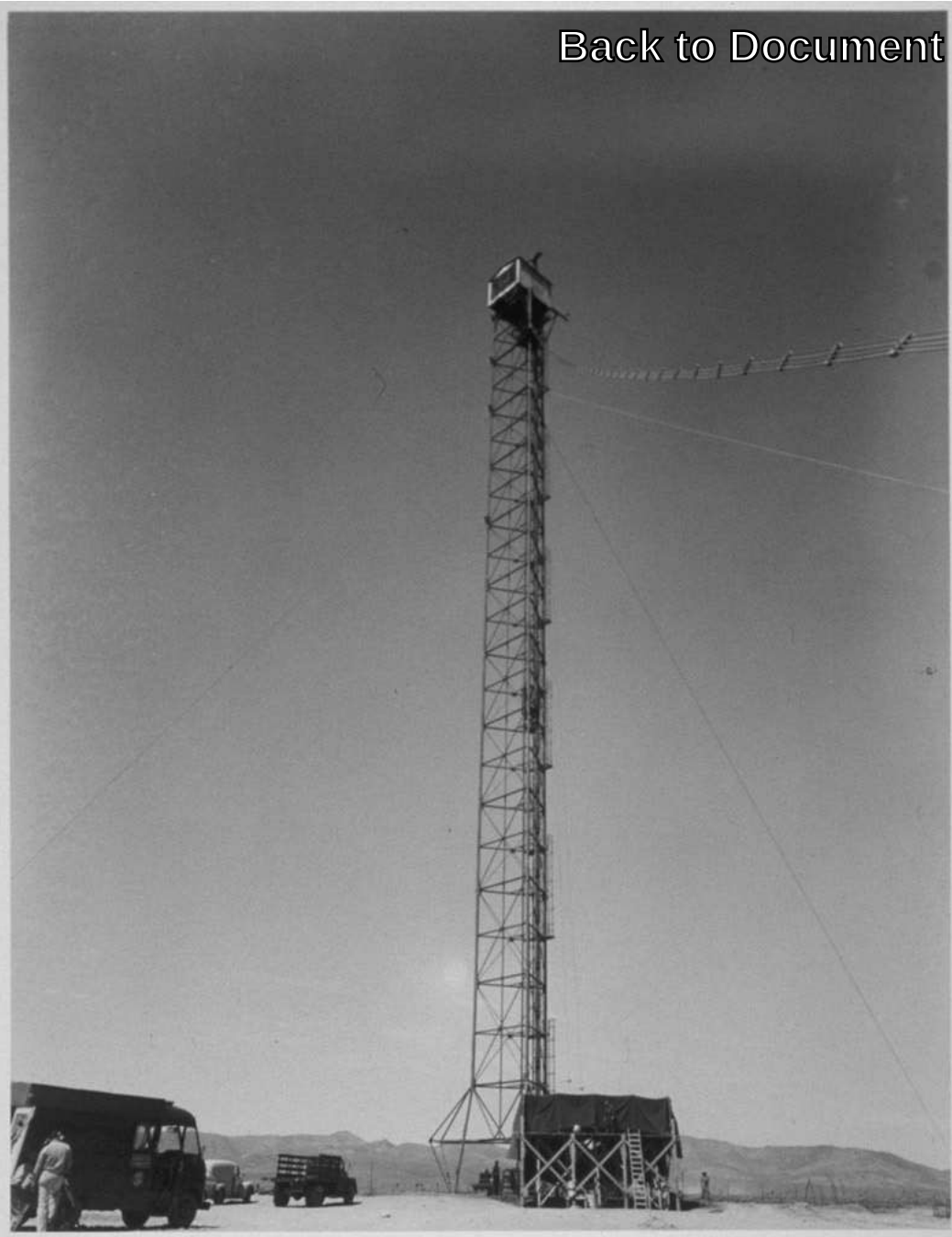
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